

Stable Vector Autoregressive Processes

In this chapter, the basic, stationary finite order vector autoregressive (VAR) model will be introduced. Some important properties will be discussed. The main uses of vector autoregressive models are forecasting and structural analysis. These two uses will be considered in Sections 2.2 and 2.3. Throughout this chapter, the model of interest is assumed to be known. Although this assumption is unrealistic in practice, it helps to see the problems related to VAR models without contamination by estimation and specification issues. The latter two aspects of an analysis will be treated in detail in subsequent chapters.

2.1 Basic Assumptions and Properties of VAR Processes

2.1.1 Stable VAR(p) Processes

The object of interest in the following is the VAR(p) model (VAR model of order p),

$$y_t = \nu + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \quad t = 0, \pm 1, \pm 2, \dots, \quad (2.1.1)$$

where $y_t = (y_{1t}, \dots, y_{Kt})'$ is a $(K \times 1)$ random vector, the A_i are fixed $(K \times K)$ coefficient matrices, $\nu = (\nu_1, \dots, \nu_K)'$ is a fixed $(K \times 1)$ vector of intercept terms allowing for the possibility of a nonzero mean $E(y_t)$. Finally, $u_t = (u_{1t}, \dots, u_{Kt})'$ is a K -dimensional *white noise* or *innovation process*, that is, $E(u_t) = 0$, $E(u_t u_t') = \Sigma_u$ and $E(u_t u_s') = 0$ for $s \neq t$. The covariance matrix Σ_u is assumed to be nonsingular if not otherwise stated.

At this stage, it may be worth thinking a little more about which process is described by (2.1.1). In order to investigate the implications of the model let us consider the VAR(1) model

$$y_t = \nu + A_1 y_{t-1} + u_t. \quad (2.1.2)$$

If this generation mechanism starts at some time $t = 1$, say, we get

$$\begin{aligned}
y_1 &= \nu + A_1 y_0 + u_1, \\
y_2 &= \nu + A_1 y_1 + u_2 = \nu + A_1(\nu + A_1 y_0 + u_1) + u_2 \\
&= (I_K + A_1)\nu + A_1^2 y_0 + A_1 u_1 + u_2, \\
&\vdots \\
y_t &= (I_K + A_1 + \cdots + A_1^{t-1})\nu + A_1^t y_0 + \sum_{i=0}^{t-1} A_1^i u_{t-i} \\
&\vdots
\end{aligned} \tag{2.1.3}$$

Hence, the vectors $y_1, \dots, y_t$ are uniquely determined by $y_0, u_1, \dots, u_t$. Also, the joint distribution of $y_1, \dots, y_t$ is determined by the joint distribution of $y_0, u_1, \dots, u_t$.

Although we will sometimes assume that a process is started in a specified period, it is often convenient to assume that it has been started in the infinite past. This assumption is in fact made in (2.1.1). What kind of process is consistent with the mechanism (2.1.1) in that case? To investigate this question we consider again the VAR(1) process (2.1.2). From (2.1.3) we have

$$\begin{aligned}
y_t &= \nu + A_1 y_{t-1} + u_t \\
&= (I_K + A_1 + \cdots + A_1^j)\nu + A_1^{j+1} y_{t-j-1} + \sum_{i=0}^j A_1^i u_{t-i}.
\end{aligned}$$

If all eigenvalues of A_1 have modulus less than 1, the sequence A_1^i , $i = 0, 1, \dots$, is absolutely summable (see Appendix A, Section A.9.1). Hence, the infinite sum

$$\sum_{i=1}^{\infty} A_1^i u_{t-i}$$

exists in mean square (Appendix C, Proposition C.9). Moreover,

$$(I_K + A_1 + \cdots + A_1^j)\nu \xrightarrow{j \rightarrow \infty} (I_K - A_1)^{-1}\nu$$

(Appendix A, Section A.9.1). Furthermore, A_1^{j+1} converges to zero rapidly as $j \rightarrow \infty$ and, thus, we ignore the term $A_1^{j+1} y_{t-j-1}$ in the limit. Hence, if all eigenvalues of A_1 have modulus less than 1, by saying that y_t is the VAR(1) process (2.1.2) we mean that y_t is the well-defined stochastic process

$$y_t = \mu + \sum_{i=0}^{\infty} A_1^i u_{t-i}, \quad t = 0, \pm 1, \pm 2, \dots, \tag{2.1.4}$$

where

$$\mu := (I_K - A_1)^{-1}\nu.$$

The distributions and joint distributions of the y_t 's are uniquely determined by the distributions of the u_t process. From Appendix C.3, Proposition C.10, the first and second moments of the y_t process are seen to be

$$E(y_t) = \mu \quad \text{for all } t \quad (2.1.5)$$

and

$$\begin{aligned} \Gamma_y(h) &:= E(y_t - \mu)(y_{t-h} - \mu)' \\ &= \lim_{n \rightarrow \infty} \sum_{i=0}^n \sum_{j=0}^n A_1^i E(u_{t-i} u'_{t-h-j}) (A_1^j)' \\ &= \lim_{i=0}^n A_1^{h+i} \Sigma_u A_1^{i'} = \sum_{i=0}^{\infty} A_1^{h+i} \Sigma_u A_1^{i'}, \end{aligned} \quad (2.1.6)$$

because $E(u_t u'_s) = 0$ for $s \neq t$ and $E(u_t u'_t) = \Sigma_u$ for all t .

Because the condition for the eigenvalues of the matrix A_1 is of importance, we call a VAR(1) process *stable* if all eigenvalues of A_1 have modulus less than 1. By Rule (7) of Appendix A.6, the condition is equivalent to

$$\det(I_K - A_1 z) \neq 0 \quad \text{for } |z| \leq 1. \quad (2.1.7)$$

It is perhaps worth pointing out that the process y_t for $t = 0, \pm 1, \pm 2, \dots$ may also be defined if the *stability condition* (2.1.7) is not satisfied. We will not do so here because we will always assume stability of processes defined for all $t \in \mathbb{Z}$.

The previous discussion can be extended easily to VAR(p) processes with $p > 1$ because any VAR(p) process can be written in VAR(1) form. More precisely, if y_t is a VAR(p) as in (2.1.1), a corresponding Kp -dimensional VAR(1)

$$Y_t = \nu + \mathbf{A}Y_{t-1} + U_t \quad (2.1.8)$$

can be defined, where

$$\begin{aligned} Y_t &:= \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}, \quad \nu := \begin{bmatrix} \nu \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \\ &\quad \quad \quad (Kp \times 1) \quad \quad \quad (Kp \times 1) \\ \mathbf{A} &:= \begin{bmatrix} A_1 & A_2 & \dots & A_{p-1} & A_p \\ I_K & 0 & \dots & 0 & 0 \\ 0 & I_K & & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & I_K & 0 \end{bmatrix}, \quad U_t := \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \\ &\quad \quad \quad (Kp \times Kp) \quad \quad \quad (Kp \times 1) \end{aligned}$$

Following the foregoing discussion, Y_t is *stable* if

$$\det(I_{Kp} - \mathbf{A}z) \neq 0 \quad \text{for } |z| \leq 1. \quad (2.1.9)$$

Its mean vector is

$$\boldsymbol{\mu} := E(Y_t) = (I_{Kp} - \mathbf{A})^{-1} \boldsymbol{\nu}$$

and the autocovariances are

$$\Gamma_Y(h) = \sum_{i=0}^{\infty} \mathbf{A}^{h+i} \Sigma_U (\mathbf{A}^i)', \quad (2.1.10)$$

where $\Sigma_U := E(U_t U_t')$. Using the $(K \times Kp)$ matrix

$$J := [I_K : 0 : \cdots : 0], \quad (2.1.11)$$

the process y_t is obtained as $y_t = JY_t$. Because Y_t is a well-defined stochastic process, the same is true for y_t . Its mean is $E(y_t) = J\boldsymbol{\mu}$ which is constant for all t and the autocovariances $\Gamma_y(h) = J\Gamma_Y(h)J'$ are also time invariant.

It is easy to see that

$$\det(I_{Kp} - \mathbf{A}z) = \det(I_K - A_1 z - \cdots - A_p z^p)$$

(see Problem 2.1). Given the definition of the characteristic polynomial of a matrix, we call this polynomial the *reverse characteristic polynomial* of the VAR(p) process. Hence, the process (2.1.1) is *stable* if its reverse characteristic polynomial has no roots in and on the complex unit circle. Formally y_t is stable if

$$\det(I_K - A_1 z - \cdots - A_p z^p) \neq 0 \quad \text{for } |z| \leq 1. \quad (2.1.12)$$

This condition is called the *stability condition*.

In summary, we say that y_t is a stable VAR(p) process if (2.1.12) holds and

$$y_t = JY_t = J\boldsymbol{\mu} + J \sum_{i=0}^{\infty} \mathbf{A}^i U_{t-i}. \quad (2.1.13)$$

Because the $U_t := (u'_t, 0, \dots, 0)'$ involve the white noise process u_t , the process y_t is seen to be determined by its white noise or innovation process. Often specific assumptions regarding u_t are made which determine the process y_t by the foregoing convention. An important example is the assumption that u_t is *Gaussian white noise*, that is, $u_t \sim \mathcal{N}(0, \Sigma_u)$ for all t and u_t and u_s are independent for $s \neq t$. In that case, it can be shown that y_t is a *Gaussian process*, that is, subcollections $y_t, \dots, y_{t+h}$ have multivariate normal distributions for all t and h .

The condition (2.1.12) provides an easy tool for checking the stability of a VAR process. Consider, for instance, the three-dimensional VAR(1) process

$$y_t = \nu + \begin{bmatrix} .5 & 0 & 0 \\ .1 & .1 & .3 \\ 0 & .2 & .3 \end{bmatrix} y_{t-1} + u_t. \quad (2.1.14)$$

For this process the reverse characteristic polynomial is

$$\begin{aligned} \det \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} .5 & 0 & 0 \\ .1 & .1 & .3 \\ 0 & .2 & .3 \end{bmatrix} z \right) \\ = \det \begin{bmatrix} 1 - .5z & 0 & 0 \\ -.1z & 1 - .1z & -.3z \\ 0 & -.2z & 1 - .3z \end{bmatrix} \\ = (1 - .5z)(1 - .4z - .03z^2). \end{aligned}$$

The roots of this polynomial are easily seen to be

$$z_1 = 2, \quad z_2 = 2.1525, \quad z_3 = -15.4858.$$

They are obviously all greater than 1 in absolute value. Therefore the process (2.1.14) is stable.

As another example consider the bivariate (two-dimensional) VAR(2) process

$$y_t = \nu + \begin{bmatrix} .5 & .1 \\ .4 & .5 \end{bmatrix} y_{t-1} + \begin{bmatrix} 0 & 0 \\ .25 & 0 \end{bmatrix} y_{t-2} + u_t. \quad (2.1.15)$$

Its reverse characteristic polynomial is

$$\det \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} .5 & .1 \\ .4 & .5 \end{bmatrix} z - \begin{bmatrix} 0 & 0 \\ .25 & 0 \end{bmatrix} z^2 \right) = 1 - z + .21z^2 - .025z^3.$$

The roots of this polynomial are

$$z_1 = 1.3, \quad z_2 = 3.55 + 4.26i, \quad \text{and} \quad z_3 = 3.55 - 4.26i.$$

Here $i := \sqrt{-1}$ denotes the imaginary unit. Note that the modulus of z_2 and z_3 is $|z_2| = |z_3| = \sqrt{3.55^2 + 4.26^2} = 5.545$. Thus, the process (2.1.15) satisfies the stability condition (2.1.12) because all roots are outside the unit circle. Although the roots for higher dimensional and higher order processes are often difficult to compute by hand, efficient computer programs exist that do the job.

To understand the implications of the stability assumption, it may be helpful to visualize time series generated by stable processes and contrast them with realizations from unstable VAR processes. In Figure 2.1 three pairs

of time series generated by three different stable bivariate (two-dimensional) VAR processes are depicted. Although they differ considerably, a common feature is that they fluctuate around constant means and their variability (variance) does not change as they wander along. In contrast, the pairs of series plotted in Figures 2.2 and 2.3 are generated by unstable, bivariate VAR processes. The time series in Figure 2.2 have a trend and those in Figure 2.3 exhibit quite pronounced seasonal fluctuations. Both shapes are typical of certain instabilities although they are quite common in practice. Hence, the stability assumption excludes many series of practical interest. We shall therefore discuss unstable processes in more detail in Part II. For that analysis understanding the stable case first is helpful.

2.1.2 The Moving Average Representation of a VAR Process

In the previous subsection we have considered the VAR(1) representation

$$Y_t = \boldsymbol{\nu} + \mathbf{A}Y_{t-1} + U_t$$

of the VAR(p) process (2.1.1). Under the stability assumption, the process Y_t has a representation

$$Y_t = \boldsymbol{\mu} + \sum_{i=0}^{\infty} \mathbf{A}^i U_{t-i}. \quad (2.1.16)$$

This form of the process is called the *moving average (MA) representation*, where Y_t is expressed in terms of past and present error or innovation vectors U_t and the mean term $\boldsymbol{\mu}$. This representation can be used to determine the autocovariances of Y_t and the mean and autocovariances of y_t can be obtained as outlined in Section 2.1.1. Moreover, an MA representation of y_t can be found by premultiplying (2.1.16) by the $(K \times Kp)$ matrix $J := [I_K : 0 : \cdots : 0]$ (defined in (2.1.11)),

$$\begin{aligned} y_t &= JY_t = J\boldsymbol{\mu} + \sum_{i=0}^{\infty} J\mathbf{A}^i J' JU_{t-i} \\ &= \mu + \sum_{i=0}^{\infty} \Phi_i u_{t-i}. \end{aligned} \quad (2.1.17)$$

Here $\mu := J\boldsymbol{\mu}$, $\Phi_i := J\mathbf{A}^i J'$ and, due to the special structure of the white noise process U_t , we have $U_t = J'JU_t$ and $JU_t = u_t$. Because the $\mathbf{A}^i$ are absolutely summable, the same is true for the Φ_i .

Later we will also consider other MA representations of a stable VAR(p) process. The unique feature of the present representation is that the zero order coefficient matrix $\Phi_0 = I_K$ and the white noise process involved consists of the error terms u_t of the VAR representation (2.1.1). In Section 2.2.2, the u_t will be seen to be the errors of optimal forecasts made in period $t - 1$. Therefore,

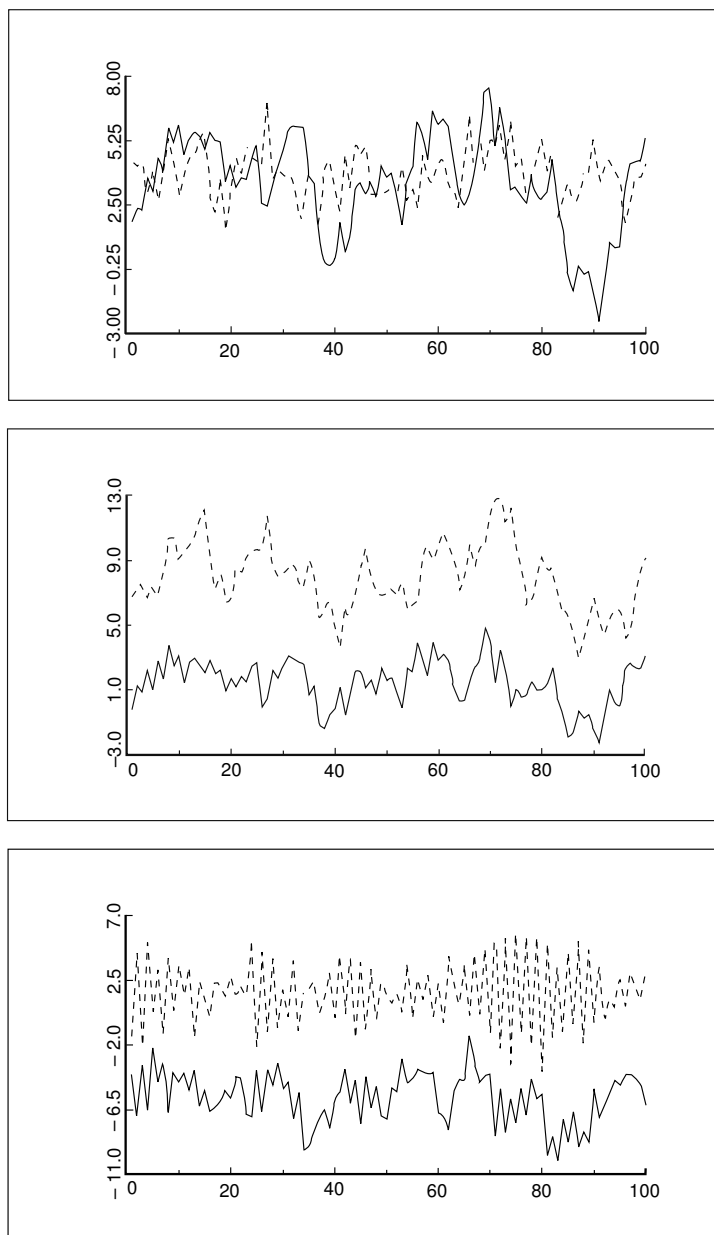


Fig. 2.1. Bivariate time series generated by stable processes.

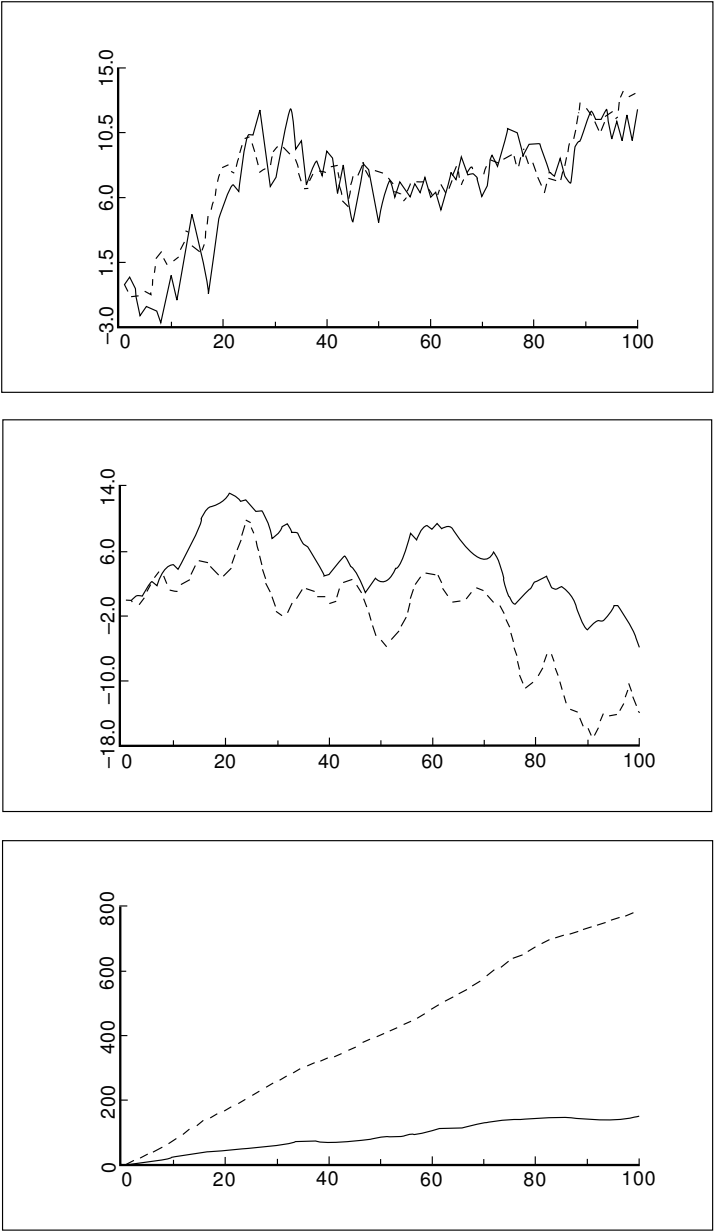


Fig. 2.2. Bivariate time series generated by unstable VAR processes.

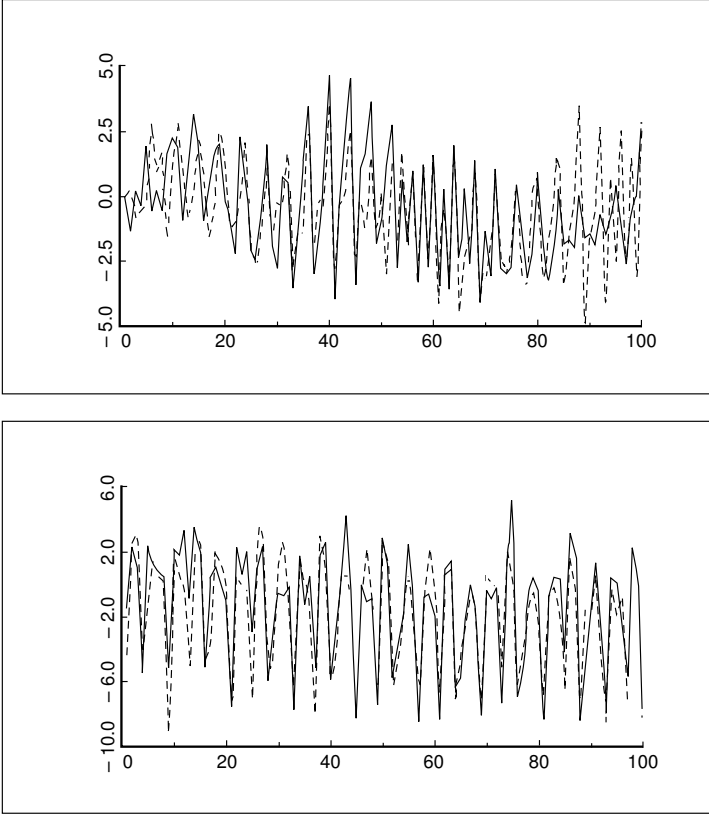


Fig. 2.3. Unstable seasonal time series.

to distinguish the present representation from other MA representations, we will sometimes refer to it as the *canonical* or *fundamental* or *prediction error representation*.

Using Proposition C.10 of Appendix C.3, the representation (2.1.17) provides a possibility for determining the mean and autocovariances of y_t :

$$E(y_t) = \mu$$

and

$$\begin{aligned}
 \Gamma_y(h) &= E[(y_t - \mu)(y_{t-h} - \mu)'] \\
 &= E \left[\left(\sum_{i=0}^{h-1} \Phi_i u_{t-i} + \sum_{i=0}^{\infty} \Phi_{h+i} u_{t-h-i} \right) \left(\sum_{i=0}^{\infty} \Phi_i u_{t-h-i} \right)' \right] \\
 &= \sum_{i=0}^{\infty} \Phi_{h+i} \Sigma_u \Phi_i'.
 \end{aligned} \tag{2.1.18}$$

There is no need to compute the MA coefficient matrices Φ_i via the VAR(1) representation corresponding to y_t as in the foregoing derivation. A more direct way for determining these matrices results from writing the VAR(p) process in *lag operator* notation. The lag operator L is defined such that $Ly_t = y_{t-1}$, that is, it lags (shifts back) the index by one period. Because of this property it is sometimes called *backshift operator*. Using this operator, (2.1.1) can be written as

$$y_t = \nu + (A_1L + \cdots + A_pL^p)y_t + u_t$$

or

$$A(L)y_t = \nu + u_t, \quad (2.1.19)$$

where

$$A(L) := I_K - A_1L - \cdots - A_pL^p.$$

Let

$$\Phi(L) := \sum_{i=0}^{\infty} \Phi_i L^i$$

be an operator such that

$$\Phi(L)A(L) = I_K. \quad (2.1.20)$$

Premultiplying (2.1.19) by $\Phi(L)$ gives

$$\begin{aligned} y_t &= \Phi(L)\nu + \Phi(L)u_t \\ &= \left(\sum_{i=0}^{\infty} \Phi_i \right) \nu + \sum_{i=0}^{\infty} \Phi_i u_{t-i}. \end{aligned} \quad (2.1.21)$$

The operator $\Phi(L)$ is the *inverse* of $A(L)$ and it is therefore sometimes denoted by $A(L)^{-1}$. Generally, we call the operator $A(L)$ *invertible* if $|A(z)| \neq 0$ for $|z| \leq 1$. If this condition is satisfied, the coefficient matrices of $\Phi(L) = A(L)^{-1}$ are absolutely summable and, hence, the process $\Phi(L)u_t = A(L)^{-1}u_t$ is well-defined (see Appendix C.3). The coefficient matrices Φ_i can be obtained from (2.1.20) using the relations

$$\begin{aligned} I_K &= (\Phi_0 + \Phi_1L + \Phi_2L^2 + \cdots)(I_K - A_1L - \cdots - A_pL^p) \\ &= \Phi_0 + (\Phi_1 - \Phi_0A_1)L + (\Phi_2 - \Phi_1A_1 - \Phi_0A_2)L^2 + \cdots \\ &\quad + \left(\Phi_i - \sum_{j=1}^i \Phi_{i-j}A_j \right) L^i + \cdots \end{aligned}$$

or

$$\begin{aligned}
I_K &= \Phi_0 \\
0 &= \Phi_1 - \Phi_0 A_1 \\
0 &= \Phi_2 - \Phi_1 A_1 - \Phi_0 A_2 \\
&\vdots \\
0 &= \Phi_i - \sum_{j=1}^i \Phi_{i-j} A_j \\
&\vdots
\end{aligned}$$

where $A_j = 0$ for $j > p$. Hence, the Φ_i can be computed recursively using

$$\begin{aligned}
\Phi_0 &= I_K, \\
\Phi_i &= \sum_{j=1}^i \Phi_{i-j} A_j, \quad i = 1, 2, \dots
\end{aligned} \tag{2.1.22}$$

The mean μ of y_t can be obtained as follows:

$$\mu = \Phi(1)\nu = A(1)^{-1}\nu = (I_K - A_1 - \dots - A_p)^{-1}\nu. \tag{2.1.23}$$

For a VAR(1) process, the recursions (2.1.22) imply that $\Phi_0 = I_K$, $\Phi_1 = A_1$, ..., $\Phi_i = A_1^i$, This result is in line with (2.1.4). For the example VAR(1) process (2.1.14), we get $\Phi_0 = I_3$,

$$\begin{aligned}
\Phi_1 &= \begin{bmatrix} .5 & 0 & 0 \\ .1 & .1 & .3 \\ 0 & .2 & .3 \end{bmatrix}, \quad \Phi_2 = \begin{bmatrix} .25 & 0 & 0 \\ .06 & .07 & .12 \\ .02 & .08 & .15 \end{bmatrix}, \\
\Phi_3 &= \begin{bmatrix} .125 & 0 & 0 \\ .037 & .031 & .057 \\ .018 & .038 & .069 \end{bmatrix},
\end{aligned} \tag{2.1.24}$$

etc. For a VAR(2), the recursions (2.1.22) result in

$$\begin{aligned}
\Phi_1 &= A_1 \\
\Phi_2 &= \Phi_1 A_1 + A_2 = A_1^2 + A_2 \\
\Phi_3 &= \Phi_2 A_1 + \Phi_1 A_2 = A_1^3 + A_2 A_1 + A_1 A_2 \\
&\vdots \\
\Phi_i &= \Phi_{i-1} A_1 + \Phi_{i-2} A_2 \\
&\vdots
\end{aligned}$$

Thus, for the example VAR(2) process (2.1.15), we get the MA coefficient matrices $\Phi_0 = I_2$,

$$\Phi_1 = \begin{bmatrix} .5 & .1 \\ .4 & .5 \end{bmatrix}, \quad \Phi_2 = \begin{bmatrix} .29 & .1 \\ .65 & .29 \end{bmatrix}, \quad \Phi_3 = \begin{bmatrix} .21 & .079 \\ .566 & .21 \end{bmatrix}, \tag{2.1.25}$$

etc. For both example processes, the Φ_i matrices approach zero as $i \rightarrow \infty$. This property is a consequence of the stability of the two processes.

It may be worth noting that the MA representation of a stable VAR(p) process is not necessarily of infinite order. That is, the Φ_i may all be zero for i greater than some finite integer q . For instance, for the bivariate VAR(1)

$$y_t = \nu + \begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix} y_{t-1} + u_t,$$

the MA representation is easily seen to be

$$y_t = \mu + u_t + \begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix} u_{t-1},$$

because

$$\begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}^i = 0$$

for $i > 1$.

2.1.3 Stationary Processes

A stochastic process is *stationary* if its first and second moments are time invariant. In other words, a stochastic process y_t is stationary if

$$E(y_t) = \mu \quad \text{for all } t \quad (2.1.26a)$$

and

$$E[(y_t - \mu)(y_{t-h} - \mu)'] = \Gamma_y(h) = \Gamma_y(-h)' \quad \text{for all } t \text{ and } h = 0, 1, 2, \dots \quad (2.1.26b)$$

Condition (2.1.26a) means that all y_t have the same finite mean vector μ and (2.1.26b) requires that the autocovariances of the process do not depend on t but just on the time period h the two vectors y_t and y_{t-h} are apart. Note that, if not otherwise stated, all quantities are assumed to be finite. For instance, μ is a vector of finite mean terms and $\Gamma_y(h)$ is a matrix of finite covariances. Other definitions of stationarity are often used in the literature. For example, the joint distribution of n consecutive vectors may be assumed to be time invariant for all n . We shall, however, use the foregoing definition in the following. We call a process *strictly stationary* if the joint distributions of n consecutive variables are time invariant and there is a reason to distinguish between our notion of stationarity and the stricter form. By our definition, the white noise process u_t used in (2.1.1) is an obvious example of a stationary process. Also, from (2.1.18) we know that a stable VAR(p) process is stationary. We state this fact as a proposition.

Proposition 2.1 (*Stationarity Condition*)

A stable VAR(p) process y_t , $t = 0, \pm 1, \pm 2, \dots$, is stationary. ■

Because stability implies stationarity, the stability condition (2.1.12) is often referred to as *stationarity condition* in the time series literature. The converse of Proposition 2.1 is not true. In other words, an unstable process is not necessarily nonstationary. Because unstable stationary processes are not of interest in the following, we will not discuss this possibility here.

At this stage, it may be worth thinking about the generality of the VAR(p) processes considered in this and many other chapters. In this context, an important result due to Wold (1938) is of interest. He has shown that every stationary process x_t can be written as the sum of two uncorrelated processes z_t and y_t ,

$$x_t = z_t + y_t,$$

where z_t is a deterministic process that can be forecast perfectly from its own past and y_t is a process with MA representation

$$y_t = \sum_{i=0}^{\infty} \Phi_i u_{t-i}, \quad (2.1.27)$$

where $\Phi_0 = I_K$, the u_t constitute a white noise process and the infinite sum is defined as a limit in mean square although the Φ_i are not necessarily absolutely summable (Hannan (1970, Chapter III)). The term “deterministic” will be explained more formally in Section 2.2. This result is often called *Wold’s Decomposition Theorem*. If we assume that in the system of interest the only deterministic component is the mean term, the theorem states that the system has an MA representation. Suppose the Φ_i are absolutely summable and there exists an operator $A(L)$ with absolutely summable coefficient matrices satisfying $A(L)\Phi(L) = I_K$. Then $\Phi(L)$ is invertible ($A(L) = \Phi(L)^{-1}$) and y_t has a VAR representation of possibly infinite order,

$$y_t = \sum_{i=1}^{\infty} A_i y_{t-i} + u_t, \quad (2.1.28)$$

where

$$A(z) := I_K - \sum_{i=1}^{\infty} A_i z^i = \left(\sum_{i=0}^{\infty} \Phi_i z^i \right)^{-1} \quad \text{for } |z| \leq 1.$$

The A_i can be obtained from the Φ_i by recursions similar to (2.1.22).

The absolute summability of the A_i implies that the VAR coefficient matrices converge to zero rapidly. In other words, under quite general conditions, every stationary, purely nondeterministic process (a process without a deterministic component) can be approximated well by a finite order VAR process.

This is a very powerful result which demonstrates the generality of the processes under study. Note that economic variables can rarely be predicted without error. Thus the assumption of having a nondeterministic system except perhaps for a mean term is not a very restrictive one. The crucial and restrictive condition is the stationarity of the system, however. We will consider nonstationary processes later. For that discussion it is useful to understand the stationary case first.

An important implication of Wold's Decomposition Theorem is worth noting at this point. The theorem implies that any *subprocess* of a purely nondeterministic, stationary process y_t consisting of any subset of the components of y_t also has an MA representation. Suppose, for instance, that interest centers on the first M components of the K -dimensional process y_t , that is, we are interested in $x_t = Fy_t$, where $F = [I_M : 0]$ is an $(M \times K)$ matrix. Then $E(x_t) = FE(y_t) = F\mu$ and $\Gamma_x(h) = F\Gamma_y(h)F'$ and, thus, x_t is stationary. Application of Wold's theorem then implies that x_t has an MA representation.

2.1.4 Computation of Autocovariances and Autocorrelations of Stable VAR Processes

Although the autocovariances of a stationary, stable VAR(p) process can be given in terms of its MA coefficient matrices as in (2.1.18), that formula is unattractive in practice, because it involves an infinite sum. For practical purposes it is easier to compute the autocovariances directly from the VAR coefficient matrices. In this section, we will develop the relevant formulas.

Autocovariances of a VAR(1) Process

In order to illustrate the computation of the autocovariances when the process coefficients are given, suppose that y_t is a stationary, stable VAR(1) process

$$y_t = \nu + A_1 y_{t-1} + u_t$$

with white noise covariance matrix $E(u_t u_t') = \Sigma_u$. Alternatively, the process may be written in mean-adjusted form as

$$y_t - \mu = A_1(y_{t-1} - \mu) + u_t, \quad (2.1.29)$$

where $\mu = E(y_t)$, as before. Postmultiplying by $(y_{t-h} - \mu)'$ and taking expectations gives

$$E[(y_t - \mu)(y_{t-h} - \mu)'] = A_1 E[(y_{t-1} - \mu)(y_{t-h} - \mu)'] + E[u_t(y_{t-h} - \mu)'].$$

Thus, for $h = 0$,

$$\Gamma_y(0) = A_1 \Gamma_y(-1) + \Sigma_u = A_1 \Gamma_y(1)' + \Sigma_u \quad (2.1.30)$$

and for $h > 0$,

$$\Gamma_y(h) = A_1 \Gamma_y(h-1). \quad (2.1.31)$$

These equations are usually referred to as *Yule-Walker equations*. If A_1 and the covariance matrix $\Gamma_y(0) = \Sigma_y$ of y_t are known, the $\Gamma_y(h)$ can be computed recursively using (2.1.31).

If A_1 and Σ_u are given, $\Gamma_y(0)$ can be determined as follows. For $h = 1$, we get from (2.1.31), $\Gamma_y(1) = A_1 \Gamma_y(0)$. Substituting $A_1 \Gamma_y(0)$ for $\Gamma_y(1)$ in (2.1.30) gives

$$\Gamma_y(0) = A_1 \Gamma_y(0) A_1' + \Sigma_u$$

or

$$\begin{aligned} \text{vec } \Gamma_y(0) &= \text{vec}(A_1 \Gamma_y(0) A_1') + \text{vec } \Sigma_u \\ &= (A_1 \otimes A_1) \text{vec } \Gamma_y(0) + \text{vec } \Sigma_u. \end{aligned}$$

(For the definition of the Kronecker product $\otimes$, the vec operator and the rules used here, see Appendix A). Hence,

$$\text{vec } \Gamma_y(0) = (I_{K^2} - A_1 \otimes A_1)^{-1} \text{vec } \Sigma_u. \quad (2.1.32)$$

Note that the invertibility of $I_{K^2} - A_1 \otimes A_1$ follows from the stability of y_t because the eigenvalues of $A_1 \otimes A_1$ are the products of the eigenvalues of A_1 (see Appendix A). Hence, the eigenvalues of $A_1 \otimes A_1$ have modulus less than 1. Consequently, $\det(I_{K^2} - A_1 \otimes A_1) \neq 0$ (see Appendix A.9.1).

Using, for instance,

$$\Sigma_u = \begin{bmatrix} 2.25 & 0 & 0 \\ 0 & 1.0 & .5 \\ 0 & .5 & .74 \end{bmatrix}, \quad (2.1.33)$$

we get for the example process (2.1.14),

$$\begin{aligned} \text{vec } \Gamma_y(0) &= (I_9 - A_1 \otimes A_1)^{-1} \text{vec } \Sigma_u \\ &= \begin{bmatrix} .75 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -.05 & .95 & -.15 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -.10 & .85 & 0 & 0 & 0 & 0 & 0 & 0 \\ -.05 & 0 & 0 & .95 & 0 & 0 & -.15 & 0 & 0 \\ -.01 & -.01 & -.03 & -.01 & .99 & -.03 & -.03 & -.03 & -.09 \\ 0 & -.02 & -.03 & 0 & -.02 & .97 & 0 & -.06 & -.09 \\ 0 & 0 & 0 & -.01 & 0 & 0 & .85 & 0 & 0 \\ 0 & 0 & 0 & -.02 & -.02 & -.06 & -.03 & .97 & -.09 \\ 0 & 0 & 0 & 0 & -.04 & -.06 & 0 & -.06 & .91 \end{bmatrix}^{-1} \begin{bmatrix} 2.25 \\ 0 \\ 0 \\ 0 \\ 1.0 \\ .5 \\ 0 \\ .5 \\ .74 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 3.000 \\ .161 \\ .019 \\ .161 \\ 1.172 \\ .674 \\ .019 \\ .674 \\ .954 \end{bmatrix}.$$

It follows that

$$\begin{aligned} \Gamma_y(0) &= \begin{bmatrix} 3.000 & .161 & .019 \\ .161 & 1.172 & .674 \\ .019 & .674 & .954 \end{bmatrix}, \\ \Gamma_y(1) &= A_1 \Gamma_y(0) = \begin{bmatrix} 1.500 & .080 & .009 \\ .322 & .335 & .355 \\ .038 & .437 & .421 \end{bmatrix}, \\ \Gamma_y(2) &= A_1 \Gamma_y(1) = \begin{bmatrix} .750 & .040 & .005 \\ .194 & .173 & .163 \\ .076 & .198 & .197 \end{bmatrix}. \end{aligned} \tag{2.1.34}$$

Note that the results are rounded after the computation. A higher precision has been used in intermediate steps.

Autocovariances of a Stable VAR(p) Process

For a higher order VAR(p) process,

$$y_t - \mu = A_1(y_{t-1} - \mu) + \cdots + A_p(y_{t-p} - \mu) + u_t, \tag{2.1.35}$$

the *Yule-Walker equations* are also obtained by postmultiplying with $(y_{t-h} - \mu)'$ and taking expectations. For $h = 0$, using $\Gamma_y(i) = \Gamma_y(-i)'$,

$$\begin{aligned} \Gamma_y(0) &= A_1 \Gamma_y(-1) + \cdots + A_p \Gamma_y(-p) + \Sigma_u \\ &= A_1 \Gamma_y(1)' + \cdots + A_p \Gamma_y(p)' + \Sigma_u, \end{aligned} \tag{2.1.36}$$

and for $h > 0$,

$$\Gamma_y(h) = A_1 \Gamma_y(h-1) + \cdots + A_p \Gamma_y(h-p). \tag{2.1.37}$$

These equations may be used to compute the $\Gamma_y(h)$ recursively for $h \geq p$, if $A_1, \dots, A_p$ and $\Gamma_y(p-1), \dots, \Gamma_y(0)$ are known.

The initial autocovariance matrices for $|h| < p$ can be determined using the VAR(1) process that corresponds to (2.1.35),

$$Y_t - \mu = \mathbf{A}(Y_{t-1} - \mu) + U_t, \tag{2.1.38}$$

where Y_t , $\mathbf{A}$, and U_t are as in (2.1.8) and $\boldsymbol{\mu} := (\mu', \dots, \mu')' = E(Y_t)$. Proceeding as in the VAR(1) case gives

$$\Gamma_Y(0) = \mathbf{A}\Gamma_Y(0)\mathbf{A}' + \Sigma_U,$$

where $\Sigma_U = E(U_t U_t')$ and

$$\begin{aligned} \Gamma_Y(0) &= E \left(\begin{bmatrix} y_t - \mu \\ \vdots \\ y_{t-p+1} - \mu \end{bmatrix} [(y_t - \mu)', \dots, (y_{t-p+1} - \mu)'] \right) \\ &= \begin{bmatrix} \Gamma_y(0) & \Gamma_y(1) & \dots & \Gamma_y(p-1) \\ \Gamma_y(-1) & \Gamma_y(0) & \dots & \Gamma_y(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma_y(-p+1) & \Gamma_y(-p+2) & \dots & \Gamma_y(0) \end{bmatrix}. \end{aligned}$$

Thus, the $\Gamma_y(h)$, $h = -p+1, \dots, p-1$, are obtained from

$$\text{vec } \Gamma_Y(0) = (I_{(Kp)^2} - \mathbf{A} \otimes \mathbf{A})^{-1} \text{vec } \Sigma_U. \quad (2.1.39)$$

For instance, for the example VAR(2) process (2.1.15) we get

$$\mathbf{A} = \begin{bmatrix} .5 & .1 & 0 & 0 \\ .4 & .5 & .25 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (2.1.40)$$

and, assuming

$$\Sigma_u = \begin{bmatrix} .09 & 0 \\ 0 & .04 \end{bmatrix}, \quad (2.1.41)$$

we have

$$\Sigma_U = \begin{bmatrix} \Sigma_u & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} .09 & 0 & 0 & 0 \\ 0 & .04 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Hence, using (2.1.39) and

$$\Gamma_Y(0) = \begin{bmatrix} \Gamma_y(0) & \Gamma_y(1) \\ \Gamma_y(1)' & \Gamma_y(0) \end{bmatrix}$$

gives

$$\Gamma_y(0) = \begin{bmatrix} .131 & .066 \\ .066 & .181 \end{bmatrix}, \quad \Gamma_y(1) = \begin{bmatrix} .072 & .051 \\ .104 & .143 \end{bmatrix},$$

$$\Gamma_y(2) = A_1\Gamma_y(1) + A_2\Gamma_y(0) = \begin{bmatrix} .046 & .040 \\ .113 & .108 \end{bmatrix}, \quad (2.1.42)$$

$$\Gamma_y(3) = A_1\Gamma_y(2) + A_2\Gamma_y(1) = \begin{bmatrix} .035 & .031 \\ .093 & .083 \end{bmatrix},$$

and so on. A method for computing $\Gamma_y(0)$ without explicitly inverting $(I - \mathbf{A} \otimes \mathbf{A})$ is given by Barone (1987).

The *autocovariance function* of a stationary VAR(p) process is *positive semidefinite*, that is,

$$\begin{aligned} & \sum_{j=0}^n \sum_{i=0}^n a'_j \Gamma_y(i-j) a_i \\ &= (a'_0, \dots, a'_n) \begin{bmatrix} \Gamma_y(0) & \Gamma_y(1) & \dots & \Gamma_y(n) \\ \Gamma_y(-1) & \Gamma_y(0) & \dots & \Gamma_y(n-1) \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma_y(-n) & \Gamma_y(-n+1) & \dots & \Gamma_y(0) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} \geq 0 \end{aligned} \quad (2.1.43)$$

for any $n \geq 0$. Here the a_i are arbitrary $(K \times 1)$ vectors. This result follows because (2.1.43) is just the variance of

$$(a'_0, \dots, a'_n) \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-n} \end{bmatrix}$$

which is always nonnegative.

Autocorrelations of a Stable VAR(p) Process

Because the autocovariances depend on the unit of measurement used for the variables of the system, they are sometimes difficult to interpret. Therefore, the autocorrelations

$$R_y(h) = D^{-1} \Gamma_y(h) D^{-1} \quad (2.1.44)$$

are usually more convenient to work with as they are scale invariant measures of the linear dependencies among the variables of the system. Here D is a diagonal matrix with the standard deviations of the components of y_t on the main diagonal. That is, the diagonal elements of D are the square roots of the diagonal elements of $\Gamma_y(0)$. Denoting the covariance between $y_{i,t}$ and $y_{j,t-h}$ by $\gamma_{ij}(h)$ (i.e., $\gamma_{ij}(h)$ is the ij -th element of $\Gamma_y(h)$) the diagonal elements $\gamma_{11}(0), \dots, \gamma_{KK}(0)$ of $\Gamma_y(0)$ are the variances of $y_{1t}, \dots, y_{Kt}$. Thus,

$$D^{-1} = \begin{bmatrix} 1/\sqrt{\gamma_{11}(0)} & & 0 \\ & \ddots & \\ 0 & & 1/\sqrt{\gamma_{KK}(0)} \end{bmatrix}$$

and the correlation between $y_{i,t}$ and $y_{j,t-h}$ is

$$\rho_{ij}(h) = \frac{\gamma_{ij}(h)}{\sqrt{\gamma_{ii}(0)}\sqrt{\gamma_{jj}(0)}} \quad (2.1.45)$$

which is just the ij -th element of $R_y(h)$.

For the VAR(1) example process (2.1.14) we get from (2.1.34),

$$D = \begin{bmatrix} \sqrt{3.000} & 0 & 0 \\ 0 & \sqrt{1.172} & 0 \\ 0 & 0 & \sqrt{.954} \end{bmatrix} = \begin{bmatrix} 1.732 & 0 & 0 \\ 0 & 1.083 & 0 \\ 0 & 0 & .977 \end{bmatrix}$$

and

$$\begin{aligned} R_y(0) &= D^{-1}\Gamma_y(0)D^{-1} = \begin{bmatrix} 1 & .086 & .011 \\ .086 & 1 & .637 \\ .011 & .637 & 1 \end{bmatrix}, \\ R_y(1) &= D^{-1}\Gamma_y(1)D^{-1} = \begin{bmatrix} .500 & .043 & .005 \\ .172 & .286 & .336 \\ .022 & .413 & .441 \end{bmatrix}, \\ R_y(2) &= D^{-1}\Gamma_y(2)D^{-1} = \begin{bmatrix} .250 & .021 & .003 \\ .103 & .148 & .154 \\ .045 & .187 & .206 \end{bmatrix}. \end{aligned} \quad (2.1.46)$$

A plot of some autocorrelations is shown in Figure 2.4. Assuming that the three variables of the system represent rates of change of investment, income, and consumption, respectively, it can, for instance, be seen that the contemporaneous and intertemporal correlations between consumption and investment are quite small, while the patterns of the autocorrelations of the individual series are similar.

2.2 Forecasting

We have argued in the introduction that forecasting is one of the main objectives of multiple time series analysis. Therefore, we will now discuss predictors based on VAR processes. Point forecasts and interval forecasts will be considered in turn. Before discussing particular predictors or forecasts (the two terms will be used interchangeably) we comment on the prediction problem in general.

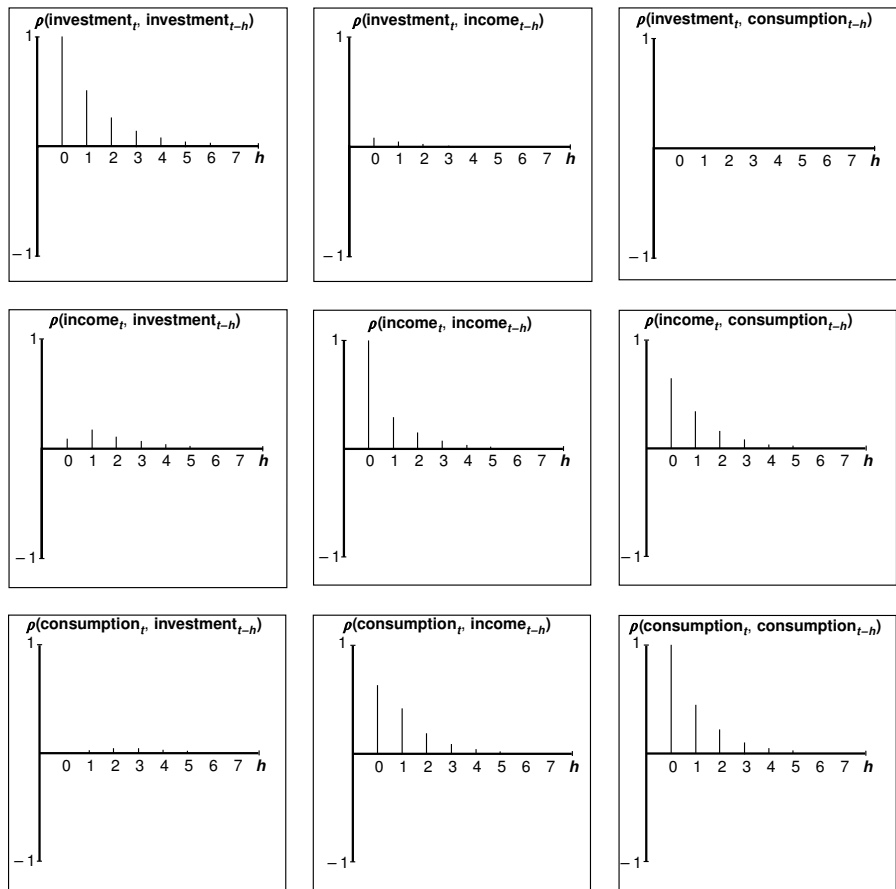


Fig. 2.4. Autocorrelations of the investment/income/consumption system.

2.2.1 The Loss Function

The forecaster usually finds himself in a situation where in a particular period t he has to make statements about the future values of variables $y_1, \dots, y_K$. For this purpose he has available a model for the data generation process and an information set, say Ω_t , containing the available information in period t . The data generation process may, for instance, be a VAR(p) process and Ω_t may contain the past and present variables of the system under consideration, that is, $\Omega_t = \{y_s | s \leq t\}$, where $y_s = (y_{1s}, \dots, y_{Ks})'$. The period t , where the forecast is made, is the *forecast origin* and the number of periods into the future for which a forecast is desired is the *forecast horizon*. A predictor, h periods ahead, is an *h -step predictor*.

If forecasts are desired for a particular purpose, a specific cost function may be associated with the forecast errors. A forecast will be optimal if it

minimizes the cost. To find a forecast that is optimal in this sense is usually too ambitious a goal to be attainable in practice. Therefore, minimizing the *expected* cost or loss is often used as an objective. In general, it will depend on the particular loss function which forecast is optimal. On the other hand, forecasts of economic variables are often published for general use. In that case, the specific cost or loss function of all potential users cannot be taken into account in computing a forecast. In this situation, the statistical properties of the forecasts and perhaps interval forecasts are of interest to enable the user to draw proper conclusions for his or her particular needs. It may also be desirable to choose the forecast such that it minimizes a wide range of plausible loss functions.

In the context of VAR models, predictors that minimize the forecast mean squared errors (MSEs) are the most widely used ones. Arguments in favor of using the MSE as loss function are given by Granger (1969b) and Granger & Newbold (1986). They show that minimum MSE forecasts also minimize a range of loss functions other than the MSE. Moreover, for many loss functions the optimal predictors are simple functions of minimum MSE predictors. Furthermore, for an unbiased predictor, the MSE is the forecast error variance which is useful in setting up interval forecasts. Therefore, minimum MSE predictors will be of major interest in the following. If not otherwise stated, the information set Ω_t is assumed to contain the variables of the system under consideration up to and including period t .

2.2.2 Point Forecasts

Conditional Expectation

Suppose $y_t = (y_{1t}, \dots, y_{Kt})'$ is a K -dimensional stable VAR(p) process as in (2.1.1). Then, the minimum MSE predictor for forecast horizon h at forecast origin t is the conditional expected value

$$E_t(y_{t+h}) := E(y_{t+h}|\Omega_t) = E(y_{t+h}|\{y_s|s \leq t\}). \quad (2.2.1)$$

This predictor minimizes the MSE of each component of y_t . In other words, if $\bar{y}_t(h)$ is any h -step predictor at origin t ,

$$\begin{aligned} \text{MSE}[\bar{y}_t(h)] &= E[(y_{t+h} - \bar{y}_t(h))(y_{t+h} - \bar{y}_t(h))'] \\ &\geq \text{MSE}[E_t(y_{t+h})] = E[(y_{t+h} - E_t(y_{t+h}))(y_{t+h} - E_t(y_{t+h}))'], \end{aligned} \quad (2.2.2)$$

where the inequality sign $\geq$ between two matrices means that the difference between the left-hand and the right-hand matrix is positive semidefinite. Equivalently, for any $(K \times 1)$ vector c ,

$$\text{MSE}[c'\bar{y}_t(h)] \geq \text{MSE}[c'E_t(y_{t+h})].$$

The optimality of the conditional expectation can be seen by noting that

$$\begin{aligned}
\text{MSE}[\bar{y}_t(h)] &= E\{[y_{t+h} - E_t(y_{t+h}) + E_t(y_{t+h}) - \bar{y}_t(h)] \\
&\quad \times [y_{t+h} - E_t(y_{t+h}) + E_t(y_{t+h}) - \bar{y}_t(h)]'\} \\
&= \text{MSE}[E_t(y_{t+h})] \\
&\quad + E\{[E_t(y_{t+h}) - \bar{y}_t(h)][E_t(y_{t+h}) - \bar{y}_t(h)]'\},
\end{aligned}$$

where $E\{[y_{t+h} - E_t(y_{t+h})][E_t(y_{t+h}) - \bar{y}_t(h)]'\} = 0$ has been used. The latter result holds because $[y_{t+h} - E_t(y_{t+h})]$ is a function of innovations after period t which are uncorrelated with the terms contained in $[E_t(y_{t+h}) - \bar{y}_t(h)]$ which are functions of y_s , $s \leq t$.

The optimality of the conditional expectation implies that

$$E_t(y_{t+h}) = \nu + A_1 E_t(y_{t+h-1}) + \cdots + A_p E_t(y_{t+h-p}) \quad (2.2.3)$$

is the optimal h -step predictor of a VAR(p) process y_t , provided u_t is *independent white noise* so that u_t and u_s are independent for $s \neq t$ and, hence, $E_t(u_{t+h}) = 0$ for $h > 0$.

The formula (2.2.3) can be used for recursively computing the h -step predictors starting with $h = 1$:

$$\begin{aligned}
E_t(y_{t+1}) &= \nu + A_1 y_t + \cdots + A_p y_{t-p+1}, \\
E_t(y_{t+2}) &= \nu + A_1 E_t(y_{t+1}) + A_2 y_t + \cdots + A_p y_{t-p+2}, \\
&\vdots
\end{aligned}$$

By these recursions we get for a VAR(1) process

$$E_t(y_{t+h}) = (I_K + A_1 + \cdots + A_1^{h-1})\nu + A_1^h y_t.$$

Assuming $y_t = (-6, 3, 5)'$ and $\nu = (0, 2, 1)'$, the following forecasts are obtained for the VAR(1) example process (2.1.14):

$$E_t(y_{t+1}) = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} .5 & 0 & 0 \\ .1 & .1 & .3 \\ 0 & .2 & .3 \end{bmatrix} \begin{bmatrix} -6 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} -3.0 \\ 3.2 \\ 3.1 \end{bmatrix}, \quad (2.2.4a)$$

$$E_t(y_{t+2}) = (I_3 + A_1)\nu + A_1^2 y_t = \begin{bmatrix} -1.50 \\ 2.95 \\ 2.57 \end{bmatrix}, \quad (2.2.4b)$$

etc. Similarly, we get for the VAR(2) process (2.1.15) with $\nu = (.02, .03)'$, $y_t = (.06, .03)'$ and $y_{t-1} = (.055, .03)'$,

$$\begin{aligned}
E_t(y_{t+1}) &= \begin{bmatrix} .02 \\ .03 \end{bmatrix} + \begin{bmatrix} .5 & .1 \\ .4 & .5 \end{bmatrix} \begin{bmatrix} .06 \\ .03 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ .25 & 0 \end{bmatrix} \begin{bmatrix} .055 \\ .03 \end{bmatrix} \\
&= \begin{bmatrix} .053 \\ .08275 \end{bmatrix},
\end{aligned}$$

$$\begin{aligned}
E_t(y_{t+2}) &= \begin{bmatrix} .02 \\ .03 \end{bmatrix} + \begin{bmatrix} .5 & .1 \\ .4 & .5 \end{bmatrix} \begin{bmatrix} .053 \\ .08275 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ .25 & 0 \end{bmatrix} \begin{bmatrix} .06 \\ .03 \end{bmatrix} \\
&= \begin{bmatrix} .0548 \\ .1076 \end{bmatrix}.
\end{aligned} \tag{2.2.5}$$

The conditional expectation has the following properties:

- (1) It is an unbiased predictor, that is, $E[y_{t+h} - E_t(y_{t+h})] = 0$.
- (2) If u_t is independent white noise, $\text{MSE}[E_t(y_{t+h})] = \text{MSE}[E_t(y_{t+h})|y_t, y_{t-1}, \dots]$, that is, the MSE of the predictor equals the conditional MSE given $y_t, y_{t-1}, \dots$.

The latter property follows by similar arguments as the optimality of the predictor $E_t(y_{t+h})$.

It must be emphasized that the prediction formula (2.2.3) relies on u_t being independent white noise. If u_t and u_s are not independent but just uncorrelated, $E_t(u_{t+h})$ will be nonzero in general. As an example consider the univariate AR(1) process $y_t = \nu + \alpha y_{t-1} + u_t$ with

$$u_t = \begin{cases} e_t & \text{for } t = 0, \pm 2, \pm 4, \dots, \\ (e_{t-1}^2 - 1)/\sqrt{2} & \text{for } t = \pm 1, \pm 3, \dots, \end{cases}$$

where the e_t are independent standard normal ($\mathcal{N}(0, 1)$) random variables (see also Fuller (1976, Chapter 2, Exercise 16)). The process u_t is easily seen to be uncorrelated but not independent white noise. For even t ,

$$\begin{aligned}
E_t(u_{t+1}) &= E[(e_t^2 - 1)/\sqrt{2}|y_t, y_{t-1}, \dots] \\
&= (e_t^2 - 1)/\sqrt{2},
\end{aligned}$$

because $e_t = y_t - \nu - \alpha y_{t-1}$.

Linear Minimum MSE Predictor

If u_t is not independent white noise, additional assumptions are usually required to find the optimal predictor (conditional expectation) of a VAR(p) process. Without such assumptions we can achieve the less ambitious goal of finding the minimum MSE predictors among those that are *linear* functions of $y_t, y_{t-1}, \dots$. Let us consider a zero mean VAR(1) process

$$y_t = A_1 y_{t-1} + u_t \tag{2.2.6}$$

first. As in (2.1.3), it follows that

$$y_{t+h} = A_1^h y_t + \sum_{i=0}^{h-1} A_1^i u_{t+h-i}.$$

Thus, for a predictor

$$y_t(h) = B_0 y_t + B_1 y_{t-1} + \cdots,$$

where the B_i 's are $(K \times K)$ coefficient matrices, we get a forecast error

$$y_{t+h} - y_t(h) = \sum_{i=0}^{h-1} A_1^i u_{t+h-i} + (A_1^h - B_0) y_t - \sum_{i=1}^{\infty} B_i y_{t-i}.$$

Using that u_{t+j} , for $j > 0$, is uncorrelated with y_{t-i} , for $i \geq 0$, we get

$$\begin{aligned} \text{MSE}[y_t(h)] &= E \left[\left(\sum_{i=0}^{h-1} A_1^i u_{t+h-i} \right) \left(\sum_{i=0}^{h-1} A_1^i u_{t+h-i} \right)' \right] \\ &\quad + E \left\{ \left[(A_1^h - B_0) y_t - \sum_{i=1}^{\infty} B_i y_{t-i} \right] \left[(A_1^h - B_0) y_t - \sum_{i=1}^{\infty} B_i y_{t-i} \right]' \right\}. \end{aligned}$$

Obviously, this MSE matrix is minimal for $B_0 = A_1^h$ and $B_i = 0$. Thus, the optimal (*linear* minimum MSE) predictor for this special case is

$$y_t(h) = A_1^h y_t = A_1 y_t(h-1).$$

The forecast error is

$$\sum_{i=0}^{h-1} A_1^i u_{t+h-i}$$

and the MSE or forecast error covariance matrix is

$$\begin{aligned} \Sigma_y(h) &:= \text{MSE}[y_t(h)] = E \left[\left(\sum_{i=0}^{h-1} A_1^i u_{t+h-i} \right) \left(\sum_{i=0}^{h-1} A_1^i u_{t+h-i} \right)' \right] \\ &= \sum_{i=0}^{h-1} A_1^i \Sigma_u (A_1^i)' = \text{MSE}[y_t(h-1)] + A_1^{h-1} \Sigma_u (A_1^{h-1})'. \end{aligned}$$

A general VAR(p) process with zero mean,

$$y_t = A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t,$$

has a VAR(1) counterpart,

$$Y_t = \mathbf{A} Y_{t-1} + U_t,$$

where Y_t , $\mathbf{A}$, and U_t are as defined in (2.1.8). Using the same arguments as above, the optimal predictor of Y_{t+h} is seen to be

$$Y_t(h) = \mathbf{A}^h Y_t = \mathbf{A} Y_t(h-1).$$

It is easily seen by induction with respect to h that

$$Y_t(h) = \begin{bmatrix} y_t(h) \\ y_t(h-1) \\ \vdots \\ y_t(h-p+1) \end{bmatrix},$$

where $y_t(j) := y_{t+j}$ for $j \leq 0$. Defining the $(K \times Kp)$ matrix $J := [I_K : 0 : \cdots : 0]$ as in (2.1.11), we get the optimal h -step predictor of the process y_t at origin t as

$$\begin{aligned} y_t(h) &= J\mathbf{A}Y_t(h-1) = [A_1, \dots, A_p]Y_t(h-1) \\ &= A_1y_t(h-1) + \cdots + A_p y_t(h-p). \end{aligned} \quad (2.2.7)$$

This formula may be used for recursively computing the forecasts. Obviously, $y_t(h)$ is the conditional expectation $E_t(y_{t+h})$ if u_t is independent white noise because the recursion in (2.2.3) is the same as the one obtained here for a zero mean process with $\nu = 0$.

If the process y_t has nonzero mean, that is,

$$y_t = \nu + A_1y_{t-1} + \cdots + A_p y_{t-p} + u_t,$$

we define $x_t := y_t - \mu$, where $\mu := E(y_t) = (I - A_1 - \cdots - A_p)^{-1}\nu$. The process x_t has zero mean and the optimal h -step predictor is

$$x_t(h) = A_1x_t(h-1) + \cdots + A_px_t(h-p).$$

Adding μ to both sides of this equation gives the optimal linear predictor of y_t ,

$$\begin{aligned} y_t(h) &= x_t(h) + \mu = \mu + A_1(y_t(h-1) - \mu) + \cdots + A_p(y_t(h-p) - \mu) \\ &= \nu + A_1y_t(h-1) + \cdots + A_p y_t(h-p). \end{aligned} \quad (2.2.8)$$

Henceforth, we will refer to $y_t(h)$ as the *optimal predictor* irrespective of the properties of the white noise process u_t , that is, even if u_t is not independent but just uncorrelated white noise.

Using

$$Y_{t+h} = \mathbf{A}^h Y_t + \sum_{i=0}^{h-1} \mathbf{A}^i U_{t+h-i}$$

for a zero mean process, we get the forecast error

$$\begin{aligned} y_{t+h} - y_t(h) &= J[Y_{t+h} - Y_t(h)] = J \left[\sum_{i=0}^{h-1} \mathbf{A}^i U_{t+h-i} \right] \\ &= \sum_{i=0}^{h-1} J\mathbf{A}^i J' J U_{t+h-i} = \sum_{i=0}^{h-1} \Phi_i u_{t+h-i}, \end{aligned} \quad (2.2.9)$$

where the Φ_i are the MA coefficient matrices from (2.1.17). The forecast error is unchanged if y_t has nonzero mean because the mean term cancels. The forecast error representation (2.2.9) shows that the predictor $y_t(h)$ can also be expressed in terms of the MA representation (2.1.17),

$$y_t(h) = \mu + \sum_{i=h}^{\infty} \Phi_i u_{t+h-i} = \mu + \sum_{i=0}^{\infty} \Phi_{h+i} u_{t-i}. \quad (2.2.10)$$

From (2.2.9) the forecast error covariance or MSE matrix is easy to obtain,

$$\Sigma_y(h) := \text{MSE}[y_t(h)] = \sum_{i=0}^{h-1} \Phi_i \Sigma_u \Phi_i' = \Sigma_y(h-1) + \Phi_{h-1} \Sigma_u \Phi_{h-1}'. \quad (2.2.11)$$

Hence, the MSEs are monotonically nondecreasing and, for $h \rightarrow \infty$, the MSE matrices approach the covariance matrix of y_t ,

$$\Gamma_y(0) = \Sigma_y = \sum_{i=0}^{\infty} \Phi_i \Sigma_u \Phi_i'$$

(see (2.1.18)). That is,

$$\Sigma_y(h) \xrightarrow{h \rightarrow \infty} \Sigma_y. \quad (2.2.12)$$

If the process mean μ is used as a forecast, the MSE matrix of that predictor is just the covariance matrix Σ_y of y_t . Hence, the optimal long range forecast ($h \rightarrow \infty$) is the process mean. In other words, the past of the process contains no information on the development of the process in the distant future. Zero mean processes with this property are *purely nondeterministic*, that is, $y_t - \mu$ is purely nondeterministic if the forecast MSEs satisfy (2.2.12).

For the example VAR(1) process (2.1.14) with Σ_u as in (2.1.33), using the MA coefficient matrices from (2.1.24), the forecast MSE matrices

$$\begin{aligned} \Sigma_y(1) &= \Sigma_u = \begin{bmatrix} 2.25 & 0 & 0 \\ 0 & 1.0 & .5 \\ 0 & .5 & .74 \end{bmatrix}, \\ \Sigma_y(2) &= \Sigma_u + \Phi_1 \Sigma_u \Phi_1' = \begin{bmatrix} 2.813 & .113 & 0 \\ .113 & 1.129 & .632 \\ 0 & .632 & .907 \end{bmatrix}, \\ \Sigma_y(3) &= \Sigma_y(2) + \Phi_2 \Sigma_u \Phi_2' = \begin{bmatrix} 2.953 & .146 & .011 \\ .146 & 1.161 & .663 \\ .011 & .663 & .943 \end{bmatrix} \end{aligned} \quad (2.2.13)$$

are obtained. Similarly, for the VAR(2) example process (2.1.15) with white noise covariance matrix (2.1.41), we get with Φ_1 from (2.1.25),

$$\Sigma_y(1) = \Sigma_u = \begin{bmatrix} .09 & 0 \\ 0 & .04 \end{bmatrix},$$

$$\Sigma_y(2) = \Sigma_u + \Phi_1 \Sigma_u \Phi_1' = \begin{bmatrix} .1129 & .02 \\ .02 & .0644 \end{bmatrix}. \quad (2.2.14)$$

2.2.3 Interval Forecasts and Forecast Regions

In order to set up interval forecasts or forecast intervals, we make an assumption about the distributions of the y_t or the u_t . It is most common to consider *Gaussian processes* where $y_t, y_{t+1}, \dots, y_{t+h}$ have a multivariate normal distribution for any t and h . Equivalently, it may be assumed that u_t is Gaussian, that is, the u_t are multivariate normal, $u_t \sim \mathcal{N}(0, \Sigma_u)$, and u_t and u_s are independent for $s \neq t$.

Under these conditions the forecast errors are also normally distributed as linear transformations of normal vectors,

$$y_{t+h} - y_t(h) = \sum_{i=0}^{h-1} \Phi_i u_{t+h-i} \sim \mathcal{N}(0, \Sigma_y(h)). \quad (2.2.15)$$

This result implies that the forecast errors of the individual components are normal so that

$$\frac{y_{k,t+h} - y_{k,t}(h)}{\sigma_k(h)} \sim \mathcal{N}(0, 1), \quad (2.2.16)$$

where $y_{k,t}(h)$ is the k -th component of $y_t(h)$ and $\sigma_k(h)$ is the square root of the k -th diagonal element of $\Sigma_y(h)$. Denoting by $z_{(\alpha)}$ the upper α 100 percentage point of the normal distribution, we get

$$\begin{aligned} 1 - \alpha &= \Pr \left\{ -z_{(\alpha/2)} \leq \frac{y_{k,t+h} - y_{k,t}(h)}{\sigma_k(h)} \leq z_{(\alpha/2)} \right\} \\ &= \Pr \left\{ y_{k,t}(h) - z_{(\alpha/2)} \sigma_k(h) \leq y_{k,t+h} \leq y_{k,t}(h) + z_{(\alpha/2)} \sigma_k(h) \right\}. \end{aligned}$$

Hence, a $(1-\alpha)$ 100% interval forecast, h periods ahead, for the k -th component of y_t is

$$y_{k,t}(h) \pm z_{(\alpha/2)} \sigma_k(h) \quad (2.2.17a)$$

or

$$[y_{k,t}(h) - z_{(\alpha/2)} \sigma_k(h), y_{k,t}(h) + z_{(\alpha/2)} \sigma_k(h)]. \quad (2.2.17b)$$

If forecast intervals of this type are computed repeatedly from a large number of time series (realizations of the considered process), then about $(1-\alpha)$ 100% of the intervals will contain the actual value of the random variable $y_{k,t+h}$.

Using (2.2.4a), (2.2.4b) and (2.2.13), 95% forecast intervals for the components of the example VAR(1) process (2.1.14) are

$$\begin{aligned}
y_{1,t}(1) \pm 1.96\sqrt{2.25} & \text{ or } -3.0 \pm 2.94, \\
y_{2,t}(1) \pm 1.96\sqrt{1.0} & \text{ or } 3.2 \pm 1.96, \\
y_{3,t}(1) \pm 1.96\sqrt{.74} & \text{ or } 3.1 \pm 1.69, \\
y_{1,t}(2) \pm 1.96\sqrt{2.813} & \text{ or } -1.50 \pm 3.29, \\
y_{2,t}(2) \pm 1.96\sqrt{1.129} & \text{ or } 2.95 \pm 2.08, \\
y_{3,t}(2) \pm 1.96\sqrt{.907} & \text{ or } 2.57 \pm 1.87.
\end{aligned} \tag{2.2.18}$$

The result in (2.2.15) can also be used to establish joint forecast regions for two or more variables. For instance, if a joint forecast region for the first N components is desired, we define the $(N \times K)$ matrix $F := [I_N : 0]$ and note that

$$[y_{t+h} - y_t(h)]' F' [F \Sigma_y(h) F']^{-1} F [y_{t+h} - y_t(h)] \sim \chi^2(N) \tag{2.2.19}$$

by a well-known result for multivariate normal vectors (see Appendix B). Hence, the $\chi^2(N)$ -distribution can be used to determine a $(1-\alpha)100\%$ forecast ellipsoid for the first N components of the process.

In practice, the construction of the ellipsoid is quite demanding if N is greater than two or three. Therefore, a more practical approach is to use *Bonferroni's method* for constructing joint confidence regions. It is based on the fact that for events $E_1, \dots, E_N$ the following probability inequality holds:

$$\Pr(E_1 \cup \dots \cup E_N) \leq \Pr(E_1) + \dots + \Pr(E_N).$$

Hence,

$$\Pr\left(\bigcap_{i=1}^N E_i\right) \geq 1 - \sum_{i=1}^N \Pr(\bar{E}_i),$$

where $\bar{E}_i$ denotes the complement of E_i . Consequently, if E_i is the event that $y_{i,t+h}$ falls within an interval H_i ,

$$\Pr(Fy_{t+h} \in H_1 \times \dots \times H_N) \geq 1 - \sum_{i=1}^N \Pr(\bar{E}_i). \tag{2.2.20}$$

In other words, if we choose a $(1 - \frac{\alpha}{N})100\%$ forecast interval for each of the N components, the resulting joint forecast region has probability at least $(1-\alpha)100\%$ of containing all N variables jointly. For instance, for the VAR(1) example process considered previously,

$$\{(y_1, y_2) | -3.0 - 2.94 \leq y_1 \leq -3.0 + 2.94, 3.2 - 1.96 \leq y_2 \leq 3.2 + 1.96\}$$

is a joint forecast region of $(y_{1,t+1}, y_{2,t+1})$ with probability content at least 90%.

By the same method joint forecast regions for different horizons h can be obtained. For instance, a joint forecast region with probability content of at least $(1 - \alpha)100\%$ for $y_{k,t+1}, \dots, y_{k,t+h}$ is

$$\{(y_{k,1}, \dots, y_{k,h}) | y_{k,t}(i) - z_{(\alpha/2h)}\sigma_k(i) \leq y_{k,i} \leq y_{k,t}(i) + z_{(\alpha/2h)}\sigma_k(i), \\ i = 1, \dots, h\}. \quad (2.2.21)$$

Thus, for the example, a joint forecast region for $y_{2,t+1}, y_{2,t+2}$ with probability content of at least 90% is given by

$$\{(y_{2,1}, y_{2,2}) | 1.24 \leq y_{2,1} \leq 5.16, .87 \leq y_{2,2} \leq 5.03\}.$$

Under our assumption of a Gaussian process, the distribution of the forecasts and forecast errors is known and, consequently, forecast intervals are easy to set up. If the underlying process has a different and potentially unknown distribution, considering the forecast distribution becomes more difficult. Even then methods are available to determine more than just point forecasts. A survey of density forecasting is given by Tay & Wallis (2002).

2.3 Structural Analysis with VAR Models

Because VAR models represent the correlations among a set of variables, they are often used to analyze certain aspects of the relationships between the variables of interest. In the following, three ways to interpret a VAR model will be discussed. They are all closely related and they are all beset with problems that will be pointed out subsequently.

2.3.1 Granger-Causality, Instantaneous Causality, and Multi-Step Causality

Definitions of Causality

Granger (1969a) has defined a concept of causality which, under suitable conditions, is fairly easy to deal with in the context of VAR models. Therefore it has become quite popular in recent years. The idea is that a cause cannot come after the effect. Thus, if a variable x affects a variable z , the former should help improving the predictions of the latter variable.

To formalize this idea, suppose that Ω_t is the information set containing all the relevant information in the universe available up to and including period t . Let $z_t(h|\Omega_t)$ be the optimal (minimum MSE) h -step predictor of the process z_t at origin t , based on the information in Ω_t . The corresponding forecast MSE

will be denoted by $\Sigma_z(h|\Omega_t)$. The process x_t is said to *cause* z_t in *Granger's sense* if

$$\Sigma_z(h|\Omega_t) < \Sigma_z(h|\Omega_t \setminus \{x_s | s \leq t\}) \quad \text{for at least one } h = 1, 2, \dots \quad (2.3.1)$$

Alternatively, we will say that x_t *Granger-causes* (or briefly *causes*) z_t or x_t is *Granger-causal* for z_t if (2.3.1) holds. In (2.3.1) $\Omega_t \setminus \{x_s | s \leq t\}$ is the set containing all the relevant information in the universe except for the information in the past and present of the x_t process. In other words, if z_t can be predicted more efficiently if the information in the x_t process is taken into account in addition to all other information in the universe, then x_t is *Granger-causal* for z_t .

The definition extends immediately to the case where z_t and x_t are M - and N -dimensional processes, respectively. In that case, x_t is said to Granger-cause z_t if

$$\Sigma_z(h|\Omega_t) \neq \Sigma_z(h|\Omega_t \setminus \{x_s | s \leq t\}) \quad (2.3.2)$$

for some t and h . Alternatively, this could be expressed by requiring the two MSEs to be different and

$$\Sigma_z(h|\Omega_t) \leq \Sigma_z(h|\Omega_t \setminus \{x_s | s \leq t\})$$

(i.e., the difference between the right-hand and the left-hand matrix is positive semidefinite). Because the null matrix is also positive semidefinite, it is necessary to require in addition that the two matrices are not identical. If x_t causes z_t and z_t also causes x_t the process $(z'_t, x'_t)'$ is called a *feedback system*.

Sometimes the term “instantaneous causality” is used in economic analyses. We say that there is *instantaneous causality* between z_t and x_t if

$$\Sigma_z(1|\Omega_t \cup \{x_{t+1}\}) \neq \Sigma_z(1|\Omega_t). \quad (2.3.3)$$

In other words, in period t , adding x_{t+1} to the information set helps to improve the forecast of z_{t+1} . We will see shortly that this concept of causality is really symmetric, that is, if there is instantaneous causality between z_t and x_t , then there is also instantaneous causality between x_t and z_t (see Proposition 2.3). Therefore we do not use the notion “instantaneous causality from x_t to z_t ” in the foregoing definition.

A possible criticism of the foregoing definitions could relate to the choice of the MSE as a measure of the forecast precision. Of course, the choice of another measure could lead to a different definition of causality. However, in the situations of interest in the following, equality of the MSEs will imply equality of the corresponding predictors. In that case a process z_t is not Granger-caused by x_t if the optimal predictor of z_t does not use information from the x_t process. This result is intuitively appealing.

A more serious practical problem is the choice of the information set Ω_t . Usually all the relevant information in the universe is not available to a forecaster and, thus, the optimal predictor given Ω_t cannot be determined. Therefore a less demanding definition of causality is often used in practice. Instead

of all the information in the universe, only the information in the past and present of the process under study is considered relevant and Ω_t is replaced by $\{z_s, x_s | s \leq t\}$. Furthermore, instead of *optimal* predictors, *optimal linear* predictors are compared. In other words, $z_t(h|\Omega_t)$ is replaced by the linear minimum MSE h -step predictor based on the information in $\{z_s, x_s | s \leq t\}$ and $z_t(h|\Omega_t \setminus \{x_s | s \leq t\})$ is replaced by the linear minimum MSE h -step predictor based on $\{z_s | s \leq t\}$. In the following, when the terms “Granger-causality” and “instantaneous causality” are used, these restrictive assumptions are implicitly used if not otherwise noted.

Characterization of Granger-Causality

In order to determine the Granger-causal relationships between the variables of the K -dimensional VAR process y_t , suppose it has the canonical MA representation

$$y_t = \mu + \sum_{i=0}^{\infty} \Phi_i u_{t-i} = \mu + \Phi(L)u_t, \quad \Phi_0 = I_K, \quad (2.3.4)$$

where u_t is a white noise process with nonsingular covariance matrix Σ_u . Suppose that y_t consists of the M -dimensional process z_t and the $(K - M)$ -dimensional process x_t and the MA representation is partitioned accordingly,

$$y_t = \begin{bmatrix} z_t \\ x_t \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \Phi_{11}(L) & \Phi_{12}(L) \\ \Phi_{21}(L) & \Phi_{22}(L) \end{bmatrix} \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}. \quad (2.3.5)$$

Using the prediction formula (2.2.10), the optimal 1-step forecast of z_t based on y_t is

$$\begin{aligned} z_t(1|\{y_s | s \leq t\}) &= [I_M : 0]y_t(1) \\ &= \mu_1 + \sum_{i=1}^{\infty} \Phi_{11,i}u_{1,t+1-i} + \sum_{i=1}^{\infty} \Phi_{12,i}u_{2,t+1-i}. \end{aligned} \quad (2.3.6)$$

Hence the forecast error is

$$z_{t+1} - z_t(1|\{y_s | s \leq t\}) = u_{1,t+1}. \quad (2.3.7)$$

As mentioned in Section 2.1.3, a subprocess of a stationary process also has a prediction error MA representation. Thus,

$$\begin{aligned} z_t &= \mu_1 + \sum_{i=0}^{\infty} \Phi_{11,i}u_{1,t-i} + \sum_{i=1}^{\infty} \Phi_{12,i}u_{2,t-i} \\ &= \mu_1 + \sum_{i=0}^{\infty} F_i v_{t-i}, \end{aligned} \quad (2.3.8)$$

where $F_0 = I_M$ and the last expression is a prediction error MA representation. Thus, the optimal 1-step predictor based on z_t only is

$$z_t(1|\{z_s|s \leq t\}) = \mu_1 + \sum_{i=1}^{\infty} F_i v_{t+1-i} \quad (2.3.9)$$

and the corresponding forecast error is

$$z_{t+1} - z_t(1|\{z_s|s \leq t\}) = v_{t+1}. \quad (2.3.10)$$

Consequently, the predictors (2.3.6) and (2.3.9) are identical if and only if $v_t = u_{1,t}$ for all t . In other words, equality of the predictors is equivalent to z_t having the MA representation

$$\begin{aligned} z_t &= \mu_1 + \sum_{i=0}^{\infty} F_i u_{1,t-i} = \mu_1 + \sum_{i=0}^{\infty} [F_i : 0] u_{t-i} \\ &= \mu_1 + \sum_{i=0}^{\infty} [\Phi_{11,i} : \Phi_{12,i}] u_{t-i} \\ &= \mu_1 + \sum_{i=0}^{\infty} \Phi_{11,i} u_{1,t-i} + \sum_{i=1}^{\infty} \Phi_{12,i} u_{2,t-i}. \end{aligned}$$

Uniqueness of the canonical MA representation implies that $F_i = \Phi_{11,i}$ and $\Phi_{12,i} = 0$ for $i = 1, 2, \dots$. Hence, we get the following proposition.

Proposition 2.2 (*Characterization of Granger-Noncausality*)

Let y_t be a VAR process as in (2.3.4)/(2.3.5) with canonical MA operator $\Phi(z)$. Then

$$z_t(1|\{y_s|s \leq t\}) = z_t(1|\{z_s|s \leq t\}) \Leftrightarrow \Phi_{12,i} = 0 \quad \text{for } i = 1, 2, \dots \quad (2.3.11)$$

■

Because we have just used the MA representation (2.3.4) and not its finite order VAR form, the proposition is not only valid for VAR processes but more generally for processes having a canonical MA representation such as (2.3.4). From (2.2.10) it is obvious that equality of the 1-step predictors implies equality of the h -step predictors for $h = 2, 3, \dots$. Hence, the proposition provides a necessary and sufficient condition for x_t being not Granger-causal for z_t , that is, z_t is not Granger-caused by x_t if and only if $\Phi_{12,i} = 0$ for $i = 1, 2, \dots$. Thus, Granger-noncausality can be checked easily by looking at the MA representation of y_t . Because we are mostly concerned with VAR processes, it is worth noting that for a stationary, stable VAR(p) process

$$\begin{aligned} y_t = \begin{bmatrix} z_t \\ x_t \end{bmatrix} &= \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix} + \begin{bmatrix} A_{11,1} & A_{12,1} \\ A_{21,1} & A_{22,1} \end{bmatrix} \begin{bmatrix} z_{t-1} \\ x_{t-1} \end{bmatrix} + \dots \\ &\quad + \begin{bmatrix} A_{11,p} & A_{12,p} \\ A_{21,p} & A_{22,p} \end{bmatrix} \begin{bmatrix} z_{t-p} \\ x_{t-p} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}, \end{aligned} \quad (2.3.12)$$

the condition (2.3.11) is satisfied if and only if

$$A_{12,i} = 0 \text{ for } i = 1, \dots, p.$$

This result follows from the recursions in (2.1.22) or, alternatively, because the inverse of

$$\begin{bmatrix} \Phi_{11}(L) & 0 \\ \Phi_{21}(L) & \Phi_{22}(L) \end{bmatrix}$$

is

$$\begin{bmatrix} \Phi_{11}(L)^{-1} & 0 \\ -\Phi_{22}(L)^{-1}\Phi_{21}(L)\Phi_{11}(L)^{-1} & \Phi_{22}(L)^{-1} \end{bmatrix}$$

Thus, we have the following result.

Corollary 2.2.1

If y_t is a stable VAR(p) process as in (2.3.12) with nonsingular white noise covariance matrix Σ_u , then

$$\begin{aligned} z_t(h|\{y_s|s \leq t\}) &= z_t(h|\{z_s|s \leq t\}), \quad h = 1, 2, \dots \\ \Leftrightarrow A_{12,i} &= 0 \quad \text{for } i = 1, \dots, p. \end{aligned} \tag{2.3.13}$$

Alternatively,

$$\begin{aligned} x_t(h|\{y_s|s \leq t\}) &= x_t(h|\{x_s|s \leq t\}), \quad h = 1, 2, \dots \\ \Leftrightarrow A_{21,i} &= 0 \quad \text{for } i = 1, \dots, p. \end{aligned} \tag{2.3.14}$$

■

This corollary implies that noncausalities can be determined by just looking at the VAR representation of the system. For instance, for the example process (2.1.14),

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \\ y_{3,t} \end{bmatrix} = \nu + \begin{bmatrix} .5 & 0 & 0 \\ .1 & .1 & .3 \\ 0 & .2 & .3 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \\ y_{3,t-1} \end{bmatrix} + u_t,$$

$x_t := (y_{2t}, y_{3t})'$ does not Granger-cause $z_t := y_{1t}$ because $A_{12,1} = 0$ if the coefficient matrix is partitioned according to (2.3.12). On the other hand, z_t Granger-causes x_t . To give this discussion economic content let us assume that the variables in the system are rates of change of investment (y_1), income (y_2), and consumption (y_3). With these specifications, the previous discussion shows that investment Granger-causes the consumption/income system whereas the converse is not true. It is also easy to check that consumption causes the income/investment system and vice versa. Note that so

far we have defined Granger-causality only in terms of two groups of variables. Therefore, at this stage, we cannot talk about the Granger-causal relationship between consumption and income in the three-dimensional investment/income/consumption system.

Let us assume that the variables in the example VAR(2) process (2.1.15),

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \nu + \begin{bmatrix} .5 & .1 \\ .4 & .5 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ .25 & 0 \end{bmatrix} \begin{bmatrix} y_{1,t-2} \\ y_{2,t-2} \end{bmatrix} + u_t,$$

represent the inflation rate (y_1), and some interest rate (y_2). Using Corollary 2.2.1, it is immediately obvious that inflation causes the interest rate and vice versa. Hence the system is a feedback system. In the following we will refer to (2.1.15) as the inflation/interest rate system.

Characterization of Instantaneous Causality

In order to study the concept of instantaneous causality in the framework of the MA process (2.3.5), it is useful to rewrite that representation. Note that the positive definite symmetric matrix Σ_u can be written as the product $\Sigma_u = PP'$, where P is a lower triangular nonsingular matrix with positive diagonal elements (see Appendix A.9.3). Thus, (2.3.5) can be represented as

$$y_t = \mu + \sum_{i=0}^{\infty} \Phi_i PP^{-1} u_{t-i} = \mu + \sum_{i=0}^{\infty} \Theta_i w_{t-i}, \quad (2.3.15)$$

where $\Theta_i := \Phi_i P$ and $w_t := P^{-1} u_t$ is white noise with covariance matrix

$$\Sigma_w = P^{-1} \Sigma_u (P^{-1})' = I_K. \quad (2.3.16)$$

Because the white noise errors w_t have uncorrelated components, they are often called *orthogonal* residuals or innovations.

Partitioning the representation (2.3.15) according to the partitioning of $y_t = (z'_t, x'_t)'$ gives

$$\begin{aligned} \begin{bmatrix} z_t \\ x_t \end{bmatrix} &= \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \Theta_{11,0} & 0 \\ \Theta_{21,0} & \Theta_{22,0} \end{bmatrix} \begin{bmatrix} w_{1,t} \\ w_{2,t} \end{bmatrix} \\ &+ \begin{bmatrix} \Theta_{11,1} & \Theta_{12,1} \\ \Theta_{21,1} & \Theta_{22,1} \end{bmatrix} \begin{bmatrix} w_{1,t-1} \\ w_{2,t-1} \end{bmatrix} + \cdots. \end{aligned}$$

Hence,

$$z_{t+1} = \mu_1 + \Theta_{11,0} w_{1,t+1} + \Theta_{11,1} w_{1,t} + \Theta_{12,1} w_{2,t} + \cdots$$

and

$$x_{t+1} = \mu_2 + \Theta_{21,0} w_{1,t+1} + \Theta_{22,0} w_{2,t+1} + \Theta_{21,1} w_{1,t} + \Theta_{22,1} w_{2,t} + \cdots.$$

The optimal 1-step predictor of x_t based on $\{y_s|s \leq t\}$ and, in addition, on z_{t+1} , is equal to the 1-step predictor of x_t based on $\{w_s|s \leq t\} \cup \{w_{1,t+1}\}$, that is,

$$\begin{aligned} x_t(1|\{y_s|s \leq t\} \cup \{z_{t+1}\}) &= x_t(1|\{w_s = (w'_{1,s}, w'_{2,s})'|s \leq t\} \cup \{w_{1,t+1}\}) \\ &= \Theta_{21,0}w_{1,t+1} + x_t(1|\{y_s|s \leq t\}). \end{aligned} \quad (2.3.17)$$

Consequently,

$$x_t(1|\{y_s|s \leq t\} \cup \{z_{t+1}\}) = x_t(1|\{y_s|s \leq t\})$$

if and only if $\Theta_{21,0} = 0$. This condition, in turn, is easily seen to hold if and only if the covariance matrix Σ_u is block diagonal with a $((K - M) \times M)$ block of zeros in the lower left-hand corner and an $(M \times (K - M))$ block of zeros in the upper right-hand corner. Of course, this means that u_{1t} and u_{2t} in (2.3.5) have to be uncorrelated, i.e., $E(u_{1t}u'_{2t}) = 0$. Thereby the following proposition is proven.

Proposition 2.3 (*Characterization of Instantaneous Causality*)

Let y_t be as in (2.3.5)/(2.3.15) with nonsingular innovation covariance matrix Σ_u . Then there is no instantaneous causality between z_t and x_t if and only if

$$E(u_{1t}u'_{2t}) = 0. \quad (2.3.18)$$

■

This proposition provides a condition for instantaneous causality which is easy to check if the process is given in MA or VAR form. For instance, for the investment/income/consumption system with white noise covariance matrix (2.1.33),

$$\Sigma_u = \begin{bmatrix} 2.25 & 0 & 0 \\ 0 & 1.0 & .5 \\ 0 & .5 & .74 \end{bmatrix},$$

there is no instantaneous causality between (income, consumption) and investment.

From Propositions 2.2 and 2.3 it follows that $y_t = (z'_t, x'_t)'$ has a representation with orthogonal innovations as in (2.3.15) of the form

$$\begin{aligned} \begin{bmatrix} z_t \\ x_t \end{bmatrix} &= \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \Theta_{11,0} & 0 \\ 0 & \Theta_{22,0} \end{bmatrix} \begin{bmatrix} w_{1,t} \\ w_{2,t} \end{bmatrix} \\ &\quad + \begin{bmatrix} \Theta_{11,1} & 0 \\ \Theta_{21,1} & \Theta_{22,1} \end{bmatrix} \begin{bmatrix} w_{1,t-1} \\ w_{2,t-1} \end{bmatrix} + \cdots \\ &= \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \Theta_{11}(L) & 0 \\ \Theta_{21}(L) & \Theta_{22}(L) \end{bmatrix} \begin{bmatrix} w_{1,t} \\ w_{2,t} \end{bmatrix}, \end{aligned} \quad (2.3.19)$$

if x_t does not Granger-cause z_t and, furthermore, there is no instantaneous causation between x_t and z_t . In the absence of instantaneous causality, a similar representation with $\Theta_{21}(L) \equiv 0$ is obtained if z_t is not Granger-causal for x_t .

Discussion of Instantaneous and Granger-Causality

At this point, some words of caution seem appropriate. The term “causality” suggests a cause and effect relationship between two sets of variables. Proposition 2.3 shows that such an interpretation is problematic with respect to instantaneous causality because this term only describes a nonzero correlation between two sets of variables. It does not say anything about the cause and effect relation. The direction of instantaneous causation cannot be derived from the MA or VAR representation of the process but must be obtained from further knowledge on the relationship between the variables. Such knowledge may exist in the form of an economic theory.

Although a direction of causation has been defined in relation with Granger-causality it is problematic to interpret the absence of causality from x_t to z_t in the sense that variations in x_t will have no effect on z_t . To see this consider, for instance, the stable bivariate VAR(1) system

$$\begin{bmatrix} z_t \\ x_t \end{bmatrix} = \begin{bmatrix} \alpha_{11} & 0 \\ \alpha_{21} & \alpha_{22} \end{bmatrix} \begin{bmatrix} z_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}. \quad (2.3.20)$$

In this system, x_t does not Granger-cause z_t by Corollary 2.2.1. However, the system may be multiplied by some nonsingular matrix

$$B = \begin{bmatrix} 1 & \beta \\ 0 & 1 \end{bmatrix}$$

so that

$$\begin{bmatrix} z_t \\ x_t \end{bmatrix} = \begin{bmatrix} 0 & -\beta \\ 0 & 0 \end{bmatrix} \begin{bmatrix} z_t \\ x_t \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} z_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} v_{1t} \\ v_{2t} \end{bmatrix}, \quad (2.3.21)$$

where $\gamma_{11} := \alpha_{11} + \alpha_{21}\beta$, $\gamma_{12} := \alpha_{22}\beta$, $\gamma_{21} := \alpha_{21}$, $\gamma_{22} := \alpha_{22}$ and $(v_{1t}, v_{2t})' := B(u_{1t}, u_{2t})'$. Note that this is just another representation of the process $(z_t, x_t)'$ and not another process. (The reader may check that the process (2.3.21) has the same means and autocovariances as the one in (2.3.20).)

In other words, the stochastic interrelationships between the random variables of the system can either be characterized by (2.3.20) or by (2.3.21) although the two representations have quite different physical interpretations. If (2.3.21) happens to represent the actual ongoing in the system, changes in x_t may affect z_t through the term with the coefficient $-\beta$ in the first equation. Thus, the lack of a Granger-causal relationship from one group of variables to the remaining variables cannot necessarily be interpreted as lack of a cause and effect relationship. It must be remembered that a VAR or MA representation characterizes the joint distribution of sets of random variables. In order to derive cause and effect relationships from it, usually requires further assumptions regarding the relationship between the variables involved. We will return to this problem in the following subsections.

Further problems related to the interpretation of Granger-causality result from restricting the information set to contain only past and present variables of the system rather than all information in the universe. Only if all other information in the universe is irrelevant for the problem at hand, the reduction of the information set is of no consequence. Some related problems will be discussed in the following.

Changing the Information Set

So far it has been assumed that the information set contains the variables or groups of variables only for which we want to analyze the causal links. Often we are interested in the causal links between two variables in a higher dimensional system. In other words, we are interested in analyzing Granger-causality in a framework where the information set contains more than just the variables of direct interest. In the bivariate framework when the information set is limited to the two variables of interest, it was seen that if the 1-step ahead forecasts of one variable cannot be improved by using the information in the other variable, the same holds for all h -step forecasts, $h = 1, 2, \dots$. This result does not hold anymore if the information set contains additional variables, as pointed out by Lütkepohl (1993).

To be more explicit, suppose the vector time series z_t, y_t, x_t with dimensions K_z, K_y, K_x , respectively, are jointly generated by a VAR(p) process

$$\begin{bmatrix} z_t \\ y_t \\ x_t \end{bmatrix} = \sum_{i=1}^p A_i \begin{bmatrix} z_{t-i} \\ y_{t-i} \\ x_{t-i} \end{bmatrix} + u_t, \quad (2.3.22)$$

where

$$A_i = \begin{bmatrix} A_{zz,i} & A_{zy,i} & A_{zx,i} \\ A_{yz,i} & A_{yy,i} & A_{yx,i} \\ A_{xz,i} & A_{xy,i} & A_{xx,i} \end{bmatrix}, \quad i = 1, \dots, p,$$

with $A_{kl,i}$ having dimension $(K_k \times K_l)$ and u_t is zero mean white noise, as usual. In this process, if $A_{zy,i} = 0$, $i = 1, 2, \dots, p$, it is not difficult to see that the information in y_t cannot be used to improve the 1-step ahead forecasts of z_t but it is still possible that it can be used to improve the h -step forecasts for $h = 2, 3, \dots$. In other words, if y_t is 1-step noncausal for z_t , it may still be h -step causal for $h > 1$. Consequently, it makes sense to define more refined concepts of causality which refer explicitly to the forecast horizon. For instance, y_t may be called *h-step noncausal* for z_t ($y_t \not\rightarrow_{(h)} z_t$) for $h = 1, 2, \dots$, if the j -step ahead forecasts of z_t cannot be improved for $j \leq h$ by taking into account the information in past and present y_t . Now the original concept of Granger-causality corresponds to infinite-step causality.

The corresponding restrictions of multi-step causality on the VAR coefficients have been considered by Dufour & Renault (1998). Unlike in the bivariate setting explored earlier, now nonlinear restrictions on the VAR coefficients

are obtained which make it more difficult to check for h -step causality if the information set is expanded by additional variables.

To state the restrictions formally, let $\mathbf{A}$ be defined as in the VAR(1) representation (2.1.8), let $J = [I_K : 0 : \dots : 0]$ be a $(K \times Kp)$ matrix as before and define $A^{(j)} = J\mathbf{A}^j$ and $\alpha^{(j)} = \text{vec}(A^{(j)})$. Dufour & Renault (1998) show that in the process (2.3.22), $y_t \not\rightarrow_{(h)} z_t$ if and only if

$$R\alpha^{(j)} = 0 \quad \text{for } j = 1, \dots, h, \quad (2.3.23)$$

and $y_t \not\rightarrow_{(\infty)} z_t$ if and only if

$$R\alpha^{(j)} = 0 \quad \text{for } j = 1, \dots, pK_x + 1. \quad (2.3.24)$$

Here the restriction matrix R is such that $R\text{vec}[A_1, \dots, A_p] = \text{vec}[A_{zy,1}, \dots, A_{zy,p}]$, that is, it collects the elements of the second block in the first row of each of the coefficient matrices.

As an example consider again the 3-dimensional VAR(1) process (2.1.14). For infinite-step causality or noncausality from y_{2t} to y_{1t} we need to check the relevant elements of the coefficient matrix and its second power:

$$A_1 = \begin{bmatrix} .5 & 0 & 0 \\ .1 & .1 & .3 \\ 0 & .2 & .3 \end{bmatrix}, \quad A_1^2 = \begin{bmatrix} .25 & 0 & 0 \\ .06 & .07 & .12 \\ .02 & .08 & .15 \end{bmatrix}.$$

Clearly, $y_{2t} \not\rightarrow_{(1)} y_{1t}$ holds because $\alpha_{12,1} = 0$ and also the restrictions for $y_{2t} \not\rightarrow_{(\infty)} y_{1t}$ are satisfied in this case because the (1,2)-th element of A_1^2 is also zero. In contrast, $y_{1t} \not\rightarrow_{(1)} y_{3t}$ holds, while $y_{1t} \not\rightarrow_{(\infty)} y_{3t}$ does not, because the lower left-hand element of A_1^2 is nonzero. Notice that the definition and characterizations of multi-step causality are given for the first two sets of subvectors with the third one containing the extra variables. For applying the definition and results in the present example, the variables may just be rearranged accordingly.

In addition to these extensions related to increasing the information set, there are also other problems which may make it difficult to interpret Granger-causal relations even in a bivariate setting. Let us discuss some of them in terms of an inflation/interest rate system. For example, it may make a difference whether the information set contains monthly, quarterly or annual data. If a quarterly system is considered and no causality is found from the interest rate to inflation it does not follow that a corresponding monthly interest rate has no impact on the monthly inflation rate. In other words, the interest rate may Granger-cause inflation in a monthly system even if it does not in a quarterly system.

Furthermore, putting seasonally adjusted variables in the information set is not the same as using unadjusted variables. Consequently, if Granger-causality is found for the seasonally adjusted variables, it is still possible that in the actual seasonal system the interest rate is not Granger-causal for inflation. Similar comments apply in the presence of measurement errors. Finally, causality

analyses are usually based on estimated rather than known systems. Additional problems result in that case. We will return to them in the next chapter.

The previous critical remarks are meant to caution the reader and multiple time series analyst against overinterpreting the evidence from a VAR model. Still, causality analyses are useful tools in practice if these critical points are kept in mind. At the very least, a Granger-causality analysis tells the analyst whether a set of variables contains useful information for improving the predictions of another set of variables. Further discussions of causality issues and many further references may be found in Geweke (1984) and Granger (1982).

2.3.2 Impulse Response Analysis

In the previous subsection, we have seen that Granger-causality may not tell us the complete story about the interactions between the variables of a system. In applied work, it is often of interest to know the *response* of *one* variable to an *impulse* in another variable in a system that involves a number of further variables as well. Thus, one would like to investigate the impulse response relationship between two variables in a higher dimensional system. Of course, if there is a reaction of one variable to an impulse in another variable we may call the latter causal for the former. In this subsection, we will study this type of causality by tracing out the effect of an exogenous shock or innovation in one of the variables on some or all of the other variables. This kind of impulse response analysis is sometimes called *multiplier analysis*. For instance, in a system consisting of an inflation rate and an interest rate, the effect of an increase in the inflation rate may be of interest. In the real world, such an increase may be induced exogenously from outside the system by events like the increase of the oil price in 1973/74 when the OPEC agreed on a joint action to raise prices. Alternatively, an increase or reduction in the interest rate may be administered by the central bank for reasons outside the simple two variable system under study.

Responses to Forecast Errors

Suppose the effect of an innovation in investment in a system containing investment (y_1), income (y_2), and consumption (y_3) is of interest. To isolate such an effect, suppose that all three variables assume their mean value prior to time $t = 0$, $y_t = \mu$ for $t < 0$, and investment increases by one unit in period $t = 0$, that is, $u_{1,0} = 1$. Now we can trace out what happens to the system during periods $t = 1, 2, \dots$ if no further shocks occur, that is, $u_{2,0} = u_{3,0} = 0$, $u_1 = 0$, $u_2 = 0, \dots$. Because we are not interested in the mean of the system in such an exercise but just in the variations of the variables around their means, we assume that all three variables have mean zero and set $\nu = 0$ in (2.1.14). Hence, $y_t = A_1 y_{t-1} + u_t$ or, more precisely,

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \\ y_{3,t} \end{bmatrix} = \begin{bmatrix} .5 & 0 & 0 \\ .1 & .1 & .3 \\ 0 & .2 & .3 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \\ y_{3,t-1} \end{bmatrix} + \begin{bmatrix} u_{1,t} \\ u_{2,t} \\ u_{3,t} \end{bmatrix}. \quad (2.3.25)$$

Tracing a unit shock in the first variable in period $t = 0$ in this system we get

$$\begin{aligned} y_0 &= \begin{bmatrix} y_{1,0} \\ y_{2,0} \\ y_{3,0} \end{bmatrix} = \begin{bmatrix} u_{1,0} \\ u_{2,0} \\ u_{3,0} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \\ y_1 &= \begin{bmatrix} y_{1,1} \\ y_{2,1} \\ y_{3,1} \end{bmatrix} = A_1 y_0 = \begin{bmatrix} .5 \\ .1 \\ 0 \end{bmatrix}, \\ y_2 &= \begin{bmatrix} y_{1,2} \\ y_{2,2} \\ y_{3,2} \end{bmatrix} = A_1 y_1 = A_1^2 y_0 = \begin{bmatrix} .25 \\ .06 \\ .02 \end{bmatrix}. \end{aligned}$$

Continuing the procedure, it turns out that $y_i = (y_{1,i}, y_{2,i}, y_{3,i})'$ is just the first column of A_1^i . An analogous line of arguments shows that a unit shock in y_{2t} (y_{3t}) at $t = 0$, after i periods, results in a vector y_i which is just the second (third) column of A_1^i . Thus, the elements of A_1^i represent the effects of unit shocks in the variables of the system after i periods. Therefore they are called impulse responses or dynamic multipliers.

Recall that $A_1^i = \Phi_i$ is just the i -th coefficient matrix of the MA representation of a VAR(1) process. Consequently, the MA coefficient matrices contain the impulse responses of the system. This result holds more generally for higher order VAR(p) processes as well. To see this, suppose that y_t is a stationary VAR(p) process as in (2.1.1) with $\nu = 0$. This process has a corresponding VAR(1) process $Y_t = \mathbf{A}Y_{t-1} + U_t$ as in (2.1.8) with $\nu = 0$. Under the assumptions of the previous example, $y_t = 0$ for $t < 0$, $u_t = 0$ for $t > 0$ and $y_0 = u_0$ is a K -dimensional unit vector e_k , say, with a one as the k -th coordinate and zeros elsewhere. It follows that $Y_0 = (e'_k, 0, \dots, 0)'$ and $Y_i = \mathbf{A}^i Y_0$. Hence, the impulse responses are the elements of the upper left-hand ($K \times K$) block of $\mathbf{A}^i$. This matrix, however, was shown to be the i -th coefficient matrix Φ_i of the MA representation (2.1.17) of y_t , i.e., $\Phi_i = J\mathbf{A}^i J'$ with $J := [I_K : 0 : \dots : 0]$ a ($K \times Kp$) matrix. In other words, $\phi_{jk,i}$, the jk -th element of Φ_i , represents the reaction of the j -th variable of the system to a unit shock in variable k , i periods ago, provided, of course, the effect is not contaminated by other shocks to the system. Because the u_t are just the 1-step ahead forecast errors of the VAR process, the shocks considered here may be regarded as forecast errors and the impulse responses are sometimes referred to as *forecast error impulse responses*.

The response of variable j to a unit shock (forecast error) in variable k is sometimes depicted graphically to get a visual impression of the dynamic interrelationships within the system. Impulse responses of the investment/income/consumption system are plotted in Figure 2.5 and the dynamic

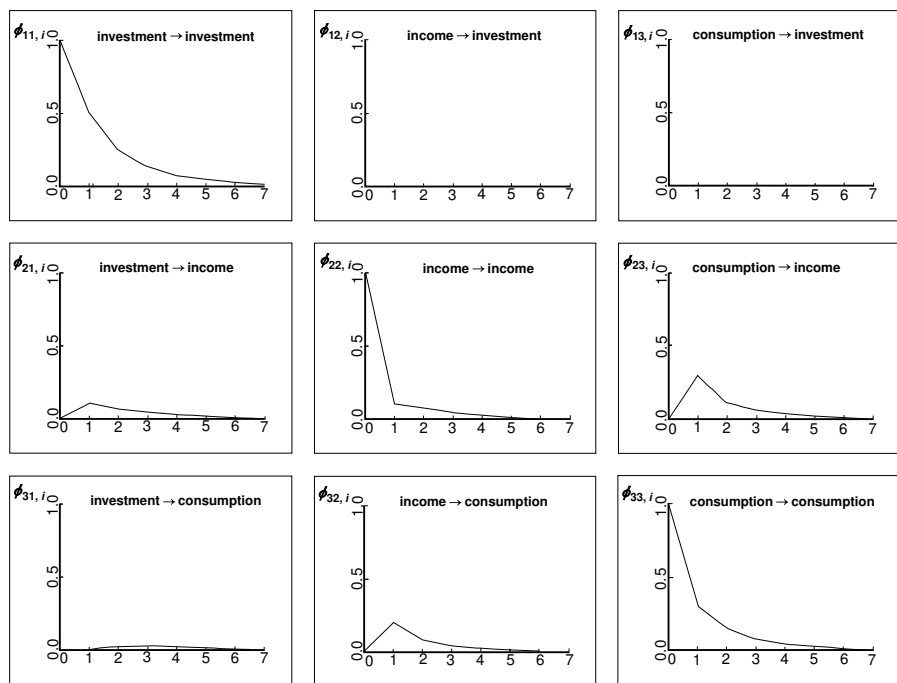


Fig. 2.5. Impulse responses of the investment/income/consumption system (impulse $\rightarrow$ response).

responses of the inflation/interest rate system are depicted in Figure 2.6. For instance, in the latter figure an inflation innovation is seen to induce the interest rate to increase for two periods and then it tapers off to zero. In both systems the effect of a unit shock in any of the variables dies away quite rapidly due to the stability of the systems.

If the variables have different scales, it is sometimes useful to consider innovations of one standard deviation rather than unit shocks. For instance, instead of tracing an unexpected unit increase in investment in the investment/income/consumption system with white noise covariance matrix (2.1.33), one may follow up on a shock of $\sqrt{2.25} = 1.5$ units because the standard deviation of u_{1t} is 1.5. Of course, this is just a matter of rescaling the impulse responses. In Figures 2.5 and 2.6, it suffices to choose the units at the vertical axes equal to the standard deviations of the residuals corresponding to the variables whose effects are considered. Such a rescaling may sometimes give a better picture of the dynamic relationships because the average size of the innovations occurring in a system depends on their standard deviation.

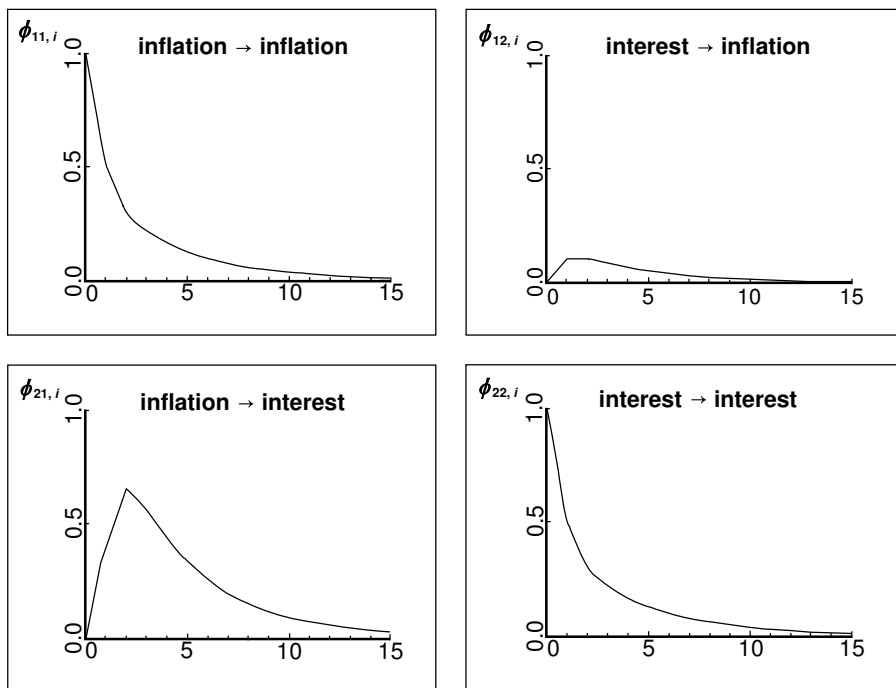


Fig. 2.6. Impulse responses of the inflation/interest rate system (impulse $\rightarrow$ response).

It follows from Proposition 2.2 that the impulse responses are zero if one of the variables does not Granger-cause the other variables taken as a group. More precisely, an innovation in variable k has no effect on the other variables if the former variable does not Granger-cause the set of the remaining variables. As we have mentioned previously, in applied work it is often of foremost interest whether one variable has an impact on a specific other variable. That is, one would like to know whether, for some $k \neq j$, $\phi_{jk,i} = 0$ for $i = 1, 2, \dots$. If the $\phi_{jk,i}$ represent the actual reactions of variable j to a unit shock in variable k , we may call the latter *noncausal* for the j -th variable if $\phi_{jk,i} = 0$ for $i = 1, 2, \dots$. In order to check the latter condition, it is not necessary to compute infinitely many Φ_i matrices. The following proposition shows that it suffices to check the first $p(K-1)$ Φ_i matrices.

Proposition 2.4 (*Zero Impulse Responses*)

If y_t is a K -dimensional stable VAR(p) process, then, for $j \neq k$,

$$\phi_{jk,i} = 0 \quad \text{for } i = 1, 2, \dots$$

is equivalent to

$$\phi_{jk,i} = 0 \quad \text{for } i = 1, \dots, p(K-1).$$

■

In other words, the proposition asserts that for a K -dimensional, stationary, stable VAR(p), if the first $pK - p$ responses of variable j to an impulse in variable k are zero, all the following responses must also be zero. For instance, in the investment/income/consumption VAR(1) system, because the responses of investment for the next two periods after a consumption impulse are zero, we know that investment will not react at all to such an impulse. Note, that in a VAR(1) system of dimension greater than 2, it does not suffice to check, say, the upper right-hand corner element of the coefficient matrix in order to determine whether the last variable is noncausal for the first variable. Notice that Proposition 2.4 is related to the conditions for multi-step causality in (2.3.23) and (2.3.24). In general, the conditions are not identical, however, because the two concepts differ. Proposition 2.4 will be helpful when testing of impulse response relations is discussed in the next chapter. We will now prove the proposition.

Proof of Proposition 2.4:

Returning to the lag operator notation of Section 2.1.2, we have

$$\Phi(L) = (\phi_{jk}(L))_{j,k} = A(L)^{-1} = A(L)^{adj} / \det(A(L)),$$

where $A(L)^{adj} = (A_{jk}(L))_{j,k}$ is the adjoint of $A(L) = I_K - A_1L - \dots - A_pL^p$ (see Appendix A.4.1). Obviously, $\phi_{jk}(L) \equiv 0$ is equivalent to $A_{jk}(L) \equiv 0$. From the definition of a cofactor of a matrix in Appendix A.3, it is easy to see that $A_{jk}(L)$ has degree not greater than $pK - p$. Defining $\gamma(L) = [\det A(L)]^{-1}$, we get for $k \neq j$,

$$\begin{aligned} \phi_{jk}(L) &= \phi_{jk,1}L + \phi_{jk,2}L^2 + \dots \\ &= A_{jk}(L)\gamma(L) \\ &= (A_{jk,1}L + \dots + A_{jk,pK-p}L^{pK-p})(1 + \gamma_1L + \dots). \end{aligned}$$

Hence,

$$\phi_{jk,1} = A_{jk,1} \quad \text{and} \quad \phi_{jk,i} = A_{jk,i} + \sum_{n=1}^{i-1} A_{jk,n}\gamma_{i-n} \quad \text{for } i > 1,$$

with $A_{jk,n} = 0$ for $n > pK - p$. Consequently, $A_{jk,i} = 0$ for $i = 1, \dots, pK - p$, is equivalent to $\phi_{jk,i} = 0$ for $i = 1, 2, \dots, pK - p$, which proves the proposition. ■

Sometimes interest centers on the accumulated effect over several or more periods of a shock in one variable. This effect may be determined by summing up the MA coefficient matrices. For instance, the k -th column of $\Psi_n := \sum_{i=0}^n \Phi_i$ contains the *accumulated responses* over n periods to a

unit shock in the k -th variable of the system. These quantities are sometimes called n -th *interim multipliers*. The total accumulated effects for all future periods are obtained by summing up all the MA coefficient matrices. $\Psi_\infty := \sum_{i=0}^{\infty} \Phi_i$ is sometimes called the matrix of *long-run effects* or *total multipliers*. Because the MA operator $\Phi(z)$ is the inverse of the VAR operator $A(z) = I_K - A_1 z - \cdots - A_p z^p$, the long-run effects are easily obtained as

$$\Psi_\infty = \Phi(1) = (I_K - A_1 - \cdots - A_p)^{-1}. \quad (2.3.26)$$

As an example, accumulated responses for the investment/income/consumption system are depicted in Figure 2.7. Similarly, interim and total multipliers of the inflation/interest rate system are shown in Figure 2.8.

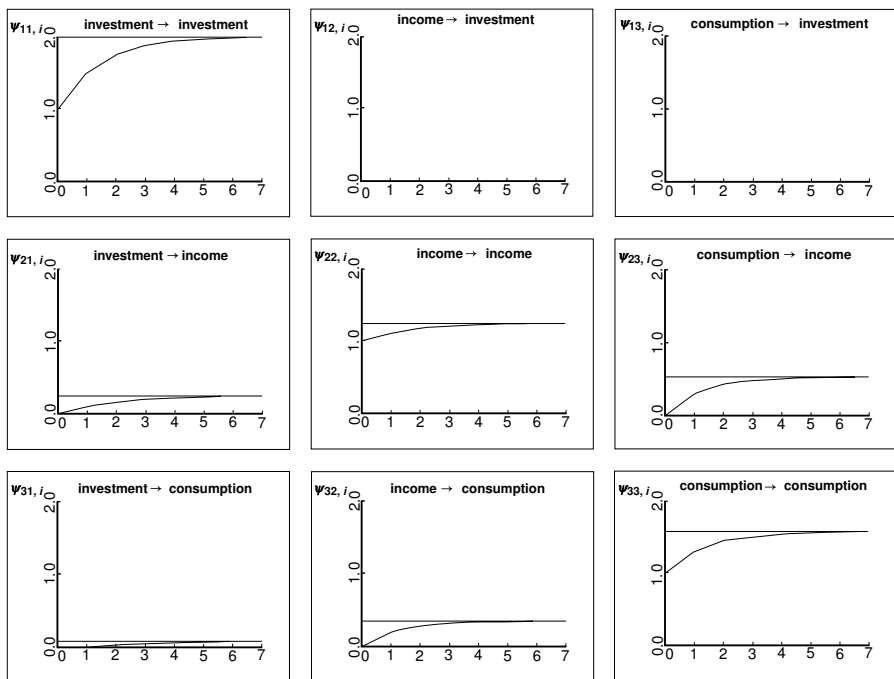


Fig. 2.7. Accumulated and long-run responses of the investment/income/consumption system (impulse $\rightarrow$ response).

Responses to Orthogonal Impulses

A problematic assumption in this type of impulse response analysis is that a shock occurs only in one variable at a time. Such an assumption may be

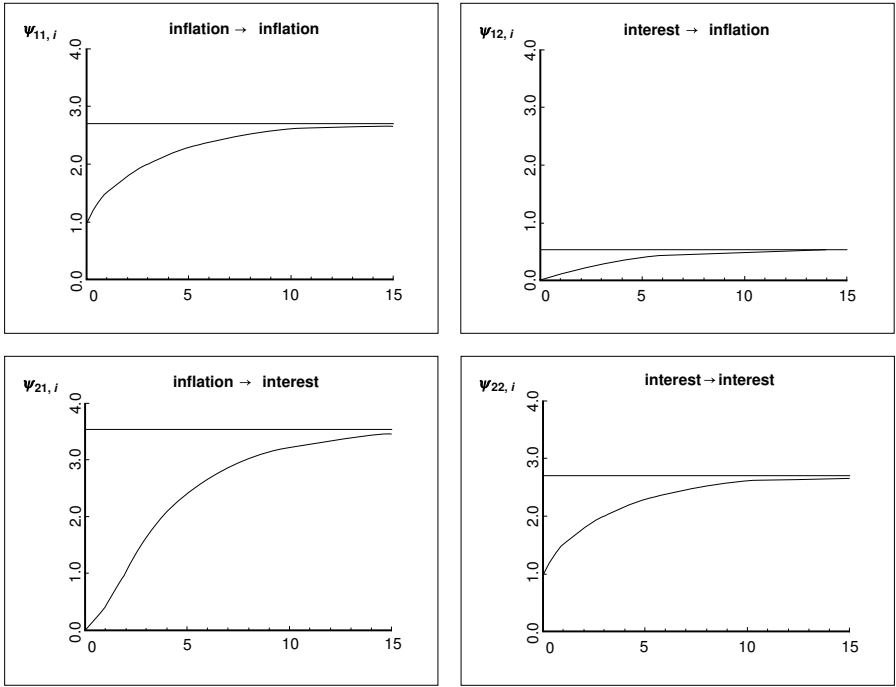


Fig. 2.8. Accumulated and total responses of the inflation/interest rate system (impulse $\rightarrow$ response).

reasonable if the shocks in different variables are independent. If they are not independent one may argue that the error terms consist of all the influences and variables that are not directly included in the set of y variables. Thus, in addition to forces that affect all the variables, there may be forces that affect variable 1, say, only. If a shock in the first variable is due to such forces it may again be reasonable to interpret the Φ_i coefficients as dynamic responses. On the other hand, correlation of the error terms may indicate that a shock in one variable is likely to be accompanied by a shock in another variable. In that case, setting all other residuals to zero may provide a misleading picture of the actual dynamic relationships between the variables. For example, in the investment/income/consumption system, the white noise or innovation covariance matrix is given in (2.1.33),

$$\Sigma_u = \begin{bmatrix} 2.25 & 0 & 0 \\ 0 & 1.0 & .5 \\ 0 & .5 & .74 \end{bmatrix}.$$

Obviously, there is a quite strong positive correlation between $u_{2,t}$ and $u_{3,t}$, the residuals of the income and consumption equations, respectively. Conse-

quently, a shock in income may be accompanied by a shock in consumption in the same period. Therefore, forcing the consumption innovation to zero when the effect of an income shock is traced, as in the previous analysis, may in fact obscure the actual relation between the variables.

This is the reason why impulse response analysis is often performed in terms of the MA representation (2.3.15),

$$y_t = \sum_{i=0}^{\infty} \Theta_i w_{t-i}, \quad (2.3.27)$$

where the components of $w_t = (w_{1t}, \dots, w_{Kt})'$ are uncorrelated and have unit variance, $\Sigma_w = I_K$. The mean term is dropped again because it is of no interest in the present analysis. Recall that the representation (2.3.27) is obtained by decomposing Σ_u as $\Sigma_u = PP'$, where P is a lower triangular matrix, and defining $\Theta_i = \Phi_i P$ and $w_t = P^{-1}u_t$. In (2.3.27) it is reasonable to assume that a change in one component of w_t has no effect on the other components because the components are orthogonal (uncorrelated). Moreover, the variances of the components are one. Thus, a unit innovation is just an innovation of size one standard deviation. The elements of the Θ_i are interpreted as responses of the system to such innovations. More precisely, the jk -th element of Θ_i is assumed to represent the effect on variable j of a unit innovation in the k -th variable that has occurred i periods ago.

To relate these impulse responses to a VAR model, we consider the zero mean VAR(p) process

$$y_t = A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t. \quad (2.3.28)$$

This process can be rewritten in such a way that the residuals of different equations are uncorrelated. For this purpose, we choose a decomposition of the white noise covariance matrix $\Sigma_u = W\Sigma_\varepsilon W'$, where Σ_ε is a diagonal matrix with positive diagonal elements and W is a lower triangular matrix with unit diagonal. This decomposition is obtained from the Choleski decomposition $\Sigma_u = PP'$ by defining a diagonal matrix D which has the same main diagonal as P and by specifying $W = PD^{-1}$ and $\Sigma_\varepsilon = DD'$.

Premultiplying (2.3.28) by $A := W^{-1}$ gives

$$Ay_t = A_1^* y_{t-1} + \dots + A_p^* y_{t-p} + \varepsilon_t, \quad (2.3.29)$$

where $A_i^* := AA_i$, $i = 1, \dots, p$, and $\varepsilon_t = (\varepsilon_{1t}, \dots, \varepsilon_{Kt})' := Au_t$ has diagonal covariance matrix,

$$\Sigma_\varepsilon = E(\varepsilon_t \varepsilon_t') = A E(u_t u_t') A' = A \Sigma_u A'.$$

Adding $(I_K - A)y_t$ to both sides of (2.3.29) gives

$$y_t = A_0^* y_t + A_1^* y_{t-1} + \dots + A_p^* y_{t-p} + \varepsilon_t, \quad (2.3.30)$$

where $A_0^* := I_K - A$. Because W is lower triangular with unit diagonal, the same is true for A . Hence,

$$A_0^* = I_K - A = \begin{bmatrix} 0 & 0 & \dots & 0 & 0 \\ \beta_{21} & 0 & \dots & 0 & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ \vdots & & \ddots & \ddots & \vdots \\ \beta_{K1} & \beta_{K2} & \dots & \beta_{K,K-1} & 0 \end{bmatrix}$$

is a lower triangular matrix with zero diagonal and, thus, in the representation (2.3.30) of our VAR(p) process, the first equation contains no instantaneous y 's on the right-hand side. The second equation may contain y_{1t} and otherwise lagged y 's on the right-hand side. More generally, the k -th equation may contain $y_{1t}, \dots, y_{k-1,t}$ and not $y_{kt}, \dots, y_{Kt}$ on the right-hand side. Thus, if (2.3.30) reflects the actual ongoing in the system, y_{st} cannot have an instantaneous impact on y_{kt} for $k < s$. In the econometrics literature such a system is called a *recursive model* (see Theil (1971, Section 9.6)). Herman Wold has advocated these models where the researcher has to specify the instantaneous "causal" ordering of the variables. This type of causality is therefore sometimes referred to as *Wold-causality*. If we trace ε_{it} innovations of size one standard error through the system (2.3.30), we just get the Θ impulse responses. This can be seen by solving the system (2.3.30) for y_t ,

$$y_t = (I_K - A_0^*)^{-1} A_1^* y_{t-1} + \dots + (I_K - A_0^*)^{-1} A_p^* y_{t-p} + (I_K - A_0^*)^{-1} \varepsilon_t.$$

Noting that $(I_K - A_0^*)^{-1} = W = PD^{-1}$ shows that the instantaneous effects of one-standard deviation shocks (ε_{it} 's of size one standard deviation) to the system are represented by the elements of $WD = P = \Theta_0$ because the diagonal elements of D are just standard deviations of the components of ε_t . The Θ_i may then be obtained by tracing these effects through the system.

The Θ_i 's may provide response functions that are quite different from the Φ_i responses. For the example VAR(1) system (2.3.25) with Σ_u as in (2.1.33) we get

$$\begin{aligned} \Theta_0 = P &= \begin{bmatrix} 1.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & .5 & .7 \end{bmatrix}, \\ \Theta_1 = \Phi_1 P &= \begin{bmatrix} .75 & 0 & 0 \\ .15 & .25 & .21 \\ 0 & .35 & .21 \end{bmatrix}, \\ \Theta_2 = \Phi_2 P &= \begin{bmatrix} .375 & 0 & 0 \\ .090 & .130 & .084 \\ .030 & .055 & .105 \end{bmatrix}, \end{aligned} \tag{2.3.31}$$

and so on. Some more innovation responses are depicted in Figure 2.9. Although they are similar to those given in Figure 2.5, there is an obvious difference in the response of consumption to an income innovation. While consumption responds with a time lag of one period in Figure 2.5, there is an instantaneous effect in Figure 2.9.

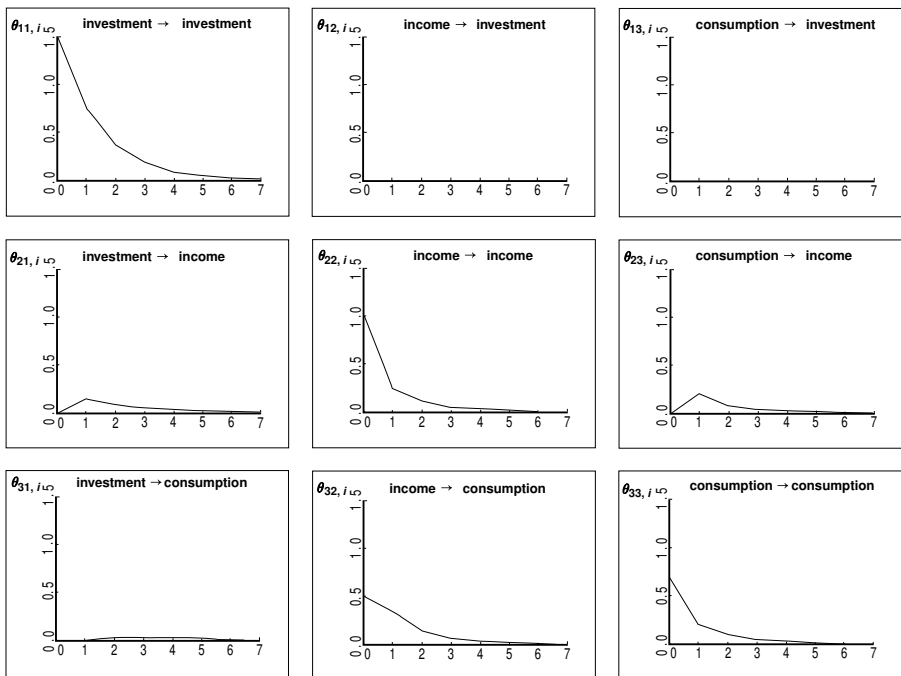


Fig. 2.9. Orthogonalized impulse responses of the investment/income/consumption system (impulse $\rightarrow$ response).

Note that $\Theta_0 = P$ is lower triangular and some elements below the diagonal will be nonzero if Σ_u has nonzero off-diagonal elements. For instance, for the investment/income/consumption example Θ_0 indicates that an income (y_2) innovation has an immediate impact on consumption (y_3). If the white noise covariance matrix Σ_u contains zeros, some components of $u_t = (u_{1t}, \dots, u_{Kt})'$ are contemporaneously uncorrelated. Suppose, for instance, that u_{1t} is uncorrelated with u_{it} for $i = 2, \dots, K$. In this case, $A = W^{-1}$ and, thus, A_0^* has a block of zeros so that y_1 has no instantaneous effect on y_i , $i = 2, \dots, K$. In the example, investment has no instantaneous impact on income and consumption because

$$\Sigma_u = \begin{bmatrix} 2.25 & 0 & 0 \\ 0 & 1.0 & .5 \\ 0 & .5 & .74 \end{bmatrix}$$

and, hence, u_{1t} is uncorrelated with u_{2t} and u_{3t} . This, of course, is reflected in the matrix of instantaneous effects Θ_0 given in (2.3.31). Because the elements of $P = \Theta_0$ represent the immediate responses of the system to unit innovations they are sometimes called *impact multipliers*.

In order to determine whether there is no response at all of one variable to an impulse in one of the other variables, it suffices to consider the first $pK - p$ response coefficients and the immediate effect. This result is stated formally in the next proposition where $\theta_{jk,i}$ denotes the jk -th element of Θ_i .

Proposition 2.5 (*Zero Orthogonalized Impulse Responses*)

If y_t is a K -dimensional stable VAR(p) process, then, for $j \neq k$,

$$\theta_{jk,i} = 0 \quad \text{for } i = 0, 1, 2, \dots$$

is equivalent to

$$\theta_{jk,i} = 0 \quad \text{for } i = 0, 1, \dots, p(K - 1).$$

■

The proof of this result is analogous to that of Proposition 2.4 and is left as an exercise (see Problem 2.2).

The fact that Θ_0 is lower triangular shows that the ordering of the variables is of importance, that is, it is important which of the variables is called y_1 and which one is called y_2 and so on. One problem with this type of impulse response analysis is that the ordering of the variables cannot be determined with statistical methods but has to be specified by the analyst. The ordering has to be such that the first variable is the only one with a potential immediate impact on all other variables. The second variable may have an immediate impact on the last $K - 2$ components of y_t but not on y_{1t} and so on. To establish such an ordering may be a quite difficult exercise in practice. The choice of the ordering, the Wold causal ordering, may, to a large extent, determine the impulse responses and is therefore critical for the interpretation of the system. Currently we are dealing with known systems only. In this situation, assuming that the ordering is known may not be a great restriction. For the investment/income/consumption example it may be reasonable to assume that an increase in income has an immediate effect on consumption while increased consumption stimulates the economy and, hence, income with some time lag.

Our interpretation of orthogonalized impulse responses is based on the representation (2.3.30) and the impulses are viewed as changes in the observed variables. Sometimes it is more plausible to focus on impulses which cannot be associated easily with changes in a specific observed variable within the

system. In that case, it may be more logical to base the interpretation on the MA representation (2.3.27) which decomposes the variables in contributions of the w_{kt} innovations. If these innovations can be associated with a specific impulse to the system, the orthogonalized impulse responses reflect the reactions of the variables to such possibly unobserved innovations. In that case, a specific impulse or shock to the system can have an instantaneous impact on several variables while some other impulse may only have an instantaneous effect on one specific variable and may effect the other variables only with some delay. By decomposing $\Sigma_u = PP'$ with some non-triangular P matrix, it is also possible that all shocks have instantaneous effects on all observed variables of the system. In this kind of interpretation, finding the decomposition matrix P and, hence, the innovations w_t which actually can be associated with shocks of interest, is often a difficult part of the analysis. We will provide a more in-depth discussion of the related problems in Chapter 9 which deals with structural VAR models.

Critique of Impulse Response Analysis

Besides specifying the relevant impulses to a system, there are a number of further problems that render the interpretation of impulse responses difficult. We have mentioned some of them in the context of Granger-causality. A major limitation of our systems is their potential incompleteness. Although in real economic systems almost everything depends on everything else, we will usually work with low-dimensional VAR systems. All effects of omitted variables are assumed to be in the innovations. If important variables are omitted from the system, this may lead to major distortions in the impulse responses and makes them worthless for structural interpretations. The system may still be useful for prediction, though.

To see the related problems more clearly, consider a system y_t which is partitioned in vectors z_t and x_t as in (2.3.5). If the z_t variables are considered only and the x_t variables are omitted from the analysis, we get a system

$$\begin{aligned} z_t &= \mu_1 + \sum_{i=0}^{\infty} \Phi_{11,i} u_{1,t-i} + \sum_{i=1}^{\infty} \Phi_{12,i} u_{2,t-i} \\ &= \mu_1 + \sum_{i=0}^{\infty} F_i v_{t-i}, \end{aligned} \tag{2.3.32}$$

as in (2.3.8). The actual reactions of the z_t components to innovations u_{1t} may be given by the $\Phi_{11,i}$ matrices. On the other hand, the F_i or corresponding orthogonalized “impulse responses” are likely to be interpreted as impulse responses if the analyst does not realize that important variables have been omitted. As we have seen in Section 2.3.1, the F_i will be equal to the $\Phi_{11,i}$ if and only if x_t does not Granger-cause z_t .

Further problems related to the interpretation of the MA coefficients as dynamic multipliers or impulse responses result from measurement errors and

the use of seasonally adjusted or temporally and/or contemporaneously aggregated variables. A detailed account of the aggregation problem is given by Lütkepohl (1987). We will discuss these problems in more detail in Chapter 11 in the context of more general models. These problems severely limit the interpretability of the MA coefficients of a VAR system as impulse responses. In the next subsection a further possibility to interpret VAR models will be considered.

2.3.3 Forecast Error Variance Decomposition

If the innovations which actually drive the system can be identified, a further tool for interpreting VAR models is available. Suppose a recursive identification scheme is available so that the MA representation (2.3.15) with orthogonal white noise innovations may be considered. In the context of the representation

$$y_t = \mu + \sum_{i=0}^{\infty} \Theta_i w_{t-i} \quad (2.3.33)$$

with $\Sigma_w = I_K$, the error of the optimal h -step forecast is

$$\begin{aligned} y_{t+h} - y_t(h) &= \sum_{i=0}^{h-1} \Phi_i u_{t+h-i} = \sum_{i=0}^{h-1} \Phi_i P P^{-1} u_{t+h-i} \\ &= \sum_{i=0}^{h-1} \Theta_i w_{t+h-i}. \end{aligned} \quad (2.3.34)$$

Denoting the mn -th element of Θ_i by $\theta_{mn,i}$ as before, the h -step forecast error of the j -th component of y_t is

$$\begin{aligned} y_{j,t+h} - y_{j,t}(h) &= \sum_{i=0}^{h-1} (\theta_{j1,i} w_{1,t+h-i} + \cdots + \theta_{jK,i} w_{K,t+h-i}) \\ &= \sum_{k=1}^K (\theta_{jk,0} w_{k,t+h} + \cdots + \theta_{jk,h-1} w_{k,t+1}). \end{aligned} \quad (2.3.35)$$

Thus, the forecast error of the j -th component potentially consists of all the innovations $w_{1t}, \dots, w_{Kt}$. Of course, some of the $\theta_{mn,i}$ may be zero so that some components may not appear in (2.3.35). Because the $w_{k,t}$'s are uncorrelated and have unit variances, the MSE of $y_{j,t}(h)$ is

$$E(y_{j,t+h} - y_{j,t}(h))^2 = \sum_{k=1}^K (\theta_{jk,0}^2 + \cdots + \theta_{jk,h-1}^2).$$

Therefore,

$$\theta_{jk,0}^2 + \theta_{jk,1}^2 + \cdots + \theta_{jk,h-1}^2 = \sum_{i=0}^{h-1} (e_j' \Theta_i e_k)^2 \quad (2.3.36)$$

is sometimes interpreted as the contribution of innovations in variable k to the forecast error variance or MSE of the h -step forecast of variable j . Here e_k is the k -th column of I_K . Dividing (2.3.36) by

$$\text{MSE}[y_{j,t}(h)] = \sum_{i=0}^{h-1} \sum_{k=1}^K \theta_{jk,i}^2$$

gives

$$\omega_{jk,h} = \sum_{i=0}^{h-1} (e_j' \Theta_i e_k)^2 / \text{MSE}[y_{j,t}(h)] \quad (2.3.37)$$

which is the proportion of the h -step forecast error variance of variable j , accounted for by w_{kt} innovations. If w_{kt} can be associated with variable k , $\omega_{jk,h}$ represents the proportion of the h -step forecast error variance accounted for by innovations in variable k . Thereby, the forecast error variance is decomposed into components accounted for by innovations in the different variables of the system. From (2.3.34), the h -step forecast MSE matrix is seen to be

$$\Sigma_y(h) = \text{MSE}[y_t(h)] = \sum_{i=0}^{h-1} \Theta_i \Theta_i' = \sum_{i=0}^{h-1} \Phi_i \Sigma_u \Phi_i'.$$

The diagonal elements of this matrix are the MSEs of the y_{jt} variables which may be used in (2.3.37).

For the investment/income/consumption example, forecast error variance decompositions of all three variables are given in Table 2.1. For instance, about 66% of the 1-step forecast error variance of consumption is accounted for by own innovations and about 34% is accounted for by income innovations. For long term forecasts, 57.5% and 42.3% of the error variance is accounted for by consumption and income innovations, respectively. For any forecast horizon, investment innovations contribute less than 1% to the forecast error variance of consumption. Moreover, only small fractions (less than 10%) of the forecast error variances of income are accounted for by innovations in the other variables of the system. This kind of analysis is sometimes called *innovation accounting*.

From Proposition 2.5, it is obvious that for a stationary, stable, K -dimensional VAR(p) process y_t all forecast error variance proportions of variable j , accounted for by innovations in variable k , will be zero if $\omega_{jk,h} = 0$ for $h = pK - p + 1$. In this context it is perhaps worth pointing out the relationship between Granger-causality and forecast error variance components. For that purpose we consider a bivariate system $y_t = (z_t, x_t)'$ first. In such a system, if z_t does not Granger-cause x_t , the proportions of forecast error variances of x_t

Table 2.1. Forecast error variance decomposition of the investment/income/consumption system

forecast error in	forecast horizon h	proportions of forecast error variance h periods ahead accounted for by innovations in		
		investment	income	consumption
investment	1	1	0	0
	2	1	0	0
	3	1	0	0
	4	1	0	0
	5	1	0	0
	10	1	0	0
	∞	1	0	0
income	1	0	1	0
	2	.020	.941	.039
	3	.026	.930	.044
	4	.029	.926	.045
	5	.030	.925	.045
	10	.030	.925	.045
	∞	.030	.925	.045
consumption	1	0	.338	.662
	2	0	.411	.589
	3	.001	.421	.578
	4	.002	.423	.576
	5	.002	.423	.575
	10	.002	.423	.575
	∞	.002	.423	.575

accounted for by innovations in z_t may still be nonzero. This property follows directly from the definition of the Θ_i in (2.3.15). Granger-noncausality, by Proposition 2.2, implies zero constraints on the Φ_i which may disappear in the Θ_i if the error covariance matrix Σ_u is not diagonal. On the other hand, if Σ_u is diagonal, so that there is no instantaneous causation between z_t and x_t and if, in addition, z_t is not Granger-causal for x_t the lower left-hand elements of the Θ_i will be zero (see (2.3.19)). Therefore, the proportion of forecast error variance of x_t accounted for by z_t innovations will also be zero.

In a higher dimensional system, suppose a set of variables z_t does not Granger-cause the remaining variables x_t and there is also no instantaneous causality between the two sets of variables. In that case, the forecast MSE proportions of all x_t variables accounted for by z_t variables will be zero.

It is important to understand, however, that Granger-causality and forecast error variance decompositions are quite different concepts because Granger-causality and instantaneous causality are different concepts. While Granger-causality is a uniquely defined property of two subsets of variables of a given process, the forecast error variance decomposition is not unique as it depends on the Θ_i matrices and, thus, on the choice of the transformation matrix

P. Therefore, the interpretation of a forecast error variance decomposition is subject to similar criticisms as the interpretation of impulse responses. In addition, all the critical points raised in the context of Granger-causality apply. That is, the forecast error variance components are conditional on the system under consideration. They may change if the system is expanded by adding further variables or if variables are deleted from the system. Also measurement errors, seasonal adjustment and the use of aggregates may contaminate the forecast error variance decompositions.

2.3.4 Remarks on the Interpretation of VAR Models

Innovation accounting and impulse response analysis in the framework of VAR models have been pioneered by Sims (1980, 1981) and others as an alternative to classical macroeconomic analyses. Sims' main criticism of the latter type of analysis is that macroeconometric models are often not based on sound economic theories or the available theories are not capable of providing a completely specified model. If economic theories are not available to specify the model, statistical tools must be applied. In this approach, a fairly loose model is set up which does not impose rigid a priori restrictions on the data generation process. Statistical tools are then used to determine possible constraints. VAR models represent a class of loose models that may be used in such an approach. Of course, in order to interpret these models, some restrictive assumptions need to be made. In particular, the ordering of the variables may be essential for interpretations of the types discussed in the previous subsections. Sims (1981) suggests to try different orderings and investigate the sensitivity of the corresponding orthogonalized impulse responses and the related conclusions to the ordering of the variables.

So far we have assumed that a VAR model is given to us. Under this assumption we have discussed forecasting and interpretation of the system. In this situation it is of course unnecessary to use statistical tools in order to determine constraints for the system because all constraints are known. In practice, we will virtually never be in such a fortunate situation but we have to determine the model from a given set of time series data. This problem will be treated in subsequent chapters. The purpose of this chapter is to identify some problems that are not related to estimation and model specification but are inherent to the types of models considered.

2.4 Exercises

Problem 2.1

Show that

$$\det(I_{Kp} - \mathbf{A}z) = \det(I_K - A_1z - \cdots - A_pz^p)$$

where A_i , $i = 1, \dots, p$, and $\mathbf{A}$ are as in (2.1.1) and (2.1.8), respectively.

Problem 2.2

Prove Proposition 2.5.

(Hint: $\Theta(L) = \Phi(L)P = A(L)^{adj}P / \det A(L)$).

Problem 2.3

In the United States of Wonderland the growth rates of income (GNP) and the money stock (M2) as well as an interest rate (IR) are related as in the following VAR(2) model:

$$\begin{bmatrix} \text{GNP}_t \\ \text{M2}_t \\ \text{IR}_t \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} .7 & .1 & 0 \\ 0 & .4 & .1 \\ .9 & 0 & .8 \end{bmatrix} \begin{bmatrix} \text{GNP}_{t-1} \\ \text{M2}_{t-1} \\ \text{IR}_{t-1} \end{bmatrix} + \begin{bmatrix} -.2 & 0 & 0 \\ 0 & .1 & .1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \text{GNP}_{t-2} \\ \text{M2}_{t-2} \\ \text{IR}_{t-2} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \end{bmatrix},$$

$$\Sigma_u = \begin{bmatrix} .26 & .03 & 0 \\ .03 & .09 & 0 \\ 0 & 0 & .81 \end{bmatrix} = PP', \quad P = \begin{bmatrix} .5 & .1 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & .9 \end{bmatrix}. \quad (2.4.1)$$

- Show that the process $y_t = (\text{GNP}_t, \text{M2}_t, \text{IR}_t)'$ is stable.
- Determine the mean vector of y_t .
- Write the process y_t in VAR(1) form.
- Compute the coefficient matrices $\Phi_1, \dots, \Phi_5$ of the MA representation (2.1.17) of y_t .

Problem 2.4

Determine the autocovariances $\Gamma_y(0), \Gamma_y(1), \Gamma_y(2), \Gamma_y(3)$ of the process defined in (2.4.1). Compute and plot the autocorrelations $R_y(0), R_y(1), R_y(2), R_y(3)$.

Problem 2.5

Consider again the process (2.4.1).

- Suppose that

$$y_{2000} = \begin{bmatrix} .7 \\ 1.0 \\ 1.5 \end{bmatrix} \quad \text{and} \quad y_{1999} = \begin{bmatrix} 1.0 \\ 1.5 \\ 3.0 \end{bmatrix}$$

and forecast y_{2001}, y_{2002} , and y_{2003} .

- Determine the MSE matrices for forecast horizons $h = 1, 2, 3$.
- Assume that y_t is a Gaussian process and construct 90% and 95% forecast intervals for $t = 2001, 2002, 2003$.
- Use the Bonferroni method to determine a joint forecast region for $\text{GNP}_{2001}, \text{GNP}_{2002}, \text{GNP}_{2003}$ with probability content at least 97%.

Problem 2.6

Answer the following questions for the process (2.4.1).

- (a) Is M2 Granger-causal for (GNP, IR)?
- (b) Is IR Granger-causal for (GNP, M2)?
- (c) Is there instantaneous causality between M2 and (GNP, IR)?
- (d) Is there instantaneous causality between IR and (GNP, M2)?
- (e) Is IR 2-step causal for GNP?

Problem 2.7

Plot the effect of a unit innovation in the interest rate (IR) on the three variables of the system (2.4.1) in terms of the MA representation (2.1.17). Consider only 5 periods following the innovation. Plot also the accumulated responses and interpret the plots.

Problem 2.8

For the system (2.4.1), derive the coefficient matrices $\Theta_0, \dots, \Theta_5$ of the MA representation (2.3.15) using the *upper* triangular P matrix given in (2.4.1). Plot the effects of a unit innovation in IR in terms of that representation. Compare to the plots obtained in Problem 2.7 and interpret. Repeat the analysis with a lower triangular P matrix and comment on the results.

Problem 2.9

Decompose the MSE of the forecast $\text{GNP}_t(5)$ into the proportions accounted for by its own innovations and innovations in M2 and IR.

Estimation of Vector Autoregressive Processes

3.1 Introduction

In this chapter, it is assumed that a K -dimensional multiple time series $y_1, \dots, y_T$ with $y_t = (y_{1t}, \dots, y_{Kt})'$ is available that is known to be generated by a stationary, stable VAR(p) process

$$y_t = \nu + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t. \quad (3.1.1)$$

All symbols have their usual meanings, that is, $\nu = (\nu_1, \dots, \nu_K)'$ is a $(K \times 1)$ vector of intercept terms, the A_i are $(K \times K)$ coefficient matrices and u_t is white noise with nonsingular covariance matrix Σ_u . In contrast to the assumptions of the previous chapter, the coefficients $\nu, A_1, \dots, A_p$, and Σ_u are assumed to be unknown in the following. The time series data will be used to estimate the coefficients. Note that notationwise we do not distinguish between the stochastic process and a time series as a realization of a stochastic process. The particular meaning of a symbol should be obvious from the context.

In the next three sections, different possibilities for estimating a VAR(p) process are discussed. In Section 3.5, the consequences of forecasting with estimated processes will be considered and, in Section 3.6, tests for causality are described. The distribution of impulse responses obtained from estimated processes is considered in Section 3.7.

3.2 Multivariate Least Squares Estimation

In this section, multivariate least squares (LS) estimation is discussed. The estimator obtained for the standard form (3.1.1) of a VAR(p) process is considered in Section 3.2.1. Some properties of the estimator are derived in Sections 3.2.2 and 3.2.4 and an example is given in Section 3.2.3.

3.2.1 The Estimator

It is assumed that a time series $y_1, \dots, y_T$ of the y variables is available, that is, we have a sample of size T for each of the K variables for the same sample period. In addition, p presample values for each variable, $y_{-p+1}, \dots, y_0$, are assumed to be available. Partitioning a multiple time series in sample and presample values is convenient in order to simplify the notation. We define

$$\begin{aligned}
 Y &:= (y_1, \dots, y_T) & (K \times T), \\
 B &:= (\nu, A_1, \dots, A_p) & (K \times (Kp + 1)), \\
 Z_t &:= \begin{bmatrix} 1 \\ y_t \\ \vdots \\ y_{t-p+1} \end{bmatrix} & ((Kp + 1) \times 1), \\
 Z &:= (Z_0, \dots, Z_{T-1}) & ((Kp + 1) \times T), \\
 U &:= (u_1, \dots, u_T) & (K \times T), \\
 \mathbf{y} &:= \text{vec}(Y) & (KT \times 1), \\
 \boldsymbol{\beta} &:= \text{vec}(B) & ((K^2p + K) \times 1), \\
 \mathbf{b} &:= \text{vec}(B') & ((K^2p + K) \times 1), \\
 \mathbf{u} &:= \text{vec}(U) & ((KT \times 1)).
 \end{aligned} \tag{3.2.1}$$

Here vec is the column stacking operator as defined in Appendix A.12.

Using this notation, for $t = 1, \dots, T$, the $\text{VAR}(p)$ model (3.1.1) can be written compactly as

$$Y = BZ + U \tag{3.2.2}$$

or

$$\begin{aligned}
 \text{vec}(Y) &= \text{vec}(BZ) + \text{vec}(U) \\
 &= (Z' \otimes I_K) \text{vec}(B) + \text{vec}(U)
 \end{aligned}$$

or

$$\mathbf{y} = (Z' \otimes I_K) \boldsymbol{\beta} + \mathbf{u}. \tag{3.2.3}$$

Note that the covariance matrix of $\mathbf{u}$ is

$$\Sigma_{\mathbf{u}} = I_T \otimes \Sigma_u. \tag{3.2.4}$$

Thus, multivariate LS estimation (or GLS estimation) of $\boldsymbol{\beta}$ means to choose the estimator that minimizes

$$\begin{aligned}
S(\beta) &= \mathbf{u}'(I_T \otimes \Sigma_u)^{-1} \mathbf{u} = \mathbf{u}'(I_T \otimes \Sigma_u^{-1}) \mathbf{u} \\
&= [\mathbf{y} - (Z' \otimes I_K) \beta]' (I_T \otimes \Sigma_u^{-1}) [\mathbf{y} - (Z' \otimes I_K) \beta] \\
&= \text{vec}(Y - BZ)' (I_T \otimes \Sigma_u^{-1}) \text{vec}(Y - BZ) \\
&= \text{tr} [(Y - BZ)' \Sigma_u^{-1} (Y - BZ)].
\end{aligned} \tag{3.2.5}$$

In order to find the minimum of this function we note that

$$\begin{aligned}
S(\beta) &= \mathbf{y}'(I_T \otimes \Sigma_u^{-1}) \mathbf{y} + \beta'(Z \otimes I_K)(I_T \otimes \Sigma_u^{-1})(Z' \otimes I_K) \beta \\
&\quad - 2\beta'(Z \otimes I_K)(I_T \otimes \Sigma_u^{-1}) \mathbf{y} \\
&= \mathbf{y}'(I_T \otimes \Sigma_u^{-1}) \mathbf{y} + \beta'(ZZ' \otimes \Sigma_u^{-1}) \beta - 2\beta'(Z \otimes \Sigma_u^{-1}) \mathbf{y}.
\end{aligned}$$

Hence,

$$\frac{\partial S(\beta)}{\partial \beta} = 2(ZZ' \otimes \Sigma_u^{-1}) \beta - 2(Z \otimes \Sigma_u^{-1}) \mathbf{y}.$$

Equating to zero gives the *normal equations*

$$(ZZ' \otimes \Sigma_u^{-1}) \hat{\beta} = (Z \otimes \Sigma_u^{-1}) \mathbf{y} \tag{3.2.6}$$

and, consequently, the LS estimator is

$$\begin{aligned}
\hat{\beta} &= ((ZZ')^{-1} \otimes \Sigma_u)(Z \otimes \Sigma_u^{-1}) \mathbf{y} \\
&= ((ZZ')^{-1} Z \otimes I_K) \mathbf{y}.
\end{aligned} \tag{3.2.7}$$

The Hessian of $S(\beta)$,

$$\frac{\partial^2 S}{\partial \beta \partial \beta'} = 2(ZZ' \otimes \Sigma_u^{-1}),$$

is positive definite which confirms that $\hat{\beta}$ is indeed a minimizing vector. Strictly speaking, for these results to hold, it has to be assumed that ZZ' is nonsingular. This result will hold with probability 1 if y_t has a continuous distribution which will always be assumed in the following.

It may be worth noting that the multivariate LS estimator $\hat{\beta}$ is identical to the ordinary LS (OLS) estimator obtained by minimizing

$$\bar{S}(\beta) = \mathbf{u}' \mathbf{u} = [\mathbf{y} - (Z' \otimes I_K) \beta]' [\mathbf{y} - (Z' \otimes I_K) \beta] \tag{3.2.8}$$

(see Problem 3.1). This result is due to Zellner (1962) who showed that GLS and LS estimation in a multiple equation model are identical if the regressors in all equations are the same.

The LS estimator can be written in different ways that will be useful later on:

$$\begin{aligned}
\hat{\beta} &= ((ZZ')^{-1} Z \otimes I_K) [(Z' \otimes I_K) \beta + \mathbf{u}] \\
&= \beta + ((ZZ')^{-1} Z \otimes I_K) \mathbf{u}
\end{aligned} \tag{3.2.9}$$

or

$$\begin{aligned}\text{vec}(\widehat{B}) &= \widehat{\beta} = ((ZZ')^{-1}Z \otimes I_K) \text{vec}(Y) \\ &= \text{vec}(YZ'(ZZ')^{-1}).\end{aligned}$$

Thus,

$$\begin{aligned}\widehat{B} &= YZ'(ZZ')^{-1} \\ &= (BZ + U)Z'(ZZ')^{-1} \\ &= B + UZ'(ZZ')^{-1}.\end{aligned}\tag{3.2.10}$$

Another possibility for deriving this estimator results from postmultiplying

$$y_t = BZ_{t-1} + u_t$$

by Z'_{t-1} and taking expectations:

$$E(y_t Z'_{t-1}) = BE(Z_{t-1} Z'_{t-1}).\tag{3.2.11}$$

Estimating $E(y_t Z'_{t-1})$ by

$$\frac{1}{T} \sum_{t=1}^T y_t Z'_{t-1} = \frac{1}{T} Y Z'$$

and $E(Z_{t-1} Z'_{t-1})$ by

$$\frac{1}{T} \sum_{t=1}^T Z_{t-1} Z'_{t-1} = \frac{1}{T} Z Z',$$

we obtain the normal equations

$$\frac{1}{T} Y Z' = \widehat{B} \frac{1}{T} Z Z'$$

and, hence, $\widehat{B} = YZ'(ZZ')^{-1}$. Note that (3.2.11) is similar but not identical to the system of Yule-Walker equations in (2.1.37). While central moments about the expectation $\mu = E(y_t)$ are considered in (2.1.37), moments about zero are used in (3.2.11).

Yet another possibility to write the LS estimator is

$$\widehat{\mathbf{b}} = \text{vec}(\widehat{B}') = (I_K \otimes (ZZ')^{-1}Z) \text{vec}(Y').\tag{3.2.12}$$

In this form, it is particularly easy to see that multivariate LS estimation is equivalent to OLS estimation of each of the K equations in (3.1.1) separately. Let b'_k be the k -th row of B , that is, b_k contains all the parameters of the k -th equation. Obviously $\mathbf{b}' = (b'_1, \dots, b'_K)$. Furthermore, let $y_{(k)} = (y_{k1}, \dots, y_{kT})'$ be the time series available for the k -th variable, so that

$$\text{vec}(Y') = \begin{bmatrix} y_{(1)} \\ \vdots \\ y_{(K)} \end{bmatrix}.$$

With this notation $\widehat{b}_k = (ZZ')^{-1}Zy_{(k)}$ is the OLS estimator of the model $y_{(k)} = Z'b_k + u_{(k)}$, where $u_{(k)} = (u_{k1}, \dots, u_{kT})'$ and $\widehat{\mathbf{b}}' = (\widehat{b}'_1, \dots, \widehat{b}'_K)$.

3.2.2 Asymptotic Properties of the Least Squares Estimator

Because small sample properties of the LS estimator are difficult to derive analytically, we focus on asymptotic properties. Consistency and asymptotic normality of the LS estimator are easily established if the following results hold:

$$\Gamma := \text{plim } ZZ'/T \text{ exists and is nonsingular} \quad (3.2.13)$$

and

$$\begin{aligned} \frac{1}{\sqrt{T}} \sum_{t=1}^T \text{vec}(u_t Z'_{t-1}) &= \frac{1}{\sqrt{T}} \text{vec}(U Z') = \frac{1}{\sqrt{T}} (Z \otimes I_K) \mathbf{u} \\ &\xrightarrow[T \rightarrow \infty]{d} \mathcal{N}(0, \Gamma \otimes \Sigma_u), \end{aligned} \quad (3.2.14)$$

where, as usual, $\xrightarrow{d}$ denotes convergence in distribution. It follows from a theorem due to Mann & Wald (1943) that these results are true under suitable conditions for u_t , if y_t is a stationary, stable VAR(p). For instance, the conditions stated in the following definition are sufficient.

Definition 3.1 (*Standard White Noise*)

A white noise process $u_t = (u_{1t}, \dots, u_{Kt})'$ is called *standard white noise* if the u_t are continuous random vectors satisfying $E(u_t) = 0$, $\Sigma_u = E(u_t u'_t)$ is nonsingular, u_t and u_s are independent for $s \neq t$, and, for some finite constant c ,

$$E|u_{it} u_{jt} u_{kt} u_{mt}| \leq c \quad \text{for } i, j, k, m = 1, \dots, K, \text{ and all } t.$$

■

The last condition means that all fourth moments exist and are bounded. Obviously, if the u_t are normally distributed (Gaussian) they satisfy the moment requirements. With this definition it is easy to state conditions for consistency and asymptotic normality of the LS estimator. The following lemma will be essential in proving these large sample results.

Lemma 3.1

If y_t is a stable, K -dimensional VAR(p) process as in (3.1.1) with standard white noise residuals u_t , then (3.2.13) and (3.2.14) hold. ■

Proof: See Theorem 8.2.3 of Fuller (1976, p. 340). ■

The lemma holds also for other definitions of standard white noise. For example, the convergence result in (3.2.14) follows from a central limit theorem for martingale differences or martingale difference arrays (see Proposition C.13) by noting that $w_t = \text{vec}(u_t Z'_{t-1})$ is a martingale difference sequence under quite general conditions. The convergence result in (3.2.13) may then be

obtained from a suitable weak law of large numbers (see Proposition C.12). In the next proposition the resulting asymptotic properties of the LS estimator are stated formally.

Proposition 3.1 (*Asymptotic Properties of the LS Estimator*)

Let y_t be a stable, K -dimensional VAR(p) process as in (3.1.1) with standard white noise residuals, $\hat{B} = YZ'(ZZ')^{-1}$ is the LS estimator of the VAR coefficients B and all symbols are as defined in (3.2.1). Then,

$$\text{plim } \hat{B} = B$$

and

$$\sqrt{T}(\hat{\beta} - \beta) = \sqrt{T} \text{vec}(\hat{B} - B) \xrightarrow{d} \mathcal{N}(0, \Gamma^{-1} \otimes \Sigma_u) \quad (3.2.15)$$

or, equivalently,

$$\sqrt{T}(\hat{\mathbf{b}} - \mathbf{b}) = \sqrt{T} \text{vec}(\hat{B}' - B') \xrightarrow{d} \mathcal{N}(0, \Sigma_u \otimes \Gamma^{-1}), \quad (3.2.16)$$

where $\Gamma = \text{plim } ZZ'/T$. ■

Proof: Using (3.2.10),

$$\text{plim}(\hat{B} - B) = \text{plim} \left(\frac{UZ'}{T} \right) \text{plim} \left(\frac{ZZ'}{T} \right)^{-1} = 0$$

by Lemma 3.1, because (3.2.14) implies $\text{plim } UZ'/T = 0$. Thus, the consistency of $\hat{B}$ is established.

Using (3.2.9),

$$\begin{aligned} \sqrt{T}(\hat{\beta} - \beta) &= \sqrt{T}((ZZ')^{-1}Z \otimes I_K)\mathbf{u} \\ &= \left(\left(\frac{1}{T}ZZ' \right)^{-1} \otimes I_K \right) \frac{1}{\sqrt{T}}(Z \otimes I_K)\mathbf{u}. \end{aligned}$$

Thus, by Proposition C.2(4) of Appendix C, $\sqrt{T}(\hat{\beta} - \beta)$ has the same asymptotic distribution as

$$\left[\text{plim} \left(\frac{1}{T}ZZ' \right)^{-1} \otimes I_K \right] \frac{1}{\sqrt{T}}(Z \otimes I_K)\mathbf{u} = (\Gamma^{-1} \otimes I_K) \frac{1}{\sqrt{T}}(Z \otimes I_K)\mathbf{u}.$$

Hence, the asymptotic distribution of $\sqrt{T}(\hat{\beta} - \beta)$ is normal by Lemma 3.1 and the covariance matrix is

$$(\Gamma^{-1} \otimes I_K)(\Gamma \otimes \Sigma_u)(\Gamma^{-1} \otimes I_K) = \Gamma^{-1} \otimes \Sigma_u.$$

The result (3.2.16) can be established with similar arguments (see Problem 3.2). ■

As mentioned previously, if u_t is *Gaussian* (normally distributed) white noise, it satisfies the conditions of Proposition 3.1 so that consistency and asymptotic normality of the LS estimator are ensured for stable Gaussian (normally distributed) VAR(p) processes y_t . Note that normality of u_t implies normality of the y_t for stable processes.

In order to assess the asymptotic dispersion of the LS estimator, we need to know the matrices Γ and Σ_u . From (3.2.13) an obvious consistent estimator of Γ is

$$\hat{\Gamma} = ZZ'/T. \quad (3.2.17)$$

Because $\Sigma_u = E(u_t u_t')$, a plausible estimator for this matrix is

$$\begin{aligned} \tilde{\Sigma}_u &= \frac{1}{T} \sum_{t=1}^T \hat{u}_t \hat{u}_t' = \frac{1}{T} \hat{U} \hat{U}' = \frac{1}{T} (Y - \hat{B}Z)(Y - \hat{B}Z)' \\ &= \frac{1}{T} [Y - YZ'(ZZ')^{-1}Z][Y - YZ'(ZZ')^{-1}Z]' \\ &= \frac{1}{T} Y[I_T - Z'(ZZ')^{-1}Z][I_T - Z'(ZZ')^{-1}Z]'Y' \\ &= \frac{1}{T} Y(I_T - Z'(ZZ')^{-1}Z)Y'. \end{aligned} \quad (3.2.18)$$

Often an adjustment for degrees of freedom is desired because in a regression with fixed, nonstochastic regressors this leads to an unbiased estimator of the covariance matrix. Thus, an estimator

$$\hat{\Sigma}_u = \frac{T}{T - Kp - 1} \tilde{\Sigma}_u \quad (3.2.19)$$

may be considered. Note that there are $Kp + 1$ parameters in each of the K equations of (3.1.1) and, hence, there are $Kp + 1$ parameters in each equation of the system (3.2.2). Of course, $\hat{\Sigma}_u$ and $\tilde{\Sigma}_u$ are asymptotically equivalent. They are consistent estimators of Σ_u if the conditions of Proposition 3.1 hold. In fact, a bit more can be shown.

Proposition 3.2 (*Asymptotic Properties of the White Noise Covariance Matrix Estimators*)

Let y_t be a stable, K -dimensional VAR(p) process as in (3.1.1) with standard white noise innovations and let $\bar{B}$ be an estimator of the VAR coefficients B so that $\sqrt{T} \text{vec}(\bar{B} - B)$ converges in distribution. Furthermore, using the symbols from (3.2.1), suppose that

$$\bar{\Sigma}_u = (Y - \bar{B}Z)(Y - \bar{B}Z)'/(T - c),$$

where c is a fixed constant. Then

$$\text{plim} \sqrt{T}(\bar{\Sigma}_u - UU'/T) = 0. \quad (3.2.20)$$

■

Proof:

$$\begin{aligned} \frac{1}{T}(Y - \bar{B}Z)(Y - \bar{B}Z)' &= (B - \bar{B}) \left(\frac{ZZ'}{T} \right) (B - \bar{B})' + (B - \bar{B}) \frac{ZU'}{T} \\ &\quad + \frac{UZ'}{T} (B - \bar{B})' + \frac{UU'}{T}. \end{aligned}$$

Under the conditions of the proposition, $\text{plim}(B - \bar{B}) = 0$. Hence, by Lemma 3.1,

$$\text{plim } (B - \bar{B})ZU'/\sqrt{T} = 0$$

and

$$\text{plim } \left[(B - \bar{B}) \frac{ZZ'}{T} \sqrt{T} (B - \bar{B})' \right] = 0$$

(see Appendix C.1). Thus,

$$\text{plim } \sqrt{T} [(Y - \bar{B}Z)(Y - \bar{B}Z)' / T - UU' / T] = 0.$$

Therefore, the proposition follows by noting that $T/(T - c) \rightarrow 1$ as $T \rightarrow \infty$. ■

The proposition covers both estimators $\hat{\Sigma}_u$ and $\tilde{\Sigma}_u$. It implies that the feasible estimators $\tilde{\Sigma}_u$ and $\hat{\Sigma}_u$ have the same asymptotic properties as the estimator

$$\frac{UU'}{T} = \frac{1}{T} \sum_{t=1}^T u_t u_t'$$

which is based on the unknown true residuals and is therefore not feasible in practice. In particular, if $\sqrt{T} \text{vec}(UU'/T - \Sigma_u)$ converges in distribution, $\sqrt{T} \text{vec}(\hat{\Sigma}_u - \Sigma_u)$ and $\sqrt{T} \text{vec}(\tilde{\Sigma}_u - \Sigma_u)$ will have the same limiting distribution (see Proposition C.2 of Appendix C.1). Moreover, it can be shown that the asymptotic distributions are independent of the limiting distribution of the LS estimator $\hat{B}$. Another immediate implication of Proposition 3.2 is that $\tilde{\Sigma}_u$ and $\hat{\Sigma}_u$ are consistent estimators of Σ_u . This result is established next.

Corollary 3.2.1

Under the conditions of Proposition 3.2,

$$\text{plim } \tilde{\Sigma}_u = \text{plim } \hat{\Sigma}_u = \text{plim } UU'/T = \Sigma_u.$$

■

Proof: By Proposition 3.2, it suffices to show that $\text{plim } UU'/T = \Sigma_u$ which follows from Proposition C.12(4) because

$$E\left(\frac{1}{T}UU'\right) = \frac{1}{T} \sum_{t=1}^T E(u_t u_t') = \Sigma_u$$

and

$$\text{Var}\left(\frac{1}{T}\text{vec}(UU')\right) = \frac{1}{T^2} \sum_{t=1}^T \text{Var}[\text{vec}(u_t u_t')] \leq \frac{T}{T^2} g \xrightarrow{T \rightarrow \infty} 0,$$

where g is a constant upper bound for $\text{Var}[\text{vec}(u_t u_t')]$. This bound exists because the fourth moments of u_t are bounded by Definition 3.1. ■

If y_t is stable with standard white noise, Proposition 3.1 and Corollary 3.2.1 imply that $(\hat{\beta}_i - \beta_i)/\hat{s}_i$ has an asymptotic standard normal distribution. Here β_i ($\hat{\beta}_i$) is the i -th component of β ($\hat{\beta}$) and $\hat{s}_i$ is the square root of the i -th diagonal element of

$$(ZZ')^{-1} \otimes \hat{\Sigma}_u. \quad (3.2.21)$$

This result means that we can use the “ t -ratios” provided by common regression programs in setting up confidence intervals and tests for individual coefficients. The critical values and percentiles may be based on the asymptotic standard normal distribution. Because it was found in simulation studies that the small sample distributions of the “ t -ratios” have fatter tails than the standard normal distribution, one may want to approximate the small sample distribution by some t -distribution. The question is then what number of degrees of freedom (d.f.) should be used. The overall model (3.2.3) may suggest a choice of d.f. = $KT - K^2p - K$ because in a standard regression model with nonstochastic regressors the d.f. of the “ t -ratios” are equal to the sample size minus the number of estimated parameters. In the present case, it seems also reasonable to use d.f. = $T - Kp - 1$ because the multivariate LS estimator is identical to the LS estimator obtained for each of the K equations in (3.2.2) separately. In a separate regression for each individual equation, we would have T observations and $Kp + 1$ parameters. If the sample size T is large and, thus, the number of degrees of freedom is large, the corresponding t -distribution will be very close to the standard normal so that the choice between the two becomes irrelevant for large samples. Before we look a little further into the problem of choosing appropriate critical values, let us illustrate the foregoing results by an example.

3.2.3 An Example

As an example, we consider a three-dimensional system consisting of first differences of the logarithms of quarterly, seasonally adjusted West German fixed investment (y_1), disposable income (y_2), and consumption expenditures (y_3) from File E1 of the data sets associated with this book. We use only

data from 1960–1978 and reserve the data for 1979–1982 for a subsequent analysis. The original data and first differences of logarithms are plotted in Figures 3.1 and 3.2, respectively. The original data have a trend and are thus considered to be nonstationary. The trend is removed by taking first differences of logarithms. We will discuss this issue in some more detail in Part II. Note that the value for 1960.1 is lost in the differenced series.

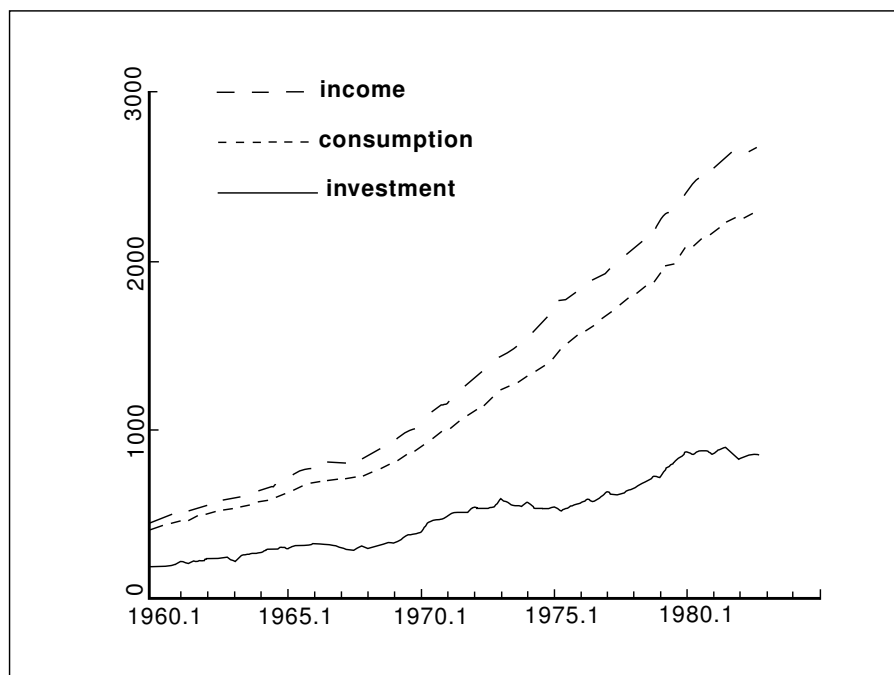


Fig. 3.1. West German investment, income, and consumption data.

Let us assume that the data have been generated by a VAR(2) process. The choice of the VAR order $p = 2$ is arbitrary at this point. In the next chapter, criteria for choosing the VAR order will be considered. Because the VAR order is two, we keep the first two observations of the differenced series as presample values and use a sample size of $T = 73$. Thus, we have a (3×73) matrix Y , $B = (\nu, A_1, A_2)$ is (3×7) , Z is (7×73) and β and $\mathbf{b}$ are both (21×1) vectors.

The LS estimates are

$$\begin{aligned} \hat{B} &= (\hat{\nu}, \hat{A}_1, \hat{A}_2) = YZ'(ZZ')^{-1} \\ &= \begin{bmatrix} -.017 & -.320 & .146 & .961 & -.161 & .115 & .934 \\ .016 & .044 & -.153 & .289 & .050 & .019 & -.010 \\ .013 & -.002 & .225 & -.264 & .034 & .355 & -.022 \end{bmatrix}. \end{aligned} \quad (3.2.22)$$

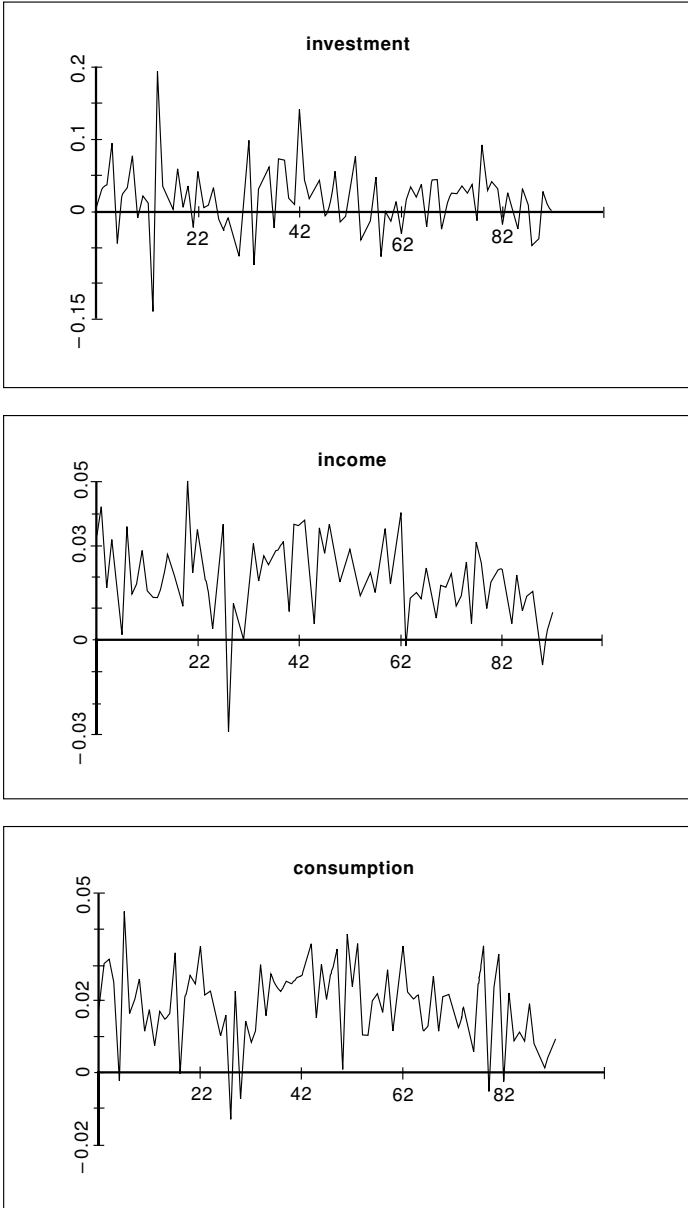


Fig. 3.2. First differences of logarithms of West German investment, income, and consumption.

To check the stability of the estimated process, we determine the roots of the polynomial $\det(I_3 - \hat{A}_1 z - \hat{A}_2 z^2)$ which is easily seen to have degree 6. Its roots are

$$z_1 = 1.753, z_2 = -2.694, z_{3/4} = -0.320 \pm 2.008i, z_{5/6} = -1.285 \pm 1.280i.$$

Note that these roots have been computed using higher precision than the three digits in (3.2.22). They all have modulus greater than 1 and, hence, the stability condition is satisfied.

We get

$$\begin{aligned}\hat{\Sigma}_u &= \frac{1}{T - Kp - 1} (YY' - YZ'(ZZ')^{-1}ZY') \\ &= \begin{bmatrix} 21.30 & .72 & 1.23 \\ .72 & 1.37 & .61 \\ 1.23 & .61 & .89 \end{bmatrix} \times 10^{-4}\end{aligned}\quad (3.2.23)$$

as estimate of the residual covariance matrix Σ_u . Furthermore,

$$\begin{aligned}\hat{I}^{-1} &= (ZZ'/T)^{-1} \\ &= T \begin{bmatrix} .14 & .17 & -.69 & -2.51 & .10 & -.67 & -2.57 \\ \bullet & 7.39 & 1.24 & -10.56 & 1.80 & 1.08 & -8.70 \\ \bullet & \bullet & 139.81 & -87.40 & -4.58 & 30.21 & -50.88 \\ \bullet & \bullet & \bullet & 207.22 & .84 & -55.35 & 73.82 \\ \bullet & \bullet & \bullet & \bullet & 7.33 & -.03 & -9.31 \\ \bullet & \bullet & \bullet & \bullet & \bullet & 134.19 & -82.64 \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & 207.71 \end{bmatrix}.\end{aligned}$$

Dividing the elements of $\hat{B}$ by square roots of the corresponding diagonal elements of $(ZZ')^{-1} \otimes \hat{\Sigma}_u$ we get the matrix of t -ratios:

$$\begin{bmatrix} -0.97 & -2.55 & 0.27 & 1.45 & -1.29 & 0.21 & 1.41 \\ 3.60 & 1.38 & -1.10 & 1.71 & 1.58 & 0.14 & -0.06 \\ 3.67 & -0.09 & 2.01 & -1.94 & 1.33 & 3.24 & -0.16 \end{bmatrix}.\quad (3.2.24)$$

We may compare these quantities with critical values from a t -distribution with d.f. = $KT - K^2p - K = 198$ or d.f. = $T - Kp - 1 = 66$. In both cases, we get critical values of approximately ± 2 for a two-tailed test with significance level 5%. Thus, the critical values are approximately the same as those from a standard normal distribution.

Apparently quite a few coefficients are not significant under this criterion. This observation suggests that the model contains unnecessarily many free parameters. In subsequent chapters, we will discuss the problem of choosing the VAR order and possible restrictions for the coefficients. Also, before an estimated model is used for forecasting and analysis purposes, the assumptions underlying the analysis should be checked carefully. Checking the model adequacy will be treated in greater detail in Chapter 4.

3.2.4 Small Sample Properties of the LS Estimator

As mentioned earlier, it is difficult to derive small sample properties of the LS estimator analytically. In such a case it is sometimes helpful to use *Monte*

Carlo methods to get some idea about the small sample properties. In a Monte Carlo analysis, specific processes are used to artificially generate a large number of time series. Then a set of estimates is computed for each multiple time series generated and the properties of the resulting empirical distributions of these estimates are studied (see Appendix D). Such an approach usually permits rather limited conclusions only because the findings may depend on the particular processes used for generating the time series. Nevertheless, such exercises give some insight into the small sample properties of estimators.

In the following, we use the bivariate VAR(2) example process (2.1.15),

$$y_t = \begin{bmatrix} .02 \\ .03 \end{bmatrix} + \begin{bmatrix} .5 & .1 \\ .4 & .5 \end{bmatrix} y_{t-1} + \begin{bmatrix} 0 & 0 \\ .25 & 0 \end{bmatrix} y_{t-2} + u_t \quad (3.2.25)$$

with error covariance matrix

$$\Sigma_u = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix} \times 10^{-4} \quad (3.2.26)$$

to investigate the small sample properties of the multivariate LS estimator. With this process we have generated 1000 bivariate time series of length $T = 30$ plus 2 presample values using independent standard normal errors, that is, $u_t \sim \mathcal{N}(0, \Sigma_u)$. Thus the 1000 bivariate time series are generated by a stable Gaussian process so that Propositions 3.1 and 3.2 provide the asymptotic properties of the LS estimators.

In Table 3.1, some empirical results are given. In particular, the empirical mean, variance, and mean squared error (MSE) of each parameter estimator are given. Obviously, the empirical means differ from the actual values of the coefficients. However, measuring the estimation precision by the empirical variance (average squared deviation from the mean in 1000 samples) or MSE (average squared deviation from the true parameter value), the coefficients are seen to be estimated quite precisely even with a sample size as small as $T = 30$. This is partly a consequence of the special properties of the process.

In Table 3.1, empirical percentiles of the t -ratios are also given together with the corresponding percentiles from the t - and standard normal distributions (d.f. = ∞). Even with the presently considered relatively small sample size the percentiles of the three distributions that might be used for inference do not differ much. Consequently, it does not matter much which of the theoretical percentiles are used, in particular, because the empirical percentiles, in many cases, differ quite a bit from the corresponding theoretical quantities. This example shows that the asymptotic results have to be used cautiously in setting up small sample tests and confidence intervals. On the other hand, this example also demonstrates that the asymptotic theory does provide some guidance for inference. For example, the empirical 95th percentiles of all coefficients lie between the 90th and the 99th percentile of the standard normal distribution given in the last row of the table. Of course, this is just one example and not a general finding.

Table 3.1. Empirical percentiles of t -ratios of parameter estimates for the example process and actual percentiles of t -distributions for sample size $T = 30$

parameter	empirical			empirical percentiles of t -ratios							
	mean	variance	MSE	1.	5.	10.	50.	90.	95.	99.	
$\nu_1 = .02$	.041	.0011	.0015	-1.91	-1.04	-0.64	0.62	1.92	2.29	3.12	
$\nu_2 = .03$	.038	.0005	.0006	-2.30	-1.40	-1.02	0.25	1.65	2.11	2.83	
$\alpha_{11,1} = .5$	.41	.041	.049	-2.78	-2.18	-1.74	-0.43	0.92	1.28	2.01	
$\alpha_{21,1} = .4$	.40	.018	.018	-2.61	-1.74	-1.28	0.04	1.28	1.71	2.65	
$\alpha_{12,1} = .1$	.10	.078	.078	-2.27	-1.67	-1.35	-0.03	1.29	1.67	2.38	
$\alpha_{22,1} = .5$	.44	.030	.034	-2.69	-1.97	-1.59	-0.35	0.89	1.30	2.06	
$\alpha_{11,2} = 0$	-.05	.056	.058	-2.75	-1.93	-1.50	-0.24	1.02	1.38	2.09	
$\alpha_{21,2} = .25$	.29	.023	.024	-1.99	-1.32	-0.99	0.20	1.45	1.81	2.48	
$\alpha_{12,2} = 0$	-.07	.053	.058	-2.48	-1.91	-1.61	-0.28	0.97	1.39	2.03	
$\alpha_{22,2} = 0$	-.01	.023	.024	-2.71	-1.72	-1.36	-0.03	1.18	1.53	2.18	
degrees of freedom(d.f.)				percentiles of t -distributions							
				1.	5.	10.	50.	90.	95.	99.	
$T - Kp - 1 = 25$				-2.49	-1.71	-1.32	0	1.32	1.71	2.49	
$K(T - Kp - 1) = 50$				-2.41	-1.68	-1.30	0	1.30	1.68	2.41	
∞				-2.33	-1.65	-1.28	0	1.28	1.65	2.33	
(normal distribution)											

In an extensive study, Nankervis & Savin (1988) investigated the small sample distribution of the “ t -statistic” for the parameter of a *univariate* AR(1) process. They found that it differs quite substantially from the corresponding t -distribution, especially if the sample size is small ($T < 100$) and the parameter lies close to the instability region. Analytical results on the bias in estimating VAR models were derived by Nicholls & Pope (1988) and Tjøstheim & Paulsen (1983). What should be learned from our Monte Carlo investigation and these remarks is that asymptotic distributions in the present context can only be used as rough guidelines for small sample inference. That, however, is much better than having no guidance at all.

3.3 Least Squares Estimation with Mean-Adjusted Data and Yule-Walker Estimation

3.3.1 Estimation when the Process Mean Is Known

Occasionally a VAR(p) model is given in *mean-adjusted form*,

$$(y_t - \mu) = A_1(y_{t-1} - \mu) + \cdots + A_p(y_{t-p} - \mu) + u_t. \tag{3.3.1}$$

Multivariate LS estimation of this model form is straightforward if the mean vector μ is known. Defining

$$\begin{aligned}
Y^0 &:= (y_1 - \mu, \dots, y_T - \mu) & (K \times T), \\
A &:= (A_1, \dots, A_p) & (K \times Kp), \\
Y_t^0 &:= \begin{bmatrix} y_t - \mu \\ \vdots \\ y_{t-p+1} - \mu \end{bmatrix} & (Kp \times 1), \\
X &:= (Y_0^0, \dots, Y_{T-1}^0) & (Kp \times T), \\
\mathbf{y}^0 &:= \text{vec}(Y^0) & (KT \times 1), \\
\boldsymbol{\alpha} &:= \text{vec}(A) & (K^2p \times 1),
\end{aligned} \tag{3.3.2}$$

we can write (3.3.1), for $t = 1, \dots, T$, compactly as

$$Y^0 = AX + U \tag{3.3.3}$$

or

$$\mathbf{y}^0 = (X' \otimes I_K) \boldsymbol{\alpha} + \mathbf{u}, \tag{3.3.4}$$

where U and $\mathbf{u}$ are defined as in (3.2.1). The LS estimator is easily seen to be

$$\hat{\boldsymbol{\alpha}} = ((XX')^{-1}X \otimes I_K) \mathbf{y}^0 \tag{3.3.5}$$

or

$$\hat{A} = Y^0 X' (X X')^{-1}. \tag{3.3.6}$$

If y_t is stable and u_t is standard white noise, it can be shown that

$$\sqrt{T}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\hat{\boldsymbol{\alpha}}}), \tag{3.3.7}$$

where

$$\Sigma_{\hat{\boldsymbol{\alpha}}} = \Gamma_Y(0)^{-1} \otimes \Sigma_u \tag{3.3.8}$$

and $\Gamma_Y(0) := E(Y_t^0 Y_t^{0'})$.

3.3.2 Estimation of the Process Mean

Usually μ will not be known in advance. In that case, it may be estimated by the vector of sample means,

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t. \tag{3.3.9}$$

Using (3.3.1), $\bar{y}$ can be written as

$$\begin{aligned}
\bar{y} &= \mu + A_1 \left[\bar{y} + \frac{1}{T}(y_0 - y_T) - \mu \right] + \cdots \\
&\quad + A_p \left[\bar{y} + \frac{1}{T}(y_{-p+1} + \cdots + y_0 - y_{T-p+1} - \cdots - y_T) - \mu \right] \\
&\quad + \frac{1}{T} \sum_{t=1}^T u_t.
\end{aligned}$$

Hence,

$$(I_K - A_1 - \cdots - A_p)(\bar{y} - \mu) = \frac{1}{T} z_T + \frac{1}{T} \sum_t u_t, \quad (3.3.10)$$

where

$$z_T = \sum_{i=1}^p A_i \left[\sum_{j=0}^{i-1} (y_{0-j} - y_{T-j}) \right].$$

Evidently,

$$E(z_T/\sqrt{T}) = \frac{1}{\sqrt{T}} E(z_T) = 0$$

and

$$\text{Var}(z_T/\sqrt{T}) = \frac{1}{T} \text{Var}(z_T) \xrightarrow{T \rightarrow \infty} 0$$

because y_t is stable. In other words, $z_T/\sqrt{T}$ converges to zero in mean square. It follows that $\sqrt{T}(I_K - A_1 - \cdots - A_p)(\bar{y} - \mu)$ has the same asymptotic distribution as $\sum u_t/\sqrt{T}$ (see Appendix C, Proposition C.2). Hence, noting that, by a central limit theorem (e.g., Fuller (1976) or Proposition C.13),

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T u_t \xrightarrow{d} \mathcal{N}(0, \Sigma_u), \quad (3.3.11)$$

if u_t is standard white noise, we get the following result:

Proposition 3.3 (*Asymptotic Properties of the Sample Mean*)

If the VAR(p) process y_t given in (3.3.1) is stable and u_t is standard white noise, then

$$\sqrt{T}(\bar{y} - \mu) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\bar{y}}), \quad (3.3.12)$$

where

$$\Sigma_{\bar{y}} = (I_K - A_1 - \cdots - A_p)^{-1} \Sigma_u (I_K - A_1 - \cdots - A_p)'^{-1}.$$

In particular, $\text{plim } \bar{y} = \mu$. ■

The proposition follows from (3.3.10), (3.3.11), and Proposition C.15 of Appendix C. The limiting distribution in (3.3.11) holds even in small samples for Gaussian white noise u_t .

Because $\mu = (I_K - A_1 - \cdots - A_p)^{-1}\nu$ (see Chapter 2, Section 2.1), an alternative estimator for the process mean is obtained from the LS estimator of the previous section:

$$\hat{\mu} = (I_K - \hat{A}_1 - \cdots - \hat{A}_p)^{-1}\hat{\nu}. \quad (3.3.13)$$

Using again Proposition C.15 of Appendix C, this estimator is also consistent and has an asymptotic normal distribution,

$$\sqrt{T}(\hat{\mu} - \mu) \xrightarrow{d} \mathcal{N}\left(0, \frac{\partial \mu}{\partial \beta'}(\Gamma^{-1} \otimes \Sigma_u) \frac{\partial \mu'}{\partial \beta}\right), \quad (3.3.14)$$

provided the conditions of Proposition 3.1 are satisfied. It can be shown that

$$\frac{\partial \mu}{\partial \beta'}(\Gamma^{-1} \otimes \Sigma_u) \frac{\partial \mu'}{\partial \beta} = \Sigma_{\bar{y}} \quad (3.3.15)$$

and, hence, the estimators $\hat{\mu}$ and $\bar{y}$ for μ are asymptotically equivalent (see Section 3.4). This result suggests that it does not matter asymptotically whether the mean is estimated separately or jointly with the other VAR coefficients. While this holds asymptotically, it will usually matter in small samples which estimator is used. An example will be given shortly.

3.3.3 Estimation with Unknown Process Mean

If the mean vector μ is unknown, it may be replaced by $\bar{y}$ in the vectors and matrices in (3.3.2) giving $\hat{X}, \hat{Y}^0$ and so on. The resulting LS estimator,

$$\hat{\hat{\alpha}} = ((\hat{X}\hat{X}')^{-1}\hat{X} \otimes I_K)\hat{\hat{y}}^0,$$

is asymptotically equivalent to $\hat{\alpha}$. More precisely, it can be shown that, under the conditions of Proposition 3.3,

$$\sqrt{T}(\hat{\hat{\alpha}} - \alpha) \xrightarrow{d} \mathcal{N}(0, \Gamma_Y(0)^{-1} \otimes \Sigma_u), \quad (3.3.16)$$

where $\Gamma_Y(0) := E(Y_t^0 Y_t^{0'})$. This result will be discussed further in the next section on maximum likelihood estimation for Gaussian processes.

3.3.4 The Yule-Walker Estimator

The LS estimator can also be derived from the Yule-Walker equations given in Chapter 2, (2.1.37). They imply

$$\Gamma_y(h) = [A_1, \dots, A_p] \begin{bmatrix} \Gamma_y(h-1) \\ \vdots \\ \Gamma_y(h-p) \end{bmatrix}, \quad h > 0,$$

or

$$\begin{aligned} [\Gamma_y(1), \dots, \Gamma_y(p)] &= [A_1, \dots, A_p] \begin{bmatrix} \Gamma_y(0) & \dots & \Gamma_y(p-1) \\ \vdots & \ddots & \vdots \\ \Gamma_y(-p+1) & \dots & \Gamma_y(0) \end{bmatrix} \\ &= A\Gamma_Y(0) \end{aligned} \tag{3.3.17}$$

and, hence,

$$A = [\Gamma_y(1), \dots, \Gamma_y(p)] \Gamma_Y(0)^{-1}.$$

Estimating $\Gamma_Y(0)$ by $\hat{X}\hat{X}'/T$ and $[\Gamma_y(1), \dots, \Gamma_y(p)]$ by $\hat{Y}^0\hat{X}'/T$, the resulting estimator is just the LS estimator,

$$\hat{\hat{A}} = \hat{Y}^0\hat{X}'(\hat{X}\hat{X}')^{-1}. \tag{3.3.18}$$

Alternatively, the moment matrices $\Gamma_y(h)$ may be estimated using as many data as are available, including the presample values. Thus, if a sample $y_1, \dots, y_T$ and p presample observations $y_{-p+1}, \dots, y_0$ are available, μ may be estimated as

$$\bar{y}^* = \frac{1}{T+p} \sum_{t=-p+1}^T y_t$$

and $\Gamma_y(h)$ may be estimated as

$$\hat{\Gamma}_y(h) = \frac{1}{T+p-h} \sum_{t=-p+h+1}^T (y_t - \bar{y}^*)(y_{t-h} - \bar{y}^*)'. \tag{3.3.19}$$

Using these estimators in (3.3.17), the so-called *Yule-Walker estimator* for A is obtained. For stable processes, this estimator has the same asymptotic properties as the LS estimator. However, it may have less attractive small sample properties (e.g., Tjøstheim & Paulsen (1983)).

The Yule-Walker estimator always produces estimates in the stability region (see Brockwell & Davis (1987, §8.1) for a discussion of the univariate case). In other words, the estimated process is always stable. This property is sometimes regarded as an advantage of the Yule-Walker estimator. It is responsible for possibly considerable bias of the estimator, however. Also, in practice, it may not be known a priori whether the data generation process of a given multiple time series is stable. In the unstable case, LS and Yule-Walker estimation are not asymptotically equivalent anymore (see also the discussion in Reinsel (1993, Section 4.4)). Therefore, enforcing stability may not be a good strategy in practice. The LS estimator is usually used in the following.

3.3.5 An Example

To illustrate the results of this section, we use again the West German investment, income, and consumption data. The variables y_1 , y_2 , and y_3 are defined as in Section 3.2.3, the sample period ranges from 1960.4 to 1978.4, that is, $T = 73$ and the data for 1960.2 and 1960.3 are used as presample values. Using only the sample values we get

$$\bar{y} = \begin{bmatrix} .018 \\ .020 \\ .020 \end{bmatrix} \quad (3.3.20)$$

which is different, though not substantially so, from

$$\hat{\mu} = (I_3 - \hat{A}_1 - \hat{A}_2)^{-1} \hat{\nu} = \begin{bmatrix} .017 \\ .020 \\ .020 \end{bmatrix} \quad (3.3.21)$$

as obtained from the LS estimates in (3.2.22).

Subtracting the sample means from the data we get, based on (3.3.18),

$$\hat{\hat{A}} = (\hat{\hat{A}}_1, \hat{\hat{A}}_2) = \begin{bmatrix} -.319 & .143 & .960 & -.160 & .112 & .933 \\ .044 & -.153 & .288 & .050 & .019 & -.010 \\ -.002 & .224 & -.264 & .034 & .354 & -.023 \end{bmatrix}. \quad (3.3.22)$$

This estimate is clearly distinct from the corresponding part of (3.2.22), although the two estimates do not differ dramatically.

If the two presample values are used in estimating the process means and moment matrices we get

$$\hat{A}_{YW} = \begin{bmatrix} -.319 & .147 & .959 & -.160 & .115 & .932 \\ .044 & -.152 & .286 & .050 & .020 & -.012 \\ -.002 & .225 & -.264 & .034 & .355 & -.022 \end{bmatrix} \quad (3.3.23)$$

which is the Yule-Walker estimate. Although the sample size is moderate, there is a slight difference between the estimates in (3.3.22) and (3.3.23).

3.4 Maximum Likelihood Estimation

3.4.1 The Likelihood Function

Assuming that the distribution of the process is known, maximum likelihood (ML) estimation is an alternative to LS estimation. We will consider ML estimation under the assumption that the $\text{VAR}(p)$ process y_t is Gaussian. More precisely,

$$\mathbf{u} = \text{vec}(U) = \begin{bmatrix} u_1 \\ \vdots \\ u_T \end{bmatrix} \sim \mathcal{N}(0, I_T \otimes \Sigma_u). \quad (3.4.1)$$

In other words, the probability density of $\mathbf{u}$ is

$$f_{\mathbf{u}}(\mathbf{u}) = \frac{1}{(2\pi)^{KT/2}} |I_T \otimes \Sigma_u|^{-1/2} \exp \left[-\frac{1}{2} \mathbf{u}' (I_T \otimes \Sigma_u^{-1}) \mathbf{u} \right]. \quad (3.4.2)$$

Moreover,

$$\begin{aligned} \mathbf{u} = & \begin{bmatrix} I_K & 0 & \cdots & 0 & \cdots & \cdots & 0 \\ -A_1 & I_K & & 0 & \cdots & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & & \vdots \\ -A_p & -A_{p-1} & \cdots & I_K & & & 0 \\ 0 & -A_p & & & \ddots & & \vdots \\ \vdots & & \ddots & & & \ddots & \vdots \\ 0 & 0 & \cdots & -A_p & \cdots & \cdots & I_K \end{bmatrix} (\mathbf{y} - \boldsymbol{\mu}^*) \\ & + \begin{bmatrix} -A_1 & -A_2 & \cdots & -A_p \\ -A_2 & -A_3 & \cdots & 0 \\ \vdots & & & \vdots \\ -A_p & 0 & \cdots & 0 \\ & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} (Y_0 - \boldsymbol{\mu}), \end{aligned} \quad (3.4.3)$$

where $\mathbf{y} := \text{vec}(Y)$ and $\boldsymbol{\mu}^* := (\mu'_1, \dots, \mu'_T)'$ are $(TK \times 1)$ vectors and $Y_0 := (y'_0, \dots, y'_{-p+1})'$ and $\boldsymbol{\mu} := (\mu'_1, \dots, \mu'_p)'$ are $(Kp \times 1)$. Consequently, $\partial \mathbf{u} / \partial \mathbf{y}'$ is a lower triangular matrix with unit diagonal which has unit determinant. Hence, using that $\mathbf{u} = \mathbf{y} - \boldsymbol{\mu}^* - (X' \otimes I_K) \boldsymbol{\alpha}$,

$$\begin{aligned} f_{\mathbf{y}}(\mathbf{y}) &= \left| \frac{\partial \mathbf{u}}{\partial \mathbf{y}'} \right| f_{\mathbf{u}}(\mathbf{u}) \\ &= \frac{1}{(2\pi)^{KT/2}} |I_T \otimes \Sigma_u|^{-1/2} \\ &\quad \times \exp \left[-\frac{1}{2} (\mathbf{y} - \boldsymbol{\mu}^* - (X' \otimes I_K) \boldsymbol{\alpha})' (I_T \otimes \Sigma_u^{-1}) \right. \\ &\quad \left. \times (\mathbf{y} - \boldsymbol{\mu}^* - (X' \otimes I_K) \boldsymbol{\alpha}) \right], \end{aligned} \quad (3.4.4)$$

where X and $\boldsymbol{\alpha}$ are as defined in (3.3.2). For simplicity, the initial values Y_0 are assumed to be given fixed numbers. Hence, we get a log-likelihood function

$$\ln l(\boldsymbol{\mu}, \boldsymbol{\alpha}, \Sigma_u)$$

$$\begin{aligned}
&= -\frac{KT}{2} \ln 2\pi - \frac{T}{2} \ln |\Sigma_u| \\
&\quad - \frac{1}{2} [\mathbf{y} - \boldsymbol{\mu}^* - (X' \otimes I_K) \boldsymbol{\alpha}]' (I_T \otimes \Sigma_u^{-1}) [\mathbf{y} - \boldsymbol{\mu}^* - (X' \otimes I_K) \boldsymbol{\alpha}] \\
&= -\frac{KT}{2} \ln 2\pi - \frac{T}{2} \ln |\Sigma_u| - \frac{1}{2} \sum_{t=1}^T \left[(y_t - \mu) - \sum_{i=1}^p A_i (y_{t-i} - \mu) \right]' \\
&\quad \times \Sigma_u^{-1} \left[(y_t - \mu) - \sum_{i=1}^p A_i (y_{t-i} - \mu) \right] \\
&= -\frac{KT}{2} \ln 2\pi - \frac{T}{2} \ln |\Sigma_u| \\
&\quad - \frac{1}{2} \sum_t \left(y_t - \sum_i A_i y_{t-i} \right)' \Sigma_u^{-1} \left(y_t - \sum_i A_i y_{t-i} \right) \\
&\quad + \mu' \left(I_K - \sum_i A_i \right)' \Sigma_u^{-1} \sum_t \left(y_t - \sum_i A_i y_{t-i} \right) \\
&\quad - \frac{T}{2} \mu' \left(I_K - \sum_i A_i \right)' \Sigma_u^{-1} \left(I_K - \sum_i A_i \right) \mu \\
&= -\frac{KT}{2} \ln 2\pi - \frac{T}{2} \ln |\Sigma_u| - \frac{1}{2} \text{tr}[(Y^0 - AX)' \Sigma_u^{-1} (Y^0 - AX)], \quad (3.4.5)
\end{aligned}$$

where $Y^0 := (y_1 - \mu, \dots, y_T - \mu)$ and $A := (A_1, \dots, A_p)$ are as defined in (3.3.2). These different expressions of the log-likelihood function will be useful in the following.

3.4.2 The ML Estimators

In order to determine the ML estimators of μ , $\boldsymbol{\alpha}$, and Σ_u , the system of first order partial derivatives is needed:

$$\begin{aligned}
\frac{\partial \ln l}{\partial \mu} &= \left(I_K - \sum_i A_i \right)' \Sigma_u^{-1} \sum_t \left(y_t - \sum_i A_i y_{t-i} \right) \\
&\quad - T \left(I_K - \sum_i A_i \right)' \Sigma_u^{-1} \left(I_K - \sum_i A_i \right) \mu \\
&= [I_K - A(\mathbf{j} \otimes I_K)]' \Sigma_u^{-1} \left[\sum_t (y_t - \mu - AY_{t-1}^0) \right], \quad (3.4.6)
\end{aligned}$$

where Y_t^0 is as defined in (3.3.2) and $\mathbf{j} := (1, \dots, 1)'$ is a $(p \times 1)$ vector of ones,

$$\begin{aligned}
\frac{\partial \ln l}{\partial \boldsymbol{\alpha}} &= (X \otimes I_K)(I_T \otimes \Sigma_u^{-1})[\mathbf{y} - \boldsymbol{\mu}^* - (X' \otimes I_K)\boldsymbol{\alpha}] \\
&= (X \otimes \Sigma_u^{-1})(\mathbf{y} - \boldsymbol{\mu}^*) - (XX' \otimes \Sigma_u^{-1})\boldsymbol{\alpha},
\end{aligned} \tag{3.4.7}$$

$$\frac{\partial \ln l}{\partial \Sigma_u} = -\frac{T}{2}\Sigma_u^{-1} + \frac{1}{2}\Sigma_u^{-1}(Y^0 - AX)(Y^0 - AX)'\Sigma_u^{-1}. \tag{3.4.8}$$

Equating to zero gives the system of normal equations which can be solved for the estimators:

$$\tilde{\boldsymbol{\mu}} = \frac{1}{T} \left(I_K - \sum_i \tilde{A}_i \right)^{-1} \sum_t \left(y_t - \sum_i \tilde{A}_i y_{t-i} \right), \tag{3.4.9}$$

$$\tilde{\boldsymbol{\alpha}} = ((\tilde{X}\tilde{X}')^{-1}\tilde{X} \otimes I_K)(\mathbf{y} - \tilde{\boldsymbol{\mu}}^*), \tag{3.4.10}$$

$$\tilde{\Sigma}_u = \frac{1}{T}(\tilde{Y}^0 - \tilde{A}\tilde{X})(\tilde{Y}^0 - \tilde{A}\tilde{X})', \tag{3.4.11}$$

where $\tilde{X}$ and $\tilde{Y}^0$ are obtained from X and Y^0 , respectively, by replacing μ with $\tilde{\mu}$.

3.4.3 Properties of the ML Estimators

Comparing these results with the LS estimators obtained in Section 3.3, it turns out that the ML estimators of μ and $\boldsymbol{\alpha}$ are identical to the LS estimators. Thus, $\tilde{\mu}$ and $\tilde{\boldsymbol{\alpha}}$ are consistent estimators if y_t is a stationary, stable Gaussian VAR(p) process and $\sqrt{T}(\tilde{\mu} - \mu)$ and $\sqrt{T}(\tilde{\boldsymbol{\alpha}} - \boldsymbol{\alpha})$ are asymptotically normally distributed. This result also follows from a more general maximum likelihood theory (see Appendix C.6). In fact, that theory implies that the covariance matrix of the asymptotic distribution of the ML estimators is the limit of T times the inverse information matrix. The information matrix is

$$\mathcal{I}(\boldsymbol{\delta}) = -E \left[\frac{\partial^2 \ln l}{\partial \boldsymbol{\delta} \partial \boldsymbol{\delta}'} \right] \tag{3.4.12}$$

where $\boldsymbol{\delta}' := (\mu', \boldsymbol{\alpha}', \boldsymbol{\sigma}')$ with $\boldsymbol{\sigma} := \text{vech}(\Sigma_u)$. Note that vech is a column stacking operator that stacks only the elements on and below the main diagonal of Σ_u . It is related to the vec operator by the $(\frac{1}{2}K(K+1) \times K^2)$ elimination matrix $\mathbf{L}_K$, that is, $\text{vech}(\Sigma_u) = \mathbf{L}_K \text{vec}(\Sigma_u)$ or, defining $\boldsymbol{\omega} := \text{vec}(\Sigma_u)$, $\boldsymbol{\sigma} = \mathbf{L}_K \boldsymbol{\omega}$ (see Appendix A.12). For instance, for $K = 3$,

$$\begin{aligned}
\boldsymbol{\omega} &= \text{vec}(\Sigma_u) = \text{vec} \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} \\
&= (\sigma_{11}, \sigma_{12}, \sigma_{13}, \sigma_{12}, \sigma_{22}, \sigma_{23}, \sigma_{13}, \sigma_{23}, \sigma_{33})'
\end{aligned}$$

and

$$\boldsymbol{\sigma} = \text{vech}(\Sigma_u) = \mathbf{L}_3 \boldsymbol{\omega} = \begin{bmatrix} \sigma_{11} \\ \sigma_{12} \\ \sigma_{13} \\ \sigma_{22} \\ \sigma_{23} \\ \sigma_{33} \end{bmatrix}. \quad (3.4.13)$$

Note that in $\boldsymbol{\delta}$ we collect only the potentially different elements of Σ_u .

The asymptotic covariance matrix of the ML estimator $\tilde{\boldsymbol{\delta}}$ is known to be

$$\lim_{T \rightarrow \infty} [\mathcal{I}(\boldsymbol{\delta})/T]^{-1}. \quad (3.4.14)$$

In order to determine this matrix, we need the second order partial derivatives of the log-likelihood. From (3.4.6) to (3.4.8) we get

$$\frac{\partial^2 \ln l}{\partial \boldsymbol{\mu} \partial \boldsymbol{\mu}'} = -T \left(I_K - \sum_i A_i \right)' \Sigma_u^{-1} \left(I_K - \sum_i A_i \right), \quad (3.4.15)$$

$$\frac{\partial^2 \ln l}{\partial \boldsymbol{\alpha} \partial \boldsymbol{\alpha}'} = -(X X' \otimes \Sigma_u^{-1}), \quad (3.4.16)$$

$$\begin{aligned} \frac{\partial^2 \ln l}{\partial \boldsymbol{\omega} \partial \boldsymbol{\omega}'} &= \frac{T}{2} (\Sigma_u^{-1} \otimes \Sigma_u^{-1}) - \frac{1}{2} (\Sigma_u^{-1} \otimes \Sigma_u^{-1} U U' \Sigma_u^{-1}) \\ &\quad - \frac{1}{2} (\Sigma_u^{-1} U U' \Sigma_u^{-1} \otimes \Sigma_u^{-1}), \end{aligned} \quad (3.4.17)$$

where $\boldsymbol{\omega} = \text{vec}(\Sigma_u)$ (see Problem 3.3),

$$\begin{aligned} \frac{\partial^2 \ln l}{\partial \boldsymbol{\mu} \partial \boldsymbol{\alpha}'} &= -[I_K - (\mathbf{j}' \otimes I_K) A'] \Sigma_u^{-1} \sum_t Y_{t-1}^{0'} \otimes I_K \\ &\quad - \left(\sum_t u_t' \Sigma_u^{-1} \otimes I_K \right) (I_K \otimes \mathbf{j}' \otimes I_K) \frac{\partial \text{vec}(A')}{\partial \boldsymbol{\alpha}'} \end{aligned} \quad (3.4.18)$$

(see Problem 3.4),

$$\frac{\partial^2 \ln l}{\partial \boldsymbol{\omega} \partial \boldsymbol{\mu}'} = \frac{1}{2} (\Sigma_u^{-1} \otimes \Sigma_u^{-1}) \left[(I_K \otimes U) \frac{\partial \text{vec}(U')}{\partial \boldsymbol{\mu}'} + (U \otimes I_K) \frac{\partial \text{vec}(U)}{\partial \boldsymbol{\mu}'} \right] \quad (3.4.19)$$

(see Problem 3.5), and

$$\frac{\partial^2 \ln l}{\partial \boldsymbol{\omega} \partial \boldsymbol{\alpha}'} = -\frac{1}{2} (\Sigma_u^{-1} \otimes \Sigma_u^{-1}) \left[(I_K \otimes U X') \frac{\partial \text{vec}(A')}{\partial \boldsymbol{\alpha}'} + (U X' \otimes I_K) \right] \quad (3.4.20)$$

(see Problem 3.6).

It is obvious from (3.4.18) that

$$\lim T^{-1} E \left(\frac{\partial^2 \ln l}{\partial \mu \partial \alpha'} \right) = 0 \quad (3.4.21)$$

because $E(\sum_t Y_{t-1}^0/T) \rightarrow 0$. Furthermore, from (3.4.19), it follows that

$$E \left(\frac{\partial^2 \ln l}{\partial \omega \partial \mu'} \right) = 0 \quad (3.4.22)$$

because $E(U) = 0$ and $\partial \text{vec}(U')/\partial \mu'$ is a matrix of constants. Moreover, from (3.4.20), we have

$$\lim T^{-1} E \left(\frac{\partial^2 \ln l}{\partial \omega \partial \alpha'} \right) = 0 \quad (3.4.23)$$

because $E(UX'/T) \rightarrow 0$. Thus, $\lim \mathcal{I}(\delta)/T$ is block diagonal and we get the asymptotic distributions of μ , α , and σ as follows.

Multiplying minus the inverse of (3.4.15) by T gives the asymptotic covariance matrix of the ML estimator for the mean vector μ , that is,

$$\sqrt{T}(\tilde{\mu} - \mu) \xrightarrow{d} \mathcal{N} \left(0, \left(I_K - \sum_{i=1}^p A_i \right)^{-1} \Sigma_u \left(I_K - \sum_{i=1}^p A'_i \right)^{-1} \right). \quad (3.4.24)$$

Hence, $\tilde{\mu}$ has the same asymptotic distribution as $\bar{y}$ (see Proposition 3.3). In other words, the two estimators for μ are asymptotically equivalent and, under the present conditions, this fact implies that $\bar{y}$ is asymptotically efficient because the ML estimator is asymptotically efficient. The asymptotic equivalence of $\tilde{\mu}$ and $\bar{y}$ can also be seen from (3.4.9) (see the argument prior to Proposition 3.3 and Problem 3.7).

Taking the limit of T^{-1} times the expectation of minus (3.4.16) gives $\Gamma_Y(0) \otimes \Sigma_u^{-1}$. Note that $E(XX'/T)$ is not strictly equal to $\Gamma_Y(0)$ because we have assumed fixed initial values $y_{-p+1}, \dots, y_0$. However, asymptotically, as T goes to infinity, the impact of the initial values vanishes. Thus, we get

$$\sqrt{T}(\tilde{\alpha} - \alpha) \xrightarrow{d} \mathcal{N}(0, \Gamma_Y(0)^{-1} \otimes \Sigma_u). \quad (3.4.25)$$

Of course, this result also follows from the equivalence of the ML and LS estimators.

Noting that $E(UU') = T\Sigma_u$, it follows from (3.4.17) that

$$E \left(\frac{\partial^2 \ln l}{\partial \omega \partial \omega'} \right) = -\frac{T}{2} (\Sigma_u^{-1} \otimes \Sigma_u^{-1}). \quad (3.4.26)$$

Denoting by $\mathbf{D}_K$ the $(K^2 \times \frac{1}{2}K(K+1))$ duplication matrix (see Appendix A.12) so that $\omega = \mathbf{D}_K \sigma$, we get

$$\frac{\partial^2 \ln l}{\partial \boldsymbol{\sigma} \partial \boldsymbol{\sigma}'} = \frac{\partial \boldsymbol{\omega}'}{\partial \boldsymbol{\sigma}} \frac{\partial^2 \ln l}{\partial \boldsymbol{\omega} \partial \boldsymbol{\omega}'} \frac{\partial \boldsymbol{\omega}}{\partial \boldsymbol{\sigma}'} = \mathbf{D}'_K \frac{\partial^2 \ln l}{\partial \boldsymbol{\omega} \partial \boldsymbol{\omega}'} \mathbf{D}_K$$

and, hence,

$$\sqrt{T}(\tilde{\boldsymbol{\sigma}} - \boldsymbol{\sigma}) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\tilde{\boldsymbol{\sigma}}}) \quad (3.4.27)$$

with

$$\begin{aligned} \Sigma_{\tilde{\boldsymbol{\sigma}}} &= -TE \left(\frac{\partial^2 \ln l}{\partial \boldsymbol{\sigma} \partial \boldsymbol{\sigma}'} \right)^{-1} = 2 [\mathbf{D}'_K (\Sigma_u^{-1} \otimes \Sigma_u^{-1}) \mathbf{D}_K]^{-1} \\ &= 2\mathbf{D}_K^+ (\Sigma_u \otimes \Sigma_u) \mathbf{D}_K^{+'}, \end{aligned} \quad (3.4.28)$$

where $\mathbf{D}_K^+ = (\mathbf{D}'_K \mathbf{D}_K)^{-1} \mathbf{D}'_K$ is the Moore-Penrose generalized inverse of the duplication matrix $\mathbf{D}_K$ and Rule (17) from Appendix A.12 has been used. In summary, we get the following proposition.

Proposition 3.4 (*Asymptotic Properties of ML Estimators*)

Let y_t be a stationary, stable Gaussian VAR(p) process as in (3.3.1). Then the ML estimators $\tilde{\boldsymbol{\mu}}$, $\tilde{\boldsymbol{\alpha}}$, and $\tilde{\boldsymbol{\sigma}} = \text{vech}(\tilde{\Sigma}_u)$ given in (3.4.9)–(3.4.11) are consistent and

$$\sqrt{T} \begin{bmatrix} \tilde{\boldsymbol{\mu}} - \boldsymbol{\mu} \\ \tilde{\boldsymbol{\alpha}} - \boldsymbol{\alpha} \\ \tilde{\boldsymbol{\sigma}} - \boldsymbol{\sigma} \end{bmatrix} \xrightarrow{d} \mathcal{N} \left(0, \begin{bmatrix} \Sigma_{\tilde{\boldsymbol{\mu}}} & 0 & 0 \\ 0 & \Sigma_{\tilde{\boldsymbol{\alpha}}} & 0 \\ 0 & 0 & \Sigma_{\tilde{\boldsymbol{\sigma}}} \end{bmatrix} \right), \quad (3.4.29)$$

so that $\tilde{\boldsymbol{\mu}}$ is asymptotically independent of $\tilde{\boldsymbol{\alpha}}$ and $\tilde{\Sigma}_u$ and $\tilde{\boldsymbol{\alpha}}$ is asymptotically independent of $\tilde{\boldsymbol{\mu}}$ and $\tilde{\Sigma}_u$. The covariance matrices are

$$\Sigma_{\tilde{\boldsymbol{\mu}}} = \left(I_K - \sum_i A_i \right)^{-1} \Sigma_u \left(I_K - \sum_i A'_i \right)^{-1},$$

$$\Sigma_{\tilde{\boldsymbol{\alpha}}} = \Gamma_Y(0)^{-1} \otimes \Sigma_u,$$

$$\Sigma_{\tilde{\boldsymbol{\sigma}}} = 2\mathbf{D}_K^+ (\Sigma_u \otimes \Sigma_u) \mathbf{D}_K^{+'}.$$

They may be estimated consistently by replacing the unknown quantities by their ML estimators and estimating $\Gamma_Y(0)$ by $\tilde{X}\tilde{X}'/T$. ■

In this section, we have chosen to consider the mean-adjusted form of a VAR(p) process. Of course, it is possible to perform a similar derivation for the standard form given in (3.1.1). In that case the ML estimators of ν and $\boldsymbol{\alpha}$ are not asymptotically independent though. Their joint asymptotic distribution is identical to that of $\hat{\boldsymbol{\beta}}$ given in Proposition 3.1. From Proposition 3.2 we know that the asymptotic distribution of $\tilde{\boldsymbol{\sigma}}$ remains unaltered. In the next section, we will investigate the consequences of forecasting with estimated rather than known processes.

3.5 Forecasting with Estimated Processes

3.5.1 General Assumptions and Results

In Chapter 2, Section 2.2, we have seen that the optimal h -step forecast of the process (3.1.1) is

$$y_t(h) = \nu + A_1 y_t(h-1) + \cdots + A_p y_t(h-p), \quad (3.5.1)$$

where $y_t(j) = y_{t+j}$ for $j \leq 0$. If the true coefficients $B = (\nu, A_1, \dots, A_p)$ are replaced by estimators $\hat{B} = (\hat{\nu}, \hat{A}_1, \dots, \hat{A}_p)$, we get a forecast

$$\hat{y}_t(h) = \hat{\nu} + \hat{A}_1 \hat{y}_t(h-1) + \cdots + \hat{A}_p \hat{y}_t(h-p), \quad (3.5.2)$$

where $\hat{y}_t(j) = y_{t+j}$ for $j \leq 0$. Thus, the forecast error is

$$\begin{aligned} y_{t+h} - \hat{y}_t(h) &= [y_{t+h} - y_t(h)] + [y_t(h) - \hat{y}_t(h)] \\ &= \sum_{i=0}^{h-1} \Phi_i u_{t+h-i} + [y_t(h) - \hat{y}_t(h)], \end{aligned} \quad (3.5.3)$$

where the Φ_i are the coefficient matrices of the canonical MA representation of y_t (see (2.2.9)). Under quite general conditions for the process y_t , the forecast errors can be shown to have zero mean, $E[y_{t+h} - \hat{y}_t(h)] = 0$, so that the forecasts are unbiased even if the coefficients are estimated. Because we do not need this result in the following, we refer to Dufour (1985) for the details and a proof. All the u_s in the first term on the right-hand side of the last equality sign in (3.5.3) are attached to periods $s > t$, whereas all the y_s in the second term correspond to periods $s \leq t$, if estimation is done with observations from periods up to time t only. Therefore, the two terms are uncorrelated. Hence, the MSE matrix of the forecast $\hat{y}_t(h)$ is of the form

$$\begin{aligned} \Sigma_{\hat{y}}(h) &:= \text{MSE}[\hat{y}_t(h)] = E\{[y_{t+h} - \hat{y}_t(h)][y_{t+h} - \hat{y}_t(h)]'\} \\ &= \Sigma_y(h) + \text{MSE}[y_t(h) - \hat{y}_t(h)], \end{aligned} \quad (3.5.4)$$

where

$$\Sigma_y(h) = \sum_{i=0}^{h-1} \Phi_i \Sigma_u \Phi_i'$$

(see (2.2.11)). In order to evaluate the last term in (3.5.4), the distribution of the estimator $\hat{B}$ is needed. Because we have not been able to derive the small sample distributions of the estimators considered in the previous sections but we have derived the asymptotic distributions instead, we cannot hope for more than an asymptotic approximation to the MSE of $y_t(h) - \hat{y}_t(h)$. Such an approximation will be derived in the following.

There are two alternative assumptions that can be made in order to facilitate the derivation of the desired result:

- (1) Only data up to the forecast origin are used for estimation.
- (2) Estimation is done using a realization (time series) of a process that is independent of the process used for prediction and has the same stochastic structure (for instance, it is Gaussian and has the same first and second moments as the process used for prediction).

The first assumption is the more realistic one from a practical point of view because estimation and forecasting are usually based on the same data set. In that case, because the sample size is assumed to go to infinity in deriving asymptotic results, either the forecast origin has to go to infinity too or it has to be assumed that more and more data at the beginning of the sample become available. Because the forecast uses only p vectors y_s prior to the forecast period, these variables will be asymptotically independent of the estimator $\hat{B}$ (they are asymptotically negligible in comparison with all the other observations going into the estimate). Thus, asymptotically the first assumption implies the same results as the second one. In the following, for simplicity, the second assumption will therefore be used. Furthermore, it will be assumed that for $\beta = \text{vec}(B)$ and $\hat{\beta} = \text{vec}(\hat{B})$ we have

$$\sqrt{T}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\hat{\beta}}). \quad (3.5.5)$$

Samaranayake & Hasza (1988) and Basu & Sen Roy (1986) give a formal proof of the result that the MSE approximation obtained in the following remains valid under assumption (1) above.

With the foregoing assumptions it follows that, conditional on a particular realization $Y_t = (y'_t, \dots, y'_{t-p+1})'$ of the process used for prediction,

$$\sqrt{T} [\hat{y}_t(h) - y_t(h) | Y_t] \xrightarrow{d} \mathcal{N} \left(0, \frac{\partial y_t(h)}{\partial \beta'} \Sigma_{\hat{\beta}} \frac{\partial y_t(h)'}{\partial \beta} \right) \quad (3.5.6)$$

because $y_t(h)$ is a differentiable function of β (see Appendix C, Proposition C.15(3)). Here T is the sample size (time series length) used for estimation. This result suggests the approximation of $\text{MSE}[\hat{y}_t(h) - y_t(h)]$ by $\Omega(h)/T$, where

$$\Omega(h) = E \left[\frac{\partial y_t(h)}{\partial \beta'} \Sigma_{\hat{\beta}} \frac{\partial y_t(h)'}{\partial \beta} \right]. \quad (3.5.7)$$

In fact, for a Gaussian process y_t ,

$$\sqrt{T} [\hat{y}_t(h) - y_t(h)] \xrightarrow{d} \mathcal{N}(0, \Omega(h)). \quad (3.5.8)$$

Hence, we get an approximation

$$\Sigma_{\hat{y}}(h) = \Sigma_y(h) + \frac{1}{T} \Omega(h) \quad (3.5.9)$$

for the MSE matrix of $\hat{y}_t(h)$.

From (3.5.7) it is obvious that $\Omega(h)$ and, thus, the approximate MSE $\Sigma_{\hat{y}}(h)$ can be reduced by using an estimator that is asymptotically more efficient than $\hat{\beta}$, if such an estimator exists. In other words, efficient estimation is of importance in order to reduce the forecast uncertainty.

3.5.2 The Approximate MSE Matrix

To derive an explicit expression for $\Omega(h)$, the derivatives $\partial y_t(h)/\partial \beta'$ are needed. They can be obtained easily by noting that

$$y_t(h) = J_1 \mathbf{B}^h Z_t, \quad (3.5.10)$$

where $Z_t := (1, y'_t, \dots, y'_{t-p+1})'$,

$$\mathbf{B} := \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ \nu & A_1 & A_2 & \dots & A_{p-1} & A_p \\ 0 & I_K & 0 & \dots & 0 & 0 \\ 0 & 0 & I_K & & 0 & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ 0 & 0 & 0 & \dots & I_K & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ & B & & \\ 0 & I_{K(p-1)} & & 0 \end{bmatrix}$$

[(Kp+1) × (Kp+1)]

and

$$J_1 := \left[\underbrace{0}_{(K \times 1)} : I_K : \underbrace{0 : \dots : 0}_{(K \times K(p-1))} \right] \quad [K \times (Kp+1)].$$

The relation (3.5.10) follows by induction (see Problem 3.8). Using (3.5.10), we get

$$\begin{aligned} \frac{\partial y_t(h)}{\partial \beta'} &= \frac{\partial \text{vec}(J_1 \mathbf{B}^h Z_t)}{\partial \beta'} = (Z'_t \otimes J_1) \frac{\partial \text{vec}(\mathbf{B}^h)}{\partial \beta'} \\ &= (Z'_t \otimes J_1) \left[\sum_{i=0}^{h-1} (\mathbf{B}')^{h-1-i} \otimes \mathbf{B}^i \right] \frac{\partial \text{vec}(\mathbf{B})}{\partial \beta'} \\ &\quad (\text{Appendix A.13, Rule (8)}) \\ &= (Z'_t \otimes J_1) \left[\sum_{i=0}^{h-1} (\mathbf{B}')^{h-1-i} \otimes \mathbf{B}^i \right] (I_{Kp+1} \otimes J'_1) \\ &\quad (\text{see the definition of } \mathbf{B}) \\ &= \sum_{i=0}^{h-1} Z'_t (\mathbf{B}')^{h-1-i} \otimes J_1 \mathbf{B}^i J'_1 \\ &= \sum_{i=0}^{h-1} Z'_t (\mathbf{B}')^{h-1-i} \otimes \Phi_i, \end{aligned} \quad (3.5.11)$$

where $\Phi_i = J_1 \mathbf{B}^i J_1'$ follows as in (2.1.17). Using the LS estimator $\hat{\beta}$ with asymptotic covariance matrix $\Sigma_{\hat{\beta}} = \Gamma^{-1} \otimes \Sigma_u$ (see Proposition 3.1), the matrix $\Omega(h)$ is seen to be

$$\begin{aligned}
 \Omega(h) &= E \left[\frac{\partial y_t(h)}{\partial \beta'} (\Gamma^{-1} \otimes \Sigma_u) \frac{\partial y_t(h)'}{\partial \beta} \right] \\
 &= \sum_{i=0}^{h-1} \sum_{j=0}^{h-1} E(Z_t'(\mathbf{B}')^{h-1-i} \Gamma^{-1} \mathbf{B}^{h-1-j} Z_t) \otimes \Phi_i \Sigma_u \Phi_j' \\
 &= \sum_i \sum_j E[\text{tr}(Z_t'(\mathbf{B}')^{h-1-i} \Gamma^{-1} \mathbf{B}^{h-1-j} Z_t)] \Phi_i \Sigma_u \Phi_j' \\
 &= \sum_i \sum_j \text{tr}[(\mathbf{B}')^{h-1-i} \Gamma^{-1} \mathbf{B}^{h-1-j} E(Z_t Z_t')] \Phi_i \Sigma_u \Phi_j' \\
 &= \sum_{i=0}^{h-1} \sum_{j=0}^{h-1} \text{tr}[(\mathbf{B}')^{h-1-i} \Gamma^{-1} \mathbf{B}^{h-1-j} \Gamma] \Phi_i \Sigma_u \Phi_j', \tag{3.5.12}
 \end{aligned}$$

provided y_t is stable so that

$$\Gamma := \text{plim}(ZZ'/T) = E(Z_t Z_t').$$

Here $Z := (Z_0, \dots, Z_{T-1})$ is the $((Kp+1) \times T)$ matrix defined in (3.2.1).

For example, for $h = 1$,

$$\Omega(1) = (Kp+1)\Sigma_u.$$

Hence, the approximation

$$\Sigma_{\hat{y}}(1) = \Sigma_u + \frac{Kp+1}{T} \Sigma_u = \frac{T+Kp+1}{T} \Sigma_u \tag{3.5.13}$$

of the MSE matrix of the 1-step forecast with estimated coefficients is obtained. This expression shows that the contribution of the estimation variability to the forecast MSE matrix $\Sigma_{\hat{y}}(1)$ depends on the dimension K of the process, the VAR order p , and the sample size T used for estimation. It can be quite substantial if the sample size is small or moderate. For instance, considering a three-dimensional process of order 8 which is estimated from 15 years of quarterly data (i.e., $T = 52$ plus 8 presample values needed for LS estimation), the 1-step forecast MSE matrix Σ_u for known processes is inflated by a factor $(T+Kp+1)/T = 1.48$. Of course, this approximation is derived from asymptotic theory so that its small sample validity is not guaranteed. We will take a closer look at this problem shortly. Obviously, the inflation factor $(T+Kp+1)/T \rightarrow 1$ for $T \rightarrow \infty$. Thus the MSE contribution due to sampling variability vanishes if the sample size gets large. This result is a consequence of estimating the VAR coefficients consistently. An expression for $\Omega(h)$ can also be derived on the basis of the mean-adjusted form of the VAR process (see Problem 3.9).

In practice, for $h > 1$, it will not be possible to evaluate $\Omega(h)$ without knowing the AR coefficients summarized in the matrix B . A consistent estimator $\hat{\Omega}(h)$ may be obtained by replacing all unknown parameters by their LS estimators, that is, $\mathbf{B}$ is replaced by $\hat{\mathbf{B}}$ which is obtained by using $\hat{B}$ for B , Σ_u is replaced by $\hat{\Sigma}_u$, Φ_i is estimated by $\hat{\Phi}_i = J_1 \hat{\mathbf{B}}^i J_1'$, and Γ is estimated by $\hat{\Gamma} = ZZ'/T$. The resulting estimator of $\Sigma_{\hat{y}}(h)$ will be denoted by $\hat{\Sigma}_{\hat{y}}(h)$ in the following.

The foregoing discussion is of importance in setting up interval forecasts. Assuming that y_t is Gaussian, an approximate $(1 - \alpha)100\%$ interval forecast, h periods ahead, for the k -th component $y_{k,t}$ of y_t is

$$\hat{y}_{k,t}(h) \pm z_{(\alpha/2)} \hat{\sigma}_k(h) \quad (3.5.14)$$

or

$$\left[\hat{y}_{k,t}(h) - z_{(\alpha/2)} \hat{\sigma}_k(h), \hat{y}_{k,t}(h) + z_{(\alpha/2)} \hat{\sigma}_k(h) \right], \quad (3.5.15)$$

where $z_{(\alpha)}$ is the upper $\alpha 100$ -th percentile of the standard normal distribution and $\hat{\sigma}_k(h)$ is the square root of the k -th diagonal element of $\hat{\Sigma}_{\hat{y}}(h)$. Using Bonferroni's inequality, approximate joint confidence regions for a set of forecasts can be obtained just as described in Section 2.2.3 of Chapter 2.

3.5.3 An Example

To illustrate the previous results, we consider again the investment/income/consumption example of Section 3.2.3. Using the VAR(2) model with the coefficient estimates given in (3.2.22) and

$$y_{T-1} = y_{72} = \begin{bmatrix} .02551 \\ .02434 \\ .01319 \end{bmatrix} \quad \text{and} \quad y_T = y_{73} = \begin{bmatrix} .03637 \\ .00517 \\ .00599 \end{bmatrix}$$

results in forecasts

$$\begin{aligned} \hat{y}_T(1) &= \hat{\nu} + \hat{A}_1 y_T + \hat{A}_2 y_{T-1} = \begin{bmatrix} -.011 \\ .020 \\ .022 \end{bmatrix}, \\ \hat{y}_T(2) &= \hat{\nu} + \hat{A}_1 \hat{y}_T(1) + \hat{A}_2 y_T = \begin{bmatrix} .011 \\ .020 \\ .015 \end{bmatrix}, \end{aligned} \quad (3.5.16)$$

and so on.

The estimated forecast MSE matrix for $h = 1$ is

$$\begin{aligned} \hat{\Sigma}_{\hat{y}}(1) &= \frac{T + Kp + 1}{T} \hat{\Sigma}_u = \frac{73 + 6 + 1}{73} \hat{\Sigma}_u \\ &= \begin{bmatrix} 23.34 & .785 & 1.351 \\ .785 & 1.505 & .674 \\ 1.351 & .674 & .978 \end{bmatrix} \times 10^{-4}, \end{aligned} \quad (3.5.17)$$

where $\widehat{\Sigma}_u$ from (3.2.23) has been used. We need $\widehat{\Phi}_1$ for evaluating

$$\widehat{\Sigma}_{\widehat{y}}(2) = \widehat{\Sigma}_y(2) + \frac{1}{T} \widehat{\Omega}(2),$$

where

$$\widehat{\Sigma}_y(2) = \widehat{\Sigma}_u + \widehat{\Phi}_1 \widehat{\Sigma}_u \widehat{\Phi}_1'$$

and

$$\begin{aligned} \widehat{\Omega}(2) &= \sum_{i=0}^1 \sum_{j=0}^1 \text{tr} \left[(\widehat{\mathbf{B}}')^{1-i} (ZZ'/T)^{-1} \widehat{\mathbf{B}}^{1-j} (ZZ'/T) \right] \widehat{\Phi}_i \widehat{\Sigma}_u \widehat{\Phi}_j' \\ &= \text{tr}[\widehat{\mathbf{B}}'(ZZ')^{-1} \widehat{\mathbf{B}}ZZ'] \widehat{\Sigma}_u + \text{tr}(\widehat{\mathbf{B}}') \widehat{\Sigma}_u \widehat{\Phi}_1' \\ &\quad + \text{tr}(\widehat{\mathbf{B}}) \widehat{\Phi}_1 \widehat{\Sigma}_u + \text{tr}(I_{K_{p+1}}) \widehat{\Phi}_1 \widehat{\Sigma}_u \widehat{\Phi}_1'. \end{aligned}$$

From (2.1.22) we know that $\Phi_1 = A_1$. Hence, we use $\widehat{\Phi}_1 = \widehat{A}_1$ from (3.2.22). Thus, we get

$$\widehat{\Sigma}_y(2) = \begin{bmatrix} 23.67 & .547 & 1.226 \\ .547 & 1.488 & .554 \\ 1.226 & .554 & .952 \end{bmatrix} \times 10^{-4}$$

and

$$\widehat{\Omega}(2) = \begin{bmatrix} 10.59 & .238 & .538 \\ .238 & .675 & .233 \\ .538 & .233 & .422 \end{bmatrix} \times 10^{-3}.$$

Consequently,

$$\widehat{\Sigma}_{\widehat{y}}(2) = \begin{bmatrix} 25.12 & .580 & 1.300 \\ .580 & 1.581 & .586 \\ 1.300 & .586 & 1.009 \end{bmatrix} \times 10^{-4}. \quad (3.5.18)$$

Assuming that the data are generated by a Gaussian process, we get the following approximate 95% interval forecasts:

$$\begin{aligned} \widehat{y}_{1,T}(1) &\pm 1.96 \widehat{\sigma}_1(1) \text{ or } -.011 \pm .095, \\ \widehat{y}_{2,T}(1) &\pm 1.96 \widehat{\sigma}_2(1) \text{ or } .020 \pm .024, \\ \widehat{y}_{3,T}(1) &\pm 1.96 \widehat{\sigma}_3(1) \text{ or } .022 \pm .019, \\ \widehat{y}_{1,T}(2) &\pm 1.96 \widehat{\sigma}_1(2) \text{ or } .011 \pm .098, \\ \widehat{y}_{2,T}(2) &\pm 1.96 \widehat{\sigma}_2(2) \text{ or } .020 \pm .025, \\ \widehat{y}_{3,T}(2) &\pm 1.96 \widehat{\sigma}_3(2) \text{ or } .015 \pm .020. \end{aligned} \quad (3.5.19)$$

In Figure 3.3, some more forecasts of the three variables with two-standard error bounds to each side are depicted. The intervals indicated by the dashed bounds may be interpreted as approximate 95% forecast intervals for the individual forecasts. If the region enclosed by the dashed lines is viewed as a joint confidence region for all 4 forecasts, a lower bound for the (approximate) probability content is $(100-4\times 5)\% = 80\%$. In the figure it can be seen that for investment and income the actually observed values for 1979 ($t = 77, \dots, 80$) are well inside the forecast regions, whereas two of the four consumption values are outside that region.

3.5.4 A Small Sample Investigation

It is not obvious that the MSE and interval forecast approximations derived in the foregoing are reasonable in small samples because the MSE modification has been based on asymptotic theory. To investigate the small sample behavior of the predictor with estimated coefficients, we have used again 1000 realizations of the bivariate VAR(2) process (3.2.25)/(3.2.26) of Section 3.2.4 and we have computed forecast intervals for the period following the last sample period. In Table 3.2, the proportions of actual values falling in these intervals are reported for sample sizes of $T = 30$ and 100.

Table 3.2. Accuracy of forecast intervals in small samples based on 1000 bivariate time series

		percent of actual values falling in the forecast interval			
MSE used in interval construction	% forecast interval	$T = 30$		$T = 100$	
		y_1	y_2	y_1	y_2
$\Sigma_y(1)$	90	86.5	85.7	89.7	89.4
	95	92.6	91.8	94.5	94.0
	99	98.1	98.0	99.0	98.5
$\Sigma_{\hat{y}}(1)$	90	89.3	88.2	90.4	90.0
	95	94.4	94.1	95.3	94.6
	99	99.0	98.4	99.3	98.8
$\hat{\Sigma}_y(1)$	90	85.2	84.2	89.6	88.5
	95	90.5	90.4	94.7	93.9
	99	98.4	96.5	98.9	98.3
$\hat{\Sigma}_{\hat{y}}(1)$	90	88.1	86.9	90.3	89.1
	95	93.4	92.7	95.2	94.0
	99	99.4	97.8	99.1	98.5

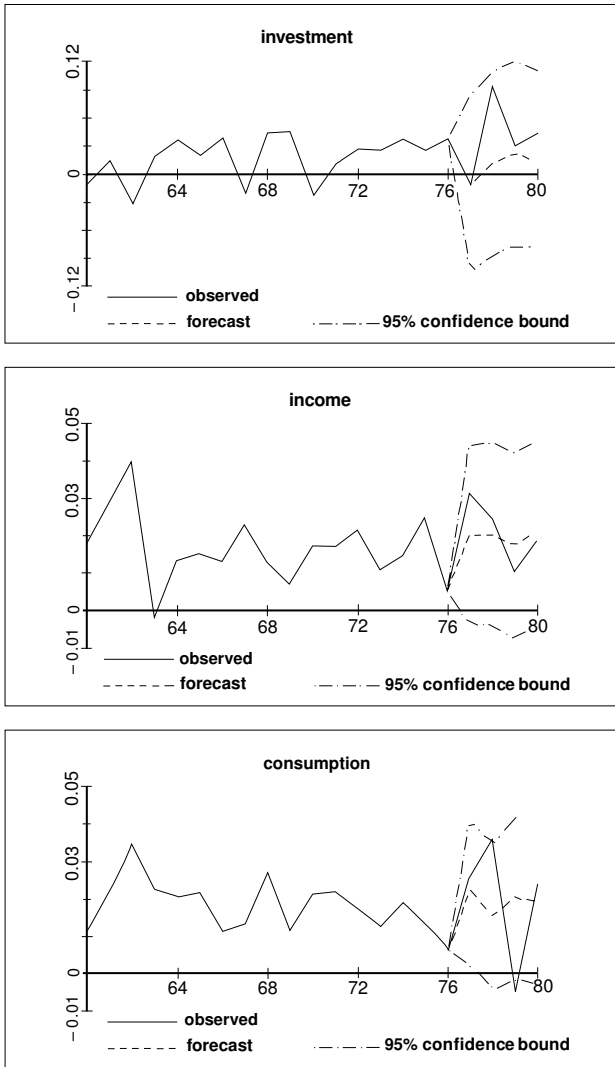


Fig. 3.3. Forecasts of the investment/income/consumption system.

Obviously, for $T = 30$, the theoretical and actual percentages are in best agreement if the approximate MSEs $\Sigma_{\hat{y}}(h)$ are used in setting up the forecast intervals. On the other hand, only forecast intervals based on $\hat{\Sigma}_y(h) = \sum_{i=0}^{h-1} \hat{\Phi}_i \hat{\Sigma}_u \hat{\Phi}_i'$ and $\hat{\Sigma}_{\hat{y}}(h)$ are feasible in practice when the actual process coefficients are unknown and have to be estimated. Comparing only the results based on these two MSE matrices shows that it pays to use the asymptotic approximation $\hat{\Sigma}_{\hat{y}}(h)$.

In Table 3.2, we also give the corresponding results for $T = 100$. Because the estimation uncertainty decreases with increasing sample size, one would expect that now the theoretical and actual percentages are in good agreement for all MSEs. This is precisely what can be observed in the table. Nevertheless, even now the use of the MSE adjustment in $\hat{\Sigma}_{\hat{y}}(1)$ gives slightly more accurate interval forecasts.

3.6 Testing for Causality

3.6.1 A Wald Test for Granger-Causality

In Chapter 2, Section 2.3.1, we have partitioned the $\text{VAR}(p)$ process y_t in subprocesses z_t and x_t , that is, $y'_t = (z'_t, x'_t)$ and we have defined Granger-causality from x_t to z_t and vice versa. We have seen that this type of causality can be characterized by specific zero constraints on the VAR coefficients (see Corollary 2.2.1). Thus, in an estimated $\text{VAR}(p)$ system, if we want to test for Granger-causality, we need to test zero constraints for the coefficients. Given the results of Sections 3.2, 3.3, and 3.4 it is straightforward to derive *asymptotic* tests of such constraints.

More generally we consider testing

$$H_0 : C\beta = c \quad \text{against} \quad H_1 : C\beta \neq c, \quad (3.6.1)$$

where C is an $(N \times (K^2p + K))$ matrix of rank N and c is an $(N \times 1)$ vector. Assuming that

$$\sqrt{T}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \Gamma^{-1} \otimes \Sigma_u) \quad (3.6.2)$$

as in LS/ML estimation, we get

$$\sqrt{T}(C\hat{\beta} - C\beta) \xrightarrow{d} \mathcal{N}[0, C(\Gamma^{-1} \otimes \Sigma_u)C'] \quad (3.6.3)$$

(see Appendix C, Proposition C.15) and, hence,

$$T(C\hat{\beta} - c)' [C(\Gamma^{-1} \otimes \Sigma_u)C']^{-1} (C\hat{\beta} - c) \xrightarrow{d} \chi^2(N). \quad (3.6.4)$$

This statistic is the *Wald statistic* (see Appendix C.7).

Replacing Γ and Σ_u by their usual estimators $\hat{\Gamma} = ZZ'/T$ and $\hat{\Sigma}_u$ as given in (3.2.19), the resulting statistic

$$\lambda_W = (C\hat{\beta} - c)' [C((ZZ')^{-1} \otimes \hat{\Sigma}_u)C']^{-1} (C\hat{\beta} - c) \quad (3.6.5)$$

still has an asymptotic χ^2 -distribution with N degrees of freedom, provided y_t satisfies the conditions of Proposition 3.2, because under these conditions $[C((ZZ')^{-1} \otimes \hat{\Sigma}_u)C']^{-1}/T$ is a consistent estimator of $[C(\Gamma^{-1} \otimes \Sigma_u)C']^{-1}$. Hence, we have the following result.

Proposition 3.5 (*Asymptotic Distribution of the Wald Statistic*)

Suppose (3.6.2) holds. Furthermore, $\text{plim}(ZZ'/T) = \Gamma$, $\text{plim } \widehat{\Sigma}_u = \Sigma_u$ are both nonsingular and $H_0 : C\beta = c$ is true, with C being an $(N \times (K^2p + K))$ matrix of rank N . Then

$$\lambda_W = (C\widehat{\beta} - c)'[C((ZZ')^{-1} \otimes \widehat{\Sigma}_u)C']^{-1}(C\widehat{\beta} - c) \xrightarrow{d} \chi^2(N).$$

■

In practice, it may be useful to make adjustments to the statistic or the critical values of the test to compensate for the fact that the matrix $\Gamma^{-1} \otimes \Sigma_u$ is unknown and has been replaced by an estimator. Working in that direction, we note that

$$NF(N, T) \xrightarrow[T \rightarrow \infty]{d} \chi^2(N), \quad (3.6.6)$$

where $F(N, T)$ denotes an F random variable with N and T degrees of freedom (d.f.) (Appendix C, Proposition C.3). Because an $F(N, T)$ -distribution has a fatter tail than the $\chi^2(N)$ -distribution divided by N , it seems reasonable to consider the test statistic

$$\lambda_F = \lambda_W / N \quad (3.6.7)$$

in conjunction with critical values from some F -distribution. The question is then what numbers of degrees of freedom should be used? From the foregoing discussion it is plausible to use N as the numerator degrees of freedom. On the other hand, any sequence that goes to infinity with the sample size qualifies as a candidate for the denominator d.f. The usual F -statistic for a regression model with nonstochastic regressors has denominator d.f. equal to the sample size minus the number of estimated parameters. Therefore we may use this number here too. Note that, in the model (3.2.3), we have a vector $\mathbf{y}$ with KT observations and β contains $K(Kp+1)$ parameters. Alternatively, we will argue shortly that $T - Kp - 1$ is also a reasonable number for the denominator d.f. Hence, we have the approximate distributions

$$\lambda_F \approx F(N, KT - K^2p - K) \approx F(N, T - Kp - 1). \quad (3.6.8)$$

3.6.2 An Example

To see how this result can be used in a test for Granger-causality, let us consider again our example system from Section 3.2.3. The null hypothesis of no Granger-causality from income/consumption (y_2, y_3) to investment (y_1) may be expressed in terms of the coefficients of the VAR(2) process as

$$H_0 : \alpha_{12,1} = \alpha_{13,1} = \alpha_{12,2} = \alpha_{13,2} = 0. \quad (3.6.9)$$

This null hypothesis may be written as in (3.6.1) by defining the (4×1) vector $c = 0$ and the (4×21) matrix

$$C = \begin{bmatrix} 000 & 000 & 100 & 000 & 000 & 000 & 000 & 000 \\ 000 & 000 & 000 & 100 & 000 & 000 & 000 & 000 \\ 000 & 000 & 000 & 000 & 000 & 000 & 100 & 000 \\ 000 & 000 & 000 & 000 & 000 & 000 & 000 & 100 \end{bmatrix}.$$

With this notation, using the estimation results from Section 3.2.3,

$$\lambda_F = \hat{\beta}' C' \left[C((ZZ')^{-1} \otimes \hat{\Sigma}_u) C' \right]^{-1} C \hat{\beta} / 4 = 1.59. \quad (3.6.10)$$

In contrast, the 95th percentile of the $F(4, 3 \cdot 73 - 9 \cdot 2 - 3) = F(4, 198) \approx F(4, 73 - 3 \cdot 2 - 1) = F(4, 66)$ -distribution is about 2.5. Thus, in a 5% level test, we cannot reject Granger-noncausality from income/consumption to investment.

In this example, the denominator d.f. are so large (namely 198 or 66) that we could just as well use λ_W in conjunction with a critical value from a $\chi^2(4)$ -distribution. The 95th percentile of that distribution is 9.49 and, thus, it is about four times that of the F -test while $\lambda_W = 4\lambda_F$.

In an example of this type it is quite reasonable to use $T - Kp - 1$ denominator d.f. for the F -test because all the restrictions are imposed on coefficients from one equation. Therefore λ_F actually reduces to an F -statistic related to one equation with $Kp + 1$ parameters which are estimated from T observations. The use of $T - Kp - 1$ d.f. may also be justified by arguments that do not rely on the restrictions being imposed on the parameters of one equation only, namely by appealing to the similarity between the λ_F statistic and Hotelling's T^2 (e.g., Anderson (1984)).

Many other tests for Granger-causality have been proposed and investigated (see, e.g., Geweke, Meese & Dent (1983)). In the next chapter, we will return to the testing of hypotheses and then an alternative test will be considered.

3.6.3 Testing for Instantaneous Causality

Tests for instantaneous causality can be developed in the same way as tests for Granger-causality because instantaneous causality can be expressed in terms of zero restrictions for $\sigma = \text{vech}(\Sigma_u)$ (see Proposition 2.3). If y_t is a stable Gaussian VAR(p) process and we wish to test

$$H_0 : C\sigma = 0 \quad \text{against} \quad H_1 : C\sigma \neq 0, \quad (3.6.11)$$

we may use the asymptotic distribution of the ML estimator given in Proposition 3.4 to set up the Wald statistic

$$\lambda_W = T \tilde{\sigma}' C' [2C \mathbf{D}_K^+ (\tilde{\Sigma}_u \otimes \tilde{\Sigma}_u) \mathbf{D}_K^{+'} C']^{-1} C \tilde{\sigma}, \quad (3.6.12)$$

where $\mathbf{D}_K^+$ is the Moore-Penrose inverse of the duplication matrix $\mathbf{D}_K$ and C is an $(N \times K(K+1)/2)$ matrix of rank N . Under H_0 , λ_W has an asymptotic χ^2 -distribution with N degrees of freedom.

Alternatively, a Wald test of (3.6.11) could be based on the lower triangular matrix P which is obtained from a Choleski decomposition of Σ_u . Noting that instantaneous noncausality implies zero elements of Σ_u that correspond to zero elements of P , we can write H_0 from (3.6.11) equivalently as

$$H_0 : C \text{vech}(P) = 0. \quad (3.6.13)$$

Because $\text{vech}(P)$ is a continuously differentiable function of σ , the asymptotic distribution of the estimator P obtained from decomposing $\tilde{\Sigma}_u$ follows from Proposition C.15(3) of Appendix C:

$$\sqrt{T} \text{vech}(\tilde{P} - P) \xrightarrow{d} \mathcal{N}(0, \bar{H} \Sigma_{\tilde{\sigma}} \bar{H}'), \quad (3.6.14)$$

where

$$\bar{H} = \frac{\partial \text{vech}(P)}{\partial \sigma'} = [\mathbf{L}_K (I_{K^2} + \mathbf{K}_{KK}) (P \otimes I_K) \mathbf{L}'_K]^{-1}$$

(see Appendix A.13, Rule (10)). Here $\mathbf{K}_{mn}$ is the commutation matrix defined such that $\text{vec}(G) = \mathbf{K}_{mn} \text{vec}(G')$ for any $(m \times n)$ matrix G and $\mathbf{L}_K$ is the $(\frac{1}{2}K(K+1) \times K^2)$ elimination matrix defined such that $\text{vech}(F) = \mathbf{L}_K \text{vec}(F)$ for any $(K \times K)$ matrix F (see Appendix A.12.2). A Wald test of (3.6.13) may therefore be based on

$$\lambda_W = T \text{vech}(\tilde{P})' C' [C \hat{H} \hat{\Sigma}_{\tilde{\sigma}} \hat{H}' C']^{-1} C \text{vech}(\tilde{P}) \xrightarrow{d} \chi^2(N), \quad (3.6.15)$$

where hats denote the usual estimators. Although the two tests based on $\tilde{\sigma}$ and $\tilde{P}$ are derived from the same asymptotic distribution, they may differ in small samples. Of course, in the previous discussion we may replace $\tilde{\Sigma}_u$ by the asymptotically equivalent estimator $\hat{\Sigma}_u$.

In our investment/income/consumption example, suppose we wish to test for instantaneous causality between (income, consumption) and investment. Following Proposition 2.3, the null hypothesis of no causality is

$$H_0 : \sigma_{21} = \sigma_{31} = 0 \quad \text{or} \quad C\sigma = 0,$$

where σ_{ij} is a typical element of Σ_u and

$$C = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}.$$

For this hypothesis, the test statistic in (3.6.12) assumes the value $\lambda_W = 5.46$. Alternatively, we may test

$$H_0 : p_{21} = p_{31} = 0 \quad \text{or} \quad C \text{vech}(P) = 0,$$

where p_{ij} is a typical element of P . The corresponding value of the test statistic from (3.6.15) is $\lambda_W = 5.70$. Both tests are based on asymptotic $\chi^2(2)$ -distributions and therefore do not reject the null hypothesis of no instantaneous causality at a 5% level. Note that the critical value for a 5% level test is 5.99.

3.6.4 Testing for Multi-Step Causality

In Section 2.3.1, we have also discussed the possibility of extending the information set and considering causality between two variables in a system that includes further variables. Using the same ideas as in the definition of Granger-causality resulted in the definition of h -step causality. This concept implies nonlinear restrictions for the VAR coefficients for which the usual application of the Wald principle does not result in a valid test. The following example from Lütkepohl & Burda (1997) illustrates the problem.

Consider a three-dimensional VAR(1) process:

$$\begin{bmatrix} z_t \\ y_t \\ x_t \end{bmatrix} = \begin{bmatrix} \alpha_{zz} & \alpha_{zy} & \alpha_{zx} \\ \alpha_{yz} & \alpha_{yy} & \alpha_{yx} \\ \alpha_{xz} & \alpha_{xy} & \alpha_{xx} \end{bmatrix} \begin{bmatrix} z_{t-1} \\ y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} u_{z,t} \\ u_{y,t} \\ u_{x,t} \end{bmatrix}. \quad (3.6.16)$$

From (2.3.24) we know that a test of ∞ -step noncausality from y_t to z_t ($y_t \not\rightarrow_{(\infty)} z_t$) needs to check $h = 2$ restrictions on the VAR coefficient vector. They are of the following nonlinear form:

$$r(\alpha) = \begin{bmatrix} R\alpha \\ R\alpha^{(2)} \end{bmatrix} = (I_2 \otimes R) \begin{bmatrix} \alpha \\ \alpha^{(2)} \end{bmatrix},$$

where

$$R = [0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0],$$

$\alpha = \text{vec}(A_1)$ and $\alpha^{(2)} = \text{vec}(A_1^2)$, with A_1 being the coefficient matrix of the process in (3.6.16). Hence,

$$r(\alpha) = \begin{bmatrix} \alpha_{zy} \\ \alpha_{zz}\alpha_{zy} + \alpha_{zy}\alpha_{yy} + \alpha_{zx}\alpha_{xy} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (3.6.17)$$

Denoting the covariance matrix of the asymptotic distribution of $\sqrt{T}(\hat{\alpha} - \alpha)$ as usual by $\Sigma_{\hat{\alpha}}$ and a consistent estimator by $\hat{\Sigma}_{\hat{\alpha}}$, the Wald statistic for testing these restrictions has the form

$$\lambda_W = \text{Tr}(\hat{\alpha})' \left(\frac{\widehat{\partial r}}{\partial \alpha'} \hat{\Sigma}_{\hat{\alpha}} \frac{\widehat{\partial r'}}{\partial \alpha} \right)^{-1} r(\hat{\alpha}),$$

where $\widehat{\partial r}/\partial \alpha'$ is an estimator of $\partial r/\partial \alpha'$ (see Appendix C.7). The statistic has an asymptotic $\chi^2(2)$ -distribution under the null hypothesis, provided the matrix

$$\frac{\partial r}{\partial \alpha'} \Sigma_{\hat{\alpha}} \frac{\partial r'}{\partial \alpha}$$

is nonsingular. In the present case, the latter condition is unfortunately not satisfied for all relevant parameter values.

To see this, note that the matrix of first order partial derivatives of the function $r(\boldsymbol{\alpha})$ is

$$\frac{\partial r}{\partial \boldsymbol{\alpha}'} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \alpha_{zy} & 0 & 0 & \alpha_{zz} + \alpha_{yy} & \alpha_{zy} & \alpha_{zx} & \alpha_{xy} & 0 & 0 \end{bmatrix}.$$

The restrictions (3.6.17) are satisfied if

$$\alpha_{zy} = \alpha_{zx} = 0, \alpha_{xy} \neq 0, \quad (3.6.18)$$

or

$$\alpha_{zy} = \alpha_{xy} = 0, \alpha_{zx} \neq 0, \quad (3.6.19)$$

or

$$\alpha_{zy} = \alpha_{zx} = \alpha_{xy} = 0. \quad (3.6.20)$$

Clearly, $\partial r / \partial \boldsymbol{\alpha}'$ has rank 1 only and, thus,

$$\text{rk} \left(\frac{\partial r}{\partial \boldsymbol{\alpha}'} \Sigma_{\tilde{\boldsymbol{\alpha}}} \frac{\partial r'}{\partial \boldsymbol{\alpha}} \right) = 1,$$

if (3.6.20) holds. Hence, the standard Wald statistic will not have its asymptotic $\chi^2(2)$ -distribution under the null hypothesis $r(\boldsymbol{\alpha}) = 0$ if (3.6.20) holds.

Lütkepohl & Burda (1997) discussed a possibility to circumvent the problem by simply drawing a random variable from a normal distribution and adding it to the second restriction. Thereby a nonsingular distribution of the modified restriction vector is obtained and a Wald type statistic can be constructed for this vector.

More generally, Lütkepohl & Burda (1997) proposed the following approach for testing the null hypothesis that the K_y -dimensional vector y_t is not h -step causal for the K_z -dimensional vector z_t ($y_t \not\rightarrow_{(h)} z_t$) if additional K_x variables x_t are present in the system of interest. Using the notation from Section 2.3.1, that is, $\mathbf{A}$ is defined as in the VAR(1) representation (2.1.8), $J := [I_K : 0 : \cdots : 0]$ is a $(K \times Kp)$ matrix, $A^{(j)} := J\mathbf{A}^j$, and $\boldsymbol{\alpha}^{(j)} := \text{vec}(A^{(j)})$, the hypotheses of interest can be stated as

$$H_0 : (I_h \otimes R) \mathbf{a}^{(h)} = 0 \quad \text{against} \quad H_1 : (I_h \otimes R) \mathbf{a}^{(h)} \neq 0, \quad (3.6.21)$$

where R is a $(pK_z K_y \times pK^2)$ matrix, as defined in (2.3.23), and

$$\mathbf{a}^{(h)} = \begin{bmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\alpha}^{(2)} \\ \vdots \\ \boldsymbol{\alpha}^{(h)} \end{bmatrix}.$$

Let $\hat{\mathbf{a}}^{(h)}$ be the estimator corresponding to $\mathbf{a}^{(h)}$ based on the multivariate LS estimator $\hat{\boldsymbol{\alpha}}$ of $\boldsymbol{\alpha}$. Furthermore, we denote by $\text{diag}(D)$ a diagonal matrix which has the diagonal elements of the square matrix D on its main diagonal and define the $(hpK_zK_y \times hpK_zK_y)$ matrix

$$\hat{\Sigma}_w(h) = \begin{bmatrix} 0 & 0 \\ 0 & I_{h-1} \otimes \text{diag}(R\hat{\Sigma}_{\hat{\boldsymbol{\alpha}}}R') \end{bmatrix}.$$

Moreover, we define a random vector $w_{\lambda}^{(h)} \sim \mathcal{N}(0, \lambda \hat{\Sigma}_w(h))$ which is drawn independently of $\hat{\boldsymbol{\alpha}}$. Here $\lambda > 0$ is some fixed real number. Lütkepohl & Burda (1997) defined the following modified Wald statistic for testing the pair of hypotheses in (3.6.21):

$$\begin{aligned} \lambda_W^{mod} &= T \left((I_h \otimes R) \hat{\mathbf{a}}^{(h)} + \frac{w_{\lambda}^{(h)}}{\sqrt{T}} \right)' \\ &\times \left[(I_h \otimes R) \hat{\Sigma}_{\hat{\mathbf{a}}}(h) (I_h \otimes R') + \lambda \hat{\Sigma}_w(h) \right]^{-1} \\ &\times \left((I_h \otimes R) \hat{\mathbf{a}}^{(h)} + \frac{w_{\lambda}^{(h)}}{\sqrt{T}} \right). \end{aligned}$$

Here $\hat{\Sigma}_{\hat{\mathbf{a}}}(h)$ is a consistent estimator of the asymptotic covariance matrix of $\sqrt{T}(\hat{\mathbf{a}}^{(h)} - \mathbf{a}^{(h)})$. It can be shown that

$$\lambda_W^{mod} \xrightarrow{d} \chi^2(hpK_zK_y)$$

under H_0 . Notice that there is no need to add anything to the first pK_zK_y components of $(I_h \otimes R)\hat{\mathbf{a}}^{(h)}$ because they are equal to $R\hat{\boldsymbol{\alpha}}$ which has a non-singular asymptotic distribution.

Clearly, adding some random term to $\hat{\mathbf{a}}^{(h)}$ reduces the efficiency of the procedure and is likely to result in a loss in power of the test relative to a procedure which does not use this device. In particular, if the noise term is substantial in relation to the estimated variance, there may be some loss in power. Therefore, the amount of noise (the variance of the noise) is linked to the variance of the estimator through $\Sigma_w(h)$. Moreover, the quantity λ may be chosen close to zero. Thereby the loss in efficiency can be made arbitrarily small.

There are in fact also other possibilities to avoid the problems related to the Wald test. One way to get around it is to impose zero restrictions directly on the VAR coefficients prior to analyzing multi-step causality. The relevant subset models will be discussed in Chapter 5.

3.7 The Asymptotic Distributions of Impulse Responses and Forecast Error Variance Decompositions

3.7.1 The Main Results

In Chapter 2, Section 2.3.2, we have seen that the coefficients of the MA representations

$$y_t = \mu + \sum_{i=0}^{\infty} \Phi_i u_{t-i}, \quad \Phi_0 = I_K, \quad (3.7.1)$$

and

$$y_t = \mu + \sum_{i=0}^{\infty} \Theta_i w_{t-i} \quad (3.7.2)$$

are sometimes interpreted as impulse responses or dynamic multipliers of the system of variables y_t . Here $\mu = E(y_t)$, the $\Theta_i = \Phi_i P$, $w_t = P^{-1}u_t$, and P is the lower triangular Choleski decomposition of Σ_u such that $\Sigma_u = PP'$. Hence, $\Sigma_w = E(w_t w_t') = I_K$. In this section, we will assume that the Φ_i 's and Θ_i 's are unknown and they are computed from the estimated VAR coefficients and error covariance matrix. We will derive the asymptotic distributions of the resulting estimated Φ_i 's and Θ_i 's. In these derivations, we will not need the existence of MA representations (3.7.1) and (3.7.2). We will just assume that the Φ_i 's are obtained from given coefficient matrices $A_1, \dots, A_p$ by recursions

$$\Phi_i = \sum_{j=1}^i \Phi_{i-j} A_j, \quad i = 1, 2, \dots,$$

starting with $\Phi_0 = I_K$ and setting $A_j = 0$ for $j > p$. Furthermore, the Θ_i 's are obtained from $A_1, \dots, A_p$, and Σ_u as $\Theta_i = \Phi_i P$, where P is as specified in the foregoing. In addition, the asymptotic distributions of the corresponding accumulated responses

$$\begin{aligned} \Psi_n &= \sum_{i=0}^n \Phi_i, & \Psi_{\infty} &= \sum_{i=0}^{\infty} \Phi_i = (I_K - A_1 - \dots - A_p)^{-1} \quad (\text{if it exists}), \\ \Xi_n &= \sum_{i=0}^n \Theta_i, & \Xi_{\infty} &= \sum_{i=0}^{\infty} \Theta_i = (I_K - A_1 - \dots - A_p)^{-1} P \quad (\text{if it exists}), \end{aligned}$$

and the forecast error variance components,

$$\omega_{jk,h} = \sum_{i=0}^{h-1} (e_j' \Theta_i e_k)^2 / \text{MSE}_j(h), \quad (3.7.3)$$

will be given. Here e_k is the k -th column of I_K and

$$\text{MSE}_j(h) = \sum_{i=0}^{h-1} e_j' \Phi_i \Sigma_u \Phi_i' e_j$$

is the j -th diagonal element of the MSE matrix $\Sigma_y(h)$ of an h -step forecast (see Chapter 2, Section 2.2.2).

The derivation of the asymptotic distributions is based on the following result from Appendix C, Proposition C.15(3). Suppose β is an $(n \times 1)$ vector of parameters and $\hat{\beta}$ is an estimator such that

$$\sqrt{T}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\hat{\beta}}),$$

where T , as usual, denotes the sample size (time series length) used for estimation. Let $g(\beta)$ be a continuously differentiable function with values in the m -dimensional Euclidean space and suppose that $\partial g_i / \partial \beta' = (\partial g_i / \partial \beta_j)$ is nonzero at the true vector β , for $i = 1, \dots, m$. Then,

$$\sqrt{T} \left[g(\hat{\beta}) - g(\beta) \right] \xrightarrow{d} \mathcal{N} \left(0, \frac{\partial g}{\partial \beta'} \Sigma_{\hat{\beta}} \frac{\partial g'}{\partial \beta} \right).$$

In writing down the asymptotic distributions formally, we use the notation

$$\alpha := \text{vec}(A_1, \dots, A_p) \quad (K^2 p \times 1),$$

$$\mathbf{A} := \begin{bmatrix} A_1 & A_2 & \dots & A_{p-1} & A_p \\ I_K & 0 & \dots & 0 & 0 \\ 0 & I_K & & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & I_K & 0 \end{bmatrix} \quad (Kp \times Kp),$$

$$\sigma := \text{vech}(\Sigma_u) \quad (\tfrac{1}{2}K(K+1) \times 1)$$

and the corresponding estimators are furnished with a hat. As before, vec denotes the column stacking operator and vech is the corresponding operator that stacks the elements on and below the main diagonal only. We also use the commutation matrix $\mathbf{K}_{mn}$, defined such that, for any $(m \times n)$ matrix G , $\mathbf{K}_{mn} \text{vec}(G) = \text{vec}(G')$, the $(m^2 \times \frac{1}{2}m(m+1))$ duplication matrix $\mathbf{D}_m$, defined such that $\mathbf{D}_m \text{vech}(F) = \text{vec}(F)$, for any symmetric $(m \times m)$ matrix F , and the $(\frac{1}{2}m(m+1) \times m^2)$ elimination matrix $\mathbf{L}_m$, defined such that, for any $(m \times m)$ matrix F , $\text{vech}(F) = \mathbf{L}_m \text{vec}(F)$ (see Appendix A.12.2). Furthermore, $J := [I_K : 0 : \dots : 0]$ is a $(K \times Kp)$ matrix. With this notation, the following proposition from Lütkepohl (1990) can be stated.

Proposition 3.6 (*Asymptotic Distributions of Impulse Responses*)

Suppose

$$\sqrt{T} \begin{bmatrix} \hat{\alpha} - \alpha \\ \hat{\sigma} - \sigma \end{bmatrix} \xrightarrow{d} \mathcal{N} \left(0, \begin{bmatrix} \Sigma_{\hat{\alpha}} & 0 \\ 0 & \Sigma_{\hat{\sigma}} \end{bmatrix} \right). \quad (3.7.4)$$

Then

$$\sqrt{T} \text{vec}(\widehat{\Phi}_i - \Phi_i) \xrightarrow{d} \mathcal{N}(0, G_i \Sigma_{\widehat{\alpha}} G_i'), \quad i = 1, 2, \dots, \quad (3.7.5)$$

where

$$G_i := \frac{\partial \text{vec}(\Phi_i)}{\partial \alpha'} = \sum_{m=0}^{i-1} J(\mathbf{A}')^{i-1-m} \otimes \Phi_m.$$

$$\sqrt{T} \text{vec}(\widehat{\Psi}_n - \Psi_n) \xrightarrow{d} \mathcal{N}(0, F_n \Sigma_{\widehat{\alpha}} F_n'), \quad n = 1, 2, \dots, \quad (3.7.6)$$

where $F_n := G_1 + \dots + G_n$.

If $(I_K - A_1 - \dots - A_p)$ is nonsingular,

$$\sqrt{T} \text{vec}(\widehat{\Psi}_\infty - \Psi_\infty) \xrightarrow{d} \mathcal{N}(0, F_\infty \Sigma_{\widehat{\alpha}} F_\infty'), \quad (3.7.7)$$

where $F_\infty := \underbrace{(\Psi'_\infty, \dots, \Psi'_\infty)}_{p \text{ times}} \otimes \Psi_\infty$.

$$\sqrt{T} \text{vec}(\widehat{\Theta}_i - \Theta_i) \xrightarrow{d} \mathcal{N}(0, C_i \Sigma_{\widehat{\alpha}} C_i' + \bar{C}_i \Sigma_{\widehat{\alpha}} \bar{C}_i'), \quad i = 0, 1, 2, \dots, \quad (3.7.8)$$

where

$$C_0 := 0, \quad C_i := (P' \otimes I_K) G_i, \quad i = 1, 2, \dots, \quad \bar{C}_i := (I_K \otimes \Phi_i) H, \quad i = 0, 1, \dots,$$

and

$$\begin{aligned} H &:= \frac{\partial \text{vec}(P)}{\partial \sigma'} = \mathbf{L}'_K \{ \mathbf{L}_K [(I_K \otimes P) \mathbf{K}_{KK} + (P \otimes I_K)] \mathbf{L}'_K \}^{-1} \\ &= \mathbf{L}'_K \{ \mathbf{L}_K (I_{K^2} + \mathbf{K}_{KK}) (P \otimes I_K) \mathbf{L}'_K \}^{-1}. \end{aligned}$$

$$\sqrt{T} \text{vec}(\widehat{\Xi}_n - \Xi_n) \xrightarrow{d} \mathcal{N}(0, B_n \Sigma_{\widehat{\alpha}} B_n' + \bar{B}_n \Sigma_{\widehat{\sigma}} \bar{B}_n'), \quad (3.7.9)$$

where $B_n := (P' \otimes I_K) F_n$ and $\bar{B}_n := (I_K \otimes \Psi_n) H$.

If $(I_K - A_1 - \dots - A_p)$ is nonsingular,

$$\sqrt{T} \text{vec}(\widehat{\Xi}_\infty - \Xi_\infty) \xrightarrow{d} \mathcal{N}(0, B_\infty \Sigma_{\widehat{\alpha}} B_\infty' + \bar{B}_\infty \Sigma_{\widehat{\sigma}} \bar{B}_\infty'), \quad (3.7.10)$$

where $B_\infty := (P' \otimes I_K) F_\infty$ and $\bar{B}_\infty := (I_K \otimes \Psi_\infty) H$.

Finally,

$$\begin{aligned} \sqrt{T}(\widehat{\omega}_{jk,h} - \omega_{jk,h}) &\xrightarrow{d} \mathcal{N}(0, d_{jk,h} \Sigma_{\widehat{\alpha}} d'_{jk,h} + \bar{d}_{jk,h} \Sigma_{\widehat{\sigma}} \bar{d}'_{jk,h}) \\ j, k &= 1, \dots, K, \quad h = 1, 2, \dots, \end{aligned} \quad (3.7.11)$$

where

$$\begin{aligned} d_{jk,h} &:= \frac{2}{\text{MSE}_j(h)^2} \sum_{i=0}^{h-1} \left[\text{MSE}_j(h) (e'_j \Phi_i P e_k) (e'_k P' \otimes e'_j) G_i \right. \\ &\quad \left. - (e'_j \Phi_i P e_k)^2 \sum_{m=0}^{h-1} (e'_j \Phi_m \Sigma_u \otimes e'_j) G_m \right] \end{aligned}$$

with $G_0 := 0$ and

$$\begin{aligned} \bar{d}_{jk,h} := & \sum_{i=0}^{h-1} \left[2 \text{MSE}_j(h) (e'_j \Phi_i P e_k) (e'_k \otimes e'_j \Phi_i) H \right. \\ & \left. - (e'_j \Phi_i P e_k)^2 \sum_{m=0}^{h-1} (e'_j \Phi_m \otimes e'_j \Phi_m) \mathbf{D}_K \right] / \text{MSE}_j(h)^2. \end{aligned}$$

■

In the next subsection, the proof of the proposition is indicated. Some remarks are worthwhile now.

Remark 1 In the proposition, some matrices of partial derivatives may be zero. For instance, if a VAR(1) model is fitted although the true order is zero, that is, y_t is white noise, then

$$G_2 = J\mathbf{A}' \otimes I_K + JI_K \otimes \Phi_1 = 0$$

because $\mathbf{A} = A_1 = 0$ and $\Phi_1 = A_1 = 0$. Hence, a degenerate asymptotic distribution with zero covariance matrix is obtained for $\sqrt{T} \text{vec}(\hat{\Phi}_2 - \Phi_2)$. As explained in Appendix B, we call such a distribution also multivariate normal. Otherwise it would be necessary to distinguish between cases with zero and nonzero partial derivatives or we have to assume that all partial derivatives are such that the covariance matrices have no zeros on the diagonal. Note that estimators of the covariance matrices obtained by replacing unknown quantities by their usual estimators may be problematic when the asymptotic distribution is degenerate. In that case, the usual t -ratios and confidence intervals may not be appropriate.

To illustrate the potential problems resulting from a degenerate asymptotic distribution, we follow Benkwitz, Lütkepohl & Neumann (2000) and consider a univariate AR(1) process $y_t = \alpha y_{t-1} + u_t$. In this case, $\Phi_i = \alpha^i$. Suppose that $\hat{\alpha}$ is an estimator of α satisfying $\sqrt{T}(\hat{\alpha} - \alpha) \xrightarrow{d} \mathcal{N}(0, \sigma_{\hat{\alpha}}^2)$ with $\sigma_{\hat{\alpha}}^2 \neq 0$. For instance, $\hat{\alpha}$ may be the LS estimator of α . Then

$$\sqrt{T}(\hat{\alpha}^2 - \alpha^2) \xrightarrow{d} \mathcal{N}(0, \sigma_{\hat{\alpha}^2}^2)$$

with $\sigma_{\hat{\alpha}^2}^2 = 4\alpha^2\sigma_{\hat{\alpha}}^2$. This quantity is, of course, zero if $\alpha = 0$. In the latter case, $\sqrt{T}\hat{\alpha}/\sigma_{\hat{\alpha}}$ has an asymptotic standard normal distribution and, hence, $T\hat{\alpha}^2/\sigma_{\hat{\alpha}}^2$ has an asymptotic $\chi^2(1)$ -distribution. Thus, it is clear that in this case $\sqrt{T}\hat{\alpha}^2$ is asymptotically degenerate.

Because the estimated $\sigma_{\hat{\alpha}^2}^2$ obtained by replacing α and $\sigma_{\hat{\alpha}}^2$ by their usual LS estimators is nonzero almost surely, it is tempting to use the quantity $\sqrt{T}(\hat{\alpha}^2 - \alpha^2)/2\hat{\alpha}\hat{\sigma}_{\hat{\alpha}}$ for constructing a confidence interval, say, for Φ_2 . However, for $\alpha = 0$, the t -ratio becomes $\sqrt{T}\hat{\alpha}/2\hat{\sigma}_{\hat{\alpha}}$ which converges to $\mathcal{N}(0, 1/4)$ asymptotically, because $\sqrt{T}\hat{\alpha}/\hat{\sigma}_{\hat{\alpha}} \xrightarrow{d} \mathcal{N}(0, 1)$. A confidence interval constructed on

the basis of the asymptotic standard normal distribution would therefore be a conservative one. In other words, asymptotic inference which ignores the possible singularity in the asymptotic distribution of the impulse responses may be misleading (see Benkwitz et al. (2000) for further discussion). ■

Remark 2 In the proposition, it is not explicitly assumed that y_t is stable. While the stability condition is partly introduced in (3.7.7) and (3.7.10) by requiring that $(I_K - A_1 - \cdots - A_p)$ be nonsingular so that

$$\det(I_K - A_1 z - \cdots - A_p z^p) \neq 0 \quad \text{for } z = 1,$$

it is not needed for the other results to hold. The crucial condition is the asymptotic distribution of the process parameters in (3.7.4). Although we have used the stationarity and stability assumptions in Sections 3.2–3.4 in order to derive the asymptotic distribution of the process parameters, we will see in later chapters that asymptotic normality is also obtained for certain nonstationary, unstable processes. Therefore, at least parts of Proposition 3.6 will be useful in a nonstationary environment. ■

Remark 3 The block-diagonal structure of the covariance matrix of the asymptotic distribution in (3.7.4) is in no way essential for the asymptotic normality of the impulse responses. In fact, the asymptotic distributions in (3.7.5)–(3.7.7) remain unchanged if the asymptotic covariance matrix of the parameter estimators is not block-diagonal. On the other hand, without the block-diagonal structure, the simple additive structure of the asymptotic covariance matrices in (3.7.8)–(3.7.11) is lost. Although these asymptotic distributions are easily generalizable to the case of a general asymptotic covariance matrix of the VAR coefficients in (3.7.4), we have not stated the more general result here because it is not needed in subsequent chapters of this text. ■

Remark 4 Under the conditions of Proposition 3.4, the covariance matrix of the asymptotic distribution of the parameters has precisely the block-diagonal structure assumed in (3.7.4) with

$$\Sigma_{\hat{\alpha}} = \Gamma_Y(0)^{-1} \otimes \Sigma_u$$

and

$$\Sigma_{\hat{\sigma}} = 2\mathbf{D}_K^+(\Sigma_u \otimes \Sigma_u)\mathbf{D}_K^{+'},$$

where $\mathbf{D}_K^+ = (\mathbf{D}_K' \mathbf{D}_K)^{-1} \mathbf{D}_K'$ is the Moore-Penrose inverse of the duplication matrix $\mathbf{D}_K$. Using these expressions in the proposition, some simplifications of the covariance matrices can be obtained. For instance, the covariance matrix in (3.7.5) becomes

$$G_i \Sigma_{\hat{\alpha}} G_i'$$

$$\begin{aligned}
&= \left[\sum_{m=0}^{i-1} J(\mathbf{A}')^{i-1-m} \otimes \Phi_m \right] (\Gamma_Y(0)^{-1} \otimes \Sigma_u) \left[\sum_{n=0}^{i-1} J(\mathbf{A}')^{i-1-n} \otimes \Phi_n \right]' \\
&= \sum_{m=0}^{i-1} \sum_{n=0}^{i-1} [J(\mathbf{A}')^{i-1-m} \Gamma_Y(0)^{-1} \mathbf{A}^{i-1-n} J'] \otimes (\Phi_m \Sigma_u \Phi_n')
\end{aligned}$$

which is computationally convenient because all matrices involved are of a relatively small size. The advantage of the general formulation is that it can be used with other $\Sigma_{\hat{\alpha}}$ matrices as well. We will see examples in subsequent chapters. ■

Remark 5 In practice, to use the asymptotic distributions for inference, the unknown quantities in the covariance matrices in Proposition 3.6 may be replaced by their usual estimators given in Sections 3.2–3.4 for the case of a stationary, stable process y_t (see, however, Remark 1). ■

Remark 6 Summing the forecast error variance components over k ,

$$\sum_{k=1}^K \omega_{jk,h} = \sum_{k=1}^K \hat{\omega}_{jk,h} = 1$$

for each j and h . These restrictions are not taken into account in the derivation of the asymptotic distributions in (3.7.11). It is easily checked, however, that for dimension $K = 1$ the standard errors obtained from Proposition 3.6 are zero as they should be, because all forecast error variance components are 1 in that case. A problem in this context is that the asymptotic distribution of $\hat{\omega}_{jk,h}$ cannot be used in the usual way for tests of significance and setting up confidence intervals if $\omega_{jk,h} = 0$. In that case, from the definitions of $d_{jk,h}$ and $\bar{d}_{jk,h}$, the variance of the asymptotic distribution is easily seen to be zero and, hence, estimating this quantity by replacing unknown parameters by their usual estimators may lead to t -ratios that are not standard normal asymptotically and, hence, cannot be used in the usual way for inference (see Remark 1). This state of affairs is unfortunate from a practical point of view because testing the significance of forecast error variance components is of particular interest in practice. Note, however, that

$$\omega_{jk,h} = 0 \iff \theta_{jk,i} = 0 \quad \text{for } i = 0, \dots, h.$$

A test of the latter hypothesis may be possible. ■

Remark 7 Joint confidence regions and test statistics for testing hypotheses that involve several of the response coefficients can be obtained from Proposition 3.6 in the usual way. However, it has to be taken into account that, for instance, the elements of $\hat{\Phi}_i$ and $\hat{\Phi}_j$ will not be independent asymptotically. If elements from two or more MA matrices are involved the joint distribution of all the matrices must be determined. This distribution can be derived easily

from the results given in the proposition. For instance, the covariance matrix of the joint asymptotic distribution of $\text{vec}(\hat{\Phi}_i, \hat{\Phi}_j)$ is

$$\frac{\partial \text{vec}(\Phi_i, \Phi_j)}{\partial \alpha'} \Sigma_{\hat{\alpha}} \frac{\partial \text{vec}(\Phi_i, \Phi_j)'}{\partial \alpha},$$

where

$$\frac{\partial \text{vec}(\Phi_i, \Phi_j)}{\partial \alpha'} = \begin{bmatrix} \frac{\partial \text{vec}(\Phi_i)}{\partial \alpha'} \\ \frac{\partial \text{vec}(\Phi_j)}{\partial \alpha'} \end{bmatrix}$$

etc. We have chosen to state the proposition for individual MA coefficient matrices because thereby all required matrices have relatively small dimensions and, hence, are easy to compute. ■

Remark 8 Denoting the jk -th elements of Φ_i and Θ_i by $\phi_{jk,i}$ and $\theta_{jk,i}$, respectively, hypotheses of obvious interest, for $j \neq k$, are

$$H_0 : \phi_{jk,i} = 0 \quad \text{for } i = 1, 2, \dots \quad (3.7.12)$$

and

$$H_0 : \theta_{jk,i} = 0 \quad \text{for } i = 0, 1, 2, \dots \quad (3.7.13)$$

because they can be interpreted as hypotheses on noncausality from variable k to variable j , that is, an impulse in variable k does not induce any response of variable j . From Chapter 2, Propositions 2.4 and 2.5, we know that (3.7.12) is equivalent to

$$H_0 : \phi_{jk,i} = 0 \quad \text{for } i = 1, 2, \dots, p(K-1) \quad (3.7.14)$$

and (3.7.13) is equivalent to

$$H_0 : \theta_{jk,i} = 0 \quad \text{for } i = 0, 1, \dots, p(K-1). \quad (3.7.15)$$

Using Bonferroni's inequality (see Chapter 2, Section 2.2.3), a test of (3.7.14) with significance level at most $100\gamma\%$ is obtained by rejecting H_0 if

$$|\sqrt{T} \hat{\phi}_{jk,i} / \hat{\sigma}_{\phi_{jk}}(i)| > z_{(\gamma/2p(K-1))} \quad (3.7.16)$$

for at least one $i \in \{1, 2, \dots, p(K-1)\}$. Here $z_{(\gamma)}$ is the upper 100γ percentage point of the standard normal distribution and $\hat{\sigma}_{\phi_{jk}}(i)$ is an estimate of the asymptotic standard deviation $\sigma_{\phi_{jk}}(i)$ of $\sqrt{T} \hat{\phi}_{jk,i}$ obtained via Proposition 3.6. In order to obtain an asymptotic standard normal distribution of the t -ratio $\sqrt{T} \hat{\phi}_{jk,i} / \hat{\sigma}_{\phi_{jk}}(i)$, the variance $\sigma_{\phi_{jk}}^2(i)$ must be nonzero, however.

A test of (3.7.15) with significance level at most γ is obtained by rejecting H_0 if

$$|\sqrt{T}\widehat{\theta}_{jk,i}/\widehat{\sigma}_{\theta_{jk}}(i)| \begin{cases} > z_{(\gamma/2(pK-p+1))} & \text{for at least one} \\ & i \in \{0, 1, 2, \dots, p(K-1)\} \text{ if } j > k \\ > z_{(\gamma/2(pK-p))} & \text{for at least one} \\ & i \in \{1, 2, \dots, p(K-1)\} \text{ if } j < k. \end{cases} \quad (3.7.17)$$

Here $\widehat{\sigma}_{\theta_{jk}}(i)$ is a consistent estimator of the standard deviation of the asymptotic distribution of $\sqrt{T}\widehat{\theta}_{jk,i}$ obtained from Proposition 3.6 and that standard deviation is assumed to be nonzero.

A test based on Bonferroni's principle may have quite low power because the actual significance level may be much smaller than the given upper bound. Therefore a test based on some χ^2 - or F -statistic would be preferable. Unfortunately, such tests are not easily available for the present situation. The problem is similar to the one discussed in Section 3.6.4 in the context of testing for multi-step causality. For more discussion of this point see also Lütkepohl (1990) and for a different approach of representing the uncertainty in estimated impulse responses see Sims & Zha (1999). ■

3.7.2 Proof of Proposition 3.6

The proof of Proposition 3.6 is a straightforward application of the matrix differentiation rules given in Appendix A.13. It is sketched here for completeness and because it is spread out over a number of publications in the literature. Readers mainly interested in applying the proposition may skip this section without loss of continuity.

To prove (3.7.5), note that $\Phi_i = J\mathbf{A}^i J'$ (see Chapter 2, Section 2.1.2) and apply Rule (8) of Appendix A.13. The expression for F_n in (3.7.6) follows because

$$\frac{\partial \text{vec}(\Psi_n)}{\partial \alpha'} = \sum_{i=1}^n \frac{\partial \text{vec}(\Phi_i)}{\partial \alpha'}$$

and

$$\begin{aligned} F_\infty &= \frac{\partial \text{vec}(\Psi_\infty)}{\partial \alpha'} = \frac{\partial \text{vec}(\Psi_\infty)}{\partial \text{vec}(\Psi_\infty^{-1})'} \frac{\partial \text{vec}(\Psi_\infty^{-1})}{\partial \alpha'} \\ &= -(\Psi_\infty' \otimes \Psi_\infty) \frac{\partial \text{vec}(I_K - A_1 - \dots - A_p)}{\partial \alpha'}. \end{aligned}$$

Furthermore,

$$C_i = \frac{\partial \text{vec}(\Theta_i)}{\partial \alpha'} = \frac{\partial \text{vec}(\Phi_i P)}{\partial \alpha'} = (P' \otimes I_K) \frac{\partial \text{vec}(\Phi_i)}{\partial \alpha'}$$

and

$$\bar{C}_i = \frac{\partial \text{vec}(\Theta_i)}{\partial \sigma'} = (I_K \otimes \Phi_i) \frac{\partial \text{vec}(P)}{\partial \sigma'},$$

where

$$\frac{\partial \text{vec}(P)}{\partial \sigma'} = \mathbf{L}'_K \frac{\partial \text{vech}(P)}{\partial \sigma'} = H,$$

follows from Appendix A.13, Rule (10). The matrices $B_n, \bar{B}_n, B_\infty$, and $\bar{B}_\infty$ are obtained in a similar manner, using the relations $\Xi_n = \Psi_n P$ and $\Xi_\infty = \Psi_\infty P$.

Finally, in (3.7.11),

$$\begin{aligned} d_{jk,h} &= \frac{\partial \omega_{jk,h}}{\partial \alpha'} \\ &= \left[2 \sum_{i=0}^{h-1} (e'_j \Phi_i P e_k) (e'_k P' \otimes e'_j) \frac{\partial \text{vec}(\Phi_i)}{\partial \alpha'} \text{MSE}_j(h) \right. \\ &\quad \left. - \sum_{i=0}^{h-1} (e'_j \Phi_i P e_k)^2 \frac{\partial \text{MSE}_j(h)}{\partial \alpha'} \right] \bigg/ \text{MSE}_j(h)^2, \end{aligned}$$

$$\begin{aligned} \frac{\partial \text{MSE}_j(h)}{\partial \alpha'} &= \sum_{m=0}^{h-1} \left[(e'_j \Phi_m \Sigma_u \otimes e'_j) \frac{\partial \text{vec}(\Phi_m)}{\partial \alpha'} \right. \\ &\quad \left. + (e'_j \otimes e'_j \Phi_m \Sigma_u) \frac{\partial \text{vec}(\Phi'_m)}{\partial \alpha'} \right] \\ &= \sum_{m=0}^{h-1} \left[(e'_j \Phi_m \Sigma_u \otimes e'_j) + (e'_j \otimes e'_j \Phi_m \Sigma_u) \mathbf{K}_{KK} \right] \frac{\partial \text{vec}(\Phi_m)}{\partial \alpha'} \\ &= \sum_{m=0}^{h-1} \left[(e'_j \Phi_m \Sigma_u \otimes e'_j) + \mathbf{K}_{11} (e'_j \Phi_m \Sigma_u \otimes e'_j) \right] G_m \\ &= 2 \sum_{m=0}^{h-1} (e'_j \Phi_m \Sigma_u \otimes e'_j) G_m, \\ &\quad \text{(see Appendix A.12.2, Rule (23))} \end{aligned}$$

$$\begin{aligned} \bar{d}_{jk,h} &= \frac{\partial \omega_{jk,h}}{\partial \sigma'} \\ &= \sum_{i=0}^{h-1} \left[2(e'_j \Phi_i P e_k) (e'_k \otimes e'_j \Phi_i) \frac{\partial \text{vec}(P)}{\partial \sigma'} \text{MSE}_j(h) \right. \\ &\quad \left. - (e'_j \Phi_i P e_k)^2 \frac{\partial \text{MSE}_j(h)}{\partial \sigma'} \right] \bigg/ \text{MSE}_j(h)^2, \end{aligned}$$

and

$$\begin{aligned}
\frac{\partial \text{MSE}_j(h)}{\partial \boldsymbol{\sigma}'} &= \sum_{m=0}^{h-1} (e'_j \Phi_m \otimes e'_j \Phi_m) \frac{\partial \text{vec}(\Sigma_u)}{\partial \boldsymbol{\sigma}'} \\
&= \sum_{m=0}^{h-1} (e'_j \Phi_m \otimes e'_j \Phi_m) \mathbf{D}_K \frac{\partial \text{vech}(\Sigma_u)}{\partial \boldsymbol{\sigma}'} .
\end{aligned}$$

Thereby Proposition 3.6 is proven. In the next section an example is discussed.

3.7.3 An Example

To illustrate the results of Section 3.7.1, we use again the investment/income/consumption example from Section 3.2.3. Because

$$\hat{\Phi}_1 = \hat{A}_1 = \begin{bmatrix} -.320 & .146 & .961 \\ .044 & -.153 & .289 \\ -.002 & .225 & -.264 \end{bmatrix},$$

the elements of $\hat{\Phi}_1$ must have the same standard errors as the elements of $\hat{A}_1$. Checking the covariance matrix in (3.7.5), it is seen that the asymptotic covariance matrix of $\hat{\Phi}_1$ is indeed the upper left-hand ($K^2 \times K^2$) block of $\Sigma_{\hat{\alpha}}$ because

$$G_1 = J \otimes I_K = [I_{K^2} : 0 : \cdots : 0].$$

Thus, the square roots of the diagonal elements of

$$G_1 \hat{\Sigma}_{\hat{\alpha}} G_1' / T = \frac{1}{T} [I_9 : 0 : \cdots : 0] (\hat{\Gamma}_Y(0)^{-1} \otimes \hat{\Sigma}_u) \begin{bmatrix} I_9 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

are estimates of the asymptotic standard errors of $\hat{\Phi}_1$. Note that here and in the following we use the LS estimators from the standard form of the VAR process (see Section 3.2) and not the mean-adjusted form. Accordingly, the estimate $\hat{\Gamma}_Y(0)^{-1}$ is obtained from $(ZZ'/T)^{-1}$ by deleting the first row and column.

From (2.1.22) we get

$$\hat{\Phi}_2 = \hat{\Phi}_1 \hat{A}_1 + \hat{A}_2 = \begin{bmatrix} -.054 & .262 & .416 \\ .029 & .114 & -.088 \\ .045 & .261 & .110 \end{bmatrix}.$$

To estimate the corresponding standard errors, we note that

$$G_2 = J \mathbf{A}' \otimes I_K + J \otimes \Phi_1.$$

Replacing the unknown quantities by the usual estimates gives

$$\begin{aligned} \frac{1}{T} \hat{G}_2 \hat{\Sigma}_{\hat{\alpha}} \hat{G}_2' &= \frac{1}{T} [J \hat{\mathbf{A}}' \hat{\Gamma}_Y(0)^{-1} \hat{\mathbf{A}} J' \otimes \hat{\Sigma}_u + J \hat{\mathbf{A}}' \hat{\Gamma}_Y(0)^{-1} J' \otimes \hat{\Sigma}_u \hat{\Phi}_1' \\ &\quad + J \hat{\Gamma}_Y(0)^{-1} \hat{\mathbf{A}} J' \otimes \hat{\Phi}_1 \hat{\Sigma}_u + J \hat{\Gamma}_Y(0)^{-1} J' \otimes \hat{\Phi}_1 \hat{\Sigma}_u \hat{\Phi}_1']. \end{aligned}$$

The square roots of the diagonal elements of this matrix are estimates of the standard deviations of the elements of $\hat{\Phi}_2$ and so on. Some $\hat{\Phi}_i$ matrices together with estimated standard errors are given in Table 3.3. In Figures 3.4 and 3.5, some impulse responses are depicted graphically along with two-standard error bounds.

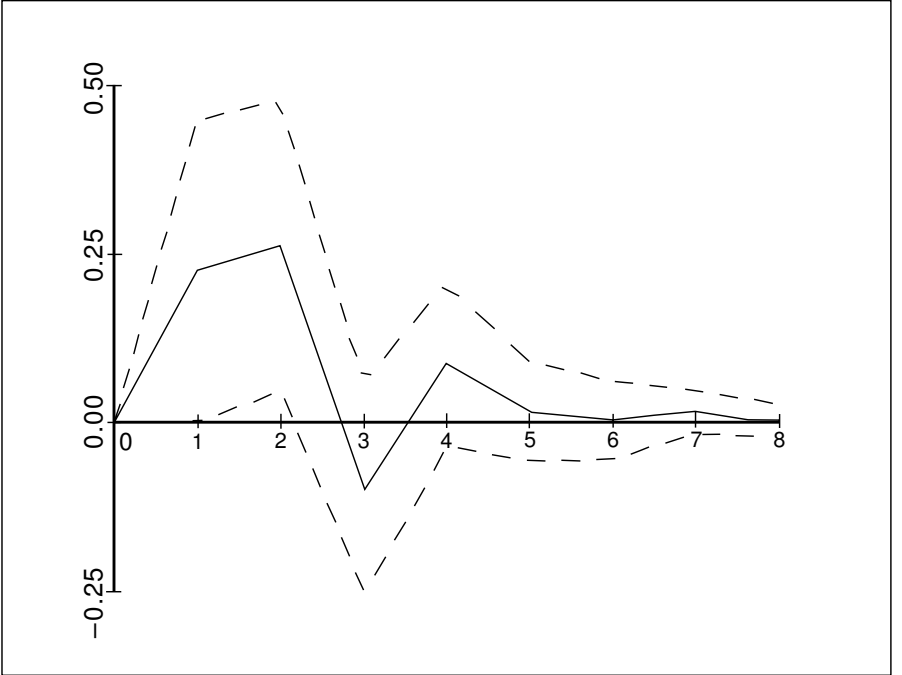


Fig. 3.4. Estimated responses of consumption to a forecast error impulse in income with estimated asymptotic two-standard error bounds.

In Figure 3.4, consumption is seen to increase in response to a unit shock in income. However, under a two-standard error criterion (approximate 95% confidence bounds) only the second response coefficient is significantly different from zero. Of course, the large standard errors of the impulse response coefficients reflect the substantial estimation uncertainty in the VAR coefficient matrices A_1 and A_2 .

To check the overall significance of the response coefficients of consumption to an income impulse, we may use the procedure described in Remark 8 of

Table 3.3. Estimates of impulse responses for the investment/income/consumption system with estimated asymptotic standard errors in parentheses

i	$\hat{\Phi}_i$	$\hat{\Psi}_i$
1	$\begin{bmatrix} -0.320 & 0.146 & 0.961 \\ (0.125) & (0.562) & (0.657) \\ 0.044 & -0.153 & 0.289 \\ (0.032) & (0.143) & (0.167) \\ -0.002 & 0.225 & -0.264 \\ (0.025) & (0.115) & (0.134) \end{bmatrix}$	$\begin{bmatrix} 0.680 & 0.146 & 0.961 \\ (0.125) & (0.562) & (0.657) \\ 0.044 & 0.847 & 0.289 \\ (0.032) & (0.143) & (0.167) \\ -0.002 & 0.225 & 0.736 \\ (0.025) & (0.115) & (0.134) \end{bmatrix}$
2	$\begin{bmatrix} -0.054 & 0.262 & 0.416 \\ (0.129) & (0.546) & (0.663) \\ 0.029 & 0.114 & -0.088 \\ (0.032) & (0.135) & (0.162) \\ 0.045 & 0.261 & 0.110 \\ (0.026) & (0.108) & (0.131) \end{bmatrix}$	$\begin{bmatrix} 0.626 & 0.408 & 1.377 \\ (0.148) & (0.651) & (0.755) \\ 0.073 & 0.961 & 0.200 \\ (0.043) & (0.192) & (0.222) \\ 0.043 & 0.486 & 0.846 \\ (0.033) & (0.144) & (0.167) \end{bmatrix}$
3	$\begin{bmatrix} 0.119 & 0.353 & -0.408 \\ (0.084) & (0.384) & (0.476) \\ -0.009 & 0.071 & 0.120 \\ (0.016) & (0.078) & (0.094) \\ -0.001 & -0.098 & 0.091 \\ (0.017) & (0.078) & (0.102) \end{bmatrix}$	$\begin{bmatrix} 0.745 & 0.761 & 0.969 \\ (0.099) & (0.483) & (0.550) \\ 0.064 & 1.033 & 0.320 \\ (0.037) & (0.176) & (0.203) \\ 0.042 & 0.388 & 0.937 \\ (0.033) & (0.156) & (0.183) \end{bmatrix}$
∞	0	$\begin{bmatrix} 0.756 & 0.836 & 1.295 \\ (0.133) & (0.661) & (0.798) \\ 0.076 & 1.076 & 0.344 \\ (0.048) & (0.236) & (0.285) \\ 0.053 & 0.505 & 0.964 \\ (0.043) & (0.213) & (0.257) \end{bmatrix}$

Section 3.7.1. That is, we have to check the significance of the first $p(K - 1) = 4$ response coefficients. Because one of them is individually significant at an asymptotic 5% level we may reject the null hypothesis of no response of consumption to income impulses at a significance level not greater than $4 \times 5\% = 20\%$. Of course, this is not a significance level we are used to in applied work. However, it becomes clear from Table 3.3 that the second response coefficient $\hat{\phi}_{32,2}$ is still significant if the individual significance levels are reduced to 2.5%. Note that the upper 1.25 percentage point of the standard normal distribution is $c_{0.0125} = 2.24$. Thus, we may reject the no-response

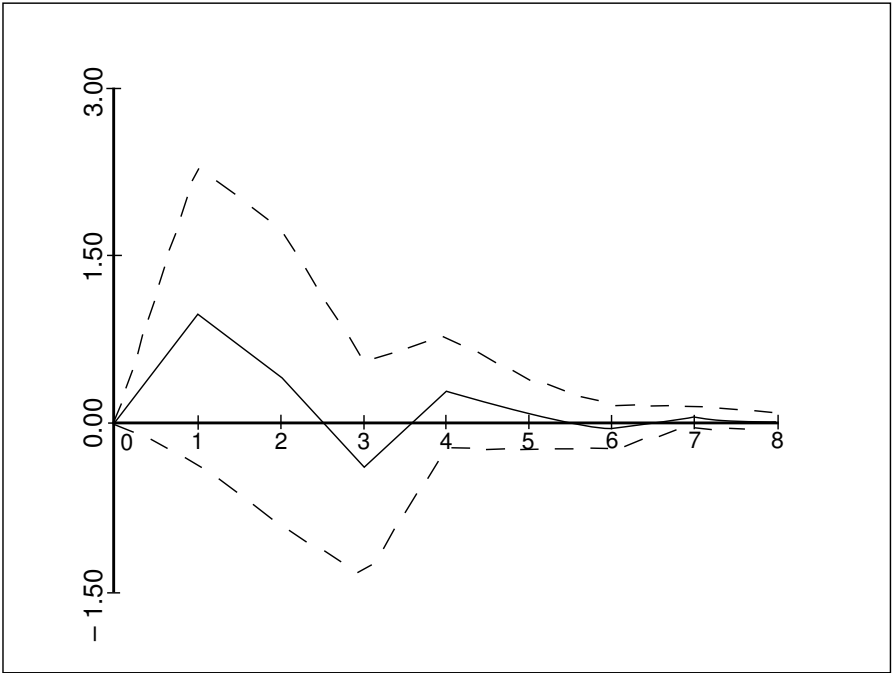


Fig. 3.5. Estimated responses of investment to a forecast error impulse in consumption with estimated asymptotic two-standard error bounds.

hypothesis at an overall $4 \times 2.5\% = 10\%$ level which is clearly a more common size for a test in applied work. Still, in this exercise, the data do not reveal strong evidence for the intuitively appealing hypothesis that consumption responds to income impulses. In later chapters, we will see how the coefficients can potentially be estimated with more precision.

In Figure 3.5, the responses of investment to consumption impulses are depicted. None of them is significant under a two-standard error criterion. This result is in line with the Granger-causality analysis in Section 3.6. In that section, we did not find evidence for Granger-causality from income/consumption to investment. Assuming that the test result describes the actual situation, the $\phi_{13,i}$ must be zero for $i = 1, 2, \dots$ (see also Chapter 2, Section 2.3.1).

The covariance matrix of

$$\hat{\Psi}_1 = I_3 + \hat{\Phi}_1 = \begin{bmatrix} .680 & .146 & .961 \\ .044 & .847 & .289 \\ -.002 & .225 & .736 \end{bmatrix}$$

is, of course, the same as that of $\hat{\Phi}_1$ and an estimate of the covariance matrix of the elements of

$$\hat{\Psi}_2 = I_3 + \hat{\Phi}_1 + \hat{\Phi}_2 = \begin{bmatrix} .626 & .408 & 1.377 \\ .073 & .961 & .200 \\ .043 & .486 & .846 \end{bmatrix}$$

is obtained as $(G_1 + \hat{G}_2) \hat{\Sigma}_{\hat{\alpha}} (G_1 + \hat{G}_2)' / T$. Some accumulated impulse responses together with estimated standard errors are also given in Table 3.3 and accumulated responses of consumption to income impulses and of investment to consumption impulses are shown in Figures 3.6 and 3.7, respectively. They reinforce the findings for the individual impulse responses in Figures 3.4 and 3.5.

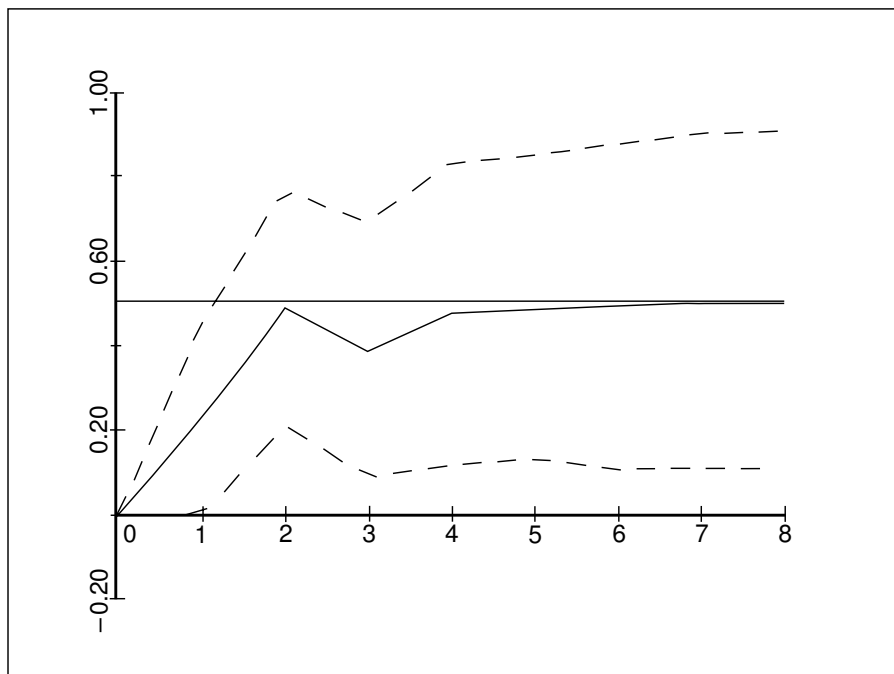


Fig. 3.6. Accumulated and long-run responses of consumption to a forecast error impulse in income with estimated asymptotic two-standard error bounds.

An estimate of the asymptotic covariance matrix of the estimated long-run responses $\hat{\Psi}_{\infty} = (I_3 - \hat{A}_1 - \hat{A}_2)^{-1}$ is

$$\frac{1}{T}([\hat{\Psi}'_{\infty} : \hat{\Psi}'_{\infty}] \otimes \hat{\Psi}_{\infty}) \hat{\Sigma}_{\hat{\alpha}} \left(\begin{bmatrix} \hat{\Psi}_{\infty} \\ \hat{\Psi}_{\infty} \end{bmatrix} \otimes \hat{\Psi}'_{\infty} \right).$$

The matrix $\hat{\Psi}_{\infty}$ together with the resulting standard errors is also given in Table 3.3. For instance, the total long-run effect $\hat{\psi}_{13,\infty}$ of a consumption impulse

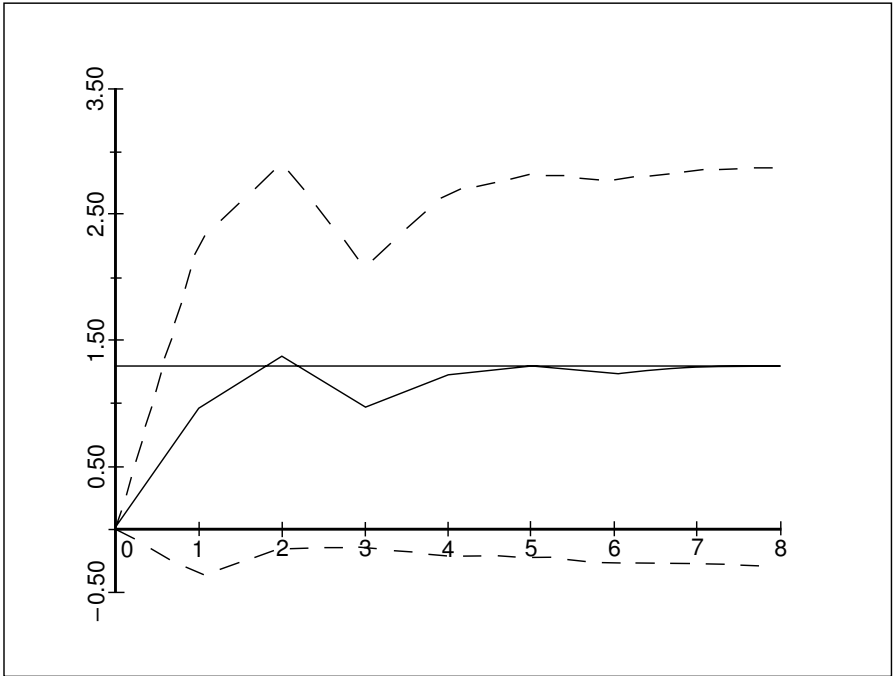


Fig. 3.7. Accumulated and long-run responses of investment to a forecast error impulse in consumption with estimated asymptotic two-standard error bounds.

on investment is 1.295 and its estimated asymptotic standard error is .798. Not surprisingly, $\hat{\psi}_{13,\infty}$ is not significantly different from zero for any common level of significance (e.g., 10%). On the other hand, $\hat{\psi}_{32,\infty}$, the long-run effect on consumption due to an impulse in income, is significant at an asymptotic 5% level.

For the interpretation of the $\hat{\Phi}_i$'s, the critical remarks at the end of Chapter 2 must be kept in mind. As explained there, the $\hat{\Phi}_i$ and $\hat{\Psi}_n$ coefficients may not reflect the actual responses of the variables in the system. As an alternative, one may want to determine the responses to orthogonal residuals. In order to obtain the asymptotic covariance matrices of the $\hat{\Theta}_i$ and $\hat{\Xi}_i$, a decomposition of $\hat{\Sigma}_u$ is needed. For our example,

$$\hat{P} = \begin{bmatrix} 4.61 & 0 & 0 \\ .16 & 1.16 & 0 \\ .27 & .49 & .76 \end{bmatrix} \times 10^{-2}$$

is the lower triangular matrix with positive diagonal elements satisfying $\hat{P}\hat{P}' = \hat{\Sigma}_u$ (Choleski decomposition). The asymptotic covariance matrix of $\text{vec}(\hat{P}) = \text{vec}(\hat{\Theta}_0)$ is a (9×9) matrix which is estimated as

$$\frac{1}{T} \widehat{C}_0 \widehat{\Sigma}_{\widehat{\theta}} \widehat{C}'_0 = \frac{2}{T} \widehat{H} \mathbf{D}_K^+ (\widehat{\Sigma}_u \otimes \widehat{\Sigma}_u) \mathbf{D}_K^{+'} \widehat{H}',$$

where, as usual, $\mathbf{D}_K^+ = (\mathbf{D}'_K \mathbf{D}_K)^{-1} \mathbf{D}'_K$ and

$$\widehat{H} = \mathbf{L}'_3 \left\{ \mathbf{L}_3 \left[(I_3 \otimes \widehat{P}) \mathbf{K}_{33} + (\widehat{P} \otimes I_3) \right] \mathbf{L}'_3 \right\}^{-1}.$$

The resulting estimated asymptotic standard errors of the elements of $\widehat{P}$ are given in Table 3.4. Note that the variances corresponding to elements above the main diagonal of $\widehat{P}$ are all zero because these elements are zero by definition and are not estimated.

The asymptotic covariance matrix of the elements of

$$\widehat{\Theta}_1 = \begin{bmatrix} -1.196 & .644 & .730 \\ .256 & -.035 & .219 \\ -.047 & .131 & -.201 \end{bmatrix} \times 10^{-2}$$

is obtained as the sum of the two matrices

$$\widehat{C}_1 \widehat{\Sigma}_{\widehat{\alpha}} \widehat{C}'_1 / T = \left[(\widehat{P}' \otimes I_3) G_1 \widehat{\Sigma}_{\widehat{\alpha}} G'_1 (\widehat{P} \otimes I_3) \right] / T$$

and

$$\widehat{C}_1 \widehat{\Sigma}_{\widehat{\theta}} \widehat{C}'_1 / T = (I_3 \otimes \widehat{\Phi}_1) \widehat{H} \widehat{\Sigma}_{\widehat{\theta}} \widehat{H}' (I_3 \otimes \widehat{\Phi}'_1) / T.$$

The resulting standard errors for the elements of $\widehat{\Theta}_1$ are given in Table 3.4 along with some more $\widehat{\Theta}_i$ and $\widehat{\Sigma}_n$ matrices.

Some responses and accumulated responses of consumption to income innovations with two-standard error bounds are depicted in Figures 3.8 and 3.9. The responses in Figures 3.4 and 3.8 are obviously a bit different. Note the (significant) immediate reaction of consumption in Figure 3.8. However, from period 1 onwards the response of consumption in both figures is qualitatively similar. The difference of scales is due to the different sizes of the shocks traced through the system. For instance, Figure 3.4 is based on a unit shock in income while Figure 3.8 is based on an innovation of size one standard deviation due to the transformation of the white noise residuals.

Again, a test of overall significance of the impulse responses in Figure 3.8 could be performed using Bonferroni's principle. Now we have to check the significance of the $\widehat{\theta}_{32,i}$'s for $i = 0, 1, \dots, 4 = p(K - 1)$. We reject the null hypothesis of no response if at least one of the coefficients is significantly different from zero. In this case, we can reject at an asymptotic 5% level of significance because $\widehat{\theta}_{32,0}$ is significant at the 1% level (see Table 3.4). Thus, we may choose individual significance levels of 1% for each of the 5 coefficients and obtain 5% as an upper bound for the overall level. Of course, all these interpretations are based on the assumption that the actual asymptotic standard errors of the impulse responses are nonzero (see Section 3.7.1, Remark 1).

Table 3.4. Estimates of responses to orthogonal innovations for the investment/income/consumption system with estimated asymptotic standard errors in parentheses

i	$\hat{\Theta}_i$	$\hat{\Xi}_i$
0	$\begin{bmatrix} 4.61 & 0 & 0 \\ (.38) & & \\ .16 & 1.16 & 0 \\ (.14) & (.10) & \\ .27 & .49 & .76 \\ (.11) & (.10) & (.06) \end{bmatrix} \times 10^{-2}$	$\begin{bmatrix} 4.61 & 0 & 0 \\ (.38) & & \\ .16 & 1.16 & 0 \\ (.14) & (.10) & \\ .27 & .49 & .76 \\ (.11) & (.10) & (.06) \end{bmatrix} \times 10^{-2}$
1	$\begin{bmatrix} -1.20 & .64 & .73 \\ (.57) & (.56) & (.50) \\ .26 & -.04 & .22 \\ (.14) & (.14) & (.13) \\ -.05 & .13 & -.20 \\ (.12) & (.12) & (.10) \end{bmatrix} \times 10^{-2}$	$\begin{bmatrix} 3.46 & .64 & .73 \\ (.63) & (.56) & (.50) \\ .41 & 1.13 & .22 \\ (.20) & (.17) & (.13) \\ .22 & .62 & .56 \\ (.15) & (.14) & (.11) \end{bmatrix} \times 10^{-2}$
2	$\begin{bmatrix} -.10 & .51 & .32 \\ (.58) & (.57) & (.50) \\ .13 & .09 & -.07 \\ (.14) & (.14) & (.12) \\ .28 & .36 & .08 \\ (.12) & (.12) & (.10) \end{bmatrix} \times 10^{-2}$	$\begin{bmatrix} 3.32 & 1.15 & 1.05 \\ (.74) & (.69) & (.58) \\ .54 & 1.22 & .15 \\ (.24) & (.22) & (.17) \\ .50 & .98 & .64 \\ (.20) & (.18) & (.14) \end{bmatrix} \times 10^{-2}$
∞	0	$\begin{bmatrix} 3.97 & 1.61 & .98 \\ (.82) & (.92) & (.61) \\ .61 & 1.42 & .26 \\ (.31) & (.34) & (.22) \\ .58 & 1.06 & .73 \\ (.28) & (.32) & (.20) \end{bmatrix} \times 10^{-2}$

We have also performed forecast error variance decompositions and we have computed the standard errors on the basis of the results given in Proposition 3.6. For some forecast horizons the decompositions are given in Table 3.5. The standard errors may be regarded as rough indications of the sampling uncertainty. It must be kept in mind, however, that they may be quite misleading if the true forecast error variance components are zero, as explained in Remark 6 of Section 3.7.1. Obviously, this qualification limits their value in

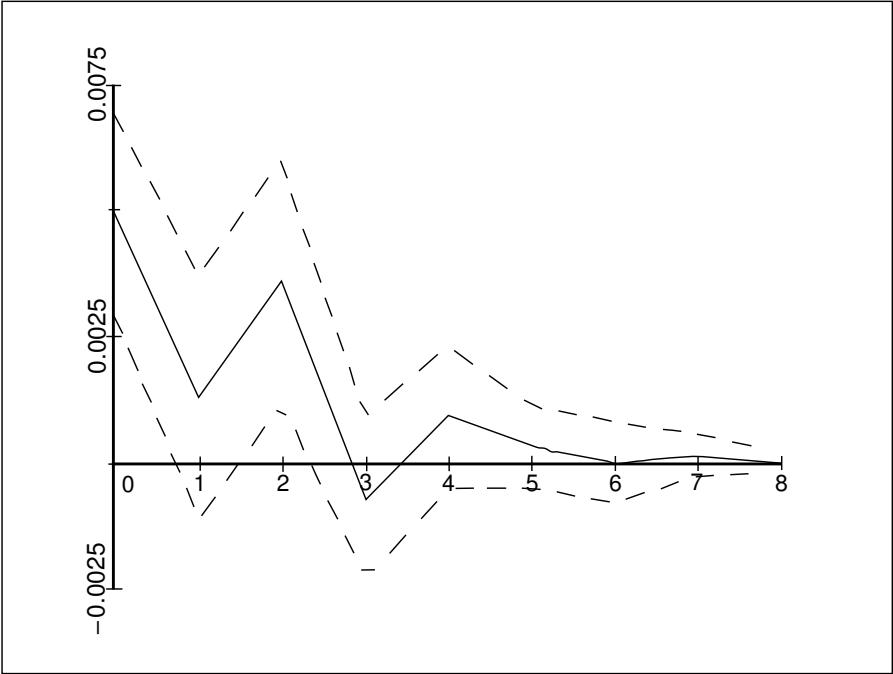


Fig. 3.8. Estimated responses of consumption to an orthogonalized impulse in income with estimated asymptotic two-standard error bounds.

the present example. Students are invited to reproduce the numbers in Table 3.5 and the previous tables of this section.

3.7.4 Investigating the Distributions of the Impulse Responses by Simulation Techniques

In the previous subsections, it was indicated repeatedly that in some cases the small sample validity of the asymptotic results is problematic. In that situation, one possibility is to use Monte Carlo or bootstrapping methods for investigating the sampling properties of the quantities of interest. Although these methods are quite expensive in terms of computer time, they were used in the past for evaluating the properties of impulse response functions (see, e.g., Runkle (1987) and Kilian (1998, 1999)). The general methodology is described in Appendix D.

In the present situation, there are different approaches to simulation. One possibility is to assume a specific distribution of the white noise process, e.g., $u_t \sim \mathcal{N}(0, \hat{\Sigma}_u)$, and generate a large number of time series realizations based on the estimated VAR coefficients. From these time series, new sets of coefficients are then estimated and the corresponding impulse responses and/or

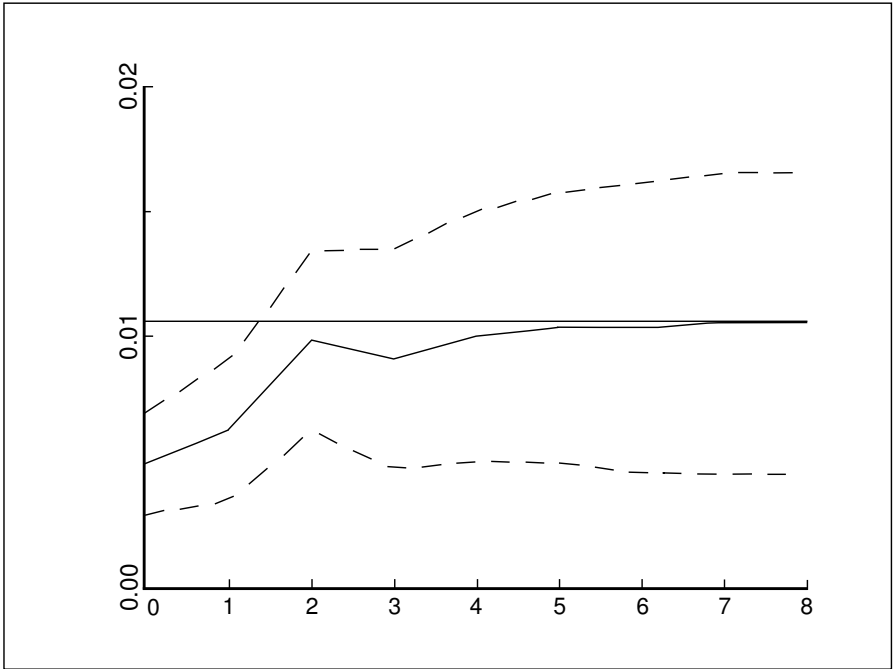


Fig. 3.9. Estimated accumulated and long-run responses of consumption to an orthogonalized impulse in income with estimated asymptotic two-standard error bounds.

forecast error variance components are computed. The empirical distributions obtained in this way may be used to investigate the actual distributions of the quantities of interest.

Alternatively, if an assumption regarding the white noise distribution cannot be made, bootstrap methods may be used and new sets of residuals may be drawn from the estimation residuals. A large number of y_t time series is generated on the basis of these sets of disturbances. The bootstrap multiple time series obtained in this way are then used to compute estimates of the quantities of interest and study their properties. Three different methods for computing bootstrap confidence intervals in the present context are described in Appendix D.3. We have used the standard and the Hall percentile methods to compute confidence intervals for the response of consumption to a forecast error impulse and an orthogonalized impulse in income for our example system. The results are shown in Figures 3.10 and 3.11, respectively.

Some interesting observations can be made. First, for the forecast error impulse responses, the two different methods for establishing confidence intervals produce quite similar results which are also at least qualitatively similar to the asymptotic confidence intervals in Figure 3.4. Second, the situation is a

Table 3.5. Forecast error variance decomposition of the investment/income/consumption system with estimated asymptotic standard errors in parentheses

forecast error in	forecast horizon h	proportions of forecast error variance, h periods ahead, accounted for by innovations in		
		investment $\hat{\omega}_{j1,h}$	income $\hat{\omega}_{j2,h}$	consumption $\hat{\omega}_{j3,h}$
investment ($j = 1$)	1	1.00(.00)	.00(.00)	.00(.00)
	2	.96(.04)	.02(.03)	.02(.03)
	3	.95(.04)	.03(.03)	.03(.03)
	4	.94(.05)	.03(.03)	.03(.03)
	8	.94(.05)	.03(.03)	.03(.04)
income ($j = 2$)	1	.02(.03)	.98(.03)	.00(.00)
	2	.06(.05)	.91(.06)	.03(.04)
	3	.07(.06)	.90(.07)	.03(.04)
	4	.07(.06)	.89(.07)	.04(.04)
	8	.07(.06)	.89(.07)	.04(.04)
consumption ($j = 3$)	1	.08(.06)	.27(.09)	.65(.09)
	2	.08(.06)	.27(.08)	.65(.09)
	3	.13(.08)	.33(.09)	.54(.09)
	4	.13(.08)	.34(.09)	.54(.09)
	8	.13(.08)	.34(.09)	.53(.09)

bit different for the orthogonalized impulse responses in Figure 3.11. Here the two different bootstrap methods produce rather different confidence intervals. These intervals are quite asymmetric in the sense that the estimated impulse responses are not in the middle between the lower and upper bound of the intervals. Thereby they also look quite differently from the asymptotic intervals shown in Figure 3.8. The latter intervals are symmetric around the estimated impulse response coefficients by construction. Again, the qualitative interpretation does not change, however. In other words, the instantaneous and the second coefficient are significantly different from zero, as before. Moreover, the confidence intervals in Figure 3.11 are consistent with a rapidly declining effect of an impulse in income.

It must be emphasized, however, that the bootstrap generally does not solve the problem of a singular asymptotic distribution of the impulse responses and the resulting potentially invalid inference. If the asymptotic distribution is singular, the bootstrap may fail to produce meaningful confidence intervals, for example. Again it may be worth considering a univariate AR(1) process $y_t = \alpha y_{t-1} + u_t$ for illustrative purposes. The second forecast error impulse response coefficient is $\Phi_2 = \alpha^2$. The corresponding estimator $\hat{\Phi}_2 = \hat{\alpha}^2$ was found to have a singular asymptotic distribution if $\alpha = 0$ (see Remark 1 in Section 3.7.1). Suppose a bootstrap is used to produce N bootstrap estimates of α , $\hat{\alpha}_{(n)}^*$, $n = 1, \dots, N$. Clearly, the corresponding bootstrap estimates

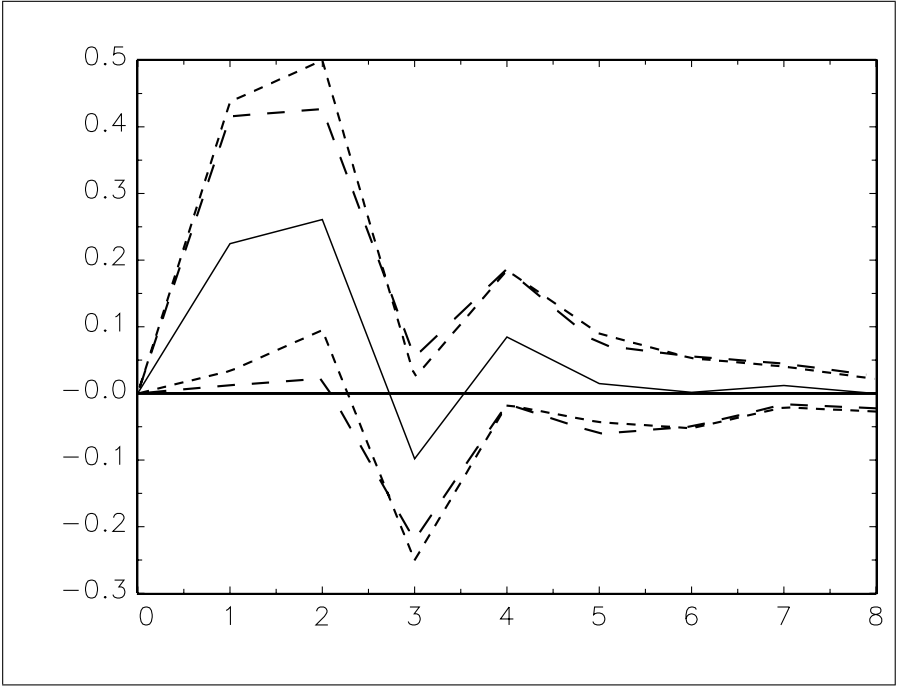


Fig. 3.10. Estimated responses (—) of consumption to a forecast error impulse in income with 95% bootstrap confidence bounds based on 2000 bootstrap replications (— — standard intervals, - - - Hall's percentile intervals).

$\hat{\Phi}_{2(n)}^* = \hat{\alpha}_{(n)}^{*2}$ will all be positive with probability one because they are squares. Thus, if the standard $(1-\gamma)100\%$ bootstrap confidence interval is constructed in the usual way by choosing $\hat{\Phi}_{2(N\gamma/2)}^*$ and $\hat{\Phi}_{2(N(1-\gamma)/2)}^*$ as lower and upper bound, respectively, the true value of zero will never be within the confidence interval. Hence, in this case the actual confidence level will be zero. Although the Hall confidence intervals may be a bit better in this case, they will also not provide the desired coverage level even in large samples. A more detailed discussion of this problem is given by Benkwitz et al. (2000), where also methods for correct asymptotic inference are considered. One possible solution is to eliminate all points where nonsingularities of the asymptotic distribution may occur by fitting subset models (see Chapter 5). Another possibility to circumvent the problem is to allow the VAR process to be of infinite order and increase the order with growing sample size. This possibility will be discussed in detail in Chapter 15.

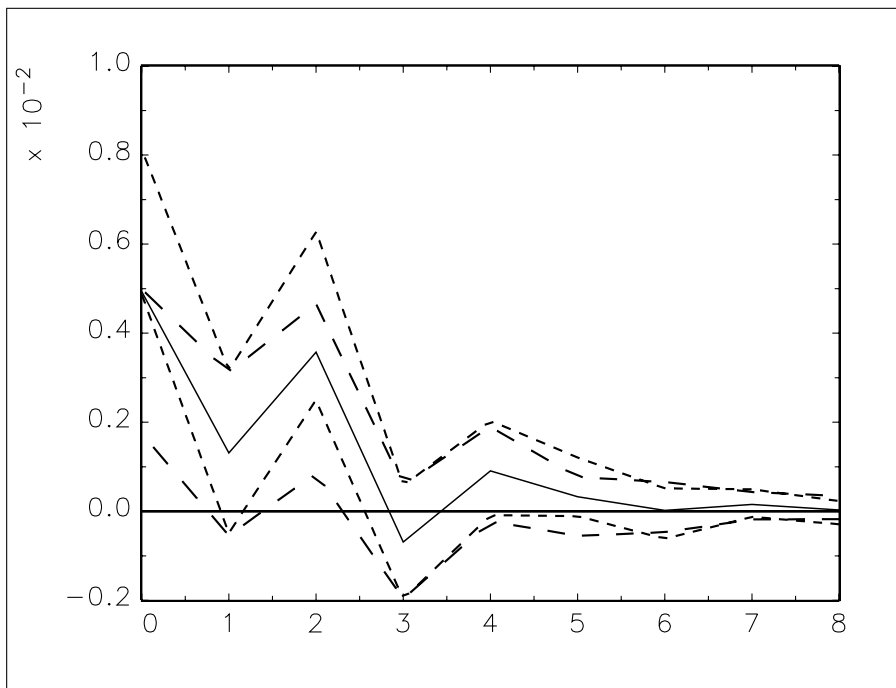


Fig. 3.11. Estimated responses (—) of consumption to an orthogonalized impulse in income with 95% bootstrap confidence bounds based on 2000 bootstrap replications (— — standard intervals, - - - Hall's percentile intervals).

3.8 Exercises

3.8.1 Algebraic Problems

The notation of Sections 3.2–3.5 is used in the following problems.

Problem 3.1

Show that $\hat{\beta} = ((ZZ')^{-1}Z \otimes I_K)\mathbf{y}$ minimizes

$$\bar{S}(\beta) = \mathbf{u}'\mathbf{u} = [\mathbf{y} - (Z' \otimes I_K)\beta]'[\mathbf{y} - (Z' \otimes I_K)\beta].$$

Problem 3.2

Prove that

$$\sqrt{T}(\hat{\mathbf{b}} - \mathbf{b}) \xrightarrow{d} \mathcal{N}(0, \Sigma_u \otimes \Gamma^{-1}),$$

if y_t is stable and

$$\frac{1}{\sqrt{T}} \text{vec}(ZU') = \frac{1}{\sqrt{T}} (I_K \otimes Z) \text{vec}(U') \xrightarrow{d} \mathcal{N}(0, \Sigma_u \otimes \Gamma).$$

Problem 3.3

Show (3.4.17). (Hint: Use the product rule for matrix differentiation and $\partial \text{vec}(\Sigma_u^{-1})/\partial \text{vec}(\Sigma_u)' = -\Sigma_u^{-1} \otimes \Sigma_u^{-1}$.)

Problem 3.4

Derive (3.4.18). (Hint: Use the last expression given in (3.4.6).)

Problem 3.5

Show (3.4.19).

Problem 3.6

Derive (3.4.20).

Problem 3.7

Prove that $\text{plim } \tilde{z}_T/\sqrt{T} = 0$, where

$$\tilde{z}_T = \sum_{i=1}^p \tilde{A}_i \sum_{j=0}^{i-1} (y_{-j} - y_{T-j}).$$

(Hint: Show that $E(\tilde{z}_T/\sqrt{T}) \rightarrow 0$ and $\text{Var}(\tilde{z}_T/\sqrt{T}) \rightarrow 0$.)

Problem 3.8

Show that Equation (3.5.10) holds.

(Hint: Define

$$Z_t(h) := \begin{bmatrix} 1 \\ y_t(h) \\ \vdots \\ y_t(h-p+1) \end{bmatrix}$$

and show $Z_t(h) = \mathbf{B}Z_t(h-1)$ by induction.)

Problem 3.9

In the context of Section 3.5, suppose that y_t is a stable Gaussian VAR(p) process which is estimated by ML in mean-adjusted form. Show that the forecast MSE correction term has the form

$$\Omega(h) = E \left(\frac{\partial y_t(h)}{\partial \mu'} \Sigma_{\tilde{\mu}} \frac{\partial y_t(h)'}{\partial \mu} \right) + E \left(\frac{\partial y_t(h)}{\partial \alpha'} \Sigma_{\tilde{\alpha}} \frac{\partial y_t(h)'}{\partial \alpha} \right),$$

with

$$\frac{\partial y_t(h)}{\partial \mu'} = I_K - J\mathbf{A}^h \begin{bmatrix} I_K \\ \vdots \\ I_K \end{bmatrix}_{(Kp \times K)}$$

and

$$\frac{\partial y_t(h)}{\partial \alpha'} = \sum_{i=0}^{h-1} (Y_t - \mu)' (\mathbf{A}')^{h-1-i} \otimes \Phi_i.$$

Here $\mu := (\mu', \dots, \mu')'$ is a $(Kp \times 1)$ vector, Y_t and $\mathbf{A}$ are as defined in (2.1.8), $J := [I_K : 0 : \dots : 0]$ is a $(K \times Kp)$ matrix, and Φ_i is the i -th coefficient matrix of the prediction error MA representation (2.1.17).

Problem 3.10

Derive the ML estimator and its asymptotic distribution for the parameter of a stable AR(1) process, $y_t = \alpha y_{t-1} + u_t$, $u_t \sim \text{i.i.d. } \mathcal{N}(0, \sigma_u^2)$.

3.8.2 Numerical Problems

The following problems require the use of a computer. They are based on the two quarterly, seasonally adjusted U.S. investment series given in File E2. Consider the variables

- y_1 – first differences of fixed investment,
- y_2 – first differences of change in business inventories,

in the following problems. Use the data from 1947 to 1968 only.

Problem 3.11

Plot the two time series y_{1t} and y_{2t} and comment on the stationarity and stability of the series.

Problem 3.12

Estimate the parameters of a VAR(1) model for $(y_{1t}, y_{2t})'$ using multivariate LS, that is, compute $\hat{B}$ and $\hat{\Sigma}_u$. Comment on the stability of the estimated process.

Problem 3.13

Use the mean-adjusted form of a VAR(1) model and estimate the coefficients. Assume that the data generation process is Gaussian and estimate the covariance matrix of the asymptotic distribution of the ML estimators.

Problem 3.14

Determine the Yule-Walker estimate of the VAR(1) coefficient matrix and compare it to the LS estimate.

Problem 3.15

Use the LS estimate and compute point forecasts $\hat{y}_{86}(1)$, $\hat{y}_{86}(2)$ (that is, the forecast origin is the last quarter of 1968) and the corresponding MSE matrices $\hat{\Sigma}_y(1)$, $\hat{\Sigma}_y(2)$, $\hat{\Sigma}_{\hat{y}}(1)$, and $\hat{\Sigma}_{\hat{y}}(2)$. Use these estimates to set up approximate 95% interval forecasts assuming that the process y_t is Gaussian.

Problem 3.16

Test the hypothesis that y_2 does not Granger-cause y_1 .

Problem 3.17

Estimate the coefficient matrices Φ_1 and Φ_2 from the LS estimates of the VAR(1) model for y_t and determine approximate standard errors of the estimates.

Problem 3.18

Determine the *upper* triangular matrix $\hat{P}$ with positive diagonal for which $\hat{P}\hat{P}' = \hat{\Sigma}_u$. Estimate the covariance matrix of the asymptotic distribution of $\hat{P}$ under the assumption that y_t is Gaussian. Test the hypothesis that the upper right-hand corner element of the underlying matrix P is zero.

Problem 3.19

Use the results of the previous problems to compute $\hat{\Theta}_0$, $\hat{\Theta}_1$, and $\hat{\Theta}_2$. Determine also estimates of the asymptotic standard errors of the elements of these three matrices.

VAR Order Selection and Checking the Model Adequacy

4.1 Introduction

In the previous chapter, we have assumed that we have given a K -dimensional multiple time series $y_1, \dots, y_T$, with $y_t = (y_{1t}, \dots, y_{Kt})'$, which is known to be generated by a VAR(p) process,

$$y_t = \nu + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t, \quad (4.1.1)$$

and we have discussed estimation of the parameters $\nu, A_1, \dots, A_p$, and $\Sigma_u = E(u_t u_t')$. In deriving the properties of the estimators, a number of assumptions were made. In practice, it will rarely be known with certainty whether the conditions hold that are required to derive the consistency and asymptotic normality of the estimators. Therefore statistical tools should be used in order to check the validity of the assumptions made. In this chapter, some such tools will be discussed.

In the next two sections, it will be discussed what to do if the VAR order p is unknown. In practice, the order will usually be unknown. In Chapter 3, we have assumed that a VAR(p) process such as (4.1.1) represents the data generation process. We have not assumed that all the A_i are nonzero. In particular A_p may be zero. In other words, p is just assumed to be an upper bound for the VAR order. On the other hand, from (3.5.13) we know that the approximate MSE matrix of the 1-step ahead predictor will increase with the order p . Thus, choosing p unnecessarily large will reduce the forecast precision of the corresponding estimated VAR(p) model. Also, the estimation precision of the impulse responses depends on the precision of the parameter estimates. Therefore it is useful to have procedures or criteria for choosing an adequate VAR order.

In Sections 4.4–4.6, possibilities are discussed for checking some of the assumptions of the previous chapters. The asymptotic distribution of the residual autocorrelations and so-called portmanteau tests are considered in Section 4.4. The latter tests are popular tools for checking the whiteness of the residuals. More precisely, they are used to test for nonzero residual autocorrelation.

In Section 4.5, tests for nonnormality are considered. The normality assumption was used in Chapter 3 in setting up forecast intervals.

One assumption underlying much of the previous analysis is the stationarity of the systems considered. Nonstationarities may have various forms. Not only trends indicate deviations from stationarity but also changes in the variability or variance of the system. Moreover, exogenous shocks may affect various characteristics of the system. Tests for structural change are presented in Section 4.6.

4.2 A Sequence of Tests for Determining the VAR Order

Obviously, there is not just one correct VAR order for the process (4.1.1). In fact, if (4.1.1) is a correct summary of the characteristics of the process y_t , then the same is true for

$$y_t = \nu + A_1 y_{t-1} + \cdots + A_p y_{t-p} + A_{p+1} y_{t-p-1} + u_t$$

with $A_{p+1} = 0$. In other words, if y_t is a VAR(p) process, in this sense it is also a VAR($p+1$) process. In the assumptions of the previous chapter, the possibility of zero coefficient matrices is not excluded. In this chapter, it is practical to have a unique number that is called the order of the process. Therefore, in the following we will call y_t a VAR(p) process if $A_p \neq 0$ and $A_i = 0$ for $i > p$ so that p is the smallest possible order. This unique number will be called the VAR order.

4.2.1 The Impact of the Fitted VAR Order on the Forecast MSE

If y_t is a VAR(p) process, it is useful to fit a VAR(p) model to the available multiple time series and not, for instance, a VAR($p+i$) because, under a mean square error measure, forecasts from the latter process will be inferior to those based on an estimated VAR(p) model. This result follows from the approximate forecast MSE matrix $\Sigma_{\hat{y}}(h)$ derived in Section 3.5.2 of Chapter 3. For instance, for $h = 1$,

$$\Sigma_{\hat{y}}(1) = \frac{T + Kp + 1}{T} \Sigma_u,$$

if a VAR(p) model is fitted to data generated by a K -dimensional VAR process with order not greater than p . Obviously, $\Sigma_{\hat{y}}(1)$ is an increasing function of the order of the model fitted to the data.

Because the approximate MSE matrix is derived from asymptotic theory, it is of interest to know whether the result remains true in small samples. To get some feeling for the answer to this question, we have generated 1000 Gaussian bivariate time series with a process similar to (3.2.25),

$$y_t = \begin{bmatrix} .02 \\ .03 \end{bmatrix} + \begin{bmatrix} .5 & .1 \\ .4 & .5 \end{bmatrix} y_{t-1} + \begin{bmatrix} 0 & 0 \\ .25 & 0 \end{bmatrix} y_{t-2} + u_t,$$

$$\Sigma_u = \begin{bmatrix} .09 & 0 \\ 0 & .04 \end{bmatrix}. \quad (4.2.1)$$

We have fitted VAR(2), VAR(4), and VAR(6) models to the generated series and we have computed forecasts with the estimated models. Then we have compared these forecasts to generated post-sample values. The resulting average squared forecasting errors for different forecast horizons h and sample sizes T are shown in Table 4.1. Obviously, the forecasts based on estimated VAR(2) models are clearly superior to the VAR(4) and VAR(6) forecasts for sample sizes $T = 30, 50$, and 100 . While the comparative advantage of the VAR(2) models is quite dramatic for $T = 30$, it diminishes with increasing sample size. This, of course, was to be expected given that the approximate forecast MSE matrix of an estimated process approaches that of the known process as the sample size increases (see Section 3.5).

Table 4.1. Average squared forecast errors for the estimated bivariate VAR(2) process (4.2.1) based on 1000 realizations

sample size T	forecast horizon h	average squared forecast errors					
		VAR(2)		VAR(4)		VAR(6)	
		y_1	y_2	y_1	y_2	y_1	y_2
30	1	.111	.052	.132	.062	.165	.075
	2	.155	.084	.182	.098	.223	.119
	3	.146	.141	.183	.166	.225	.202
50	1	.108	.043	.119	.048	.129	.054
	2	.132	.075	.144	.083	.161	.093
	3	.142	.120	.150	.130	.168	.145
100	1	.091	.044	.095	.046	.098	.049
	2	.120	.064	.125	.067	.130	.069
	3	.130	.108	.135	.113	.140	.113

Of course, the process considered in this example is a very special one. To see whether a similar result is obtained for other processes as well, we have also generated 1000 three-dimensional time series with the VAR(1) process (2.1.14),

$$y_t = \begin{bmatrix} .01 \\ .02 \\ 0 \end{bmatrix} + \begin{bmatrix} .5 & 0 & 0 \\ .1 & .1 & .3 \\ 0 & .2 & .3 \end{bmatrix} y_{t-1} + u_t \quad \text{with} \quad \Sigma_u = \begin{bmatrix} 2.25 & 0 & 0 \\ 0 & 1.0 & .5 \\ 0 & .5 & .74 \end{bmatrix}. \quad (4.2.2)$$

We have fitted VAR(1), VAR(3), and VAR(6) models to these data and we have computed forecasts and forecast errors. Some average squared forecast errors are presented in Table 4.2. Again forecasts from lower order models are clearly superior to higher order models. In fact, in a large scale simulation study involving many more processes, similar results were found (see Lütkepohl (1985)). Thus, it is useful to avoid fitting VAR models with unnecessarily large orders.

Table 4.2. Average squared forecast errors for the estimated three-dimensional VAR(1) process (4.2.2) based on 1000 realizations

sample size T	forecast horizon h	average squared forecast errors								
		VAR(1)			VAR(3)			VAR(6)		
		y_1	y_2	y_3	y_1	y_2	y_3	y_1	y_2	y_3
30	1	.87	1.14	2.68	1.14	1.52	3.62	2.25	2.78	6.82
	2	1.09	1.21	3.21	1.44	1.67	4.12	2.54	2.98	7.85
	3	1.06	1.31	3.32	1.35	1.58	4.23	2.59	2.79	8.63
50	1	.81	1.03	2.68	.96	1.22	2.97	1.18	1.53	3.88
	2	1.01	1.23	2.92	1.20	1.40	3.47	1.48	1.68	4.38
	3	1.01	1.29	3.11	1.12	1.44	3.48	1.42	1.77	4.66
100	1	.73	.93	2.35	.77	1.00	2.62	.86	1.12	2.91
	2	.94	1.15	2.86	1.00	1.24	3.12	1.12	1.38	3.53
	3	.90	1.15	3.02	.93	1.20	3.23	1.03	1.35	3.51

The question is then what to do if the true order is unknown and an upper bound, say M , for the order is known only. One possibility to check whether certain coefficient matrices may be zero is to set up a significance test. For our particular problem of determining the correct VAR order, we may set up a sequence of tests. First $H_0 : A_M = 0$ is tested. If this null hypothesis cannot be rejected, we test $H_0 : A_{M-1} = 0$ and so on until we can reject a null hypothesis. Before we discuss this procedure in more detail, we will now introduce a possible test statistic.

4.2.2 The Likelihood Ratio Test Statistic

Because we just need to test zero restrictions on the coefficients of a VAR model, we may use the Wald statistic discussed in Section 3.6 in the context of causality tests. To shed some more light on this type of statistic, it may be instructive to consider the likelihood ratio testing principle. It is based on comparing the maxima of the log-likelihood function over the unrestricted and restricted parameter space. Specifically, the likelihood ratio statistic is

$$\lambda_{LR} = 2[\ln l(\tilde{\boldsymbol{\delta}}) - \ln l(\tilde{\boldsymbol{\delta}}_r)], \tag{4.2.3}$$

where $\tilde{\delta}$ is the unrestricted ML estimator for a parameter vector δ obtained by maximizing the likelihood function over the full feasible parameter space and $\tilde{\delta}_r$ is the restricted ML estimator which is obtained by maximizing the likelihood function over that part of the parameter space where the restrictions of interest are satisfied (see Appendix C.7). For the case of interest here, where we have linear constraints for the coefficients of a VAR process, λ_{LR} can be shown to have an asymptotic χ^2 -distribution with as many degrees of freedom as there are distinct linear restrictions.

To obtain this result, let us assume for the moment that y_t is a stable Gaussian (normally distributed) VAR(p) process as in (4.1.1). Using the notation of Section 3.2.1 (as opposed to the mean-adjusted form considered in Section 3.4), the log-likelihood function is

$$\begin{aligned} \ln l(\beta, \Sigma_u) = & -\frac{KT}{2} \ln 2\pi - \frac{T}{2} \ln |\Sigma_u| \\ & - \frac{1}{2} [\mathbf{y} - (Z' \otimes I_K)\beta]' (I_T \otimes \Sigma_u^{-1}) [\mathbf{y} - (Z' \otimes I_K)\beta] \end{aligned} \quad (4.2.4)$$

(see (3.4.5)). The first order partial derivatives with respect to β are

$$\frac{\partial \ln l}{\partial \beta} = (Z \otimes \Sigma_u^{-1})\mathbf{y} - (ZZ' \otimes \Sigma_u^{-1})\beta. \quad (4.2.5)$$

Equating to zero and solving for β gives the unrestricted ML/LS estimator

$$\tilde{\beta} = ((ZZ')^{-1}Z \otimes I_K)\mathbf{y}. \quad (4.2.6)$$

Suppose the restrictions for β are given in the form

$$C\beta = c, \quad (4.2.7)$$

where C is a known $(N \times (K^2p + K))$ matrix of rank N and c is a known $(N \times 1)$ vector. Then the restricted ML estimator may be found by a Lagrangian approach (see Appendix A.14). The Lagrange function is

$$\mathcal{L}(\beta, \gamma) = \ln l(\beta) + \gamma'(C\beta - c), \quad (4.2.8)$$

where γ is an $(N \times 1)$ vector of Lagrange multipliers. Of course, $\mathcal{L}$ also depends on Σ_u . Because these parameters are not involved in the restrictions (4.2.7), we have skipped them there. The restricted maximum of the log-likelihood function with respect to β is known to be attained at a point where the first order partial derivatives of $\mathcal{L}$ are zero.

$$\frac{\partial \mathcal{L}}{\partial \beta} = (Z \otimes \Sigma_u^{-1})\mathbf{y} - (ZZ' \otimes \Sigma_u^{-1})\beta + C'\gamma, \quad (4.2.9a)$$

$$\frac{\partial \mathcal{L}}{\partial \gamma} = C\beta - c. \quad (4.2.9b)$$

Equating to zero and solving gives

$$\tilde{\beta}_r = \tilde{\beta} + [(ZZ')^{-1} \otimes \Sigma_u] C' [C((ZZ')^{-1} \otimes \Sigma_u)C']^{-1} (c - C\tilde{\beta}) \quad (4.2.10)$$

(see Problem 4.1).

Because for any given coefficient matrix B^0 the maximum of $\ln l$ with respect to Σ_u is obtained for

$$\Sigma_u^0 = \frac{1}{T}(Y - B^0 Z)(Y - B^0 Z)'$$

(see Section 3.4.2, (3.4.8) and (3.4.11)), the maximum for the unrestricted case is attained for

$$\tilde{\Sigma}_u = \frac{1}{T}(Y - \tilde{B}Z)(Y - \tilde{B}Z)' \quad (4.2.11)$$

and for the restricted case we get

$$\tilde{\Sigma}_u^r = \frac{1}{T}(Y - \tilde{B}_r Z)(Y - \tilde{B}_r Z)' \quad (4.2.12)$$

Here $\tilde{B}$ and $\tilde{B}_r$ are the coefficient matrices corresponding to $\tilde{\beta}$ and $\tilde{\beta}_r$, respectively, that is, $\tilde{\beta} = \text{vec}(\tilde{B})$ and $\tilde{\beta}_r = \text{vec}(\tilde{B}_r)$. Thus, for this particular situation, the likelihood ratio statistic becomes

$$\lambda_{LR} = 2[\ln l(\tilde{\beta}, \tilde{\Sigma}_u) - \ln l(\tilde{\beta}_r, \tilde{\Sigma}_u^r)].$$

This statistic can be shown to have an asymptotic $\chi^2(N)$ -distribution. In fact, this result also holds if y_t is not Gaussian, but has a distribution from a larger family. If y_t is not Gaussian, the estimators obtained by maximizing the Gaussian likelihood function in (4.2.4) are called *quasi ML estimators*. We will now state the previous results formally and then present a proof.

Proposition 4.1 (*Asymptotic Distribution of the LR Statistic*)

Let y_t be a stationary, stable VAR(p) process as in (4.1.1) with standard white noise u_t (see Definition 3.1). Suppose the true parameter vector β satisfies linear constraints $C\beta = c$, where C is an $(N \times (K^2p + K))$ matrix of rank N and c is an $(N \times 1)$ vector. Moreover, let $\ln l$ denote the Gaussian log-likelihood function and let $\tilde{\beta}$ and $\tilde{\beta}_r$ be the (quasi) ML and restricted (quasi) ML estimators, respectively, with corresponding estimators $\tilde{\Sigma}_u$ and $\tilde{\Sigma}_u^r$ of the white noise covariance matrix Σ_u given in (4.2.11) and (4.2.12). Then

$$\begin{aligned} \lambda_{LR} &= 2 \left[\ln l(\tilde{\beta}, \tilde{\Sigma}_u) - \ln l(\tilde{\beta}_r, \tilde{\Sigma}_u^r) \right] \\ &= T(\ln |\tilde{\Sigma}_u^r| - \ln |\tilde{\Sigma}_u|) \end{aligned} \quad (4.2.13a)$$

$$= (\tilde{\beta}_r - \tilde{\beta})'(ZZ' \otimes \tilde{\Sigma}_u^{-1})(\tilde{\beta}_r - \tilde{\beta}) \quad (4.2.13b)$$

$$= (\tilde{\beta}_r - \tilde{\beta})'(ZZ' \otimes (\tilde{\Sigma}_u^r)^{-1})(\tilde{\beta}_r - \tilde{\beta}) + o_p(1) \quad (4.2.13c)$$

$$= (C\tilde{\beta} - c)' \left[C((ZZ')^{-1} \otimes \tilde{\Sigma}_u)C' \right]^{-1} (C\tilde{\beta} - c) + o_p(1) \quad (4.2.13d)$$

$$= (C\tilde{\beta} - c)' \left[C((ZZ')^{-1} \otimes \tilde{\Sigma}_u^r)C' \right]^{-1} (C\tilde{\beta} - c) + o_p(1) \quad (4.2.13e)$$

and

$$\lambda_{LR} \xrightarrow{d} \chi^2(N).$$

Here T is the sample size (time series length) and $Z := (Z_0, \dots, Z_{T-1})$ with $Z'_t := (1, y'_t, \dots, y'_{t-p+1})$. ■

In this proposition, the quantity $o_p(1)$ denotes a sequence which converges to zero in probability when the sample size $T \rightarrow \infty$ (see Appendix C.2). Note that y_t is not assumed to be Gaussian (normally distributed) in the proposition. It is just assumed that u_t is independent white noise with bounded fourth moments. Thus, $\ln l$ may not really be the log-likelihood function of $\mathbf{y} := \text{vec}(y_1, \dots, y_T)$. It will only be the actual log-likelihood if $\mathbf{y}$ happens to be multivariate normal. In that case, $\tilde{\beta}$ and $\tilde{\beta}_r$ are actual ML and restricted ML estimators. Otherwise they are quasi ML estimators.

The second form of the LR statistic in (4.2.13a) is sometimes convenient for computing the actual test value. It is also useful for comparing the likelihood ratio tests to other procedures for VAR order selection, as we will see in Section 4.3. The expression in (4.2.13b) shows the similarity of the LR statistic to the LM statistic given in (4.2.13c). Using (4.2.5) and

$$\frac{\partial^2 \ln l}{\partial \beta \partial \beta'} = -(ZZ' \otimes \Sigma_u^{-1})$$

gives

$$\lambda_{LM} = \frac{\partial \ln l(\tilde{\beta}_r)}{\partial \beta'} \frac{\partial^2 \ln l(\tilde{\beta}_r)}{\partial \beta \partial \beta'} \frac{\partial \ln l(\tilde{\beta}_r)}{\partial \beta} = (\tilde{\beta}_r - \tilde{\beta})'(ZZ' \otimes (\tilde{\Sigma}_u^r)^{-1})(\tilde{\beta}_r - \tilde{\beta})$$

(see Appendix C.7 and Problem 4.5). Notice that in the present case we may ignore the part of the parameter vector which corresponds to Σ_u because its ML estimator is asymptotically independent of the other parameters and the asymptotic covariance matrix is block-diagonal. Therefore, at least asymptotically, the terms related to scores of the covariance parameters vanish from the LM statistic.

Comparing (4.2.13d) to (3.6.5) shows that, for the special case considered here, the LR statistic is also similar to the Wald statistic. In fact, the important difference between the Wald and LR statistics is that the former involves only estimators of the unrestricted model while both unrestricted and restricted estimators enter into λ_{LR} (see also (4.2.13a)). The final form of the statistic given in (4.2.13e) provides another useful expression which is close to both the LR and the LM statistic. It shows that we may use the covariance matrix estimator from the restricted model instead of the unrestricted one.

As in the case of the Wald test, one may consider using the statistic λ_{LR}/N in conjunction with the $F(N, T - Kp - 1)$ -distribution in small samples. Another adjustment was suggested by Hannan (1970, p. 341).

Proof of Proposition 4.1:

We first show the equivalence of the various forms of the LR statistic given in the proposition. The equality in (4.2.13a) follows by noting that

$$\begin{aligned} [\mathbf{y} - (Z' \otimes I_K)\boldsymbol{\beta}]' (I_T \otimes \Sigma_u^{-1}) [\mathbf{y} - (Z' \otimes I_K)\boldsymbol{\beta}] \\ = \text{tr}[(Y - BZ)' \Sigma_u^{-1} (Y - BZ)] \\ = \text{tr}[\Sigma_u^{-1} (Y - BZ)(Y - BZ)']. \end{aligned}$$

Replacing the matrices B and Σ_u by $\tilde{B}$ and $\tilde{\Sigma}_u$, respectively, gives

$$\ln l(\tilde{\boldsymbol{\beta}}, \tilde{\Sigma}_u) = \text{constant} - \frac{T}{2} \ln |\tilde{\Sigma}_u|.$$

Similarly,

$$\ln l(\tilde{\boldsymbol{\beta}}_r, \tilde{\Sigma}_u^r) = \text{constant} - \frac{T}{2} \ln |\tilde{\Sigma}_u^r|,$$

which gives the desired result.

In order to prove (4.2.13b), we observe that $\ln l$ is a quadratic function in $\boldsymbol{\beta}$. Thus, by Taylor's theorem (Appendix A.13, Proposition A.3), for an arbitrary fixed vector $\boldsymbol{\beta}^0$,

$$\begin{aligned} \ln l(\boldsymbol{\beta}) &= \ln l(\boldsymbol{\beta}^0) + \frac{\partial \ln l(\boldsymbol{\beta}^0)}{\partial \boldsymbol{\beta}'} (\boldsymbol{\beta} - \boldsymbol{\beta}^0) \\ &\quad + \frac{1}{2} (\boldsymbol{\beta} - \boldsymbol{\beta}^0)' \frac{\partial^2 \ln l(\boldsymbol{\beta}^0)}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} (\boldsymbol{\beta} - \boldsymbol{\beta}^0). \end{aligned}$$

Choosing $\tilde{\boldsymbol{\beta}}$ for $\boldsymbol{\beta}^0$ and $\tilde{\boldsymbol{\beta}}_r$ for $\boldsymbol{\beta}$, $\partial \ln l(\tilde{\boldsymbol{\beta}})/\partial \boldsymbol{\beta}' = 0$ so that

$$\lambda_{LR} = 2 \left[\ln l(\tilde{\boldsymbol{\beta}}) - \ln l(\tilde{\boldsymbol{\beta}}_r) \right] = -(\tilde{\boldsymbol{\beta}}_r - \tilde{\boldsymbol{\beta}})' \frac{\partial^2 \ln l(\tilde{\boldsymbol{\beta}})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} (\tilde{\boldsymbol{\beta}}_r - \tilde{\boldsymbol{\beta}}). \quad (4.2.14)$$

As in Section 3.4, we can derive

$$\frac{\partial^2 \ln l}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} = -(ZZ' \otimes \Sigma_u^{-1}).$$

Hence, (4.2.13b) follows and (4.2.13c) is an immediate consequence of the fact that the restricted and unrestricted covariance matrix estimators are consistent by Proposition 3.2. Thus, $\text{plim}(\tilde{\Sigma}_u - \tilde{\Sigma}_u^r) = 0$ which can be used to show (4.2.13c).

Using (4.2.10) and (4.2.14) gives

$$\begin{aligned} \lambda_{LR} &= (C\tilde{\boldsymbol{\beta}} - c)' [C((ZZ')^{-1} \otimes \Sigma_u)C']^{-1} \\ &\quad \times C((ZZ')^{-1} \otimes \Sigma_u)(ZZ' \otimes \tilde{\Sigma}_u^{-1})((ZZ')^{-1} \otimes \Sigma_u)C' \\ &\quad \times [C((ZZ')^{-1} \otimes \Sigma_u)C']^{-1} (C\tilde{\boldsymbol{\beta}} - c). \end{aligned}$$

The result (4.2.13d) follows by replacing Σ_u with $\tilde{\Sigma}_u$ and noting that this is a consistent estimator of Σ_u . Again by consistency of $\tilde{\Sigma}_u^r$, using this estimator instead of $\tilde{\Sigma}_u$ changes the statistic only by a term which vanishes in probability as the sample size increases. Hence, we have (4.2.13e).

The asymptotic $\chi^2(N)$ -distribution of λ_{LR} now follows from Proposition C.15(5) of Appendix C because $[C((ZZ'/T)^{-1} \otimes \tilde{\Sigma}_u^r)C']^{-1}$ is a consistent estimator of $[C(\Gamma^{-1} \otimes \Sigma_u)C']^{-1}$. ■

In the next subsection a sequential testing scheme based on LR tests is discussed.

4.2.3 A Testing Scheme for VAR Order Determination

Assuming that M is known to be an upper bound for the VAR order, the following sequence of null and alternative hypotheses may be tested using LR tests:

$$\begin{array}{llll}
 H_0^1 : A_M = 0 & \text{versus} & H_1^1 : A_M \neq 0 & \\
 H_0^2 : A_{M-1} = 0 & \text{versus} & H_1^2 : A_{M-1} \neq 0 & | A_M = 0 \\
 \vdots & & & \\
 H_0^i : A_{M-i+1} = 0 & \text{versus} & H_1^i : A_{M-i+1} \neq 0 & | A_M = \cdots = A_{M-i+2} = 0 \\
 \vdots & & & \\
 H_0^M : A_1 = 0 & \text{versus} & H_1^M : A_1 \neq 0 & | A_M = \cdots = A_2 = 0.
 \end{array} \tag{4.2.15}$$

In this scheme, each null hypothesis is tested conditionally on the previous ones being true. The procedure terminates and the VAR order is chosen accordingly, if one of the null hypotheses is rejected. That is, if H_0^i is rejected, $\hat{p} = M - i + 1$ will be chosen as estimate of the autoregressive order.

The likelihood ratio statistic for testing the i -th null hypothesis is

$$\lambda_{LR}(i) = T[\ln|\tilde{\Sigma}_u(M-i)| - \ln|\tilde{\Sigma}_u(M-i+1)|], \tag{4.2.16}$$

where $\tilde{\Sigma}_u(m)$ denotes the ML estimator of Σ_u when a VAR(m) model is fitted to a time series of length T . By Proposition 4.1, this statistic has an asymptotic $\chi^2(K^2)$ -distribution if H_0^i and all previous null hypotheses are true. Note that K^2 parameters are set to zero in H_0^i . Hence, we have to test K^2 restrictions and we use $\lambda_{LR}(i)$ in conjunction with critical values from a $\chi^2(K^2)$ -distribution. Alternatively, one may use $\lambda_{LR}(i)/K^2$ in conjunction with the $F(K^2, T - K(M - i + 1) - 1)$ -distribution.

Of course, the order chosen for a particular process will depend on the significance levels used in the tests. In this procedure, it is important to realize that the significance levels of the individual tests must be distinguished from

the Type I error of the whole procedure because rejection of H_0^i means that $H_0^{i+1}, \dots, H_0^M$ are automatically rejected too. Thus, denoting by D_j the event that H_0^j is rejected in the j -th test when it is actually true, the probability of a Type I error for the i -th test in the sequence is

$$\epsilon_i = \Pr(D_1 \cup D_2 \cup \dots \cup D_i).$$

Because D_j is the event that $\lambda_{LR}(j)$ falls in the rejection region, although H_0^j is true, $\gamma_j = \Pr(D_j)$ is just the significance level of the j -th individual test. It can be shown that for $m \neq j$ and $m, j \leq i$, $\lambda_{LR}(m)$ and $\lambda_{LR}(j)$ are asymptotically independent statistics if $H_0^1, \dots, H_0^i$ are true (see Paulsen & Tjøstheim (1985, pp. 223–224)). Hence, D_m and D_j are independent events in large samples so that

$$\begin{aligned} \epsilon_i &= \Pr(D_1 \cup \dots \cup D_{i-1}) + \Pr(D_i) - \Pr\{(D_1 \cup \dots \cup D_{i-1}) \cap D_i\} \\ &= \epsilon_{i-1} + \gamma_i - \epsilon_{i-1}\gamma_i = \epsilon_{i-1} + \gamma_i(1 - \epsilon_{i-1}), \quad i = 2, 3, \dots, M. \end{aligned} \quad (4.2.17)$$

Of course, $\epsilon_1 = \gamma_1$. Thus, it is easily seen by induction that

$$\epsilon_i = 1 - (1 - \gamma_1) \dots (1 - \gamma_i), \quad i = 1, 2, \dots, M. \quad (4.2.18)$$

If, for example, a 5% significance level is chosen for each individual test ($\gamma_i = .05$), then

$$\epsilon_1 = .05, \quad \epsilon_2 = 1 - .95 \times .95 = .0975, \quad \epsilon_3 = .142625.$$

Hence, the actual rejection probability will become quite substantial if the sequence of null hypotheses to be tested is long.

It is difficult to decide on appropriate significance levels in the testing scheme (4.2.15). Whatever significance levels the researcher decides to use, she or he should keep in mind the distinction between the overall and the individual significance levels. Also, it must be kept in mind that we know the asymptotic distributions of the LR statistics only. Thus, the significance levels chosen will be approximate probabilities of Type I errors only.

Finally, in the literature another testing scheme was also suggested and used. In that scheme the first set of hypotheses ($i = 1$) is as in (4.2.15) and for $i > 1$ the following hypotheses are tested:

$$H_0^i : A_M = \dots = A_{M-i+1} = 0 \text{ versus } H_1^i : A_M \neq 0 \text{ or } \dots \text{ or } A_{M-i+1} \neq 0.$$

Here H_0^i is not tested conditionally on the previous null hypotheses being true but it is tested against the full VAR(M) model. Unfortunately, the LR statistics to be used in such a sequence will not be independent so that the overall significance level (probability of Type I error) is difficult to determine.

4.2.4 An Example

To illustrate the sequential testing procedure described in the foregoing, we use the investment/income/consumption example from Section 3.2.3. The variables y_1 , y_2 , and y_3 represent first differences of the logarithms of the investment, income, and consumption data. We assume an upper bound of $M = 4$ for the VAR order and therefore we set aside the first 4 values as presample values. The data up to 1978.4 are used for estimation so that the sample size is $T = 71$ in each test. The estimated error covariance matrices and their determinants are given in Table 4.3. The corresponding χ^2 - and F -test values are summarized in Table 4.4. Because the denominator degrees of freedom for the F -statistics are quite large (ranging from 62 to 70), the F -tests are qualitatively similar to the χ^2 -tests. Using individual significance levels of .05 in each test, $H_0^3 : A_2 = 0$ is the first null hypothesis that is rejected. Thus, the estimated order from both tests is $\hat{p} = 2$. This supports the order chosen in the example in Chapter 3. Alternative procedures for choosing VAR orders are considered in the next section.

Table 4.3. ML estimates of the error covariance matrix of the investment/income/consumption system

VAR order m	$\tilde{\Sigma}_u(m) \times 10^4$	$ \tilde{\Sigma}_u(m) \times 10^{11}$
0	$\begin{bmatrix} 21.83 & .410 & 1.228 \\ \cdot & 1.420 & .571 \\ \cdot & \cdot & 1.084 \end{bmatrix}$	2.473
1	$\begin{bmatrix} 20.14 & .493 & 1.173 \\ \cdot & 1.318 & .625 \\ \cdot & \cdot & 1.018 \end{bmatrix}$	1.782
2	$\begin{bmatrix} 19.18 & .617 & 1.126 \\ \cdot & 1.270 & .574 \\ \cdot & \cdot & .821 \end{bmatrix}$	1.255
3	$\begin{bmatrix} 19.08 & .599 & 1.126 \\ \cdot & 1.235 & .543 \\ \cdot & \cdot & .784 \end{bmatrix}$	1.174
4	$\begin{bmatrix} 16.96 & .573 & 1.252 \\ \cdot & 1.234 & .544 \\ \cdot & \cdot & .765 \end{bmatrix}$	.958

Table 4.4. LR statistics for the investment/income/consumption system

<i>i</i>	H_0^i	VAR order		
		under H_0^i	λ_{LR}^a	$\lambda_{LR}/9^b$
1	$A_4 = 0$	3	14.44	1.60
2	$A_3 = 0$	2	4.76	.53
3	$A_2 = 0$	1	24.90	2.77
4	$A_1 = 0$	0	23.25	2.58

^a Critical value for individual 5% level test: $\chi^2(9)_{.95} = 16.92$.
^b Critical value for individual 5% level test: $F(9, 71 - 3(5 - i) - 1)_{.95} \approx 2$

4.3 Criteria for VAR Order Selection

Although performing statistical tests is a common strategy for detecting nonzero parameters, the approach described in the previous section is not completely satisfactory if a model is desired for a specific purpose. For instance, a VAR model is often constructed for prediction of the variables involved. In such a case, we are not so much interested in finding the correct order of the underlying data generation process but we are interested in obtaining a good model for prediction. Hence, it seems useful to take the objective of the analysis into account when choosing the VAR order. In the next subsection, we will discuss criteria based on the forecasting objective.

If we really want to know the exact order of the data generation process (e.g., for analysis purposes) it is still questionable whether a testing procedure is the optimal strategy because that strategy has a positive probability of choosing an incorrect order even if the sample size (time series length) is large (see Section 4.3.3). In Section 4.3.2 we will present estimation procedures that choose the correct order with probability 1 at least in large samples.

4.3.1 Minimizing the Forecast MSE

If forecasting is the objective, it makes sense to choose the order such that a measure of forecast precision is minimized. The forecast MSE (mean squared error) is such a measure. Therefore Akaike (1969, 1971) suggested to base the VAR order choice on the approximate 1-step ahead forecast MSE given in Chapter 3, (3.5.13),

$$\Sigma_{\hat{y}}(1) = \frac{T + Km + 1}{T} \Sigma_u,$$

where m denotes the order of the VAR process fitted to the data, T is the sample size, and K is the dimension of the time series. To make this criterion operational, the white noise covariance matrix Σ_u has to be replaced by an estimate. Also, to obtain a unique solution we would like to have a scalar criterion rather than a matrix. Akaike suggested using the LS estimator with degrees of freedom adjustment,

$$\widehat{\Sigma}_u(m) = \frac{T}{T - Km - 1} \widetilde{\Sigma}_u(m),$$

for Σ_u and taking the determinant of the resulting expression. Here $\widetilde{\Sigma}_u(m)$ is the ML estimator of Σ_u obtained by fitting a VAR(m) model, as in the previous section. The resulting criterion is called the final prediction error (FPE) criterion, that is,

$$\begin{aligned} \text{FPE}(m) &= \det \left[\frac{T + Km + 1}{T} \frac{T}{T - Km - 1} \widetilde{\Sigma}_u(m) \right] \\ &= \left[\frac{T + Km + 1}{T - Km - 1} \right]^K \det \widetilde{\Sigma}_u(m). \end{aligned} \quad (4.3.1)$$

We have written the criterion in terms of the ML estimator of the covariance matrix because in this form the FPE criterion has intuitive appeal. If the order m is increased, $\det \widetilde{\Sigma}_u(m)$ declines while the multiplicative term $(T + Km + 1)/(T - Km - 1)$ increases. The VAR order estimate is obtained as that value for which the two forces are balanced optimally. Note that the determinant of the LS estimate $\widehat{\Sigma}_u(m)$ may increase with increasing m . On the other hand, it is quite obvious that $|\widetilde{\Sigma}_u(m)|$ cannot become larger when m increases because the maximum of the log-likelihood function is proportional to $-\ln|\widetilde{\Sigma}_u(m)|$ apart from an additive constant and, for $m < n$, a VAR(m) model may be interpreted as a restricted VAR(n) model. Thus, $-\ln|\widetilde{\Sigma}_u(m)| \leq -\ln|\widetilde{\Sigma}_u(n)|$ or $|\widetilde{\Sigma}_u(m)| \geq |\widetilde{\Sigma}_u(n)|$.

Based on the FPE criterion, the estimate $\widehat{p}(\text{FPE})$ of p is chosen such that

$$\text{FPE}[\widehat{p}(\text{FPE})] = \min\{\text{FPE}(m) | m = 0, 1, \dots, M\}.$$

That is, VAR models of orders $m = 0, 1, \dots, M$ are estimated and the corresponding $\text{FPE}(m)$ values are computed. The order minimizing the FPE values is then chosen as estimate for p .

Akaike (1973, 1974), based on a quite different reasoning, derived a very similar criterion usually abbreviated by AIC (*Akaike's Information Criterion*). For a VAR(m) process the criterion is defined as

$$\begin{aligned} \text{AIC}(m) &= \ln|\widetilde{\Sigma}_u(m)| + \frac{2}{T} (\text{number of freely estimated parameters}) \\ &= \ln|\widetilde{\Sigma}_u(m)| + \frac{2mK^2}{T}. \end{aligned} \quad (4.3.2)$$

The estimate $\widehat{p}(\text{AIC})$ for p is chosen so that this criterion is minimized. Here the constants in the VAR model may be ignored as freely estimated parameters because counting them would just add a constant to the criterion which does not change the minimizing order.

The similarity of the criteria AIC and FPE can be seen by noting that, for a constant N ,

$$\frac{T + N}{T - N} = 1 + \frac{2N}{T} + O(T^{-2}).$$

The quantity $O(T^{-2})$ denotes a sequence of order T^{-2} , that is, a sequence indexed by T that remains bounded if multiplied by T^2 (see Appendix C.2). Thus, the sequence goes to zero rapidly when $T \rightarrow \infty$. Hence,

$$\begin{aligned} \ln \text{FPE}(m) &= \ln |\tilde{\Sigma}_u(m)| + K \ln [(T + Km + 1)/(T - Km - 1)] \\ &= \ln |\tilde{\Sigma}_u(m)| + K \ln [1 + 2(Km + 1)/T + O(T^{-2})] \\ &= \ln |\tilde{\Sigma}_u(m)| + K \frac{2(Km + 1)}{T} + O(T^{-2}) \\ &= \text{AIC}(m) + 2K/T + O(T^{-2}). \end{aligned} \tag{4.3.3}$$

The third equality sign follows from a Taylor series expansion of $\ln(1 + x)$ around 1. The term $2K/T$ does not depend on the order m and, hence, $\text{AIC}(m) + 2K/T$ and $\text{AIC}(m)$ assume their minimum for the same value of m . Consequently, AIC and $\ln \text{FPE}$ differ essentially by a term of order $O(T^{-2})$ and, thus, the two criteria will be about equivalent for moderate and large T .

To illustrate these procedures for VAR order selection, we use again the investment/income/consumption example. The determinants of the residual covariance matrices are given in Table 4.3. Using these determinants, the FPE and AIC values presented in Table 4.5 are obtained. Both criteria reach their minimum for $\hat{p} = 2$, that is, $\hat{p}(\text{FPE}) = \hat{p}(\text{AIC}) = 2$. The other quantities given in the table will be discussed shortly.

Table 4.5. Estimation of the VAR order of the investment/income/consumption system

VAR order				
m	$\text{FPE}(m) \times 10^{11}$	$\text{AIC}(m)$	$\text{HQ}(m)$	$\text{SC}(m)$
0	2.691	-24.42	-24.42*	-24.42*
1	2.500	-24.50	-24.38	-24.21
2	2.272*	-24.59*	-24.37	-24.02
3	2.748	-24.41	-24.07	-23.55
4	2.910	-24.36	-23.90	-23.21

* Minimum.

4.3.2 Consistent Order Selection

If interest centers on the correct VAR order, it makes sense to choose an estimator that has desirable sampling properties. One problem of interest in this context is to determine the statistical properties of order estimators such as $\hat{p}(\text{FPE})$ and $\hat{p}(\text{AIC})$. Consistency is a desirable *asymptotic* property of an estimator. As usual, an estimator $\hat{p}$ of the VAR order p is called *consistent* if

$$\text{plim}_{T \rightarrow \infty} \hat{p} = p \quad \text{or, equivalently,} \quad \lim_{T \rightarrow \infty} \Pr\{\hat{p} = p\} = 1. \tag{4.3.4}$$

The latter definition of the plim may seem to differ slightly from the one given in Appendix C. However, it is easily checked that the two definitions are equivalent for integer valued random variables. Of course, a reasonable estimator for p should be integer valued. The estimator $\hat{p}$ is called *strongly consistent* if

$$\Pr\{\lim \hat{p} = p\} = 1. \quad (4.3.5)$$

Accordingly, a VAR order selection criterion will be called consistent or strongly consistent if the resulting estimator has these properties. The following proposition due to Hannan & Quinn (1979), Quinn (1980), and Paulsen (1984) is useful for investigating the consistency of criteria for order determination.

Proposition 4.2 (*Consistency of VAR Order Estimators*)

Let y_t be a K -dimensional stationary, stable VAR(p) process with standard white noise (that is, u_t is independent white noise with bounded fourth moments). Suppose the maximum order $M \geq p$ and $\hat{p}$ is chosen so as to minimize a criterion

$$\text{Cr}(m) = \ln|\tilde{\Sigma}_u(m)| + mc_T/T \quad (4.3.6)$$

over $m = 0, 1, \dots, M$. Here $\tilde{\Sigma}_u(m)$ denotes the (quasi) ML estimator of Σ_u obtained for a VAR(m) model and c_T is a nondecreasing sequence of real numbers that depends on the sample size T . Then $\hat{p}$ is consistent if and only if

$$c_T \rightarrow \infty \quad \text{and} \quad c_T/T \rightarrow 0 \quad \text{as} \quad T \rightarrow \infty. \quad (4.3.7a)$$

The estimator $\hat{p}$ is a strongly consistent estimator if and only if (4.3.7a) holds and

$$c_T/2 \ln \ln T > 1 \quad (4.3.7b)$$

eventually, as $T \rightarrow \infty$. ■

We will not prove this proposition here but refer the reader to Quinn (1980) and Paulsen (1984) for proofs. The basic idea of the proof is to show that, for $p > m$, the quantity $\ln|\tilde{\Sigma}_u(m)|/\ln|\tilde{\Sigma}_u(p)|$ will be greater than one in large samples because $\ln|\tilde{\Sigma}_u(m)|$ is essentially the minimum of minus the Gaussian log-likelihood function for a VAR(m) model. Consequently, because the penalty terms mc_T/T and pc_T/T go to zero as $T \rightarrow \infty$, $\text{Cr}(m) > \text{Cr}(p)$ for large T . Thus, the probability of choosing too small an order goes to zero as $T \rightarrow \infty$. Similarly, if $m > p$, $\ln|\tilde{\Sigma}_u(m)|/\ln|\tilde{\Sigma}_u(p)|$, approaches 1 in probability if $T \rightarrow \infty$ and the penalty term of the lower order model is smaller than that of a larger order process. Thus the lower order p will be chosen if the sample size is large. The following corollary is an easy consequence of the proposition.

Corollary 4.2.1

Under the conditions of Proposition 4.2, if $M > p$, $\hat{p}(\text{FPE})$ and $\hat{p}(\text{AIC})$ are not consistent. ■

Proof: Because FPE and AIC are asymptotically equivalent (see (4.3.3)), it suffices to prove the corollary for $\hat{p}(\text{AIC})$. Equating $\text{AIC}(m)$ and $\text{Cr}(m)$ given in (4.3.6) shows that

$$2mK^2/T = mc_T/T$$

or $c_T = 2K^2$. Obviously, this sequence does not satisfy (4.3.7a). ■

We will see shortly that the limiting probability for underestimating the VAR order is zero for both $\hat{p}(\text{AIC})$ and $\hat{p}(\text{FPE})$ so that asymptotically they overestimate the true order with positive probability. However, Paulsen & Tjøstheim (1985, p. 224) argued that the limiting probability for overestimating the order declines with increasing dimension K and is negligible for $K \geq 5$. In other words, asymptotically AIC and FPE choose the correct order almost with probability one if the underlying multiple time series has large dimension K .

Before we continue the investigation of AIC and FPE, we shall introduce two *consistent* criteria that have been quite popular in recent applied work. The first one is due to Hannan & Quinn (1979) and Quinn (1980). It is often denoted by HQ (*Hannan-Quinn criterion*):

$$\begin{aligned} \text{HQ}(m) &= \ln |\tilde{\Sigma}_u(m)| + \frac{2 \ln \ln T}{T} (\# \text{ freely estimated parameters}) \\ &= \ln |\tilde{\Sigma}_u(m)| + \frac{2 \ln \ln T}{T} mK^2. \end{aligned} \quad (4.3.8)$$

The estimate $\hat{p}(\text{HQ})$ is the order that minimizes $\text{HQ}(m)$ for $m = 0, 1, \dots, M$. Comparing this criterion to (4.3.6) shows that $c_T = 2K^2 \ln \ln T$ and, thus, by (4.3.7a), HQ is consistent for univariate processes and by (4.3.7b) it is strongly consistent for $K > 1$, if the conditions of Proposition 4.2 are satisfied for y_t .

Using Bayesian arguments Schwarz (1978) derived the following criterion:

$$\begin{aligned} \text{SC}(m) &= \ln |\tilde{\Sigma}_u(m)| + \frac{\ln T}{T} (\# \text{ freely estimated parameters}) \\ &= \ln |\tilde{\Sigma}_u(m)| + \frac{\ln T}{T} mK^2. \end{aligned} \quad (4.3.9)$$

Again the order estimate $\hat{p}(\text{SC})$ is chosen so as to minimize the value of the criterion. A comparison with (4.3.6) shows that for this criterion $c_T = K^2 \ln T$. Because

$$K^2 \ln T / 2 \ln \ln T$$

approaches infinity for $T \rightarrow \infty$, (4.3.7b) is satisfied and SC is seen to be strongly consistent for any dimension K .

Corollary 4.2.2

Under the conditions of Proposition 4.2, SC is strongly consistent and HQ is consistent. If the dimension K of the process is greater than one, both criteria are strongly consistent. ■

In Table 4.5, the values of HQ and SC for the investment/income/consumption example are given. Both criteria assume the minimum for $m = 0$, that is, $\hat{p}(\text{HQ}) = \hat{p}(\text{SC}) = 0$.

4.3.3 Comparison of Order Selection Criteria

It is worth emphasizing that the foregoing results do not necessarily mean that AIC and FPE are inferior to HQ and SC. Only if consistency is the yardstick for evaluating the criteria, the latter two are superior under the conditions of the previous section. So far we have not considered the small sample properties of the estimators. In small samples, AIC and FPE may have better properties (choose the correct order more often) than HQ and SC. Also, the former two criteria are designed for minimizing the forecast error variance. Thus, in small as well as large samples, models based on AIC and FPE may produce superior forecasts although they may not estimate the orders correctly. In fact, Shibata (1980) derived asymptotic optimality properties of AIC and FPE for univariate processes. He showed that, under suitable conditions, they indeed minimize the 1-step ahead forecast MSE asymptotically.

Although it is difficult in general to derive *small sample properties* of the criteria, some such properties can be obtained. The following proposition states small sample relations between the criteria.

Proposition 4.3 (*Small Sample Comparison of AIC, HQ, and SC*)

Let $y_{-M+1}, \dots, y_0, y_1, \dots, y_T$ be any K -dimensional multiple time series and suppose that VAR(m) models, $m = 0, 1, \dots, M$, are fitted to $y_1, \dots, y_T$. Then the following relations hold:

$$\hat{p}(\text{SC}) \leq \hat{p}(\text{AIC}) \quad \text{if } T \geq 8, \quad (4.3.10)$$

$$\hat{p}(\text{SC}) \leq \hat{p}(\text{HQ}) \quad \text{for all } T, \quad (4.3.11)$$

$$\hat{p}(\text{HQ}) \leq \hat{p}(\text{AIC}) \quad \text{if } T \geq 16. \quad (4.3.12)$$

■

Note that we do not require stationarity of y_t . In fact, we do not even require that the multiple time series is generated by a VAR process. Moreover, the proposition is valid in small samples and not just asymptotically. The proof is an easy consequence of the following lemma.

Lemma 4.1

Let $a_0, a_1, \dots, a_M, b_0, b_1, \dots, b_M$ and $c_0, c_1, \dots, c_M$ be real numbers. If

$$b_{m+1} - b_m < a_{m+1} - a_m, \quad m = 0, 1, \dots, M-1, \quad (4.3.13a)$$

holds and if nonnegative integers n and k are chosen such that

$$c_n + a_n = \min\{c_m + a_m | m = 0, 1, \dots, M\} \quad (4.3.13b)$$

and

$$c_k + b_k = \min\{c_m + b_m | m = 0, 1, \dots, M\}, \quad (4.3.13c)$$

then $k \geq n$.¹ ■

The proof of this lemma is left as an exercise (see Problem 4.2). It is now easy to prove Proposition 4.3.

Proof of Proposition 4.3:

Let $c_m = \ln |\hat{\Sigma}_u(m)|$, $b_m = 2mK^2/T$ and $a_m = mK^2 \ln T/T$. Then $\text{AIC}(m) = c_m + b_m$ and $\text{SC}(m) = c_m + a_m$. The sequences a_m , b_m , and c_m satisfy the conditions of the lemma if

$$\begin{aligned} 2K^2/T &= 2(m+1)K^2/T - 2mK^2/T = b_{m+1} - b_m \\ &< a_{m+1} - a_m = (m+1)K^2 \ln T/T - mK^2 \ln T/T = K^2 \ln T/T \end{aligned}$$

or, equivalently, if $\ln T > 2$ or $T > e^2 = 7.39$. Hence, choosing $k = \hat{p}(\text{AIC})$ and $n = \hat{p}(\text{SC})$ gives $\hat{p}(\text{SC}) \leq \hat{p}(\text{AIC})$ if $T \geq 8$. The relations (4.3.11) and (4.3.12) can be shown analogously. ■

An immediate consequence of Corollary 4.2.1 and Proposition 4.3 is that AIC and FPE asymptotically overestimate the true order with positive probability and underestimate the true order with probability zero.

Corollary 4.3.1

Under the conditions of Proposition 4.2, if $M > p$,

$$\lim_{T \rightarrow \infty} \Pr\{\hat{p}(\text{AIC}) < p\} = 0 \quad \text{and} \quad \lim_{T \rightarrow \infty} \Pr\{\hat{p}(\text{AIC}) > p\} > 0 \quad (4.3.14)$$

and the same holds for $\hat{p}(\text{FPE})$. ■

Proof: By (4.3.10) and Corollary 4.2.2,

$$\Pr\{\hat{p}(\text{AIC}) < p\} \leq \Pr\{\hat{p}(\text{SC}) < p\} \rightarrow 0.$$

Because AIC is not consistent by Corollary 4.2.1, $\lim_{T \rightarrow \infty} \Pr\{\hat{p}(\text{AIC}) = p\} < 1$. Hence (4.3.14) follows. The same holds for FPE because this criterion is asymptotically equivalent to AIC (see (4.3.3)). ■

The limitations of the asymptotic theory for the order selection criteria can be seen by considering the criterion obtained by setting c_T equal to $2 \ln \ln T$ in (4.3.6). This results in a criterion

¹ I am grateful to Prof. K. Schürger, Universität Bonn, for pointing out the present improvement of the corresponding lemma stated in Lütkepohl (1991).

$$C(m) = \ln |\tilde{\Sigma}_u(m)| + 2m \ln \ln T/T. \quad (4.3.15)$$

Under the conditions of Proposition 4.2, it is consistent. Yet, using Lemma 4.1 and the same line of reasoning as in the proof of Proposition 4.3, $\hat{p}(\text{AIC}) \leq \hat{p}(C)$ if $2 \ln \ln T \leq 2K^2$ or, equivalently, if $T \leq \exp(\exp K^2)$. For instance, for a bivariate process ($K = 2$), $\exp(\exp K^2) \approx 5.14 \times 10^{23}$. Consequently, if $T < 5.14 \times 10^{23}$, the consistent criterion (4.3.15) chooses an order greater than or equal to $\hat{p}(\text{AIC})$ which in turn has a positive limiting probability for exceeding the true order. This example shows that large sample results sometimes are good approximations only if extreme sample sizes are available. The foregoing result was used by Quinn (1980) as an argument for making c_T a function of the dimension K of the process in the HQ criterion.

It is also of interest to compare the order selection criteria to the sequential testing procedure discussed in the previous section. We have mentioned in Section 4.2 that the order chosen in a sequence of tests will depend on the significance levels used. As a consequence, a testing sequence may give the same order as a selection criterion if the significance levels are chosen accordingly. For instance, AIC chooses an order smaller than the maximum order M if $\text{AIC}(M-1) < \text{AIC}(M)$ or, equivalently, if

$$\lambda_{LR}(1) = T(\ln |\tilde{\Sigma}_u(M-1)| - \ln |\tilde{\Sigma}_u(M)|) < 2MK^2 - 2(M-1)K^2 = 2K^2.$$

For $K = 2$, $2K^2 = 8 \approx \chi^2(4)_{.90}$. Thus, for a bivariate process, in order to ensure that AIC chooses an order less than M whenever the LR testing procedure does, we may use approximately a 10% significance level in the first test of the sequence, provided the distribution of $\lambda_{LR}(1)$ is well approximated by a $\chi^2(4)$ -distribution.

The sequential testing procedure will not lead to a consistent order estimator if the sequence of individual significance levels is held constant. To see this, note that for $M > p$ and a fixed significance level γ , the null hypothesis $H_0 : A_M = 0$ is rejected with probability γ . In other words, in the testing scheme, M is incorrectly chosen as VAR order with probability γ . Thus, there is a positive probability of choosing too high an order. This problem can be circumvented by letting the significance level go to zero as $T \rightarrow \infty$.

4.3.4 Some Small Sample Simulation Results

As mentioned previously, many of the small sample properties of interest in the context of VAR order selection are difficult to derive analytically. Therefore we have performed a small Monte Carlo experiment to get some feeling for the small sample behavior of the estimators. Some results will now be reported.

We have simulated 1000 realizations of the VAR(2) process (4.2.1) and we have recorded the orders chosen by FPE, AIC, HQ, and SC for time series lengths of $T = 30$ and 100 and a maximum VAR order of $M = 6$. In addition, we have determined the order by the sequence of LR tests described in Section

4.2 using a significance level of 5% in each individual test and corresponding critical values from χ^2 -distributions. That is, we have used χ^2 - rather than F -tests. The frequency distributions obtained with the five different procedures are displayed in Table 4.6. Obviously, for the sample sizes reported, none of the criteria is very successful in estimating the order $p = 2$ correctly. This may be due to the fact that A_2 contains only a single, small nonzero element. The similarity of AIC and FPE derived in (4.3.3) becomes evident for $T = 100$. The orders chosen by the LR testing procedures show that the actual significance levels are quite different from their asymptotic approximations, especially for sample size $T = 30$. If λ_{LR} really had a $\chi^2(4)$ -distribution the order $\hat{p} = M = 6$ should be chosen in about 5% of the cases while in the simulation experiment $\hat{p} = 6$ is chosen for 25.4% of the realizations. Hence, the $\chi^2(4)$ -distribution is hardly a good small sample approximation to the actual distribution of λ_{LR} .

In Table 4.6, we also present the sum of normalized mean squared forecast errors of y_1 and y_2 obtained from post-sample forecasts with the estimated processes. The quantities shown in the table are

$$\frac{1}{N} \sum_{i=1}^N (y_{T+h(i)} - \hat{y}_T(h)_{(i)})' \Sigma_y(h)^{-1} (y_{T+h(i)} - \hat{y}_T(h)_{(i)}), \quad h = 1, 2, 3,$$

where N is the number of replications, that is, in this case $N = 1000$, $y_{T+h(i)}$ is the realization in the i -th repetition and $\hat{y}_T(h)_{(i)}$ is the corresponding forecast. Normalizing with the inverse of the h -step forecast error variance $\Sigma_y(h)$ is useful to standardize the forecast errors in such a way so as to have roughly the same variability and, thus, comparable quantities are averaged. For large sample size T and a large number of replications N , the average normalized squared forecast errors should be roughly equal to the dimension of the process, that is, for the present bivariate process they should be close to 2.

Although in Table 4.6 SC often underestimates the true VAR order $p = 2$, the forecasts obtained with the SC models are generally the best for $T = 30$. The reason is that not restricting the single nonzero coefficient in A_2 to zero does not sufficiently improve the forecasts to offset the additional sampling variability introduced by estimating all four elements of the A_2 coefficient matrix. For $T = 100$, corresponding forecast MSEs obtained with the different criteria and procedures are very similar, although SC chooses the correct order much less often than the other criteria. This result indicates that choosing the correct VAR order and selecting a good forecasting model are objectives that may be reached by different VAR order selection procedures. Specifically, in this example, slight underestimation of the VAR order is not harmful to the forecast precision. In fact, for $T = 30$, the most parsimonious criterion which underestimates the true VAR order in more than 80% of the realizations of our VAR(2) process provides forecasts with the smallest normalized average squared forecast errors. In fact, the LR tests which choose larger orders quite frequently, produce clearly the worst forecasts.

Table 4.6. Simulation results based on 1000 realizations of the bivariate VAR(2) process (4.2.1)

VAR	FPE	AIC	HQ	SC	LR
order	$T = 30$ frequency distributions of estimated VAR orders in %				
0	0.1	0.1	0.6	2.6	0.1
1	46.1	42.0	60.4	81.2	29.8
2	33.3	32.2	28.5	14.4	16.5
3	8.3	9.0	5.0	1.1	6.5
4	3.8	4.1	2.2	0.5	8.1
5	3.9	5.0	1.5	0.1	13.6
6	4.5	7.6	1.8	0.1	25.4
forecast					
horizon	normalized average squared forecast errors				
1	2.63	2.68	2.52	2.37	3.09
2	2.66	2.72	2.51	2.41	3.04
3	2.58	2.67	2.45	2.35	3.05

VAR	$T = 100$ frequency distributions of estimated VAR orders in %				
order					
0	0.0	0.0	0.0	0.0	0.0
1	17.6	17.4	42.7	73.1	20.8
2	69.5	69.5	55.5	26.7	53.6
3	8.4	8.4	1.7	0.2	5.3
4	2.8	2.8	0.1	0.0	6.2
5	1.0	1.0	0.0	0.0	5.4
6	0.7	0.9	0.0	0.0	8.7
forecast					
horizon	normalized average squared forecast errors				
1	2.15	2.15	2.15	2.17	2.22
2	2.20	2.20	2.20	2.22	2.25
3	2.12	2.12	2.13	2.12	2.17

It must be emphasized, however, that these results are very special and hold for the single bivariate VAR(2) process used in the simulations. Different results may be obtained for other processes. To substantiate this statement, we have also simulated 1000 time series based on the VAR(1) process (4.2.2). Some results are given in Table 4.7. While for sample size $T = 30$ again none of the criteria and procedures is very successful in detecting the correct VAR order $p = 1$, all four criteria FPE, AIC, HQ, and SC select the correct order in more than 90% of the replications for $T = 100$. The poor approximation of the small sample distribution of the LR statistic by a $\chi^2(9)$ -distribution is evident. Note that we have used the critical values for 5% level individual tests from the χ^2 -distribution. As in the VAR(2) example, the prediction performance of the SC models is best for $T = 30$, although the criterion underestimates

the true order in more than 80% of the replications. For both sample sizes, the worst forecasts are obtained with the sequential testing procedure which overestimates the true order quite often.

Table 4.7. Simulation results based on 1000 realizations of the three-dimensional VAR(1) process (4.2.2)

VAR	FPE	AIC	HQ	SC	LR
order	$T = 30$ frequency distributions of estimated VAR orders in %				
0	24.3	17.5	44.5	81.3	0.7
1	50.7	35.3	39.4	18.0	2.2
2	7.5	4.7	3.0	0.3	1.3
3	3.0	2.2	0.9	0.2	1.5
4	1.8	1.7	0.4	0.0	3.7
5	2.9	4.2	1.5	0.0	14.9
6	9.8	34.4	10.3	0.2	75.7
forecast					
horizon	normalized average squared forecast errors				
1	4.60	6.06	4.43	3.94	8.35
2	4.12	5.42	3.98	3.33	7.87
3	3.87	5.11	3.75	3.19	7.49

VAR	FPE	AIC	HQ	SC	LR
order	$T = 100$ frequency distributions of estimated VAR orders in %				
0	0.0	0.0	0.3	8.1	0.0
1	94.1	93.8	99.6	91.9	61.2
2	5.0	5.1	0.1	0.0	5.4
3	0.7	0.7	0.0	0.0	4.3
4	0.2	0.3	0.0	0.0	7.3
5	0.0	0.0	0.0	0.0	9.3
6	0.0	0.1	0.0	0.0	12.5
forecast					
horizon	normalized average squared forecast errors				
1	3.08	3.08	3.06	3.12	3.24
2	3.12	3.12	3.11	3.12	3.24
3	3.11	3.11	3.10	3.10	3.20

After these two simulation experiments, we still do not have a clear answer to the question which criterion to use in small sample situations. One conclusion that emerges from the two examples is that, in very small samples, slight underestimation of the true order is not necessarily harmful to the forecast precision. Moreover, both examples clearly demonstrate that the χ^2 -approximation to the small sample distribution of the LR statistics is a poor one. In a simulation study based on many other processes, Lütkepohl (1985) obtained similar results. In that study, for low order VAR processes,

the most parsimonious SC criterion was found to do quite well in terms of choosing the correct VAR order and providing good forecasting models. Unfortunately, in practice we often don't even know whether the underlying data generation law is of finite order VAR type. Sometimes we may just approximate an infinite order VAR process by a finite order model. In that case, for moderate sample sizes, some less parsimonious criterion like AIC may give superior results in terms of forecast precision. Therefore, it may be a good strategy to compare the order estimates obtained with different criteria and possibly perform analyses with different VAR orders.

4.4 Checking the Whiteness of the Residuals

In the previous sections, we have considered procedures for choosing the order of a VAR model for the generation process of a given multiple time series. These procedures may be interpreted as methods for determining a filter that transforms the given data into a white noise series. In this context, the criteria for model choice may be regarded as criteria for deciding whether the residuals are close enough to white noise to satisfy the investigator. Of course, if, for example, forecasting is the objective, it may not be of prime importance whether the residuals are really white noise as long as the model forecasts well. There are, however, situations where checking the white noise (whiteness) assumption for the residuals of a particular model is of interest. For instance, if the model order is chosen by nonstatistical methods (for example, on the basis of some economic theory) it may be useful to have statistical tools available for investigating the properties of the residuals. Moreover, because different criteria emphasize different aspects of the data generation process and may therefore all provide useful information for the analyst, it is common not to rely on just one procedure or criterion for model choice but use a number of different statistical tools. Therefore, in this section, we shall discuss statistical tools for checking the autocorrelation properties of the residuals of a given VAR model.

In Sections 4.4.1 and 4.4.2, the asymptotic distributions of the residual autocovariances and autocorrelations are given under the assumption that the model residuals are indeed white noise. In Sections 4.4.3 and 4.4.4, two popular statistics for checking the overall significance of the residual autocorrelations are discussed. The results of this section are adapted from Chitturi (1974), Hosking (1980, 1981a), Li & McLeod (1981), and Ahn (1988).

4.4.1 The Asymptotic Distributions of the Autocovariances and Autocorrelations of a White Noise Process

It is assumed that u_t is a K -dimensional white noise process with nonsingular covariance matrix Σ_u . For instance, u_t may represent the residuals of a

VAR(p) process. Let $U := (u_1, \dots, u_T)$. The autocovariance matrices of u_t are estimated as

$$C_i := \hat{F}_u(i) := \frac{1}{T} \sum_{t=i+1}^T u_t u'_{t-i} = \frac{1}{T} U F_i U', \quad i = 0, 1, \dots, h < T. \quad (4.4.1)$$

The $(T \times T)$ matrix F_i is defined in the obvious way. For instance, for $i = 2$,

$$\begin{aligned} F_i &:= \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & & 0 & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ 0 & 0 & \cdots & 0 \end{bmatrix}'. \end{aligned}$$

Of course, for $i = 0$, $F_0 = I_T$. In the following, the precise form of F_i is not important. It is useful, though, to remember that F_i is defined such that

$$U F_i U' = \sum_{t=i+1}^T u_t u'_{t-i}.$$

Let

$$\mathbf{C}_h := (C_1, \dots, C_h) = U F (I_h \otimes U'), \quad (4.4.2)$$

where $F := (F_1, \dots, F_h)$ is a $(T \times hT)$ matrix that is understood to depend on h and T without this being indicated explicitly. Furthermore, let

$$\mathbf{c}_h := \text{vec}(\mathbf{C}_h). \quad (4.4.3)$$

The estimated autocorrelation matrices of the u_t are denoted by R_i , that is,

$$R_i := D^{-1} C_i D^{-1}, \quad i = 0, 1, \dots, h, \quad (4.4.4)$$

where D is a $(K \times K)$ diagonal matrix, the diagonal elements being the square roots of the diagonal elements of C_0 . In other words, a typical element of R_i is

$$r_{mn,i} = \frac{c_{mn,i}}{\sqrt{c_{mm,0}} \sqrt{c_{nn,0}}},$$

where $c_{mn,i}$ is the mn -th element of C_i . The matrix R_i in (4.4.4) is an estimator of the true autocorrelation matrix $R_u(i) = 0$ for $i \neq 0$. We use the notation

$$\mathbf{R}_h := (R_1, \dots, R_h) \quad \text{and} \quad \mathbf{r}_h := \text{vec}(\mathbf{R}_h) \quad (4.4.5)$$

and we denote by R_u the *true* correlation matrix corresponding to Σ_u . Now we can give the asymptotic distributions of $\mathbf{r}_h$ and $\mathbf{c}_h$.

Proposition 4.4 (*Asymptotic Distributions of White Noise Autocovariances and Autocorrelations*)

Let u_t be a K -dimensional identically distributed standard white noise process, that is, u_t and u_s have the same multivariate distribution with nonsingular covariance matrix Σ_u and corresponding correlation matrix R_u . Then, for $h \geq 1$,

$$\sqrt{T}\mathbf{c}_h \xrightarrow{d} \mathcal{N}(0, I_h \otimes \Sigma_u \otimes \Sigma_u) \quad (4.4.6)$$

and

$$\sqrt{T}\mathbf{r}_h \xrightarrow{d} \mathcal{N}(0, I_h \otimes R_u \otimes R_u). \quad (4.4.7)$$

■

Proof: The result (4.4.6) follows from an appropriate central limit theorem. The i.i.d. assumption for the u_t implies that

$$w_t = \text{vec}(u_t u'_{t-1}, \dots, u_t u'_{t-h})$$

is a stationary white noise process with covariance matrix $E(w_t w'_t) = I_h \otimes \Sigma_u \otimes \Sigma_u$ so that the result (4.4.6) may, e.g., be obtained from the central limit theorem for stationary processes given in Proposition C.13 of Appendix C. Proofs can also be found in Fuller (1976, Chapter 6) and Hannan (1970, Chapter IV, Section 4) among others.

The result in (4.4.7) is a quite easy consequence of (4.4.6). From Proposition 3.2, we know that C_0 is a consistent estimator of Σ_u . Hence,

$$\sqrt{T} \text{vec}(R_i) = \sqrt{T}(D^{-1} \otimes D^{-1}) \text{vec}(C_i) \xrightarrow{d} \mathcal{N}(0, R_u \otimes R_u)$$

by Proposition C.15(1) of Appendix C and (4.4.6), because

$$\begin{aligned} & \text{plim}(D^{-1} \otimes D^{-1})(\Sigma_u \otimes \Sigma_u)(D^{-1} \otimes D^{-1}) \\ &= \text{plim}(D^{-1} \Sigma_u D^{-1} \otimes D^{-1} \Sigma_u D^{-1}) = R_u \otimes R_u. \end{aligned}$$

■

The result in (4.4.6) means that $\sqrt{T} \text{vec}(C_i)$ has the same asymptotic distribution as $\sqrt{T} \text{vec}(C_j)$, namely,

$$\sqrt{T} \operatorname{vec}(C_i), \sqrt{T} \operatorname{vec}(C_j) \xrightarrow{d} \mathcal{N}(0, \Sigma_u \otimes \Sigma_u).$$

Moreover, for $i \neq j$, the two estimators are asymptotically independent. By (4.4.7), the same holds for $\sqrt{T} \operatorname{vec}(R_i)$ and $\sqrt{T} \operatorname{vec}(R_j)$.

In practice, the u_t and hence U will usually be unknown and the reader may wonder about the relevance of Proposition 4.4. The result is not only useful in proving other propositions but can also be used to check whether a given time series is white noise. Before we explain that procedure, we mention that Proposition 4.4 remains valid if the considered white noise process is allowed to have nonzero mean and the mean vector is estimated by the sample mean vector. That is, we consider covariance matrices

$$C_i = \frac{1}{T} \sum_{t=i+1}^T (u_t - \bar{u})(u_{t-i} - \bar{u})',$$

where

$$\bar{u} = \frac{1}{T} \sum_{t=1}^T u_t.$$

Next we observe that the diagonal elements of $R_u \otimes R_u$ are all ones. Consequently, the variances of the asymptotic distributions of the elements of $\sqrt{T} \mathbf{r}_h$ are all unity. Hence, in large samples the $\sqrt{T} r_{mn,i}$ for $i > 0$ have approximate standard normal distributions. Denoting by $\rho_{mn}(i)$ the true correlation coefficients corresponding to the $r_{mn,i}$, a test, with level approximately 5%, of the null hypothesis

$$H_0 : \rho_{mn}(i) = 0 \quad \text{against} \quad H_1 : \rho_{mn}(i) \neq 0$$

rejects H_0 if $|\sqrt{T} r_{mn,i}| > 2$ or, equivalently, $|r_{mn,i}| > 2/\sqrt{T}$.

Now we have a test for checking the null hypothesis that a given multiple time series is generated by a white noise process. We simply compute the correlations of the original data (possibly after some stationarity transformation) and compare their absolute values with $2/\sqrt{T}$. In Section 4.3.2, we found that the SC and HQ estimate of the order for the generation process of the investment/income/consumption example data is $\hat{p} = 0$. Therefore, one may want to check the white noise hypothesis for this example. The first two correlation matrices for the data from 1960.4 to 1978.4 are

$$R_1 = \begin{bmatrix} -.197 & .103 & .128 \\ .190 & .020 & .228 \\ -.047 & .150 & -.089 \end{bmatrix} \quad \text{and} \quad R_2 = \begin{bmatrix} -.045 & .067 & .097 \\ .119 & .079 & .009 \\ .255 & .355 & .279 \end{bmatrix}. \quad (4.4.8)$$

Comparing these quantities with $2/\sqrt{T} = 2/\sqrt{73} = .234$, we find that some are significantly different from zero and, hence, we reject the white noise hypothesis on the basis of this test.

In applied work, the estimated autocorrelations are sometimes plotted and $\pm 2/\sqrt{T}$ -bounds around zero are indicated. The white noise hypothesis is then rejected if any of the estimated correlation coefficients reach out of the area between the $\pm 2/\sqrt{T}$ -bounds. In Figure 4.1, plots of some autocorrelations are provided for the example data. Some autocorrelations at lags 2, 4, 8, 11, and 12 are seen to be significant under the aforementioned criterion.

There are several points that must be kept in mind in such a procedure. First, in an exact 5% level test, on average the test will reject one out of twenty times it is performed independently, even if the null hypothesis is correct. Thus, one would expect that one out of twenty autocorrelation estimates exceeds $2/\sqrt{T}$ in absolute value even if the underlying process is indeed white noise. Note, however, that although R_i and R_j are asymptotically independent for $i \neq j$, the same is not necessarily true for the elements of R_i . Thus, considering the individual correlation coefficients may provide a misleading picture of their significance as a group. Tests for overall significance of groups of autocorrelations are discussed in Sections 4.4.3 and 4.4.4.

Second, the tests we have considered here are just asymptotic tests. In other words, the actual sizes of the tests may differ from their nominal sizes. In fact, it has been shown by Dufour & Roy (1985) and others that in small samples the variances of the correlation coefficients may differ considerably from $1/T$. They will often be smaller so that the tests are conservative in that they reject the null hypothesis less often than is indicated by the significance level chosen.

Despite this criticism, this check for whiteness of a time series enjoys much popularity as it is very easy to carry out. It is a good idea, however, not to rely on this criterion exclusively.

4.4.2 The Asymptotic Distributions of the Residual Autocovariances and Autocorrelations of an Estimated VAR Process

Theoretical Results

If a VAR(p) model has been fitted to the data, a procedure similar to that described in the previous subsection is often used to check the whiteness of the residuals. Instead of the actual u_t 's, the estimation residuals are used, however. We will now consider the consequences of that approach. For that purpose, we assume that the model has been estimated by LS and, using the notation of Section 3.2, the coefficient estimator is denoted by $\hat{B}$ and the corresponding residuals are $\hat{U} = (\hat{u}_1, \dots, \hat{u}_T) := Y - \hat{B}Z$. Furthermore,

$$\begin{aligned}\hat{C}_i &:= \frac{1}{T} \hat{U}' F_i \hat{U}, \quad i = 0, 1, \dots, h, \\ \hat{C}_h &:= (\hat{C}_1, \dots, \hat{C}_h) = \frac{1}{T} \hat{U}' F(I_h \otimes \hat{U}'),\end{aligned}\tag{4.4.9}$$

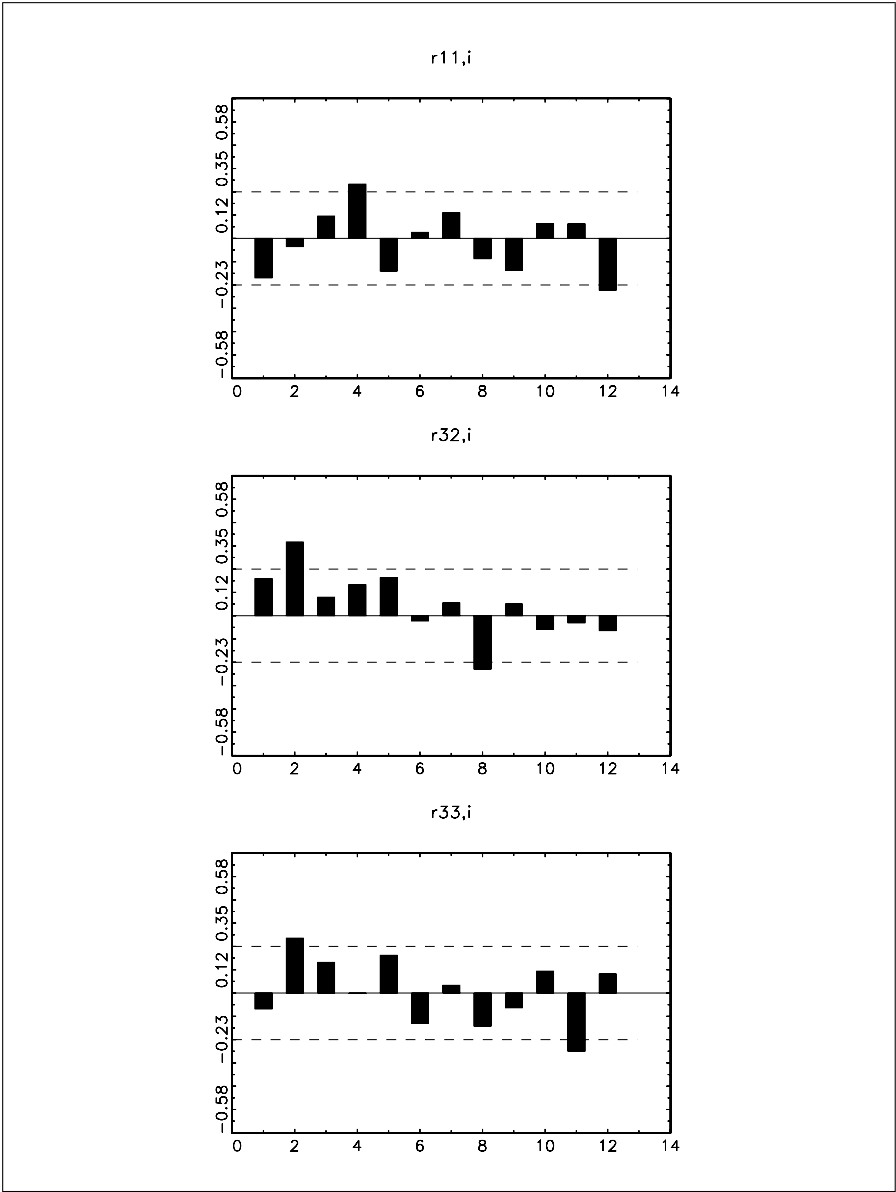


Fig. 4.1. Some estimated autocorrelations of the investment/income/consumption system.

$$\begin{aligned} \hat{\mathbf{c}}_h &:= \text{vec}(\hat{\mathbf{C}}_h), \\ \text{and, correspondingly,} \\ \hat{R}_i &:= \hat{D}^{-1} \hat{C}_i \hat{D}^{-1}, \quad \hat{\mathbf{R}}_h := (\hat{R}_1, \dots, \hat{R}_h), \quad \hat{\mathbf{r}}_h := \text{vec}(\hat{\mathbf{R}}_h), \end{aligned} \tag{4.4.10}$$

where $\widehat{D}$ is a diagonal matrix with the square roots of the diagonal elements of $\widehat{C}_0$ on the main diagonal. We will consider the asymptotic distribution of $\sqrt{T}\widehat{\mathbf{c}}_h$ first. For that purpose the following lemma is helpful.

Lemma 4.2

Let y_t be a stationary, stable VAR(p) process as in (4.1.1) with identically distributed standard white noise u_t and let $\widehat{B}$ be a consistent estimator of $B = [\nu, A_1, \dots, A_p]$ such that $\sqrt{T} \text{vec}(\widehat{B} - B)$ has an asymptotic normal distribution. Then $\sqrt{T}\widehat{\mathbf{c}}_h$ has the same asymptotic distribution as

$$\sqrt{T}\mathbf{c}_h - \sqrt{T}G \text{vec}(\widehat{B} - B), \quad (4.4.11)$$

where $G := \widetilde{G}' \otimes I_K$ with

$$\widetilde{G} := \begin{bmatrix} 0 & 0 & \cdots & 0 \\ \Sigma_u & \Phi_1 \Sigma_u & \cdots & \Phi_{h-1} \Sigma_u \\ 0 & \Sigma_u & \cdots & \Phi_{h-2} \Sigma_u \\ \vdots & & & \vdots \\ 0 & 0 & \cdots & \Phi_{h-p} \Sigma_u \end{bmatrix} \quad ((Kp+1) \times Kh). \quad (4.4.12)$$

■

Proof: Using the notation $Y = BZ + U$,

$$\widehat{U} = Y - \widehat{B}Z = BZ + U - \widehat{B}Z = U - (\widehat{B} - B)Z.$$

Hence,

$$\begin{aligned} & \widehat{U}F(I_h \otimes \widehat{U}') \\ &= UF(I_h \otimes U') - UF \left[I_h \otimes Z'(\widehat{B} - B)' \right] \\ & \quad - (\widehat{B} - B)ZF(I_h \otimes U') + (\widehat{B} - B)ZF \left[I_h \otimes Z'(\widehat{B} - B)' \right]. \end{aligned} \quad (4.4.13)$$

Dividing by T and applying the vec operator, this expression becomes $\widehat{\mathbf{c}}_h$. In order to obtain the expression in (4.4.11), we consider the terms on the right-hand side of (4.4.13) in turn. The first term becomes $\sqrt{T}\mathbf{c}_h$ upon division by $\sqrt{T}$ and application of the vec operator.

Dividing the second and last terms by $\sqrt{T}$ they can be shown to converge to zero in probability, that is,

$$\text{plim} \sqrt{T}UF \left[I_h \otimes Z'(\widehat{B} - B)' \right] / T = 0 \quad (4.4.14)$$

and

$$\text{plim} \sqrt{T}(\widehat{B} - B)ZF \left[I_h \otimes Z'(\widehat{B} - B)' \right] / T = 0 \quad (4.4.15)$$

(see Problem 4.3). Thus, it remains to show that dividing the third term in (4.4.13) by $\sqrt{T}$ and applying the vec operator yields an expression which is asymptotically equivalent to the last term in (4.4.11). To see this, consider

$$ZF(I_h \otimes U') = (ZF_1 U', \dots, ZF_h U')$$

and

$$\begin{aligned} ZF_i U' &= \sum_{t=i+1}^T Z_{t-1} u'_{t-i} = \sum_{t=i+1}^T \begin{bmatrix} 1 \\ y_{t-1} \\ \vdots \\ y_{t-p} \end{bmatrix} u'_{t-i} \\ &= \sum_t \begin{bmatrix} 1 \\ \sum_{j=0}^{\infty} \Phi_j u_{t-1-j} \\ \vdots \\ \sum_{j=0}^{\infty} \Phi_j u_{t-p-j} \end{bmatrix} u'_{t-i}, \end{aligned}$$

where the Φ_i are the coefficient matrices of the canonical MA representation of y_t (see (2.1.17)). Upon division by T and application of the plim we get

$$\text{plim} \frac{1}{T} ZF_i U' = \begin{bmatrix} 0 \\ \Phi_{i-1} \Sigma_u \\ \vdots \\ \Phi_{i-p} \Sigma_u \end{bmatrix},$$

where $\Phi_j = 0$ for $j < 0$. Hence,

$$\text{plim} \frac{1}{T} ZF(I_h \otimes U') = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ \Sigma_u & \Phi_1 \Sigma_u & \cdots & \Phi_{h-1} \Sigma_u \\ 0 & \Sigma_u & \cdots & \Phi_{h-2} \Sigma_u \\ \vdots & & & \vdots \\ 0 & 0 & \cdots & \Phi_{h-p} \Sigma_u \end{bmatrix} = \tilde{G}.$$

((Kp+1) × Kh)

The lemma follows by noting that

$$\text{vec} \left[(\hat{B} - B) ZF(I_h \otimes U') \right] = ([ZF(I_h \otimes U')] \otimes I_K) \text{vec}(\hat{B} - B).$$

■

The next lemma is also helpful later.

Lemma 4.3

If y_t is a stable VAR(p) process as in (4.1.1) with identically distributed standard white noise, then

$$\left[\begin{array}{c} \frac{1}{\sqrt{T}} \text{vec}(UZ') \\ \sqrt{T} \mathbf{c}_h \end{array} \right] \xrightarrow{d} \mathcal{N} \left(0, \left[\begin{array}{cc} \Gamma & \tilde{G} \\ \tilde{G}' & I_h \otimes \Sigma_u \end{array} \right] \otimes \Sigma_u \right), \quad (4.4.16)$$

where $\Gamma := \text{plim} ZZ'/T$ and $\tilde{G}$ is as defined in (4.4.12). ■

For the two terms $\text{vec}(UZ')/\sqrt{T}$ and $\sqrt{T} \mathbf{c}_h$ separately, the asymptotic distributions are already known from Lemma 3.1 and Proposition 4.4, respectively. So the joint asymptotic distribution is the new result here. The reader is referred to Ahn (1988) for a proof. Now the asymptotic distribution of the residual autocovariances is easily obtained.

Proposition 4.5 (*Asymptotic Distributions of Residual Autocovariances*)

Let y_t be a stationary, stable, K -dimensional VAR(p) process as in (4.1.1) with identically distributed standard white noise process u_t and let the coefficients be estimated by multivariate LS or an asymptotically equivalent procedure. Then

$$\sqrt{T} \hat{\mathbf{c}}_h \xrightarrow{d} \mathcal{N}(0, \Sigma_{\mathbf{c}}(h)),$$

where

$$\begin{aligned} \Sigma_{\mathbf{c}}(h) &= (I_h \otimes \Sigma_u - \tilde{G}' \Gamma^{-1} \tilde{G}) \otimes \Sigma_u \\ &= (I_h \otimes \Sigma_u \otimes \Sigma_u) - \tilde{G} [\Gamma_Y(0)^{-1} \otimes \Sigma_u] \tilde{G}'. \end{aligned} \quad (4.4.17)$$

Here $\tilde{G}$ and Γ are the same matrices as in Lemma 4.3, $\Gamma_Y(0)$ is the covariance matrix of $Y_t = (y'_t, \dots, y'_{t-p+1})'$ and $\tilde{G} := \tilde{G}' \otimes I_K$, where $\tilde{G}$ is a $(Kp \times Kh)$ matrix which has the same form as $\tilde{G}$ except that the first row of zeros is eliminated. ■

Proof: Using Lemma 4.2, $\sqrt{T} \hat{\mathbf{c}}_h$ is known to have the same asymptotic distribution as

$$\begin{aligned} &\sqrt{T} \mathbf{c}_h - \sqrt{T} G \text{vec}(\hat{B} - B) \\ &= \left[-\tilde{G}' \otimes I_K : I \right] \left[\begin{array}{c} \sqrt{T} \text{vec}(\hat{B} - B) \\ \sqrt{T} \mathbf{c}_h \end{array} \right] \\ &= \left[-\tilde{G}' \otimes I_K : I \right] \left[\begin{array}{cc} \left(\frac{ZZ'}{T} \right)^{-1} \otimes I_K & 0 \\ 0 & I \end{array} \right] \left[\begin{array}{c} \frac{1}{\sqrt{T}} \text{vec}(UZ') \\ \sqrt{T} \mathbf{c}_h \end{array} \right]. \end{aligned}$$

Noting that $\text{plim}(ZZ'/T)^{-1} = \Gamma^{-1}$, the desired result follows from Lemma 4.3 and Proposition C.15(l) of Appendix C because

$$\begin{aligned} &\left[-\tilde{G}' \Gamma^{-1} \otimes I_K : I \right] \left(\left[\begin{array}{cc} \Gamma & \tilde{G} \\ \tilde{G}' & I_h \otimes \Sigma_u \end{array} \right] \otimes \Sigma_u \right) \left[\begin{array}{c} -\Gamma^{-1} \tilde{G} \otimes I_K \\ I \end{array} \right] \\ &= (I_h \otimes \Sigma_u - \tilde{G}' \Gamma^{-1} \tilde{G}) \otimes \Sigma_u \end{aligned}$$

$$\begin{aligned}
&= I_h \otimes \Sigma_u \otimes \Sigma_u - (\tilde{G}' \otimes I_K)(\Gamma^{-1} \otimes \Sigma_u)(\tilde{G} \otimes I_K) \\
&= I_h \otimes \Sigma_u \otimes \Sigma_u - \tilde{G}[I_Y(0)^{-1} \otimes \Sigma_u]\tilde{G}'.
\end{aligned}$$

■

The form (4.4.17) shows that the variances are smaller than (not greater than) the diagonal elements of $I_h \otimes \Sigma_u \otimes \Sigma_u$. In other words, the variances of the asymptotic distribution of the white noise autocovariances are greater than or equal to the corresponding quantities of the estimated residuals. A similar result can also be shown for the autocorrelations of the estimated residuals.

Proposition 4.6 (*Asymptotic Distributions of Residual Autocorrelations*)

Let D be the $(K \times K)$ diagonal matrix with the square roots of Σ_u on the diagonal and define $G_0 := \tilde{G}(I_h \otimes D^{-1})$. Then, under the conditions of Proposition 4.5,

$$\sqrt{T}\hat{\mathbf{r}}_h \xrightarrow{d} \mathcal{N}(0, \Sigma_{\mathbf{r}}(h)),$$

where

$$\Sigma_{\mathbf{r}}(h) = [(I_h \otimes R_u) - G_0' \Gamma^{-1} G_0] \otimes R_u. \quad (4.4.18)$$

Specifically,

$$\sqrt{T} \text{vec}(\hat{R}_j) \xrightarrow{d} \mathcal{N}(0, \Sigma_R(j)), \quad j = 1, 2, \dots,$$

where

$$\Sigma_R(j) = \left(R_u - D^{-1} \Sigma_u [0 : \Phi'_{j-1} : \dots : \Phi'_{j-p}] \Gamma^{-1} \begin{bmatrix} 0 \\ \Phi_{j-1} \\ \vdots \\ \Phi_{j-p} \end{bmatrix} \Sigma_u D^{-1} \right) \otimes R_u \quad (4.4.19)$$

with $\Phi_i = 0$ for $i < 0$. ■

Proof: Noting that

$$\begin{aligned}
\hat{\mathbf{r}}_h &= \text{vec}(\hat{\mathbf{R}}_h) = \text{vec} \left[\hat{D}^{-1} \hat{\mathbf{C}}_h (I_h \otimes \hat{D}^{-1}) \right] \\
&= (I_h \otimes \hat{D}^{-1} \otimes \hat{D}^{-1}) \hat{\mathbf{c}}_h
\end{aligned}$$

and $\hat{D}^{-1}$ is a consistent estimator of D^{-1} , we get from Proposition 4.5 that $\sqrt{T}\hat{\mathbf{r}}_h$ has an asymptotic normal distribution with mean zero and covariance matrix

$$\begin{aligned}
&(I_h \otimes D^{-1} \otimes D^{-1}) \{ (I_h \otimes \Sigma_u - \tilde{G}' \Gamma^{-1} \tilde{G}) \otimes \Sigma_u \} (I_h \otimes D^{-1} \otimes D^{-1}) \\
&= [(I_h \otimes R_u) - G_0' \Gamma^{-1} G_0] \otimes R_u,
\end{aligned}$$

where $D^{-1} \Sigma_u D^{-1} = R_u$ has been used. ■

From (4.4.19), it is obvious that the diagonal elements of the asymptotic covariance matrix are not greater than 1 because a positive semidefinite matrix is subtracted from R_u . Hence, if estimated residual autocorrelations are used in a white noise test in a similar fashion as the autocorrelations of the original data, we will get a conservative test that rejects the null hypothesis less often than is indicated by the significance level, provided the asymptotic distribution is a correct indicator of the small sample behavior of the test. In particular, for autocorrelations at small lags the variances will be less than 1, while the asymptotic variances approach one for elements of $\sqrt{T}\hat{R}_j$ with large j . This conclusion follows because $\Phi_{j-1}, \dots, \Phi_{j-p}$ approach zero as $j \rightarrow \infty$. As a consequence, the matrix subtracted from R_u goes to zero as $j \rightarrow \infty$.

In practice, all unknown quantities are replaced by estimates in order to obtain standard errors of the residual autocorrelations and tests of specific hypotheses regarding the autocorrelations. It is perhaps worth noting, though, that if Γ is estimated by ZZ'/T , we have to use the ML estimator $\hat{\Sigma}_u$ for Σ_u to ensure positive variances.

An Illustrative Example

As an example, we consider the VAR(2) model for the investment/income/consumption system estimated in Section 3.2.3. For $j = 1$, we get

$$\hat{R}_1 = \begin{bmatrix} .015 & -.011 & -.010 \\ (.026) & (.033) & (.049) \\ -.007 & -.002 & -.068 \\ (.026) & (.033) & (.049) \\ -.024 & -.045 & -.096 \\ (.026) & (.033) & (.049) \end{bmatrix},$$

where the estimated standard errors are given in parentheses. Obviously, the standard errors of the elements of $\hat{R}_1$ are much smaller than $1/\sqrt{T} = .117$ which would be obtained if the variances of the elements of $\sqrt{T}\hat{R}_1$ were 1. In contrast, for $j = 6$, we get

$$\hat{R}_6 = \begin{bmatrix} .053 & -.008 & -.062 \\ (.117) & (.116) & (.117) \\ .165 & .030 & -.051 \\ (.117) & (.116) & (.117) \\ .068 & .026 & .020 \\ (.117) & (.116) & (.117) \end{bmatrix},$$

where the standard errors are very close to .117.

In Figure 4.2, we have plotted the residual autocorrelations and *twice* their asymptotic standard errors (approximate 95% confidence bounds) around

zero. It is apparent that the confidence bounds grow with increasing lag length. For a rough check of 5% level significance of autocorrelations at higher lags, we may use the $\pm 2/\sqrt{T}$ -bounds in practice, which is convenient from a computational viewpoint.

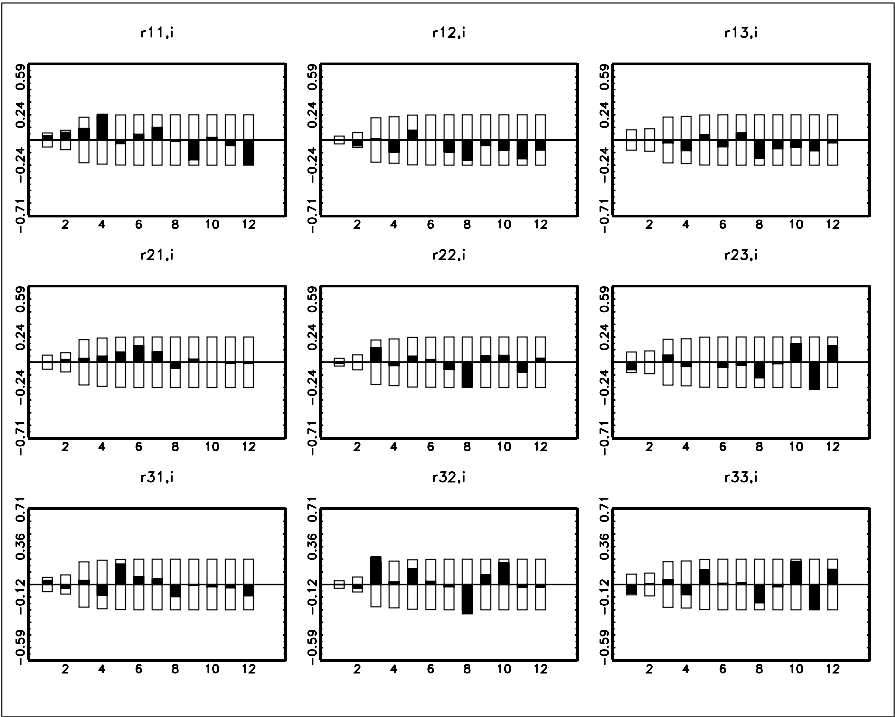


Fig. 4.2. Estimated residual autocorrelations with two-standard error bounds for the investment/income/consumption VAR(2) model.

There are significant residual autocorrelations at lags 3, 4, 8, and 11. While the significant values at lags 3 and 4 may be a reason for concern, one may not worry too much about the higher order lags because one may not be willing to fit a high order model if forecasting is the objective. As we have seen in Section 4.3.4, slight underfitting may even improve the forecast performance. In order to remove the significant residual autocorrelations at low lags, it may help to fit a VAR(3) or VAR(4) model. Of course, this conflicts with choosing the model order on the basis of the model selection criteria. Thus, it has to be decided which criterion is given priority.

It may be worth noting that a plot like that in Figure 4.2 may give a misleading picture of the overall significance of the residual autocorrelations because they are not asymptotically independent. In particular, at low lags there will not only be nonzero correlation between the elements of a specific $\hat{R}_j$

but also between $\hat{R}_j$ and $\hat{R}_i$ for $i \neq j$. Therefore, it is desirable to have tests for overall significance of the residual autocorrelations of a VAR(p) model. Such tests are discussed in the next subsections.

4.4.3 Portmanteau Tests

The foregoing results may also be used to construct a popular test for the overall significance of the residual autocorrelations up to lag h . This test is commonly called *portmanteau test*. It is designed for testing

$$H_0 : \mathbf{R}_h = (R_1, \dots, R_h) = 0 \quad \text{against} \quad H_1 : \mathbf{R}_h \neq 0. \quad (4.4.20)$$

The test statistic is

$$\begin{aligned} Q_h &:= T \sum_{i=1}^h \text{tr}(\hat{R}_i' \hat{R}_u^{-1} \hat{R}_i \hat{R}_u^{-1}) \\ &= T \sum_{i=1}^h \text{tr}(\hat{R}_i' \hat{R}_u^{-1} \hat{R}_i \hat{R}_u^{-1} \hat{D}^{-1} \hat{D}) \\ &= T \sum_{i=1}^h \text{tr}(\hat{D} \hat{R}_i' \hat{D} \hat{D}^{-1} \hat{R}_u^{-1} \hat{D}^{-1} \hat{D} \hat{R}_i \hat{D} \hat{D}^{-1} \hat{R}_u^{-1} \hat{D}^{-1}) \\ &= T \sum_{i=1}^h \text{tr}(\hat{C}_i' \hat{C}_0^{-1} \hat{C}_i \hat{C}_0^{-1}). \end{aligned} \quad (4.4.21)$$

Obviously, this statistic is very easy to compute from the estimated residuals. By Proposition 4.5, it has an approximate asymptotic χ^2 -distribution.

Proposition 4.7 (*Approximate Distribution of the Portmanteau Statistic*)
Under the conditions of Proposition 4.5, we have, approximately, for large T and h ,

$$\begin{aligned} Q_h &= T \sum_{i=1}^h \text{tr}(\hat{C}_i' \hat{C}_0^{-1} \hat{C}_i \hat{C}_0^{-1}) \\ &= T \text{vec}(\hat{\mathbf{C}}_h)' (I_h \otimes \hat{C}_0^{-1} \otimes \hat{C}_0^{-1}) \text{vec}(\hat{\mathbf{C}}_h) \approx \chi^2(K^2(h-p)). \end{aligned} \quad (4.4.22)$$

■

Sketch of the proof: By Proposition C.15(5) of Appendix C, Q_h has the same asymptotic distribution as

$$T \hat{\mathbf{c}}_h' (I_h \otimes \Sigma_u^{-1} \otimes \Sigma_u^{-1}) \hat{\mathbf{c}}_h.$$

Defining the $(K \times K)$ matrix P such that $PP' = \Sigma_u$ and

$$\tilde{\mathbf{c}}_h := (I_h \otimes P \otimes P)^{-1} \hat{\mathbf{c}}_h,$$

it is easily seen that Q_h has the same asymptotic distribution as $T\tilde{\mathbf{c}}'_h\tilde{\mathbf{c}}_h$. Hence, by Proposition C.15(6), it suffices to show that $\sqrt{T}\tilde{\mathbf{c}}_h \xrightarrow{d} \mathcal{N}(0, \Omega)$, where Ω is an idempotent matrix of rank $K^2h - K^2p$. Because an approximate limiting χ^2 -distribution of Q_h is claimed only, we just show that Ω is approximately equal to an idempotent matrix with rank $K^2(h - p)$.

Using Proposition 4.5, we get

$$\begin{aligned}\Omega &= (I_h \otimes P^{-1} \otimes P^{-1})\Sigma_{\mathbf{c}}(h)(I_h \otimes P'^{-1} \otimes P'^{-1}) \\ &= I_{hK^2} - \mathbf{P}\bar{G}[I_Y(0)^{-1} \otimes \Sigma_u]\bar{G}'\mathbf{P}',\end{aligned}$$

where $\mathbf{P} = I_h \otimes P^{-1} \otimes P^{-1}$ and $\bar{G}$ is defined in Proposition 4.5. Noting that the ij -th block of $I_Y(0)$ is

$$\text{Cov}(y_{t-i}, y_{t-j}) = \Gamma_y(j-i) = \sum_{n=0}^{\infty} \Phi_{n-i}\Sigma_u\Phi'_{n-j},$$

with $\Phi_k = 0$ for $k < 0$, we get approximately,

$$\begin{aligned}\Gamma_y(0) \otimes \Sigma_u^{-1} &\approx \left[\sum_{n=1}^h \Phi_{n-i}\Sigma_u\Phi'_{n-j} \right]_{i,j=1,\dots,p} \otimes \Sigma_u^{-1} \\ &= \left[\sum_{n=1}^h \Phi_{n-i}\Sigma_u P'^{-1} P^{-1} \Sigma_u \Phi'_{n-j} \right]_{i,j} \otimes \Sigma_u^{-1} \\ &= \bar{G}'\mathbf{P}'\bar{G}.\end{aligned}$$

Hence, if h is such that $\Phi_i \approx 0$ for $i > h - p$,

$$\Omega \approx I_{hK^2} - \mathbf{P}\bar{G}(\bar{G}'\mathbf{P}'\bar{G})^{-1}\bar{G}'\mathbf{P}'.$$

Thus, Ω is approximately equal to an idempotent matrix with rank

$$\text{tr}(I_{hK^2} - \mathbf{P}\bar{G}(\bar{G}'\mathbf{P}'\bar{G})^{-1}\bar{G}'\mathbf{P}') = hK^2 - pK^2,$$

as was to be shown. ■

Of course, these arguments do not fully prove Proposition 4.7 because we have not shown that an approximately idempotent matrix Ω leads to an approximate χ^2 -distribution. To actually obtain the limiting χ^2 -distribution, we have to assume that h goes to infinity with the sample size. Because the sketch of the proof should suffice to show in what sense the result is approximate, we do not pursue this issue further and refer the reader to Ahn (1988) for details. For practical purposes, it is important to remember that the χ^2 -approximation to the distribution of the test statistic may be misleading for small values of h .

Like in previous sections we have discussed *asymptotic* distributions in this section. Not knowing the small sample distribution is clearly a shortcoming

because, in practice, infinite samples are not available. Using Monte Carlo techniques, it was found by some researchers that in small samples the nominal size of the portmanteau test tends to be lower than the significance level chosen (Davies, Triggs & Newbold (1977), Ljung & Box (1978), Hosking (1980)). As a consequence the test has low power against many alternatives. Therefore it has been suggested to use the modified test statistic

$$\bar{Q}_h := T^2 \sum_{i=1}^h (T-i)^{-1} \text{tr}(\hat{C}'_i \hat{C}_0^{-1} \hat{C}_i \hat{C}_0^{-1}). \quad (4.4.23)$$

The modification may be regarded as an adjustment for the number of terms in the sum in

$$\hat{C}_i = \frac{1}{T} \sum_{t=i+1}^T \hat{u}_t \hat{u}'_{t-i}.$$

For $T \rightarrow \infty$, $T/[T^2(T-i)^{-1}] \rightarrow 1$ and, thus, $\bar{Q}_h$ has the same asymptotic distribution as Q_h , that is, approximately in large samples and for large h ,

$$\bar{Q}_h \approx \chi^2(K^2(h-p)). \quad (4.4.24)$$

For our example model, we obtained $\bar{Q}_{12} = 81.9$. Comparing this value with $\chi^2(K^2(h-p))_{.95} = \chi^2(90)_{.95} \approx 113$ shows that we cannot reject the white noise hypothesis for the residuals at a 5% level.

As mentioned in the introduction to this section, these tests can also be used in a model selection/order estimation procedure. A sequence of hypotheses as in (4.2.15) is tested in such a procedure by checking whether the residuals are white noise. In the following, Lagrange multiplier tests for residual autocorrelation will be presented.

4.4.4 Lagrange Multiplier Tests

Another way of testing a VAR model for residual autocorrelation is to assume a VAR model for the error vector, $u_t = D_1 u_{t-1} + \dots + D_h u_{t-h} + v_t$, where v_t is white noise. It is equal to u_t if there is no residual autocorrelation. Therefore, we wish to test the pair of hypotheses

$$\begin{aligned} H_0 : D_1 = \dots = D_h = 0 \quad \text{against} \\ H_1 : D_j \neq 0 \text{ for at least one } j \in \{1, \dots, h\}. \end{aligned} \quad (4.4.25)$$

In this case, it is convenient to use the LM principle for constructing a test because we then only need to estimate the restricted model where $u_t = v_t$. We determine the test statistic with the help of the auxiliary regression model (see also Appendix C.7)

$$\hat{u}_t = \nu + A_1 y_{t-1} + \dots + A_p y_{t-p} + D_1 \hat{u}_{t-1} + \dots + D_h \hat{u}_{t-h} + \varepsilon_t$$

or, for $t = 1, \dots, T$,

$$\hat{U} = BZ + D\hat{U} + \mathcal{E},$$

where $D := [D_1 : \dots : D_h]$ is $(K \times Kh)$, $\hat{U} := (I_h \otimes \hat{U})F'$ with F as in (4.4.2), $\mathcal{E} := [\varepsilon_1, \dots, \varepsilon_T]$ is a $(K \times T)$ matrix and the other symbols are defined as before. In particular, the $\hat{u}_t$ are the residuals from LS estimation of the original VAR(p) model and $\hat{u}_t = 0$ for $t \leq 0$. The LS estimator of $[B : D]$ from the auxiliary model is

$$\begin{aligned} [\hat{B} : \hat{D}] &= \hat{U}[Z' : \hat{U}'] \left(\begin{bmatrix} Z \\ \hat{U} \end{bmatrix} [Z' : \hat{U}'] \right)^{-1} \\ &= [\hat{U}Z' : \hat{U}\hat{U}'] \begin{bmatrix} ZZ' & Z\hat{U}' \\ \hat{U}Z' & \hat{U}\hat{U}' \end{bmatrix}^{-1} \\ &= [0 : \hat{U}\hat{U}'] \begin{bmatrix} ZZ' & Z\hat{U}' \\ \hat{U}Z' & \hat{U}\hat{U}' \end{bmatrix}^{-1}, \end{aligned}$$

where $\hat{U}Z' = 0$ from the first order conditions for computing the LS estimator has been used. Thus, applying the rules for the partitioned inverse (Appendix A.10, Rule (2)) gives

$$\hat{D} = \hat{U}\hat{U}'[\hat{U}\hat{U}' - \hat{U}Z'(ZZ')^{-1}Z\hat{U}']^{-1}. \quad (4.4.26)$$

The standard χ^2 -statistic for testing $D = 0$ then becomes

$$\begin{aligned} \lambda_{LM}(h) &= \text{vec}(\hat{D})' \left([\hat{U}\hat{U}' - \hat{U}Z'(ZZ')^{-1}Z\hat{U}'] \otimes \hat{\Sigma}_u^{-1} \right) \text{vec}(\hat{D}) \\ &= \text{vec}(\hat{U}\hat{U}')' \left([\hat{U}\hat{U}' - \hat{U}Z'(ZZ')^{-1}Z\hat{U}']^{-1} \otimes \hat{\Sigma}_u^{-1} \right) \text{vec}(\hat{U}\hat{U}'), \end{aligned}$$

where

$$\text{vec}(\hat{D}) = \left([\hat{U}\hat{U}' - \hat{U}Z'(ZZ')^{-1}Z\hat{U}']^{-1} \otimes I_K \right) \text{vec}(\hat{U}\hat{U}')$$

has been used. Noting that $\hat{U}\hat{U}' = \hat{U}F(I_h \otimes \hat{U}')$ shows that $T^{-1} \text{vec}(\hat{U}\hat{U}') = \hat{\mathbf{c}}_h$. Moreover, from results in Section 4.4.2 we get

$$\text{plim} \frac{1}{T} \hat{U}\hat{U}' = \text{plim} \frac{1}{T} (I_h \otimes \hat{U})F'F(I_h \otimes \hat{U}') = I_h \otimes \Sigma_u$$

and

$$\text{plim} \frac{1}{T} \hat{U}Z'(ZZ')^{-1}Z\hat{U}' = \tilde{G}\Gamma^{-1}\tilde{G}$$

(see the proof of Lemma 4.2). Hence,

$$\hat{\Sigma}_{\mathbf{c}}(h) = \frac{1}{T} [\hat{U}\hat{U}' - \hat{U}Z'(ZZ')^{-1}Z\hat{U}'] \otimes \hat{\Sigma}_u$$

is a consistent estimator of $\Sigma_{\mathbf{c}}(h)$ and, because the foregoing results imply that

$$\lambda_{LM}(h) = T \widehat{\mathbf{c}}_h' \widehat{\Sigma}_{\mathbf{c}}(h)^{-1} \widehat{\mathbf{c}}_h,$$

the asymptotic χ^2 -distribution of this statistic follows from Propositions 4.5 and C.15(5).

Proposition 4.8 (*Asymptotic Distribution of the LM Statistic for Residual Autocorrelation*)

Under the conditions of Proposition 4.5,

$$\lambda_{LM}(h) \xrightarrow{d} \chi^2(hK^2).$$

■

The LM test for residual autocorrelation is sometimes called Breusch-Godfrey test because it was proposed by Breusch (1978) and Godfrey (1978) (see also Godfrey (1988)). Unfortunately, the χ^2 -distribution was found to be a poor approximation of the actual null distribution of $\lambda_{LM}(h)$ in many situations (Edgerton & Shukur (1999) and Doornik (1996)). Even a standard F -approximation is unsatisfactory. However, Doornik (1996) finds that the following statistic derived from considerations in Rao (1973, §8c.5) provides satisfactory results in small samples, if it is used with critical values from an $F(hK^2, Ns - \frac{1}{2}K^2h + 1)$ -distribution:

$$F_{Rao}(h) = \left[\left(\frac{\det(\widetilde{\Sigma}_u)}{\det(\widetilde{\Sigma}_\varepsilon)} \right)^{1/s} - 1 \right] \frac{Ns - \frac{1}{2}K^2h + 1}{K^2h}.$$

Here

$$s = \left(\frac{K^4h^2 - 4}{K^2 + K^2h^2 - 5} \right)^{1/2}, \quad N = T - Kp - 1 - Kh - \frac{1}{2}(K - Kh + 1),$$

and $\widetilde{\Sigma}_\varepsilon$ is the residual covariance estimator from an unrestricted LS estimation of the auxiliary model $\widehat{U} = BZ + D\widehat{U} + \mathcal{E}$.

We have also applied these tests to our example data and give some results in Table 4.4.4. It turns out that neither of the tests finds strong evidence for remaining residual autocorrelation. All p -values exceed 10%. Recall that a p -value represents the probability of getting a test value greater than the observed one, if the null hypothesis is true. Therefore, even at a significance level of 10%, the null hypothesis of no residual autocorrelation cannot be rejected.

In contrast to the portmanteau tests which should be used for reasonably large h only, the LM tests are more suitable for small values of h . For large h , the degrees of freedom in the auxiliary regression model will be exhausted

Table 4.8. Autocorrelation tests for investment/income/consumption example VAR(2) model, estimation period 1960.4-1978.4

test	h	test value	approximate distribution	p -value
$\lambda_{LM}(h)$	1	6.37	$\chi^2(9)$	0.70
	2	15.52	$\chi^2(18)$	0.62
	3	32.81	$\chi^2(27)$	0.20
	4	46.60	$\chi^2(36)$	0.11
$F_{Rao}(h)$	1	0.62	$F(9, 148)$	0.78
	2	0.76	$F(18, 164)$	0.75
	3	1.14	$F(27, 161)$	0.30
	4	1.26	$F(36, 154)$	0.17

and the statistic cannot be computed in the way described in the foregoing. An LM test for higher order residual autocorrelation may be based on the auxiliary model

$$\widehat{u}_t = \nu + A_1 y_{t-1} + \cdots + A_p y_{t-p} + D_h \widehat{u}_{t-h} + \varepsilon_t$$

and on a test

$$H_0 : D_h = 0 \quad \text{versus} \quad H_1 : D_h \neq 0.$$

The relevant LM statistic can be shown to have an asymptotic $\chi^2(K^2)$ -distribution under H_0 .

4.5 Testing for Nonnormality

Normality of the underlying data generating process is needed, for instance, in setting up forecast intervals. Nonnormal residuals can also indicate more generally that the model is not a good representation of the data generation process (see Chapter 16 for models for nonnormal data). Therefore, testing this distributional assumption is desirable. We will present tests for multivariate normality of a white noise process first. In Subsection 4.5.2, it is then demonstrated that the tests remain valid if the true residuals are replaced by the residuals of an estimated VAR(p) process.

4.5.1 Tests for Nonnormality of a Vector White Noise Process

The tests developed in the following are based on the third and fourth central moments (skewness and kurtosis) of the normal distribution. If x is a univariate random variable with standard normal distribution, i.e., $x \sim \mathcal{N}(0, 1)$, its third and fourth moments are known to be $E(x^3) = 0$ and $E(x^4) = 3$. Let u_t

be a K -dimensional Gaussian white noise process with $u_t \sim \mathcal{N}(\mu_u, \Sigma_u)$ and let P be a matrix satisfying $PP' = \Sigma_u$. For example, P may be obtained by a Choleski decomposition of Σ_u . Then

$$w_t = (w_{1t}, \dots, w_{Kt})' := P^{-1}(u_t - \mu_u) \sim \mathcal{N}(0, I_K).$$

In other words, the components of w_t are independent standard normal random variables. Hence,

$$E \begin{bmatrix} w_{1t}^3 \\ \vdots \\ w_{Kt}^3 \end{bmatrix} = 0 \quad \text{and} \quad E \begin{bmatrix} w_{1t}^4 \\ \vdots \\ w_{Kt}^4 \end{bmatrix} = \begin{bmatrix} 3 \\ \vdots \\ 3 \end{bmatrix} =: \mathbf{3}_K. \quad (4.5.1)$$

This result will be utilized in checking the normality of the white noise process u_t . The idea is to compare the third and fourth moments of the transformed process with the theoretical values in (4.5.1) obtained for a Gaussian process. For the univariate case, the corresponding test is known as the *Jarque-Bera* or *Lomnicki-Jarque-Bera test* (see Jarque & Bera (1987) and Lomnicki (1961)).

For constructing the test, we assume to have observations $u_1, \dots, u_T$ and define

$$\bar{u} := \frac{1}{T} \sum_{t=1}^T u_t, \quad S_u := \frac{1}{T-1} \sum_t (u_t - \bar{u})(u_t - \bar{u})',$$

and P_s is a matrix for which $P_s P_s' = S_u$ and such that $\text{plim}(P_s - P) = 0$. Moreover,

$$v_t := (v_{1t}, \dots, v_{Kt})' = P_s^{-1}(u_t - \bar{u}), \quad t = 1, \dots, T,$$

$$b_1 := (b_{11}, \dots, b_{K1})' \quad \text{with} \quad b_{k1} = \frac{1}{T} \sum_t v_{kt}^3, \quad k = 1, \dots, K, \quad (4.5.2)$$

and

$$b_2 := (b_{12}, \dots, b_{K2})' \quad \text{with} \quad b_{k2} = \frac{1}{T} \sum_t v_{kt}^4, \quad k = 1, \dots, K. \quad (4.5.3)$$

Thus, b_1 and b_2 are estimators of the vectors in (4.5.1). In the next proposition, the asymptotic distribution of b_1 and b_2 is given.

Proposition 4.9 (*Asymptotic Distribution of Skewness and Kurtosis*)

If u_t is Gaussian white noise with nonsingular covariance matrix Σ_u and expectation μ_u , $u_t \sim \mathcal{N}(\mu_u, \Sigma_u)$, then

$$\sqrt{T} \begin{bmatrix} b_1 \\ b_2 - \mathbf{3}_K \end{bmatrix} \xrightarrow{d} \mathcal{N} \left(0, \begin{bmatrix} 6I_K & 0 \\ 0 & 24I_K \end{bmatrix} \right).$$

In other words, b_1 and b_2 are asymptotically independent and normally distributed. The proposition implies that

$$\lambda_s := Tb'_1 b_1 / 6 \xrightarrow{d} \chi^2(K) \quad (4.5.4)$$

and

$$\lambda_k := T(b_2 - \mathbf{3}_K)'(b_2 - \mathbf{3}_K)/24 \xrightarrow{d} \chi^2(K). \quad (4.5.5)$$

The first statistic can be used to test

$$H_0 : E \begin{bmatrix} w_{1t}^3 \\ \vdots \\ w_{Kt}^3 \end{bmatrix} = 0 \quad \text{against} \quad H_1 : E \begin{bmatrix} w_{1t}^3 \\ \vdots \\ w_{Kt}^3 \end{bmatrix} \neq 0 \quad (4.5.6)$$

and λ_k may be used to test

$$H_0 : E \begin{bmatrix} w_{1t}^4 \\ \vdots \\ w_{Kt}^4 \end{bmatrix} = \mathbf{3}_K \quad \text{against} \quad H_1 : E \begin{bmatrix} w_{1t}^4 \\ \vdots \\ w_{Kt}^4 \end{bmatrix} \neq \mathbf{3}_K. \quad (4.5.7)$$

Furthermore,

$$\lambda_{sk} := \lambda_s + \lambda_k \xrightarrow{d} \chi^2(2K), \quad (4.5.8)$$

which may be used for a joint test of the null hypotheses in (4.5.6) and (4.5.7).

Proof of Proposition 4.9

We state a helpful lemma first.

Lemma 4.4

Let $z_t = (z_{1t}, \dots, z_{Kt})'$ be a Gaussian white noise process with mean μ_z and covariance matrix I_K , i.e., $z_t \sim \mathcal{N}(\mu_z, I_K)$. Furthermore, let

$$\bar{z} = (\bar{z}_1, \dots, \bar{z}_K)' := \frac{1}{T} \sum_{t=1}^T z_t,$$

$$b_{1,z} \text{ a } (K \times 1) \text{ vector with } k\text{-th component } b_{k1,z} := \frac{1}{T} \sum_{t=1}^T (z_{kt} - \bar{z}_k)^3,$$

and

$$b_{2,z} \text{ a } (K \times 1) \text{ vector with } k\text{-th component } b_{k2,z} := \frac{1}{T} \sum_{t=1}^T (z_{kt} - \bar{z}_k)^4.$$

Then

$$\sqrt{T} \begin{bmatrix} b_{1,z} \\ b_{2,z} - \mathbf{3}_K \end{bmatrix} \xrightarrow{d} \mathcal{N} \left(0, \begin{bmatrix} 6I_K & 0 \\ 0 & 24I_K \end{bmatrix} \right). \quad (4.5.9)$$

■

The proof of this lemma is easily obtained, for instance, from results of Gasser (1975). Proposition 4.9 follows by noting that P_s is a consistent estimator of P (defined such that $PP' = \Sigma_u$) and by defining $z_t = P^{-1}u_t$. Hence,

$$\begin{aligned} & \sqrt{T}(P_s^{-1} \otimes P_s^{-1} \otimes P_s^{-1}) \frac{1}{T} \sum_t (u_t - \bar{u}) \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u}) \\ & \quad - \sqrt{T} \left[\frac{1}{T} \sum_t (z_t - \bar{z}) \otimes (z_t - \bar{z}) \otimes (z_t - \bar{z}) \right] \\ & = (P_s^{-1} \otimes P_s^{-1} \otimes P_s^{-1} - P^{-1} \otimes P^{-1} \otimes P^{-1}) \\ & \quad \times \frac{1}{\sqrt{T}} \sum_t (u_t - \bar{u}) \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u}) \xrightarrow{p} 0. \end{aligned}$$

An analogous result is obtained for the fourth moments. Consequently,

$$\sqrt{T} \begin{bmatrix} b_1 - b_{1,z} \\ b_2 - b_{2,z} \end{bmatrix} \xrightarrow{p} 0$$

and the proposition follows from Proposition C.2(2) of Appendix C. ■

Remark 1 In Proposition 4.9, the white noise process is not required to have zero mean. Thus, tests based on λ_s , λ_k , or λ_{sk} may be applied if the original observations are generated by a VAR(0) process. ■

Remark 2 It is known that in the univariate case tests based on the skewness and kurtosis (third and fourth moments) have small sample distributions that differ substantially from their asymptotic counterparts (see, e.g., White & MacDonald (1980), Jarque & Bera (1987) and the references given there). Therefore, tests based on λ_s , λ_k , and λ_{sk} , in conjunction with the asymptotic χ^2 -distributions in (4.5.4), (4.5.5), and (4.5.8), must be interpreted cautiously. They should be regarded as rough checks of normality only. ■

Remark 3 Tests based on λ_s , λ_k , and λ_{sk} cannot be expected to possess power against distributions having the same first four moments as the normal distribution. Thus, if higher order moment characteristics are of interest, these tests cannot be recommended. Other tests for multivariate normality are described by Mardia (1980), Baringhaus & Henze (1988), and others. ■

4.5.2 Tests for Nonnormality of a VAR Process

A stationary, stable VAR(p) process, say

$$y_t - \mu = A_1(y_{t-1} - \mu) + \cdots + A_p(y_{t-p} - \mu) + u_t, \quad (4.5.10)$$

is Gaussian (normally distributed) if and only if the white noise process u_t is Gaussian. Therefore, the normality of the y_t 's may be checked via the u_t 's. In

practice, the u_t 's are replaced by estimation residuals. In the following we will demonstrate that this is of no consequence for the *asymptotic* distributions of the λ statistics considered in the previous subsection.

The reader may wonder why normality tests are based on the residuals rather than the original observations y_t . The reason is that tests based on the latter may be less powerful than those based on the estimation residuals. For the univariate case this point was demonstrated by Lütkepohl & Schneider (1989). It is also worth recalling that the forecast errors used in the construction of forecast intervals are weighted sums of the u_t 's. Therefore, checking the normality of these quantities makes sense if the aim is to establish interval forecasts. The next result states that Proposition 4.9 remains valid if the true white noise innovations u_t are replaced by estimation residuals.

Proposition 4.10 (*Asymptotic Distribution of Residual Skewness and Kurtosis*)

Let y_t be a K -dimensional stationary, stable Gaussian VAR(p) process as in (4.5.10), where u_t is zero mean white noise with nonsingular covariance matrix Σ_u and let $\hat{A}_1, \dots, \hat{A}_p$ be consistent and asymptotically normally distributed estimators of the coefficients based on a sample $y_1, \dots, y_T$ and possibly some presample values. Define

$$\hat{u}_t := (y_t - \bar{y}) - \hat{A}_1(y_{t-1} - \bar{y}) - \dots - \hat{A}_p(y_{t-p} - \bar{y}), \quad t = 1, \dots, T,$$

$$\hat{\Sigma}_u := \frac{1}{T - Kp - 1} \sum_{t=1}^T \hat{u}_t \hat{u}_t',$$

and let $\hat{P}$ be a matrix satisfying $\hat{P}\hat{P}' = \hat{\Sigma}_u$ such that $\text{plim}(\hat{P} - P) = 0$. Furthermore, define

$$\hat{w}_t = (\hat{w}_{1t}, \dots, \hat{w}_{Kt})' := \hat{P}^{-1} \hat{u}_t,$$

$$\hat{b}_1 = (\hat{b}_{11}, \dots, \hat{b}_{K1})' \quad \text{with} \quad \hat{b}_{k1} := \frac{1}{T} \sum_{t=1}^T \hat{w}_{kt}^3, \quad k = 1, \dots, K,$$

and

$$\hat{b}_2 = (\hat{b}_{12}, \dots, \hat{b}_{K2})' \quad \text{with} \quad \hat{b}_{k2} := \frac{1}{T} \sum_{t=1}^T \hat{w}_{kt}^4, \quad k = 1, \dots, K.$$

Then

$$\sqrt{T} \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 - \mathbf{3}_K \end{bmatrix} \xrightarrow{d} \mathcal{N} \left(0, \begin{bmatrix} 6I_K & 0 \\ 0 & 24I_K \end{bmatrix} \right).$$

■

Although the proposition is formulated in terms of the mean-adjusted form (4.5.10) of the process, it also holds if estimation residuals from the standard intercept form are used instead. The parameter estimators may be unconstrained ML or LS estimators. However, the proposition does not require this. In other words, the proposition remains valid if, for instance, restricted LS or generalized LS estimators are used, as discussed in the next chapter. The following lemma will be helpful in proving Proposition 4.10.

Lemma 4.5

Under the conditions of Proposition 4.10,

$$\text{plim} \left[\frac{1}{\sqrt{T}} \sum_{t=1}^T \hat{u}_t \otimes \hat{u}_t \otimes \hat{u}_t - \frac{1}{\sqrt{T}} \sum_{t=1}^T (u_t - \bar{u}) \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u}) \right] = 0 \quad (4.5.11)$$

and

$$\begin{aligned} \text{plim} \left[\frac{1}{\sqrt{T}} \sum_{t=1}^T \hat{u}_t \otimes \hat{u}_t \otimes \hat{u}_t \otimes \hat{u}_t \right. \\ \left. - \frac{1}{\sqrt{T}} \sum_{t=1}^T (u_t - \bar{u}) \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u}) \right] = 0 \end{aligned} \quad (4.5.12)$$

■

Proof: A proof for the special case of a VAR(1) process y_t is given and the generalization is left to the reader. Also, we just show the first result. The second one follows with analogous arguments. For the special VAR(1) case,

$$\begin{aligned} \hat{u}_t &= (y_t - \bar{y}) - \hat{A}_1(y_{t-1} - \bar{y}) \\ &= (u_t - \bar{u}) + (A_1 - \hat{A}_1)(y_{t-1} - \bar{y}) + a_T, \end{aligned}$$

where $a_T = A_1(y_T - y_0)/T$. Hence,

$$\frac{1}{\sqrt{T}} \sum_t [\hat{u}_t \otimes \hat{u}_t \otimes \hat{u}_t] = \frac{1}{\sqrt{T}} \sum_t [(u_t - \bar{u}) \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u})] + d_T,$$

where d_T is a sum of expressions of the type

$$\begin{aligned} \frac{1}{\sqrt{T}} \sum_t \left[(A_1 - \hat{A}_1)(y_{t-1} - \bar{y}) + a_T \right] \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u}) \\ = \sqrt{T} \left[(A_1 - \hat{A}_1) \otimes I_{2K} \right] \frac{1}{T} \sum_t [(y_{t-1} - \bar{y}) \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u})] \\ + \sqrt{T} a_T \otimes \frac{1}{T} \sum_t [(u_t - \bar{u}) \otimes (u_t - \bar{u})], \end{aligned} \quad (4.5.13)$$

that is, d_T consists of sums of Kronecker products involving $(A_1 - \hat{A}_1)(y_{t-1} - \bar{y})$, $(u_t - \bar{u})$, and a_T . Therefore, $d_T = o_p(1)$. For instance, (4.5.13) goes to zero in probability because

$$\text{plim} \frac{1}{T} \sum_t (u_t - \bar{u}) \otimes (u_t - \bar{u}) \quad \text{exists and} \quad \sqrt{T} a_T = o_p(1)$$

so that the last term in (4.5.13) vanishes. Moreover, the elements of $\sqrt{T}(A_1 - \hat{A}_1)$ converge in distribution and

$$\text{plim} \frac{1}{T} \sum_t (y_{t-1} - \bar{y}) \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u}) = 0 \quad (4.5.14)$$

(see Problem 4.4). Hence the first term in (4.5.13) vanishes. ■

Proof of Proposition 4.10

By Proposition C.2(2) of Appendix C and Proposition 4.9, it suffices to show that

$$\begin{aligned} & (\hat{P}^{-1} \otimes \hat{P}^{-1} \otimes \hat{P}^{-1}) \frac{1}{\sqrt{T}} \sum_t \hat{u}_t \otimes \hat{u}_t \otimes \hat{u}_t \\ & - (P_s^{-1} \otimes P_s^{-1} \otimes P_s^{-1}) \frac{1}{\sqrt{T}} \sum_t (u_t - \bar{u}) \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u}) \xrightarrow{P} 0 \end{aligned} \quad (4.5.15)$$

and the fourth moments possess a similar property. The result (4.5.15) follows from Lemma 4.5 by noting that $\hat{P}$ and P_s are both consistent estimators of P and, for stochastic vectors h_T, g_T and stochastic matrices H_T, G_T with

$$\text{plim}(h_T - g_T) = 0, \quad h_T \xrightarrow{d} h,$$

and

$$\text{plim } H_T = \text{plim } G_T = H,$$

we get

$$H_T h_T - G_T g_T = (H_T - H)h_T + H(h_T - g_T) + (H - G_T)g_T \xrightarrow{P} 0. \quad \blacksquare$$

Proposition 4.10 implies that

$$\hat{\lambda}_s := T \hat{b}'_1 \hat{b}_1 / 6 \xrightarrow{d} \chi^2(K), \quad (4.5.16)$$

$$\hat{\lambda}_k := T(\hat{b}_2 - \mathbf{3}_K)'(\hat{b}_2 - \mathbf{3}_K) / 24 \xrightarrow{d} \chi^2(K), \quad (4.5.17)$$

and

$$\hat{\lambda}_{sk} := \hat{\lambda}_s + \hat{\lambda}_k \xrightarrow{d} \chi^2(2K). \quad (4.5.18)$$

Thus, all three statistics may be used for testing nonnormality.

As we have seen, the results hold for any matrix satisfying $\hat{P}\hat{P}' = \hat{\Sigma}_u$. For example, $\hat{P}$ may be a lower triangular matrix with positive diagonal obtained by a Choleski decomposition of $\hat{\Sigma}_u$. Clearly, in this case $\hat{P}$ is a consistent estimator of the corresponding matrix P (see Proposition 3.6). Doornik & Hansen (1994) point out that with this choice the test results will depend on the ordering of the variables. Therefore they suggest using a matrix based on the square root of the correlation matrix corresponding to $\hat{\Sigma}_u$ instead. In any case, the matrix $\hat{P}$ is not unique and, hence, the tests will depend to some extent on its choice. Strictly speaking, if one particular $\hat{P}$ is found for which the null hypothesis can be rejected, this result provides evidence against the normality of the process. Thus, different $\hat{P}$ matrices could be applied in principle.

For illustrative purposes we consider our standard investment/income/consumption example from Section 3.2.3. Using the least squares residuals from the VAR(2) model with intercepts and a Choleski decomposition of $\hat{\Sigma}_u$ yields

$$\hat{\lambda}_s = 3.15 \quad \text{and} \quad \hat{\lambda}_k = 4.69$$

which are both smaller than $\chi^2(3)_{.90} = 6.25$, the critical value of an asymptotic 10% level test. Also

$$\hat{\lambda}_{sk} = 7.84 < \chi^2(6)_{.90} = 10.64.$$

Thus, based on these asymptotic tests we cannot reject the null hypothesis of a Gaussian data generation process.

It was pointed out by Kilian & Demiroglu (2000) that the small sample distributions of the test statistics may differ substantially from their asymptotic approximations. Thus, the tests may not be very reliable in practice. Kilian & Demiroglu (2000) proposed bootstrap versions to alleviate the problem.

4.6 Tests for Structural Change

Time invariance or stationarity of the data generation process is an important condition that was used in deriving the properties of estimators and in computing forecasts and forecast intervals. Recall that stationarity is a property that ensures constant means, variances, and autocovariances of the process through time. As we have seen in the investment/income/consumption example, economic time series often have characteristics that do not conform with the assumption of stationarity of the underlying data generation process. For instance, economic time series often have trends or pronounced seasonal components and time varying variances. While these components can sometimes

be eliminated by simple transformations, there remains another important source of nonstationarity, namely events that cause turbulence in economic systems in particular time periods. For instance, wars usually change the economic conditions in some areas or countries markedly. Also new tax legislation may have a major impact on some economic variables. Furthermore, the oil price shocks in 1973/74 and 1979/80 are events that have caused drastic changes in some variables (notably the price for gasoline). Such events may be sources of structural change in economic systems.

Because stability and, hence, stationarity is an important assumption in our analysis, it is desirable to have tools for checking this assumed property of the data generation process. In this section, we consider two types of tests that can be used for this purpose. The first set of tests checks whether a change in the parameters has occurred at some point in time by comparing the estimated parameters before and after the possible break date. These tests are known as *Chow tests*. The second set of tests is based on comparing forecasts with actually observed values. More precisely, forecasts are made prior to a period of possible structural change and are compared to the values actually observed during that period. The stability or stationarity hypothesis is rejected if the forecasts differ too much from the actually observed values. These tests are presented in Sections 4.6.1 and 4.6.2. Other tests will be considered in later chapters.

4.6.1 Chow Tests

Suppose a change in the parameters of the $\text{VAR}(p)$ process (4.1.1) is suspected after period $T_1 < T$. Given a sample $y_1, \dots, y_T$ plus the required presample values, the model can be set up as follows for estimation purposes:

$$[Y_{(1)} : Y_{(2)}] = [B_1 : B_2] \mathbf{Z} + [U_{(1)} : U_{(2)}] = \mathbf{BZ} + U,$$

where $Y_{(1)} := [y_1, \dots, y_{T_1}]$, $Y_{(2)} := [y_{T_1+1}, \dots, y_T]$, U is partitioned accordingly, $B_1 := [\nu_1, A_{11}, \dots, A_{p1}]$ and $B_2 := [\nu_2, A_{12}, \dots, A_{p2}]$ are the $(K \times (pK + 1))$ dimensional parameter matrices associated with the first ($t = 1, \dots, T_1$) and last ($t = T_1 + 1, \dots, T$) subperiods, respectively, $\mathbf{B} := [B_1 : B_2]$ is $(K \times 2(Kp + 1))$ dimensional and

$$\mathbf{Z} := \begin{bmatrix} Z_{(1)} & 0 \\ 0 & Z_{(2)} \end{bmatrix}.$$

Here $Z_{(1)} := [Z_0, \dots, Z_{T_1-1}]$ and $Z_{(2)} := [Z_{T_1}, \dots, Z_{T-1}]$ with $Z'_t := (1, y'_t, \dots, y'_{t-p+1})$, as usual.

In this model setup, a test for parameter constancy checks

$$H_0 : B_1 = B_2 \text{ or } [I : -I] \text{vec}(\mathbf{B}) = 0 \quad \text{versus} \quad H_1 : B_1 \neq B_2.$$

Clearly, this is just a linear hypothesis which can be handled easily within our LS or ML framework under standard assumptions. For example, the LS estimator of $\mathbf{B}$ is

$$\begin{aligned}\hat{\mathbf{B}} &= [Y_{(1)} : Y_{(2)}] \mathbf{Z}'(\mathbf{Z}\mathbf{Z}')^{-1} \\ &= \left[Y_{(1)} Z'_{(1)} (Z_{(1)} Z'_{(1)})^{-1} : Y_{(2)} Z'_{(2)} (Z_{(2)} Z'_{(2)})^{-1} \right].\end{aligned}$$

It has an asymptotic normal distribution under the assumptions of Proposition 3.1. To appeal to that proposition it has to be ensured, however, that $T^{-1}\mathbf{Z}\mathbf{Z}'$ converges in probability to a nonsingular matrix. In other words,

$$\text{plim} \frac{T_i}{T} \frac{1}{T_i} Z_{(i)} Z'_{(i)}, \quad i = 1, 2,$$

has to exist and be nonsingular. Hence, T_i/T must not go to zero when T goes to ∞ , so that both subperiods before and after the break must be assumed to increase with T . If the assumptions for asymptotic normality can be justified, a Wald test can, for example, be used to test the stability hypothesis. Alternatively, an LR or quasi LR test may be applied. This type of test is often given the label *Chow test* in the literature.

There are some practical matters in applying these tests in the present context that are worth noting. If the possible break date is very close to the sample beginning or the sample end, the LS/ML estimators of B_i may not be available due to lack of degrees of freedom. While at the sample beginning one may be ready to delete a few observations to eliminate the structural break, this option is often undesirable at the end of the sample. For example, if forecasting is the objective of the analysis, a break towards the end of the sample would clearly be problematic. Therefore, the so-called *Chow forecast tests* have been proposed which also work for break dates close to the sample end. In the next subsection, we present a slightly different set of forecast tests which may be applied instead.

Even if the suspected break point is well inside the sample period so that the application of the standard Chow test is unproblematic in principle, in practice, the break may not occur in one period. If there is a longer time phase in which a parameter shift to a new level takes place, it may be useful to eliminate a few observations around the break date and use only the remaining ones in estimating the parameters. One may also argue that using some observations from periods up to T_1 in $Z_{(2)}$ may be problematic and may result in reduced power because observations from both subperiods are mixed in estimating B_2 . Under the null hypothesis of parameter constancy, this should be no problem, however, because, under H_0 , the same process is in operation before and after T_1 . Still, from the point of view of maximizing power, deleting some observations around the possible break point may be a good idea.

Other practical problems may result from multiple structural breaks within the sample period. In principle, it is no problem to test multiple break points simultaneously. Also, to improve power, one may only test some of the parameters or one may wish to test for a changing white noise covariance matrix which is implicitly assumed to be time invariant in the foregoing discussion. Details of such extensions will be discussed in Chapter 17.

So far we have considered asymptotic results only. Unfortunately, it was found by Candelon & Lütkepohl (2001) that asymptotic theory may be an extremely poor guide for the small sample properties of Chow tests, in particular, if models with many parameters are under consideration. To improve the reliability of the tests, these authors proposed to use bootstrapped versions. Bootstrapped p -values may be obtained as described in Appendix D.3.

For the German investment/income/consumption data we have fitted a VAR(2) model to data up to 1982.4 and we have performed a Chow test for a break in period 1979.1. The test value is 30.5. Comparing that to 29.6, the 10% critical value of a $\chi^2(21)$ distribution, stability is rejected at the 10% level. A bootstrapped p -value based on 2000 bootstrap replications turns out to be 0.21, however. Thus, based on the bootstrapped test, stability is not rejected. It is typical for the test based on the asymptotic χ^2 -distribution that it rejects more often in small samples than the specified nominal significance level, even if the model is stable. This distortion is at least partly corrected by the bootstrap.

4.6.2 Forecast Tests for Structural Change

A Test Statistic Based on one Forecast Period

Suppose y_t is a K -dimensional stationary, stable *Gaussian* VAR(p) process as in (4.1.1). The optimal h -step forecast at time T is denoted by $y_T(h)$ and the corresponding forecast error is

$$e_T(h) := y_{T+h} - y_T(h) = \sum_{i=0}^{h-1} \Phi_i u_{T+h-i} = [\Phi_{h-1} : \cdots : \Phi_1 : I_K] \mathbf{u}_{T,h} \quad (4.6.1)$$

where $\mathbf{u}_{T,h} := (u'_{T+1}, \dots, u'_{T+h})'$, the Φ_i are the coefficient matrices of the canonical MA representation (see Section 2.2.2). Because $\mathbf{u}_{T,h} \sim \mathcal{N}(0, I_h \otimes \Sigma_u)$, the forecast error is a linear transformation of a multivariate normal distribution and, consequently (see Appendix B),

$$e_T(h) \sim \mathcal{N}(0, \Sigma_y(h)), \quad (4.6.2)$$

where

$$\Sigma_y(h) = \sum_{i=0}^{h-1} \Phi_i \Sigma_u \Phi_i'$$

is the forecast MSE matrix (see (2.2.11)). Hence,

$$\tau_h := e_T(h)' \Sigma_y(h)^{-1} e_T(h) \sim \chi^2(K) \quad (4.6.3)$$

by Proposition B.3 of Appendix B.

This derivation assumes that y_{T+h} is generated by the same VAR(p) process that has generated the y_t for $t \leq T$. If this process does not prevail in period $T+h$, the statistic τ_h will, in general, not have a central χ^2 -distribution. Hence, τ_h may be used to test the null hypothesis

H_0 : (4.6.2) is true, that is, y_{T+h} is generated by the same Gaussian VAR(p) process that has generated $y_1, \dots, y_T$.

The alternative hypothesis is that y_{T+h} is not generated by the same process as $y_1, \dots, y_T$. The null hypothesis is rejected if the forecast errors are large so that τ_h exceeds a prespecified critical value from the $\chi^2(K)$ -distribution. Such a test may be performed for $h = 1, 2, \dots$

It may be worth noting that in these tests we also check the normality assumption for y_t . Even if the same process has generated y_{T+h} and $y_1, \dots, y_T$, (4.6.2) will not hold if that process is not Gaussian. Thus, the normality assumption for y_t is part of H_0 . Other possible deviations from the null hypothesis include changes in the mean and changes in the variance of the process.

In practice, the tests are not feasible in their present form because τ_h involves unknown quantities. The forecast errors $e_T(h)$ and the MSE matrix $\Sigma_y(h)$ are both unknown and must be replaced by estimators. For the forecast errors, we use

$$\hat{e}_T(h) := y_{T+h} - \hat{y}_t(h) = \sum_{i=0}^{h-1} \hat{\Phi}_i \hat{u}_{T+h-i}, \quad (4.6.4)$$

where the $\hat{\Phi}_i$ are obtained from the coefficient estimators $\hat{A}_i$ in the usual way (see Section 3.5.2) and

$$\hat{u}_t := y_t - \hat{\nu} - \hat{A}_1 y_{t-1} - \dots - \hat{A}_p y_{t-p}.$$

The MSE matrix may be estimated by

$$\hat{\Sigma}_y(h) := \sum_{i=0}^{h-1} \hat{\Phi}_i \hat{\Sigma}_u \hat{\Phi}_i', \quad (4.6.5)$$

where $\hat{\Sigma}_u$ is the LS estimator of Σ_u . As usual, we use only data up to period T for estimation and not the data from the forecast period. If the conditions for consistency of the estimators are satisfied, that is,

$$\text{plim } \hat{\nu} = \nu, \quad \text{plim } \hat{A}_i = A_i, i = 1, \dots, p, \quad \text{and} \quad \text{plim } \hat{\Sigma}_u = \Sigma_u,$$

then $\text{plim } \hat{\Phi}_i = \Phi_i$, $\text{plim } \hat{\Sigma}_y(h) = \Sigma_y(h)$ and

$$\begin{aligned} \text{plim}(\hat{u}_t - u_t) &= \text{plim}(\nu - \hat{\nu}) + \text{plim}(A_1 - \hat{A}_1)y_{t-1} + \dots \\ &\quad + \text{plim}(A_p - \hat{A}_p)y_{t-p} \\ &= 0. \end{aligned}$$

Hence, defining

$$\hat{\tau}_h := \hat{e}_T(h)' \hat{\Sigma}_y(h) \hat{e}_T(h),$$

we get $\text{plim } (\hat{\tau}_h - \tau_h) = 0$ and, thus, by Proposition C.2(2) of Appendix C,

$$\hat{\tau}_h \xrightarrow{d} \chi^2(K). \quad (4.6.6)$$

In other words, if the unknown coefficients are replaced by consistent estimators, the resulting test statistics $\hat{\tau}_h$ have the same *asymptotic* distributions as the τ_h .

Of course, it is desirable to know whether the $\chi^2(K)$ -distribution is a good approximation to the distribution of $\hat{\tau}_h$ in small samples. This, however, is not likely because in Section 3.5.1,

$$\Sigma_{\hat{y}}(h) = \Sigma_y(h) + \frac{1}{T} \Omega(h) \quad (4.6.7)$$

was found to be a better approximation to the MSE matrix than $\Sigma_y(h)$, if the forecasts are based on an estimated process. While asymptotically, as $T \rightarrow \infty$, the term $\Omega(h)/T$ vanishes, it seems plausible to include this term in small samples. For univariate processes, it was confirmed in a simulation study by Lütkepohl (1988b) that inclusion of the term results in a better agreement between the small sample and asymptotic distributions. For multivariate vector processes, the simulation results of Section 3.5.4 point in the same direction. Thus, in small samples a statistic of the type

$$\hat{e}_T(h)' \hat{\Sigma}_{\hat{y}}(h)^{-1} \hat{e}_T(h)$$

is more plausible than $\hat{\tau}_h$. Here $\hat{\Sigma}_{\hat{y}}(h)$ is the estimator given in Section 3.5.2. In addition to this adjustment, it is useful to adjust the statistic for using an estimated rather than known forecast error covariance matrix. Such an adjustment is often done by dividing by the degrees of freedom and using the statistic in conjunction with critical values from an F -distribution. That is, we may use

$$\bar{\tau}_h := \hat{e}_T(h)' \hat{\Sigma}_{\hat{y}}(h)^{-1} \hat{e}_T(h) / K \approx F(K, T - Kp - 1). \quad (4.6.8)$$

The approximate F -distribution follows from Proposition C.3(2) of Appendix C and the denominator degrees of freedom are chosen by analogy with a result due to Hotelling (e.g., Anderson (1984)). Other choices are possible. Proposition C.3(2) requires, however, that the denominator degrees of freedom go to infinity with the sample size T .

A Test Based on Several Forecast Periods

Another set of stationarity tests is obtained by observing that the errors of forecasts 1- to h -steps ahead are also jointly normally distributed under the null hypothesis of structural stability,

$$\mathbf{e}_T(h) := \begin{bmatrix} e_T(1) \\ \vdots \\ e_T(h) \end{bmatrix} = \mathbf{\Phi}_h \mathbf{u}_{T,h} \sim \mathcal{N}(0, \mathbf{\Sigma}_y(h)), \quad (4.6.9)$$

where

$$\mathbf{\Phi}_h := \begin{bmatrix} I_K & 0 & \dots & 0 \\ \Phi_1 & I_K & & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \Phi_{h-1} & \Phi_{h-2} & \dots & I_K \end{bmatrix} \quad (4.6.10)$$

so that

$$\mathbf{\Sigma}_y(h) := \mathbf{\Phi}_h (I_h \otimes \mathbf{\Sigma}_u) \mathbf{\Phi}_h'. \quad (4.6.11)$$

Using again Proposition B.3 of Appendix B,

$$\begin{aligned} \lambda_h &:= \mathbf{e}_T(h)' \mathbf{\Sigma}_y(h)^{-1} \mathbf{e}_T(h) = \mathbf{u}_{T,h}' (I_h \otimes \mathbf{\Sigma}_u^{-1}) \mathbf{u}_{T,h} \\ &= \sum_{i=1}^h u_{T+i}' \mathbf{\Sigma}_u^{-1} u_{T+i} = \lambda_{h-1} + u_{T+h}' \mathbf{\Sigma}_u^{-1} u_{T+h} \sim \chi^2(hK). \end{aligned} \quad (4.6.12)$$

Thus, λ_h may be used to check whether a structural change has occurred during the periods $T+1, \dots, T+h$.

To make this test feasible, it is necessary to replace unknown quantities by estimators just as in the case of the τ -tests. Denoting the test statistics based on estimated VAR processes by $\hat{\lambda}_h$,

$$\hat{\lambda}_h \xrightarrow{d} \chi^2(hK) \quad (4.6.13)$$

follows with the same arguments used for $\hat{\tau}_h$, provided consistent parameter estimators are used.

Again it seems plausible to make small sample adjustments to the statistics to take into account the fact that estimated quantities are used. The last expression in (4.6.12) suggests that a closer look at the terms

$$u_{T+i}' \mathbf{\Sigma}_u^{-1} u_{T+i} \quad (4.6.14)$$

is useful in searching for a small sample adjustment. This expression involves the 1-step ahead forecast errors $u_{T+i} = y_{T+i} - y_{T+i-1}(1)$. If estimated coefficients are used in the 1-step ahead forecast, the MSE or forecast error covariance matrix is approximately inflated by a factor $(T + Kp + 1)/T$ (see (3.5.13)). Because λ_h is the sum of terms of the form (4.6.14), it may be suitable to replace $\mathbf{\Sigma}_u$ by $(T + Kp + 1)\hat{\mathbf{\Sigma}}_u/T$ when estimated quantities are used. Note, however, that such an adjustment ignores possible dependencies between the estimated $\hat{u}_{T+i}$ and $\hat{u}_{T+j}$. Nevertheless, it leads to a computationally extremely simple form and was therefore proposed in the literature

(Lütkepohl (1989b)). Furthermore, it was suggested to divide by the degrees of freedom of the asymptotic χ^2 -distribution and, by appeal to Proposition C.3(2) of Appendix C, use the resulting statistic, $\bar{\lambda}_h$ say, in conjunction with critical values from F -distributions to adjust for the fact that Σ_u is replaced by an estimator. In other words,

$$\bar{\lambda}_h := T \sum_{i=1}^h \hat{u}'_{T+i} \hat{\Sigma}_u^{-1} \hat{u}_{T+i} / [(T+Kp+1)Kh] \approx F(Kh, T-Kp-1). \quad (4.6.15)$$

The denominator degrees of freedom are chosen by the same arguments used in (4.6.8). Obviously, $\bar{\lambda}_1 = \bar{\tau}_1$.

Now we have different sets of stationarity tests and the question arises which ones to use in practice. To answer this question, it would be useful to know the power characteristics of the tests because it is desirable to use the most powerful test available. For some alternatives the τ - and λ -statistics have noncentral χ^2 -distributions (Lütkepohl (1988b, 1989)). In these cases it is possible to investigate and compare their powers. It turns out that for some alternatives the τ -tests are more powerful than the λ -tests and for other alternatives the opposite is true. Because we usually do not know the exact form of the alternative (the exact form of the structural change) it may be a good idea to apply both tests in practice. In addition, a Chow test may be used.

An Example

To illustrate the use of the two tests for stationarity, we use the first differences of logarithms of the West German investment, income, and consumption data and test for a possible structural change caused by the oil price shocks in 1973/74 and 1979/80. Because the first drastic price increase occurred in late 1973, we have estimated a VAR(2) model using the sample period 1960.4–1973.2 and presample values from 1960.2 and 1960.3. Thus $T = 51$. It is important to note that the data from the forecast period are not used for estimation. We have used the estimated process to compute the $\bar{\tau}_h$ and $\bar{\lambda}_h$ for $h = 1, \dots, 8$. The results are given in Table 4.9 together with the p -values of the tests. The p -value is the probability that the test statistic assumes a value greater than the observed test value, if the null hypothesis is true. Thus, p -values smaller than .10 or .05 would be of concern. Obviously, in this case none of the test values is statistically significant at the 10% level. Thus, the tests do not give rise to concern about the stationarity of the underlying data generation process during the period in question. Although we have given the $\bar{\tau}_h$ and $\bar{\lambda}_h$ values for various forecast horizons h in Table 4.9, we emphasize that the tests are not independent for different h . Thus, the evidence from the set of tests should not lead to overrating the confidence we may have in this result.

Table 4.9. Stability tests for the investment/income/consumption system for 1973–1975

quarter	forecast horizon	$\bar{\tau}_h$	p -value	$\bar{\lambda}_h$	p -value
	h				
1973.3	1	.872	.46	.872	.46
4	2	.271	.85	.717	.64
1974.1	3	.206	.89	.517	.85
2	4	.836	.48	.627	.81
3	5	.581	.63	.785	.69
4	6	.172	.91	.832	.65
1975.1	7	.126	.94	.863	.63
2	8	1.450	.24	1.041	.44

To check the possibility of a structural instability due to the 1979/80 oil price increases, we used the VAR(2) model of Section 3.2.3 which is based on data up to the fourth quarter of 1978. The resulting values of the test statistics for $h = 1, \dots, 8$ are presented in Table 4.10. Again none of the values is significant at the 10% level. However, in Section 3.5.2, we found that the observed consumption values in 1979 fall outside a 95% forecast interval. Hence, looking at the three series individually, a possible nonstationarity would be detected by a prediction test. This possible instability in 1979 was a reason for using only data up to 1978 in the examples of previous chapters and sections. The example indicates what can also be demonstrated theoretically, namely that the power of a test based on joint forecasts of various variables may be lower than the power of a test based on forecasts for individual variables (see Lütkepohl (1989b)).

4.7 Exercises

4.7.1 Algebraic Problems

Problem 4.1

Show that the restricted ML estimator $\tilde{\beta}_r$ can be written in the form (4.2.10).

Problem 4.2

Prove Lemma 4.1.

[Hint: Suppose $k < n$. Then

Table 4.10. Stability tests for the investment/income/consumption system for 1979–1980

quarter	forecast horizon		$\bar{\tau}_h$	p -value	$\bar{\lambda}_h$	p -value
	h					
1979.1	1		.277	.84	.277	.84
2	2		2.003	.12	1.077	.38
3	3		2.045	.12	1.464	.18
4	4		.203	.89	1.245	.27
1980.1	5		.630	.60	1.339	.20
2	6		1.898	.86	1.374	.17
3	7		.188	.90	1.204	.28
4	8		.535	.66	1.124	.34

$$\begin{aligned} c_n + a_n &= c_n + (a_n - a_{n-1}) + \cdots + (a_{k+1} - a_k) + a_k \\ &> c_n + (b_n - b_{n-1}) + \cdots + (b_{k+1} - b_k) + a_k \\ &\geq c_k + b_k - b_k + a_k \\ &= c_k + a_k \end{aligned}$$

which contradicts (4.3.13b).^{2]}

Problem 4.3

Show (4.4.14) and (4.4.15).

[Hint:

$$\begin{aligned} &\text{vec}(\sqrt{T}UF[I_h \otimes Z'(\hat{B} - B)']/T) \\ &= \sqrt{T} \text{vec} \left[\frac{1}{T}UF(I_h \otimes Z')(I_h \otimes (\hat{B} - B)') \right] \\ &= \left[I_K \otimes \frac{1}{T}UF(I_h \otimes Z') \right] \sqrt{T} \text{vec}(I_h \otimes (\hat{B} - B)') \end{aligned}$$

and

$$\begin{aligned} &\sqrt{T} \text{vec} \left(\frac{1}{T}(\hat{B} - B)ZF \left[I_h \otimes Z'(\hat{B} - B)' \right] \right) \\ &= \left\{ \left[I_h \otimes (\hat{B} - B) \right] \frac{(I_h \otimes Z)F'Z'}{T} \otimes I_K \right\} \sqrt{T} \text{vec}(\hat{B} - B). \end{aligned}$$

Problem 4.4

Show (4.5.14).

² I thank Prof. K. Schürger, Universität Bonn, for pointing out this proof.

[Hint: Note that

$$\begin{aligned} (y_{t-1} - \bar{y}) \otimes (u_t - \bar{u}) \otimes (u_t - \bar{u}) &= (y_{t-1} - \mu) \otimes u_t \otimes u_t \\ &\quad - (y_{t-1} - \mu) \otimes u_t \otimes \bar{u} + \cdots, \end{aligned}$$

define new variables of the type

$$z_t = (y_{t-1} - \mu) \otimes u_t \otimes u_t$$

and use that

$$\text{plim} \frac{1}{T} \sum_t z_t = E(z_t) = 0.]$$

Problem 4.5

Using the notation and assumptions from Proposition 4.1, show that

$$\frac{\partial \ln l(\tilde{\beta}_r)}{\partial \beta'} \frac{\partial^2 \ln l(\tilde{\beta}_r)}{\partial \beta \partial \beta'} \frac{\partial \ln l(\tilde{\beta}_r)}{\partial \beta} = (\tilde{\beta}_r - \tilde{\beta})' (ZZ' \otimes (\tilde{\Sigma}_u^r)^{-1}) (\tilde{\beta}_r - \tilde{\beta}).$$

4.7.2 Numerical Problems

The following problems require the use of a computer. They refer to the bivariate series $y_t = (y_{1t}, y_{2t})'$ of first differences of the U.S. investment data in File E2, available from the author's webpage.

Problem 4.6

Set up a sequence of tests for the correct VAR order of the data generating process using a maximum order of $M = 4$. Compute the required χ^2 and F likelihood ratio statistics. Which order would you choose?

Problem 4.7

Determine VAR order estimates on the basis of the four criteria FPE, AIC, HQ, and SC. Use a maximum VAR order of $M = 4$ in a first estimation round and $M = 8$ in a second estimation round. Compare the results.

Problem 4.8

Compute the residual autocorrelations $\hat{R}_1, \dots, \hat{R}_{12}$ and estimate their standard errors using the VAR(1) model obtained in Problem 3.12. Interpret your results.

Problem 4.9

Compute LM test values $\lambda_{LM}(1)$, $\lambda_{LM}(2)$, and $\lambda_{LM}(4)$ and portmanteau test values Q_h and $\bar{Q}_h$ for $h = 10$ and 12 for the VAR(1) model of the previous problem. Test the whiteness of the residuals.

Problem 4.10

On the basis of a VAR(1) model, perform a test for nonnormality of the example data.

Problem 4.11

Investigate whether there was a structural change in U.S. investment after 1965 (possibly due to the increasing U.S. engagement in Vietnam).